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Welcome to DPH 2026, the 11th International Conference on Digital Public Health!

Since launching in London in 2008, DPH has travelled everywhere from Istanbul to Seoul, Florence to Madeira, building a global community of researchers, policymakers, clinicians, and innovators united by one mission: advancing digital public health. DPH 2026 continues that momentum, hosted by the UCL Centre for Digital Public Health in Emergencies (dPHE), which is now officially a WHO Collaborating Centre for Digital Public Health and Pandemic Preparedness!

Now, more than ever, the digital public health community must lead the conversation on how AI, data, and innovation are harnessed responsibly to advance health equity worldwide.

This year's programme spans three plenary sessions, oral and poster presentations, workshops, tutorials, innovation sessions, and panels. The plenaries tackle European sovereignty in health data governance, health systems resilience and One Health approaches six years after COVID-19, and the frontiers

of predictive modelling, IoT, and digital tools for population health. Keynote speakers include Prof. Esteve Fernandez (Secretary for Public Health, Catalan Health Department), Jacobo Mendioroz (Public Health Agency of Catalonia), Jordi Piera Jiménez (openEHR International), Aron Aregay (WHO Europe), and Prof. Breda Smyth (RCSI), among other amazing practitioners in the field.

Innovation and inclusion are at the heart of DPH. The Digital Health Innovation Awards return alongside dedicated Innovation Sessions connecting start-ups, SMEs, and researchers, along with curated 1:1 business matchmaking. The student day is taking place for ECRs to jumpstart their networks, while LMIC's and students are brought to the forefront of the agenda, programming, and funding.

All accepted full papers will be submitted for inclusion in the IEEE Xplore Digital Library, with extended abstracts published in Frontiers in Digital Public Health.

We sincerely thank the DPH team, authors, keynote speakers, committee members, session chairs, and partners for making this edition possible, with special thanks to DPH 2026 Managers Ava Sullivan and Iva Pencheva for their outstanding work in bringing it all together.

So happy to be here with you in Barcelona,

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The DPH2026 Conference Partners



Main track

A protocol for developing and evaluating inclusive multi-color digital interface palettes for low vision and color vision deficiency users

Author

Palak Katiyar, Suraj Senjam, Sandeep Singh – National Center for Assistive Health Technology, AIIMS New Delhi

Citation

Katiyar, P., Senjam, S., Singh, S. A protocol for developing and evaluating inclusive multi-color digital interface palettes for low vision and color vision deficiency users.

Abstract

As health systems continue to adopt complex digital user interfaces, current accessibility solutions reduce semantic clarity for people with vision impairments, adding a barrier to equitable healthcare access. Current solutions use two-tone, high-contrast filters that meet basic standards but may not accommodate for dashboards and data visualizations. This study develops eight multi-color palettes designed in OKLCH color space and proposes a protocol for their systematic evaluation against existing filters across the low vision and color vision deficiency spectrums. The protocol outlines a within-subjects comparative study measuring task accuracy, completion time, and user preference. Expected outcomes include

personalisation guidelines and design frameworks enabling equitable access to complex digital health interfaces for diverse visual capabilities.

Background

As systems, including healthcare systems, increasingly adopt digital technologies, they create barriers to equitable access for people with visual impairments [1]. Recent reviews demonstrate that older adults with chronic diseases face substantial digital health adoption barriers, including limited digital literacy, infrastructural deficits, and usability challenges, with visual accessibility remaining critically under-addressed. These changes further impact 2.2 billion people globally who have near or distance vision impairment [2], with 246 million experiencing low vision and Color vision deficiency (CVD) affecting approximately 2.59% of the global population.

The Web Content Accessibility Guidelines (WCAG) [3] establish minimum contrast ratios of

4.5:1 for normal text and 3:1 for user interface components such as icons, yet these standards were designed for simpler text-on-background scenarios. Modern digital health applications feature complex visualizations with multiple interactive elements, data dashboards, graphs, and semantic color coding that extend beyond binary color relationships. Current accommodation strategies primarily rely on two tone high-contrast filters [4] such as yellow-black, yellow-blue, and white-black combinations. Although these binary approaches ensure distinguishability through color simplification, they compromise the semantic information density essential to health applications.

Purpose

This study systematically designs multi-color palettes based on industry standards for modern interfaces, consisting of; background color, primary text color, secondary text color, highlight color, semantic (CTA) color. This will be followed with evaluation against existing two-tone filters to determine

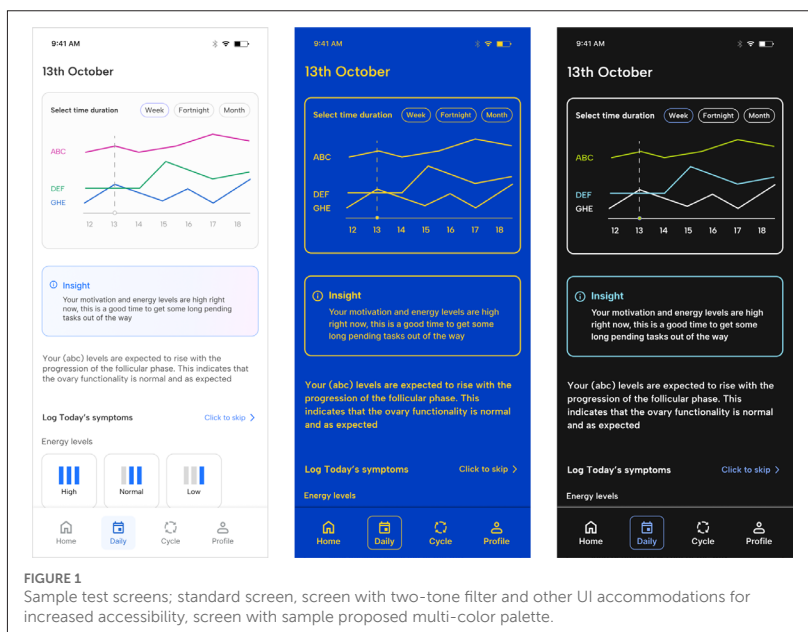
the effectiveness of maintaining semantic information in complex digital interfaces for users with low vision and CVD. The secondary objective is to characterize relationships between specific visual conditions [5], including acuity levels, light sensitivity, CVD type and severity, and individual contrast preferences with palette performance, enabling evidence-based personalization of digital interfaces for the low-vision and CVD spectrum.

Methods

Eight palette variations have been developed using the OKLCH color space [6], which enables systematic construction of accessible color schemes with increasingly uniform perception across hues. The design framework consists of two foundational color schemes, each with four preference variations accommodating light-mode versus dark-mode and high-contrast versus low-contrast preferences across the low vision spectrum. Each palette contains five distinct functional colors serving specific interface roles: call-to-action elements, highlights, backgrounds, primary text, and secondary text, duplicating existing industry standards for interfaces. All palette colors maintain WCAG AA compliance with minimum 4.5:1 text contrast and 3:1 UI component contrast. Each palette has been validated through CVD simulations for protanopia, deuteranopia, and tritanopia using the model established by Machado et al. (2009) [7] to ensure optimal color separation and distinguishability across all deficiency types.

Planned Evaluation

A mixed-methods comparative evaluation with participants representing both the low vision spectrum and CVD types. The within-subjects study design will compare task performance across our eight multi-color palettes, three established two-tone filters, and standard screens for baseline conditions. (Fig. 1) Participants will be stratified by visual acuity levels, CVD type, light sensitivity, age groups including older adults managing chronic conditions, and prior digital health interface experience.



Participants will complete standardised tasks using prototype digital interfaces representing real-world scenarios including dashboard comprehension of multi-variable metrics, data visualisation tasks requiring interpretation of graphs, interactive task completion navigating color-coded interface elements, and information retention assessments. Primary outcomes include task completion accuracy, task completion time, and error rates. Secondary outcomes include subjective preference ratings, perceived visual comfort, semantic information retention scores, and qualitative feedback on palette utility. Mixed-effects models will account for repeated measures, with participant characteristics as co-variables. Our analysis will identify which palette variations optimise accessibility for specific user profiles, establishing evidence-based guidelines for accommodating 95% of the target population.

Expected Contribution

This protocol establishes a methodology for building and evaluating color palette systems that preserve semantic complexity in digital health interfaces while maintaining accessibility. Unlike binary accommodation strategies that reduce interface functionality, our multi-color approach addresses the critical requirement for information dense interfaces to remain both accessible and functionally complete. Expected contributions include an evidence-based design framework with systematic palette construction principles for accessible digital interfaces, personalisation guidelines [8] mapping visual condition profiles to optimal palette configurations, clinical applicability assessment determining whether multi-color palettes can support complex information visualisation without compromising accessibility, and generalisability assessment establishing whether a finite set of palette variations can accommodate the vast majority of users with low vision and CVD.

As healthcare delivery increasingly relies on digital interfaces for disease management, tele-health, and patient engagement, it becomes necessary to ensure that these tools serve all users regardless of visual capability. Our findings will inform design standards for accessible digital health technologies, supporting equitable access to healthcare for aging populations and individuals with visual impairments.

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A streaming “pancreas twin”: Online regression and behavioral clustering from CGM data

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Citation

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Abstract

Digital public health deployments increasingly rely on remote monitoring of chronic diseases at scale. When a control center must continuously supervise thousands of patients, batch analytics become impractical: models and statistics must be updated incrementally as data arrive, while keeping latency and compute bounded. This paper presents a stream-first “pancreas twin” pipeline combining (i) online descriptive monitoring of continuous glucose monitoring (CGM) streams, (ii) online regression for one-step-ahead CGM forecasting with safety-oriented alerts, and (iii) a two-stage clustering workflow that discovers interpretable patient behavioral profiles from insulin bolus and carbohydrate intake patterns. Using a public wearable diabetes dataset the online regression model achieves an overall mean absolute error of 3.92 mg/dL under prequential evaluation. Behavioral clustering identifies K

= 5 patient profiles (Silhouette = 0.5180, Davies–Bouldin = 0.5219), including rare/outlier patterns relevant to population-level risk stratification.

Index Terms

digital public health, remote patient monitoring, streaming analytics, online learning, continuous glucose monitoring, regression, clustering

I. Introduction

Wearable sensing and connected medical devices are shifting chronic-disease management toward continuous, remote monitoring. This transition is not merely a “more data” problem: it is a throughput and latency problem. In operational settings, a monitoring hub may need to supervise thousands of patients simultaneously, each producing high-frequency CGM measurements plus therapy-related events (boluses, carbohydrate logs). If analytics require repeatedly reloading histories and retraining models, compute and latency scale poorly, and system performance degrades.

Streaming methods address these limitations by processing each new event once, updating sufficient statistics and model parameters incrementally, and keeping memory bounded.

We propose a practical, reproducible “pancreas twin” pipeline designed for streaming data processing:

- 1) Online descriptive monitoring for real-time summaries and quality checks;
- 2) Online regression forecasting for one-step-ahead CGM prediction and alerting;
- 3) Two-stage behavioral clustering to derive patient-level profiles from therapy behaviors.

II. Data and Preprocessing

A. Data sources

We use a publicly available dataset collected from advanced diabetes technologies [1].

B. Event streams and time alignment

For each subject, CGM readings are temporally aligned with the most recent insulin bolus and carbohydrate entry within a 30-minute lookback window.

III. Methods

A. Overall streaming pipeline

For each subject, we process CGM records sequentially (time order). At each time step, we:

- 1) Update online descriptive statistics with the current CGM value.
- 2) Construct features from lagged CGM and recent event inputs.
- 3) Produce a one-step-ahead prediction and evaluate it (prequential protocol).
- 4) Trigger alerts based on the prediction.
- 5) Update the regression model with the true observation.

B. Online descriptive monitoring

We maintain incremental estimators (streaming count, mean, variance/standard deviation, min, max) using a streaming ML toolkit [2].

C. Online regression with prequential evaluation

- 1) *Feature vector:* Let g_t be CGM at time t . We predict g_t using features available before observing it:

$$x_t = [g_{t-1}, g_{t-2}, g_{t-3}, I_t, C_t]$$

where I_t is the most recent insulin bolus value (units) within W_I minutes before t , and C_t is the most recent carbohydrate entry (grams) within W_C minutes before t . We require at least three previous CGM readings to start predictions; otherwise, we only fill the lag buffer.

- 2) *Scaling*: Each feature is standardized online using streaming mean and standard deviation estimates [2].
- 3) *Model and update rule*: We use incremental linear regression updated via SGD with fixed hyperparameters [2].
- 4) *Evaluation protocol*: We adopt prequential (test-then-train) evaluation.

D. Risk alerting logic

We generate alerts from predictions to mimic an operational monitoring setting. We use thresholds aligned with CGM clinical targets commonly used to define time-in-range (70–180 mg/dL) [3]:

$$\text{hypo if } \hat{g}_t < 70 \text{ mg/dL, hyper if } \hat{g}_t > 180 \text{ mg/dL.}$$

E. Behavioral clustering (two-stage)

- 1) *Feature engineering from bolus records*: For each bolus event, we compute the ratio between actual and device-recommended carbohydrate and insulin values.
- 2) *Stage 1: within-subject streaming clustering*: For each subject, we cluster bolus events into $K_1 = 3$ modes using MiniBatchKMeans [4]:

The output is a cluster assignment for each bolus event, capturing recurring behavioral modes and potential anomalies.

- 3) *Stage 2: across-subject profiling*: We summarize each subject s into a fixed-length vector.

We standardize z_s across subjects and run KMeans for $K \in \{2, \dots, 8\}$. We choose K by maximizing Silhouette [5] and minimizing Davies–Bouldin [6], selecting $K = 5$

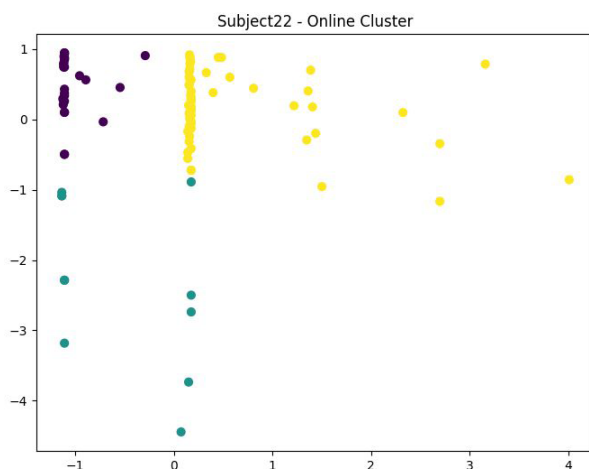


FIGURE 1

Within-subject MiniBatchKMeans clustering of bolus events using standardized carbohydrate- and insulin-ratio features for an example subject.

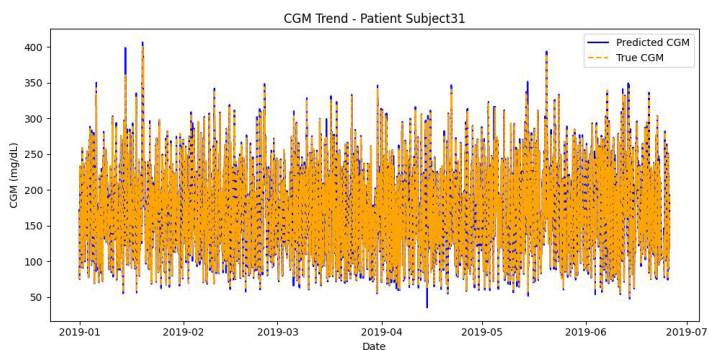


FIGURE 2

Example of one-step-ahead CGM forecasting under streaming operation: predicted vs. observed CGM over time for an example subject.

IV. Results

A. Online regression performance

Under prequential evaluation, the online regression model achieves an overall MAE of **3.92 mg/dL** on the public wearable diabetes dataset [1]. An example of predicted vs. true CGM trajectory is shown in Fig. 2.

B. Behavioral clustering quality and profiles

Across-subject profiling selects $K = 5$ clusters with:

- Silhouette Score = 0.5180 [5]
- Davies–Bouldin Score = 0.5219 [6]

These results indicate that a monitoring center can focus on a small set of dominant behavioral modes while identifying rare or outlier profiles for targeted review.

V. Discussion, Limitations and Conclusion

This work introduces a streaming “pancreas twin” pipeline that processes CGM measurements and therapy-related events incrementally, enabling bounded-memory monitoring at low latency. By combining online descriptive monitoring with one-step-ahead CGM forecasting and risk-oriented alerting, the system supports continuous safety surveillance and population-level oversight without repeated history reloads and retraining. In parallel, the two-stage behavioral clustering provides interpretable patient-level profiles from therapy behaviors, which can support stratification and targeted follow-up in remote monitoring hubs. The main limitations are that results are obtained on retrospective data and the approach depends on the quality and completeness of device-recorded events (e.g., boluses and carbohydrate logs) and on handling missing or invalid CGM values without imputation; moreover, the forecasting component relies on a simple online regression model, and alert performance and clinical impact are not validated in a prospective deployment. Future work should evaluate the pipeline in operational settings, assess alert utility and workload effects, and explore

more expressive online models and profiling features while preserving interpretability and streaming constraints.

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Adapting the garcia family personas to specific contexts of three middle income countries | the MULTIPULM experience

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Abstract

Multimorbidity presents growing challenges for health systems worldwide, particularly in low and middle income countries, where social, economic and infrastructural conditions shape care pathways in distinct ways (1).

To support the development and evaluation of innovative technologies for managing multimorbidity, this study adapts the Garcia Family personas, originally developed within the SHAFE Foundation (2), to three Middle Income Countries (MIC) contexts: Brazil, Serbia and Türkiye.

Using a participatory, human-centred design approach, we conducted structured workshops with relevant stakeholders in each country. Participants reviewed the original Garcia Family personas and collaboratively re-contextualised them, identifying locally relevant factors, socio-economic, cultural, among others. Through this iterative process, we are developing three alternative families that reflect the lived realities, priorities and care challenges specific to each country-setting.

The resulting families will nurture the development of personas that capture both shared and context-specific patterns of multimorbidity, care seeking in chronic respiratory diseases, informal caregiving and readiness for digital health.

Background

Personas are widely used to represent archetypal users and their interactions with services, technologies and environments (3); however, their effectiveness is contingent upon careful contextualisation (4). When personas are transferred across cultural and socioeconomic settings without appropriate adaptation, they risk misrepresentation and reduced validity (5). To address this challenge, the present study adapts the People–Activities–Context–Technology (PACT) methodology, previously applied in the support of multidisciplinary scenario construction in the design of telemedicine interventions (6).

The aim of this study is to undertake a systematic cultural adaptation of the SHAFE Foundation's illustrative personas, specifically the Garcia Family, to ensure their relevance and applicability within middle-income country contexts, namely Brazil, Türkiye and Serbia.

Methods

In its original formulation, the PACT framework provides a structured approach to scenario development by systematically analysing four interdependent dimensions: the characteristics and needs of the people involved, the activities they perform, the contexts in which these activities take place, and the technologies that mediate or support them. The framework emphasises collaboration across disciplines and active stakeholder involvement, using scenarios as boundary objects that facilitate shared understanding between researchers, practitioners and end-users. This study extends the PACT approach beyond its original telehealth application to support the cultural adaptation of personas, treating them and their associated scenarios as dynamic constructs that must be grounded in local socio-cultural realities.

Within the adapted framework, the **People dimension** focuses on capturing culturally specific demographic, social and behavioural characteristics of family units in Brazil, Türkiye and Serbia. This involves the collection and synthesis of qualitative and quantitative data relating to household composition, socio-economic and health status, caregiving roles and intergenerational relationships. Particular attention is paid to cultural values, norms and expectations shaping family life, ageing, health behaviours and social support structures.

The **Activities dimension** addresses the everyday practices and routines that characterise daily life within each country context. These include, but are not limited to, health-related behaviours, informal caregiving activities, social participation, use of services and interactions with digital or assistive technologies. By examining how activities are organised and prioritised within families, the study identifies culturally salient patterns of behaviour that must be reflected in the adapted Garcia Family personas.

Context is treated as a critical determinant of persona validity, encompassing sociocultural, economic and environmental conditions that influence both activities and technology use. The adaptation process situates each persona within country-specific contexts, taking into account factors such as healthcare systems, community infrastructure, levels of digital literacy, urban–rural differences and cultural attitudes towards ageing, disability and care.

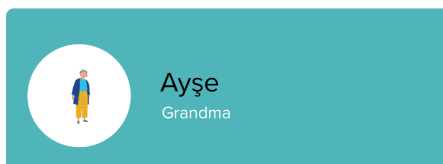
The **Technology dimension** focuses on understanding the availability, accessibility and perception of relevant digital and assistive technologies in Brazil, Türkiye and Serbia.

Rather than assuming uniform technology uptake, the adapted PACT approach examines local patterns of technology use, barriers to access and culturally informed attitudes towards innovation and digital solutions.

Results

The initial workshop round included 14 participants including 3 representatives from Brazil, 3 from Serbia and 2 from Türkiye. The online workshop was held in Teams and conveyed a plenary session for overall introduction, discussion and final conclusions, and two breakout rooms for each country co-creation workshop, supported by a collaborative online tool.

In the session, the definition of the family name (e.g. Silva family in Brazil, Yılmaz in Türkiye, Petrovich in Serbia) and the definition of the main patient characteristics were concluded. In both Brazil and Türkiye, a woman between 60 and 65 years of age was considered as the most relevant within this clinical context, while in Serbia it was an older man. It is relevant to consider that, while men have higher COPD prevalence rates, women were reported as usually being more in need of health care, due to multiple clinical and contextual reasons, and are therefore the ones representing the higher system burden.



People: Demographics and Health Profile

- What are their age, gender, family situation, education, livelihood, and where do they live (for example, urban or rural, formal or informal setting)?
- Which health conditions, risks, or vulnerabilities do they have, and are these long-term, recurring, or linked to their living or working conditions?
- In what ways do physical, mental, or social health challenges affect what they can do, the choices they make, and the support they need?

Female 65	Urban area Primary school Housewife Living with husband / or son(s) (living alone)	Diabetes Hypertension Non-smoker Long-term conditions Overweight	Peer support (family, neighbourhood)	Financially dependent on husband's salary. Free PHC
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Activities: Daily Routines and Interactions

- What are their main daily activities at home, at work, and in the community?
- Which people or groups are part of their everyday life (for example, family members, peers, community leaders, service providers), and in what situations do these interactions happen?
- When and how do they seek care, manage their health, or support others' health as part of their normal day?

House chores (laundry, cooking, shopping) Care of grandchildren and or parents (informal caregiver)	Unstructured voluntary work	Secretary (lifestyle in general)	Usually socialises at others' homes (sometimes in a park)	Elder adults are concerned about their health. Easy access to the hospital and the bus. Compliant with the doctor's appointments
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Context: Socio-cultural and environmental conditions

- What norms, beliefs, gender roles, or community expectations influence their decisions, relationships, and access to opportunities or services?
- In what ways do culture, community dynamics, and the physical environment create barriers or support for staying healthy and accessing care?

Religion: Muslim	Migration background	Overwhelmed (burden family chores)	Alternative medicine / herbal - they are open to this	
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Technology: Use of Digital and Assistive Technologies

- What tools or devices do they rely on for communication, work, learning, or accessing services (for example, mobile phones, radios, internet, or community-based technologies)?
- What role do technologies play in managing their health or accessing care (such as appointment systems, health apps, teleconsultations, digital records, or community health tools)?
- If they have functional limitations, what devices or supports help them in daily life (for example, mobility aids, hearing or vision supports), what barriers or facilitators affect their use?

Mobile phones (smartphones may be difficult for most older adults, who often seek help from family)	In Turkey they have the Digital Care Service - but it would be hard for older adults to use them without support	Glasses (not often to use mobility aids)	Main information tool: TV On health - mobile phones are recognised as main source	
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multipulm

FIGURE 1

Extract of the co-creation Mural for the Turkish persona.

AQ1

Conclusions and Future Work

This methodology emphasises an iterative scenario development and validation process that reflects the participatory principles of the original PACT framework. During March, scenario-based workshops will be conducted with interdisciplinary researchers and local stakeholders to co-construct and refine persona narratives. Feedback will be incorporated through successive iterations further to the workshops, to enhance credibility and reduce cultural bias. This process ensures that the adapted personas are not only culturally sensitive but also analytically robust and transferable within each national context.

The adapted personas provide practical tools for co-designing, testing and evaluating the innovative technologies of the MULTIPULM project (7), that will operate within diverse sociotechnical environments. By grounding innovation in local contexts while maintaining conceptual comparability across countries, this work supports a scalable and inclusive framework for designing digital solutions for multimorbidity management while enhancing empathy and understanding in the project team.

The outcome of this adapted PACT methodology will be a set of culturally grounded Garcia Family personas for Brazil, Türkiye and Serbia, accompanied by detailed scenario narratives that illustrate everyday life, care practices and technology interactions. In addition, the study demonstrates the value of culturally-adapted personas for MICs and produces a documented methodological approach that can be applied to future cross-cultural persona adaptation efforts.

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Assessing the strategic importance of health determinants in the digital era for health planning in catalonia: Results of a modified delphi study

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Citation

Cano, M., Fuster, A., Davins, M., Carrion, C. Assessing the strategic importance of health determinants in the digital era for health planning in Catalonia: results of a modified Delphi study.

Background

Digital health technologies are increasingly central to healthcare transformation, offering new opportunities to improve access, efficiency and quality of care. At the same time, a growing body of evidence shows that digitalisation can unintentionally reproduce or exacerbate existing social and health inequalities, as benefits are unevenly distributed across population groups. In response to these concerns, the concept of Digital Determinants of Health (DDoH) has emerged to refer to any factor rooted

in, contingent on or inextricably linked to the digital world that can directly or indirectly influence health or well-being, including factors that change how health care is delivered, modify existing relationships between social, political or commercial determinants and health, or create entirely new pathways through which individual or population health is shaped. Although international frameworks and global prioritisation exercises have begun to identify key DDoH, their translation into regional health planning remains limited. As a result, health systems often lack context-sensitive tools to identify which health determinants (digital, social, economical and political) are most relevant locally and how they interact with existing social, territorial and system-level inequalities.

Objective

This study aimed to (1) identify which health determinants are considered most important for health planning in the Catalan context, and (2) examine how these priorities relate to existing regional strategies and policy frameworks, with particular attention to their implications for health equity.

Methods

We conducted a modified Delphi study with experts from the Catalan health and digital ecosystem, replicating and adapting the Delphi methodology previously developed in a study published in the Bulletin of the World Health Organization. The starting list comprised 127 determinants identified in that study as emerging or evolving with digitalisation across digital, social, commercial/economic and political domains. Experts living and working in Catalonia, with experience in digital health, public health, healthcare delivery or health equity, were recruited using snowball sampling. Determinants were rated on a 5-point Likert scale over two Delphi rounds, followed by a final online consensus meeting. Median scores, interquartile ranges and the proportion of ratings ≥ 4 were calculated, and determinants were classified as having high, moderate or low relevance using the same consensus criteria applied in the original WHO study.

Results

Forty-three experts participated in the first Delphi round and 31 in the second. Overall, 20 determinants (15.7%) were classified as highly important for health planning in Catalonia. These included internet access and connectivity; device and software availability; digital literacy; data and digital capacity; interoperability; digital public infrastructure; privacy and security policy; data consent policy; access and data-sharing policy; data governance and ethics; model accuracy and algorithmic validation; transparency, accountability and ethics of artificial intelligence; health literacy; inclusive design; good practice design; and urbanicity. Although many of these determinants overlap with those prioritised in global exercises, the Catalan panel placed particular emphasis on factors linked to territorial inequalities, uneven digital maturity and the organisational complexity of the regional health system.

Conclusions

This study offers the first systematic and context-specific prioritisation of health determinants to inform strategic health planning in Catalonia, providing empirical evidence in a field that has so far been largely dominated by conceptual work. The original WHO list of 127 determinants provided an adequate and replicable foundation for this regional adaptation, though several determinants were accompanied by predominantly US-centric examples that were removed or replaced to better reflect the European and regional context. To facilitate a coherent and integrated interpretation, the results were examined through a set of analytical axes capturing key dimensions of digital transformation in health. Overall, the findings suggest that digital transformation cannot be understood simply as a technological or infrastructural endeavour, but rather as a complex, multi-level process shaped by the interaction of digital, social, commercial and political determinants.

The results underscore that even in a region with high digital health maturity and long-standing investment in public digital infrastructure, substantial equity-relevant challenges persist. Disparities arise through the interaction

of multiple structural factors operating across different levels of the health system.

- Axis 1 — Determinants of effective digital health use. In Catalonia, effective digital health use is shaped by the interaction between widespread digital service availability, uneven digital and health literacy, and marked urban–rural territorial differences. Inequalities emerge when digital services increasingly substitute in-person care in areas with fewer alternatives, transforming individual skill gaps into structural barriers for older adults and rural populations.
- Axis 2 — System-level digital capacity and infrastructure. Despite Catalonia's high level of digital maturity, interoperable systems and long-standing public digital infrastructure, limitations in organisational readiness, workforce capacity and routine integration persist. These constraints unevenly affect providers and territories, influencing whether digital investments translate into equitable and system-wide health benefits.
- Axis 3 — Governance and validation of AI-based digital health tools. In the Catalan public health system, the rapid expansion of AI-based initiatives highlights the need for governance frameworks that go beyond technical performance. Ensuring continuous validation, transparency and accountability is essential to prevent algorithmic tools from reinforcing existing territorial, social and organisational inequalities embedded in health data.
- Axis 4 — Design quality and inclusiveness of digital health solutions. In Catalonia, the growing reliance on standardised public digital health platforms underscores the role of design as a determinant of equity. Limited personalisation and insufficiently embedded inclusive design practices risk excluding users with diverse abilities, literacy levels and linguistic backgrounds, despite compliance with accessibility standards.

- Axis 5 — Data governance, privacy and access. Within Catalonia's highly interconnected public health data ecosystem, data governance arrangements shape trust, autonomy and participation. Beyond regulatory compliance, gaps in transparency and user agency over data access and use may undermine public trust in increasingly data-driven care models.

From a strategic planning standpoint, these findings suggest the need to move beyond expansion-oriented digital health strategies focused primarily on scaling technologies or services. Integrating a comprehensive perspective from health determinants into health planning can support a shift towards more reflexive, anticipatory and equity-sensitive governance of digital transformation. Lastly, the study also highlights the importance of contextualising international frameworks to regional and health specific settings.

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Behavioral drivers of antimicrobial prescribing among primary health care prescribers in imo state, Nigeria: Implications for value-based stewardship and digital health readiness

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Overview

Antimicrobial resistance (AMR) is a serious health threat all around the world, particularly in low- and middle-income countries where prescribing decisions are frequently made in the face of uncertainty, limited diagnostic options, and competing clinical and social pressures. While traditional antimicrobial stewardship initiatives have concentrated on enhancing education, new evidence reveals that prescribing behavior is influenced by a complex interplay of attitudes, contextual restrictions, and value-based decision-making processes.

This study examines the behavioral drivers of antimicrobial prescribing among primary health care (PHC) prescribers in Imo State, Nigeria, by assessing knowledge, attitudes, and practices (KAP) alongside key indicators of digital health readiness. It further explores how these factors influence prescribing behaviour and the implications for value-based antimicrobial stewardship and the effective integration of digital decision-support systems.

Background

AMR is responsible for an estimated 4.95 million deaths globally each year, with 1.27 million deaths directly attributable to drug-resistant infections. Nigeria bears a significant burden, ranking among the top countries in AMR-related mortality, with approximately 64,500 attributable and 263,400 associated deaths reported in 2019. Beyond health impacts, AMR poses a substantial economic threat, with projections suggesting potential losses of 5–7% of Nigeria's gross domestic product by 2050.

At the primary healthcare level, where the majority of antibiotics are prescribed, clinicians always operate in resource constrained settings, balancing patient's expectations, professional norms and uncertainty in clinical decision-making. This highlights the need to move beyond traditional knowledge-focused approaches and better understand the behavioral determinants of prescribing practices.

This is why understanding what prescribers know isn't enough, but how they make decisions at the point of care. In such scenarios, prescribing is not entirely technical, but reflects underlying values, perceived responsibilities, and risk management strategies. While digital health tools like electronic guidelines, decision-support systems, and online training platforms, offer opportunities to strengthen Antimicrobial Stewardship (AMS), their efficiency and effectiveness depends on behavioural readiness and system-level integration.

Methods

A quantitative cross-sectional study was conducted among PHC prescribers across the three senatorial zones (Owerri, Orlu, and Okigwe) and 27 Local Government Areas of Imo State, Nigeria. A purposive sampling approach targeted all 572 Officers-in-Charge (OICs) of PHC facilities, given their central role in prescribing and facility leadership. A total of 547 prescribers participated (478 OICs and 69 other prescribers), representing an 84% response rate among OICs.

Data were collected using a structured, self-administered online questionnaire.

Knowledge was assessed using items covering awareness, causes, transmission, consequences, and prevention of AMR, as well as understanding of key concepts such as antibiograms and antibiotic spectrum. Scores were categorized as adequate ($\geq 70\%$) or inadequate ($< 70\%$). Attitudes were measured using Likert-scale items assessing perceptions of AMR severity, professional responsibility, training needs, and policy gaps, categorized as positive ($\geq 50\%$) or negative ($< 50\%$). Practices were assessed based on self-reported prescribing behaviors, including frequency of antibiotic use, adherence to guidelines, patient education, use of laboratory results, and antibiotic selection, categorized as good ($\geq 75\%$) or poor ($< 75\%$).

Digital readiness indicators included use of online medical resources, access to digital or print facility-level antibiograms, participation in virtual or physical

antimicrobial stewardship (AMS) training, and engagement with digital learning platforms.

Data were analyzed using SPSS version 21. Descriptive statistics such as frequencies, percentages, means, and standard deviations were used to summarize variables. Bivariate associations were assessed using chi-square tests. Binary logistic regression estimated adjusted odds ratios (AORs) for predictors of good practice. Statistical significance was set at $p < 0.05$.

Results

The study included 547 respondents, the majority of whom were female (98.7%), had over 11 years of professional experience (85.2%), and served as Officers-in-Charge (87.4%).

Overall, 77.1% of respondents demonstrated adequate knowledge of AMR, while only 32.7% exhibited positive attitudes. Despite relatively high knowledge levels, 88.5% reported poor prescribing practices.

Years of experience was significantly associated with both knowledge and practice ($p = 0.021$). After adjusting for knowledge and years of experience, attitude remained the strongest independent predictor of good prescribing practice (AOR = 0.044; 95% CI: 0.011-0.186; $p < 0.001$). Each additional category of experience increased the likelihood of good practice (AOR = 1.65; 95% CI: 1.20-2.28; $p = 0.002$). Knowledge was not independently associated with good practice in the adjusted model (AOR = 0.85; 95% CI: 0.50-1.45; $p = 0.548$).

This finding suggests that prescribing behaviour is barely shaped by knowledge alone but more by attitude, experiential learning, and contextual pressures. Evidence from practice patterns suggests that patient's expectation and diagnostic uncertainties are some competing demands that influences clinicians prescribing decisions.

Digital readiness indicators revealed notable gaps. Although 69.5% of respondents had heard of AMR, only 13.3% reported online medical resources as a source of information. Access to facility-level antibiograms was limited (21.2%), and only 31.3% reported regularly updating their knowledge using digital platforms such as online courses or webinars. Furthermore, 44.2% had never participated in AMS training, including virtual formats.

In practice, fewer than half (41%) consistently followed national or international guidelines when prescribing antimicrobials, and only a minority routinely consulted antimicrobial susceptibility test results before prescribing. A substantial proportion of respondents (43.9%) reported encountering suspected or confirmed treatment failure due to AMR in their clinical practice.

Findings indicate a substantial gap between knowledge and appropriate prescribing practices, occurring alongside limited access to real-time clinical decision-support resources and digital learning opportunities.

Conclusions

This study demonstrates a significant knowledge-practice gap among PHC prescribers in Imo State, Nigeria. It shows that antimicrobial prescribing is fundamentally shaped by behavioural and contextual factors rather than knowledge alone. The strong influence of attitude on prescribing behaviour underscores the need to invest in stewardship strategies that addresses motivations, underlying values, and decision-making processes of prescribers.

At the same time, limited digital readiness, including low access to antibiograms sabotages the confidence and motivation of prescribers and frustrates ethical considerations and professional decision making at the point of care.

These findings suggest that improving antimicrobial stewardship requires an integrated approach that combines behavioural interventions with accessible, context-appropriate digital tools that support clinical decision-making at the point of care.

Recommendations

To address these gaps, we recommend the integration of point-of-care digital decision-support tools, including mobile-accessible antibiotic guidelines and local antibiograms, expansion of digitally-enabled continuous professional development, such as e-learning platforms and virtual AMS training programs, and increased investment in digital infrastructure and access within PHC settings to support the adoption and effective use of these tools. Such interventions should be tailored to the local context to ensure feasibility, acceptability, and sustainability.

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TABLE 1: AMR KAP levels of Respondents

Variable	Frequency (547)	Percent
Knowledge Level		
Inadequate Knowledge (<70%)	125	22.9
Adequate Knowledge (>=70%)	422	77.1
Attitude Rating		
Negative Attitude (<50%)	368	67.3
Positive Attitude (>=50%)	179	32.7
Practice Level		
Poor Practice (<75%)	484	88.5
Good Practice (>=75%)	63	11.5

TABLE 2: Summary of Bi-variate and Logistic Regression Analyses of Predictors of Knowledge, Attitude, and Practice Towards AMR

Predictor	Category	Knowl- edge Ade- quate n (%)	Attitude Positive n (%)	Practice Good n (%)	Chi- square (X ²)	p-value	Adjust- ed OR (Practice) [95% CI]	p-value
	≤10	8 (1.9%)	3 (1.7%)	13 (2.7%)	11.503	0.021	1.652 [1.199– 2.275]	0.002
Years of Experi- ence	11–20	197 (46.7%)	82 (45.8%)	225 (46.5%)	–	–	–	–
	21–30	169 (40.0%)	79 (44.1%)	195 (40.3%)	–	–	–	–
	31–40	43 (10.2%)	12 (6.7%)	47 (9.7%)	–	–	–	–
	≥40	5 (1.2%)	3 (1.7%)	4 (0.8%)	–	–	–	–
Knowl- edge	Adequate Vs Inade- quate	–	177 (98.9%)/ 123 (33.4%)	48 (23.1%)/ 11 (17.0%)	71.293/ 7.532	<0.001/ 0.006	0.850 [0.499– 1.446]	0.548
Attitude	Positive vs Nega- tive	–	–	–	–	–	0.044 [0.011– 0.186]	<0.001

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Beyond algorithms: A qualitative study on AI as a relational space for organ transplant patients

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Karataş, H. Beyond algorithms: A qualitative study on AI as a relational space for organ transplant patients.

Aim

This study explores how organ transplant recipients engage with AI-based conversational tools as doctor-like figures in their everyday lives. It questions how these interactions reshape patients' understandings of medical authority, trust, and care in the post-transplant period. Moving beyond a purely technological perspective, the study conceptualizes AI as a relational space through which patients negotiate uncertainty, responsibility, and emotional needs within structurally constrained healthcare systems.

Background

Organ transplantation is not a one-time medical intervention but a lifelong process requiring continuous monitoring, strict adherence to medication, and sustained vigilance toward bodily changes. While transplantation offers patients a "second life," this new phase is marked by a fragile balance between recovery and risk. Patients inhabit a liminal state in which they are neither fully ill nor completely healthy, and where bodily sensations, laboratory results, and subjective well-being often remain in tension. While patients' need for continuous medical engagement, access to healthcare

remains structurally limited. Transplant recipients face long waiting times, geographically centralized services, and brief consultations with physicians who balance heavy clinical and academic workloads. These conditions create both informational and emotional gaps in healthcare. Patients frequently find themselves alone when interpreting test results or assessing the significance of bodily sensations. While existing literature on Artificial Intelligence in Medicine has largely focused on efficiency, diagnostic accuracy, and clinical decision-making, relatively little attention has been paid to patients as active users of AI in everyday healthcare practices. However, emerging evidence suggests that patients increasingly turn to AI tools not only for information but also for emotional reassurance and guidance. This study addresses this gap by questioning how transplant recipients interpret, personalize, and integrate AI into their post-transplant lives.

Methods

This study employs a qualitative research design informed by an ethnographic sensibility. Data were collected through in-depth, semi-structured interviews with 24 organ transplant recipients in Türkiye, including 19 kidney and 5 liver transplant patients. All participants had undergone transplantation at least one year prior to the study and actively used an AI-based conversational tool (ChatGPT) for health-related purposes. Participants were selected through purposive sampling from a larger group of 120 informants whom the researcher had followed across pre-, peri-, and post-transplant stages. Additional inclusion criteria included at least one year of post-transplant experience and access to AI technologies via smartphone use. Interviews were conducted online between August 2022 and November 2023, audio-recorded, and transcribed verbatim. Data analysis was carried out using an inductive thematic approach supported by MAXQDA 2024. Coding proceeded iteratively, beginning with line-by-line coding and advancing toward the development of broader themes through constant comparison across the dataset. Ethical approval was obtained from the Bartın University Social and Humanities Research Ethics Board (Approval No. 2022 SBB-0055) prior to data collection, and all participants provided informed consent. To ensure confidentiality, all identifying information was anonymized.

Findings

The analysis revealed five interrelated themes that shows how AI is embedded within the everyday practices of post-transplant care. The first theme, *access without burden*, highlights the role of AI as an immediately accessible and low-threshold resource. Although transplant care in Türkiye is State-funded, accessing physicians often requires significant time and logistical effort. Many participants reported using AI to interpret laboratory results obtained from local clinics before deciding whether to seek medical attention. In this sense, AI reduces unnecessary hospital visits while enabling timely responses to potential risks. The second theme, *AI as a proxy clinician*, illustrates how patients position AI alongside—but not in place of—medical professionals. Participants used AI to interpret test results, seek second opinions, and prepare for consultations. Importantly, none of the participants relied on AI for independent diagnosis or treatment decisions. Instead, AI functioned as a supplementary and preparatory tool within an existing framework of medical authority. The third theme, *being taken seriously without being judged*, emphasizes the emotional and communicative dimensions of AI use. Due to time constraints and hierarchical dynamics in clinical encounters, patients often refrain from discussing issues perceived as trivial, embarrassing, or morally sensitive. AI provides a private and non-judgmental space where such concerns can be expressed freely. This function reveals the limitations of current clinical communication and highlights unmet relational needs. The fourth theme, *naming, familiarity, and trust*, demonstrates how patients anthropomorphize AI systems by assigning them common Turkish names such as “Cemile” or “Cevriye.” This practice transforms AI from a technical tool into a familiar and relatable presence. Trust in AI emerges not only from its informational capacity but also from its perceived emotional neutrality and accessibility. The fifth theme, *redistribution of care responsibility*, reflects broader transformations in post-transplant life. Following transplantation, responsibility for maintaining health shifts increasingly toward the patient. In this context, AI becomes a companion in self-management, offering guidance on diet, physical activity, and emotional well-being. Patients integrate AI into their daily routines as part of a broader ecology of care that extends beyond clinical settings.

Conclusions

This study demonstrates that transplant patients' engagement with AI-based conversational tools cannot be reduced to a simple search for medical information. Rather, AI functions as a relational space that mediates between structural constraints in healthcare systems and the emotional, moral, and practical challenges of post-transplant life. Patients do not position AI as a replacement for physicians but as a complementary actor that fills gaps left by time-limited and hierarchical clinical encounters. AI enables patients to interpret bodily signals, articulate concerns, and manage uncertainty without fear of judgment or burdening healthcare professionals. This study contributes to broader debates on Artificial Intelligence in Medicine. It suggests that future research should move beyond questions of technological replacement and instead focus on the relational dimensions of AI use. Understanding what patients seek in AI interactions offers critical insights into the gaps, silences, and unmet needs within contemporary healthcare systems.

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Building trust in ChatGPT: Accuracy and reliability in healthcare research an umbrella review

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Overview

Large language models (LLMs), and ChatGPT in particular, are being rapidly adopted across patient care, clinical decision-making, medical education, and biomedical research (1, 2). Their capacity to support clinician reasoning, accelerate information retrieval, and assist with documentation has generated optimism about their potential to improve health-system

efficiency (3). However, the evidence supporting their reliability and accuracy in clinical settings has not been systematically consolidated, and no higher-order synthesis has yet examined the methodological quality of the literature evaluating their use.

Background

Since the public release of ChatGPT in late 2022, the volume of academic literature evaluating its applications in medicine has expanded at a pace that has outstripped the development of robust evaluative frameworks. Topic-specific systematic reviews and meta-analyses have proliferated across clinical specialties, but their findings have not been integrated, and the methodological rigour of this secondary literature has not been formally appraised. This evidentiary fragmentation has practical consequences: clinicians, researchers, journal editors, and policymakers are being asked to make implementation decisions in the absence of a coherent appraisal of the evidence on which such decisions should rest. The present umbrella review was therefore undertaken to synthesise the available evidence on the reliability and accuracy of ChatGPT in healthcare, to critically appraise the methodological quality of existing systematic reviews and meta-analyses, and to identify the evidentiary gaps that must be resolved before responsible clinical translation can be supported.

Methods

The review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (4). The Cochrane Library, PubMed, Scopus, Web of Science, and Google Scholar were searched for systematic reviews and meta-analyses published up to February 2024, without restrictions on language or publication year. Search strings combined controlled vocabulary and free-text terms relating to ChatGPT, large language models, and healthcare. Two reviewers independently screened titles, abstracts, and full texts against pre-specified eligibility criteria, with discrepancies resolved by consensus and adjudication by a third reviewer where required. Eligible studies were peer-reviewed systematic reviews and meta-analyses focused on the application of ChatGPT in healthcare contexts, including patient care, health education,

clinical decision support, and research workflows. Methodological quality was assessed using the AMSTAR-2 checklist (5), which evaluates 16 domains, including protocol registration, justification of study designs, duplicate study selection and data extraction, risk-of-bias assessment, handling of heterogeneity, investigation of publication bias, and disclosure of funding and conflicts of interest. Each review was rated as high, moderate, low, or critically low quality. One reviewer (ARR) conducted the initial appraisal and a second (AT) independently verified all ratings; full agreement was reached without the need for adjudication. Data were extracted using a structured proforma capturing review objectives, ChatGPT version evaluated, clinical domain, reported strengths, reliability and accuracy concerns, and recommended future directions. Findings were synthesised narratively given the substantial heterogeneity in scope and outcome reporting; quantitative pooling was not undertaken.

Results

Seventeen reviews met the inclusion criteria 15 systematic reviews and 2 meta-analyses all published between 2023 and early 2024, reflecting the rapid emergence of academic interest following the public release of ChatGPT (6-22). AMSTAR-2 appraisal identified substantial methodological limitations across the body of evidence: 5 reviews (29.4%) were rated as moderate quality and 12 (70.6%) were rated as low quality; none achieved a high-quality rating. The most frequent deficiencies were inadequate justification of included study designs, incomplete reporting of funding sources of the primary studies, limited consideration of risk of bias when interpreting findings, and insufficient explanation of observed heterogeneity. Pre-registration of review protocols and formal investigation of publication bias were also infrequently reported. The included reviews spanned a heterogeneous range of topics, encompassing general healthcare practice, clinical research methodology, health professional education, dental research, ophthalmology, gastroenterology, neurosurgery, radiology, and the conduct of systematic reviews. Reported strengths of ChatGPT clustered around time efficiency in scientific writing, code generation, data analysis, clinical reasoning, the drafting of patient-facing educational material, and the preparation of clinical documentation. Reliability and accuracy concerns

were strikingly consistent across reviews and clinical domains, and included the generation of incorrect or fabricated citations, plagiarism, lack of originality in synthesised content, inconsistent performance across model versions, and the production of inaccurate or contextually inappropriate clinical content. Performance was reported to vary by ChatGPT version and by task complexity, with version 4.0 generally outperforming 3.5 on complex tasks but not uniformly so.

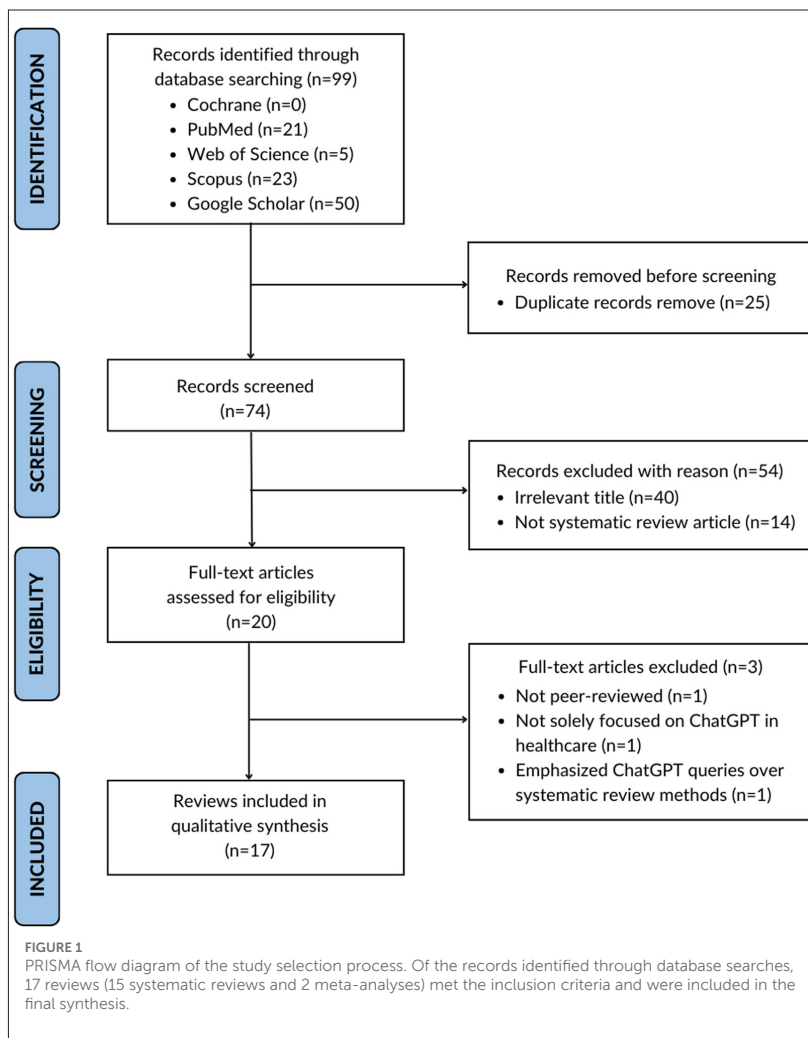
Discussion

The methodological quality of the existing evidence base on ChatGPT in healthcare is low, and the absence of any high-quality review materially constrains the conclusions that can be drawn about its reliability and accuracy. Future systematic reviews should pre-register protocols, justify the inclusion of specific study designs, disclose funding sources, apply formal risk-of-bias assessments, and explicitly investigate heterogeneity and publication bias. Future primary research should evaluate ChatGPT using larger and more clinically and demographically diverse datasets to improve external validity and should assess performance across non-English and resource-limited settings, where deployment will have the greatest equity implications. Two further priorities warrant focused investigation: the integration of longitudinal patient data including electronic health records, imaging, laboratory parameters, and patient-reported outcomes into predictive frameworks that leverage ChatGPT's natural language capabilities; and the rigorous evaluation of its role in delivering emotional and psychosocial support in patient care, with explicit attention to therapeutic appropriateness and the ethical limits of machine-generated empathy. Strengthening regulatory, validation, and reporting frameworks will be essential to translating early promise into demonstrable clinical value.

Conclusion

ChatGPT shows potential to improve efficiency in healthcare delivery and research, but accuracy and reliability concerns persist, and the systematic reviews evaluating its use are predominantly of low methodological quality. Strengthening the evidence base will require larger and more diverse datasets, multilingual evaluation, longitudinal study designs,

methodologically rigorous reviews, and targeted investigation of ChatGPT's role in delivering emotional support within patient care.



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Clinical applications of ChatGPT in healthcare

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Nkhoma, D., Tanweer, A., Zahra, M., Haider, S.M., Iqbal, U., Redfern, J. Clinical applications of ChatGPT in healthcare.

Background

Large language models (LLMs) are increasingly being used in healthcare. ChatGPT, a transformer-based natural language processing model, can produce human-like, contextually relevant responses by leveraging extensive training data rather than predefined rules(1, 2). This capability positions it as a versatile tool across healthcare domains, including medical education, research support, and clinical practice. Evidence suggests that ChatGPT can assist with generating educational materials, supporting clinical queries, and improving communication between patients and healthcare

providers. In clinical settings, the model demonstrates potential to streamline documentation, enhance workflow efficiency, and support diagnostic reasoning. It has also shown promise in specialised areas such as radiology and pathology by assisting with image interpretation and treatment planning (3). However, despite these benefits, concerns remain regarding data privacy, algorithmic bias, and the reliability of generated outputs, which may include inaccuracies or fabricated content (4, 5). These uncertainties highlight the need for comprehensive evidence synthesis to evaluate the scope, applications, and limitations of ChatGPT in healthcare. This umbrella review synthesised evidence on how ChatGPT is currently being applied across healthcare settings from patient and clinician perspectives.

Methods

An umbrella review methodology was employed to synthesise findings from existing systematic reviews and meta-analyses, providing a high-level overview of ChatGPT applications in healthcare. The review followed Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines and established umbrella review frameworks (5, 6).

Seventeen systematic reviews and meta-analyses published up to February 2024 were identified through database searches (Cochrane Library, PubMed, Scopus, Web of Science, and Google Scholar). The review included systematic reviews and meta-analyses focusing on use of ChatGPT for consumer health, published in English and available as full texts. Two reviewers independently conducted study selection and screening, with disagreements resolved through consensus. The selected reviewers were assessed using the AMSTAR-2 tool to evaluate methodological rigor and risk of bias (7). Themes related to clinical applications, patient education, and use in special care were extracted and synthesised.

Results

Seventeen studies (15 systematic reviews and 2 meta-analyses) were included, reflecting a rapidly expanding research field. Most studies were published between 2023 and early 2024, indicating growing interest in

ChatGPT applications. Five reviews (29.4%) were of moderate quality, while the remaining 12 (70.6%) had low quality.

The most frequently reported theme (16/17 reviews) was the role of ChatGPT in supporting patient and caregiver interactions. Eleven reviews described its use in patient education and health information seeking, helping patients in terms of general information gathering about disease. ChatGPT was reported to assist clinicians with documentation, discharge summaries, responding to patient queries, and decision support. Specialty-specific applications were described in radiology, neurosurgery, gastroenterology, dentistry, and ophthalmology for support in image interpretation, differential diagnosis, and treatment planning. Reviews noted improved workflow efficiency and potential reduction in administrative burden. However, concerns regarding accuracy, hallucinations, bias, and need for human verification were consistently reported across studies. Variability in performance across ChatGPT versions was also noted, with newer models better handling complex tasks but not always outperforming earlier versions in simpler contexts.

Discussion

The findings demonstrate that ChatGPT has broad and evolving applications across healthcare, with potential to enhance patient care, streamline clinical workflows, and support research and education. In clinical practice, it can assist with documentation, patient communication, and decision support, thereby reducing administrative burden and improving efficiency (8, 9).

In medical education, ChatGPT offers interactive learning opportunities and has demonstrated the ability to answer medical examination questions with reasonable accuracy, highlighting its role as a supplementary educational tool (10). Its use in specialised domains, such as neurosurgery and radiology, further underscores its versatility and potential to support complex clinical processes (8, 11, 12).

However, critical limitations must be addressed. The reliability of ChatGPT remains inconsistent, with concerns about inaccuracies, fabricated

information, and limited understanding of specialised medical terminology (13, 14).

Effective implementation of ChatGPT will require interdisciplinary collaboration among developers, clinicians, and policymakers, alongside robust validation frameworks and regulatory oversight (15, 16). For further adoption in healthcare, the necessary large pragmatic real-world studies will be required to validate their use across various settings and disciplines. In addition, such studies will have to extended for all LLMs in use within the healthcare industry. And then for higher-order evidence, researchers will have to adhere to methodological standards in order to improve quality of evidence.

Conclusion

ChatGPT demonstrates broad potential to support clinical workflows, patient communication, and specialty care. While early applications appear promising, concerns regarding reliability necessitate cautious and supervised use. Future clinical studies are required to validate these applications in real-world healthcare settings, as well as extending the scope with other relevant LLMs.

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Delphi study as a method in the conceptualization of a new digital wellbeing measure for university students and staff across Europe

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Citation

Bezmelnitsina Collé, C., Aymerich, M., Leontopoulou, S., Jacobs, N. Delphi study as a method in the conceptualization of a new digital wellbeing measure for university students and staff across Europe.

Aim

To explore expert conceptualizations of digital wellbeing and inform the content and structure of a new measure for university students and staff across Europe.

Round 1 of a two-round Delphi questionnaire was sent to 24 experts in 6 countries. Open-ended questions provided rich data on digital wellbeing in higher education from 17 completed responses. Experts agreed the new measure should use a Likert scale with 5-8 dimensions and require under 15 minutes to complete. Over 250 coding items generated seven dimensions: emotional and mental wellbeing, digital balance and time management, privacy and security, digital literacy and competence, social wellbeing and relationships, self-awareness and personalization, and physical wellbeing.

These dimensions will inform Round 2 statements. Round 1 demonstrated that a Delphi study using open-ended questions effectively explores the context-dependent concept of digital wellbeing for measure development.

Background

Current digital wellbeing measures assess general digital wellbeing (1,2,3), digital stress and technostress (4,5,6), or digital technology use and experience (7) across populations including adolescents (2), students and young adults (3,8), and adults (1,4,5,6). However, the literature lacks a clear definition of digital wellbeing. Scoping and systematic reviews (9,10,11) highlight both definitional inconsistency and the absence of a gold-standard measure. Existing measures were developed in single countries - Turkey (1), Slovenia (2), the UK (7,8), and the US (3,8) - lacking cross-cultural validation and context-specific tailoring. Besides, the digital wellbeing scale was validated on participants younger than 25 years (2,8) while the current digitalization of higher education extends the student age profile and needs.

To address gaps, the Delphi methodology was used as an initial step in the new digital wellbeing measure development across European universities in the OpenEU consortium, aiming to reach a consensus on conceptualization and basic measurement characteristics. The broader project also includes focus groups with students, academic and administrative staff; statistical analysis to select and refine items; and cross-cultural validation and linguistic adaptation of the final tool into the partner universities' languages.

Methods

The Delphi technique is a structured way to gather and refine expert opinions on a complex issue, typically over several rounds, to reach a reliable consensus (12). It proved to be a rigorous method for framework development, from theoretical construct to practical application in the measurement tool construction (13). This extended abstract reports findings from Round 1 of an ongoing Delphi study.

In the present study, experts included researchers/professors, university staff, psychologists, and predoctoral researchers, with primary expertise in

mental health, digital health, cyberpsychology, and educational technology. Geographically, the panelists represent the Netherlands, Spain, Iceland, Luxembourg, Greece, and Belgium. The panelists had been chosen based on their expertise and experience by the OpenEU Lead in each participating university. Personal contacts were used as a recruitment channel. Additionally, an open call for experts was launched via LinkedIn following the philosophy of open research.

The Delphi questionnaire comprised two blocks covering conceptualization and questionnaire construction. Block one asked panelists to describe the concept, its key components, core domains for a new measure, cultural influences, indicators and behaviors of digital wellbeing decline, priority aspects for university students and staff, and positive and negative influences. Block two collected opinions on whether different questionnaires were needed for different groups (students, academic and administrative staff), the optimal number of dimensions, response scale type, and timeframes. Round 1 also invited participants to suggest additional content and considerations. After piloting with five experts and minor revisions, invitations were sent to 24 experts. Round 1 used primarily open questions to build Round 2 and capture rich perceptions of digital wellbeing. Responses were analyzed using thematic and content analysis. An inductive coding approach was used. The first code book was developed manually, the second with the help of qualitative data analysis software (Atlas.ti) and the third - with Claude (GenAI by Anthropic, model Sonnet 4.5). Final codes were assigned by majority rule (the code occurs in two out of three sources) and revised manually. Identified key domains structured the statements for Round 2, where content analysis will examine expert agreement levels. For ordinal responses, consensus required at least 70% agreement or disagreement. Dimensions achieving consensus will be excluded from Round 2.

Results

Of 24 invited experts, 10 responded to Round 1, plus two from an open call, totaling 12 responses. Since the pilot version required no essential changes, pilot data were included, yielding 17 total responses. Panelists represented

the Netherlands (n=7), Spain (n=3), Iceland (n=2), Greece (n=2), Luxembourg (n=2), and Belgium (n=1).

Experts primarily conceptualized digital wellbeing as emotional and psychological wellbeing, healthy technology use, and digital literacy. Digital literacy appeared in responses to nearly all questions, with self-regulation coding occurring exclusively alongside it, suggesting either a latent factor or shared dimension. Digital literacy emerged as important for both students and staff, while work-life balance was identified as a key aspect for staff only.

Experts suggested dimensions “emotional and mental wellbeing”, “digital balance and time management”, “privacy and security”, “digital literacy and competence”, “social wellbeing and relationships”, “self-awareness and personalization”, and “physical wellbeing” to be included in the new measure.

All experts agreed that cultural values shape individual understanding of and relationship with technology, affecting digital habits and attitudes. This underscores the importance of cultural context in conceptualizing digital wellbeing.

Questions on digital wellbeing decline and cultural influences generated the most coded units, suggesting this emerging concept can be explored through decline or related domains. Key themes included usage intensity, overuse, and digital dependency, indicating that duration and loss of control are crucial factors in decline.

Round 1 results align with literature treating digital wellbeing as subjective, balanced, healthy tech engagement (14,15), enabled by digital literacy and self-regulatory skills (16), and context-dependent (14).

Consensus emerged on 5-8 dimensions (70.59%) and Likert scale use (Table 1). While no consensus on timeframe was reached, all agreed completion should take under 15 minutes (under 10 min., 59%; 10-15 min., 41%).

TABLE 1: Expert Consensus on the questionnaire construction

Item	Not at all n (%)	Slightly n (%)	Moder- ately n (%)	Very n (%)	Ex- tremely n (%)	Consensus		
						+ve	-ve	None
How important is it to you that... - there is a students only version of a new Digital Wellbeing Measure (DWM)	3 (17.65)	-	3 (17.65)	9 (52.94)	2 (11.76)			✓
- there is an academic and administrative staff version of a new DWM	4 (23.53)	4 (23.53)	3 (17.65)	4 (23.53)	2 (11.76)			✓
- there is an academic staff only version of a new DWM	3 (17.65)	3 (17.65)	4 (23.53)	6 (35.29)	1 (5.88)			✓
- there is an administrative staff only version of a new DWM	3 (17.65)	4 (23.53)	5 (29.41)	5 (29.41)	-			✓
- there is a general version of a new DWM for all higher education actors	3 (17.65)	3 (17.65)	3 (17.65)	4 (23.53)	4 (23.53)			✓
How useful it is in a new digital wellbeing measure ... - to use Likert scale	-	1 (5.88)	3 (17.65)	8 (47.06)	5 (29.41)	✓		
- to use illustrations	4 (23.53)	3 (17.65)	5 (29.41)	4 (23.53)	1 (5.88)			✓
- to use dichotomic (yes\ no) responses	4 (23.53)	4 (23.53)	7 (41.18)	2 (11.76)	-			✓

"Consensus" columns (+ve, -ve, None) are applied based on the >70% rule, combining the "Very" and "Extremely" (for +ve consensus) and "Not at all" and "Slightly" (for -ve consensus) categories

Conclusions

Results of Round 1 of Delphi's study provided rich data for the Round 2 statements formulation and future research on the conceptual foundation of digital wellbeing. Experts developed more than 250 coding items across nine topic areas.

Subjective feelings, inner state, harmony, and social and psychological outcomes of technology use are core elements of the multidimensional digital wellbeing construct. Experts emphasized control, limits, and intentional use as central to achieving a balanced state.

These findings represent an initial step toward a conceptual and operational framework for a new digital wellbeing measurement tool development.

Statements

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Ethics Statement

The study was approved by the University of Catalonia Ethics Committee. The participants provided informed consent to participate in this study.

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Generative AI statement

The author(s) declared that Generative AI was used in the creation of this manuscript for improving clarity and reducing the initial manuscript to fit the length limit of the extended abstract. The author acknowledges full responsibility for the content, analysis, and conclusions presented in the

paper. The use of AI tools for coding as an additional support for expert-coded results is stated in the Method section. The final manuscript has been thoroughly reviewed and approved by the author.

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Detecting medical misinformation: Methods, gaps, and next steps

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Citation

Mammadaliyev, V., Shahbazov, V. Detecting medical misinformation: Methods, gaps, and next steps.

Abstract

One of the threats to the medical-demographic security of the population is misinformation being spread online in the medical domain. To understand the current state and gaps regarding automatic misinformation detection mechanisms, this work surveys recent studies. Surveyed studies cover traditional ML approaches, deep learning, knowledge graphs, large language models, and fact-checking systems. The key identified gaps are that systems which are forced to make binary classification cannot explain the decision; whereas systems which are able to prove their decisions by supporting evidence cannot output a deterministic verdict when there is not enough evidence. We propose a two-stage hybrid architecture, addressing those issues. The proposed architecture achieves 95.09% accuracy on the evaluated dataset, with the ability to ground its decisions in evidence in 24.3% of cases.

1. Background

Medical-demographic security is a protected state of the population from various threats, including misinformation being spread online in the medical domain. This necessitates automatic misinformation detection systems. During COVID-19, this was demonstrated clearly. Falsely promoted

information led to skewing of health-related decisions, such as refusing vaccination [1]. Since 2020, significant research on automatic misinformation detection has been conducted [2], [3]. Nevertheless, challenges remain. This work presents a narrative review covering five major method families and a hybrid architecture with empirical evaluation, addressing the gaps identified by the survey.

2. Survey Overview

Studies for review were gathered from various sources: Scopus, Google Scholar, PubMed, ACM Digital Library, ACL Anthology, and arXiv. Candidate papers were selected by filtering publishing year from 2020 onward and by using keywords such as “medical misinformation detection” and “fake medical news detection.” Resulting papers were ranked for relevance with GPT-4 assistance, then manually reviewed against criteria requiring a proposed methodology, quantitative evaluation, and a reusable artifact. This process resulted in 27 studies, and a representative subset is cited in this paper.

All of the reviewed methods fall into one of five families: traditional ML classifiers (e.g., SVM, ~93% F1 [4]), deep learning and transformers [5], knowledge-graph approaches [6], large language model (LLM) approaches [7], and fact-checking pipelines [8], [9].

Traditional ML and deep learning approaches lack medical grounding in the decisions they make, although among both of them deep learning is better at generalizing and therefore more usable in real-world scenarios, at the cost of deployability due to hardware requirements. On the other hand, knowledge-graph approaches are domain-aware, although this requires completeness of the graph. LLM approaches avoid this dependency, as they are trained on large corpora including medical data, but introduce the problem of hallucination, meaning confident incorrect outputs, and depend on the reasoning ability of the LLM itself. To solve the hallucination problem, fact-checking pipelines use evidence to ground their decisions, and when evidence is in place, outperform text-only classifiers. In cases when evidence

is not available, systems face a “not enough information” (NEI) gap, which is basically abstaining from providing a classification.

To sum up the findings across all studies, when trying to introduce evidence-based decisions or any ability to explain the decision, the NEI gap is met. Conversely, approaches which are forced to make binary classifications cannot provide sufficient ground for their decisions.

3. Proposed Hybrid Architecture

We propose a two-stage architecture (Fig. 1), which addresses both gaps.

The first stage decomposes an input article into claims and for each tries to find supporting or refuting evidence from a knowledge base. Found evidence is passed as context to the LLM, which makes a verdict: “supported,” “refuted,” or NEI. While “supported” and “refuted” are considered final, NEI triggers the second stage of the pipeline.

The datasets used are cancer and diabetes related articles from DETERRENT [6]. We merged both datasets and split 80/20, yielding 1,628 test articles. Each article was decomposed into claims by openai/gpt-oss-20b (open-weight 20B LLM), vectorized using the BAAI/bge-large-en-v1.5 embedding model, and stored along with its reference label (misinformation or real).

To train the Random Forest classifier, the same training split was used. The same LLM extracted semantic features from each article, such as fear and anger scores, rhetorical indicators, and sentiment polarity. The classifier was trained on these features.

For each test article, claims are extracted and matched against the KB by cosine similarity (top-k = 10). The matched evidence is passed to the LLM, which returns one of three verdicts. If no relevant evidence is found or the evidence is insufficient, the Random Forest classifier outputs a binary decision.

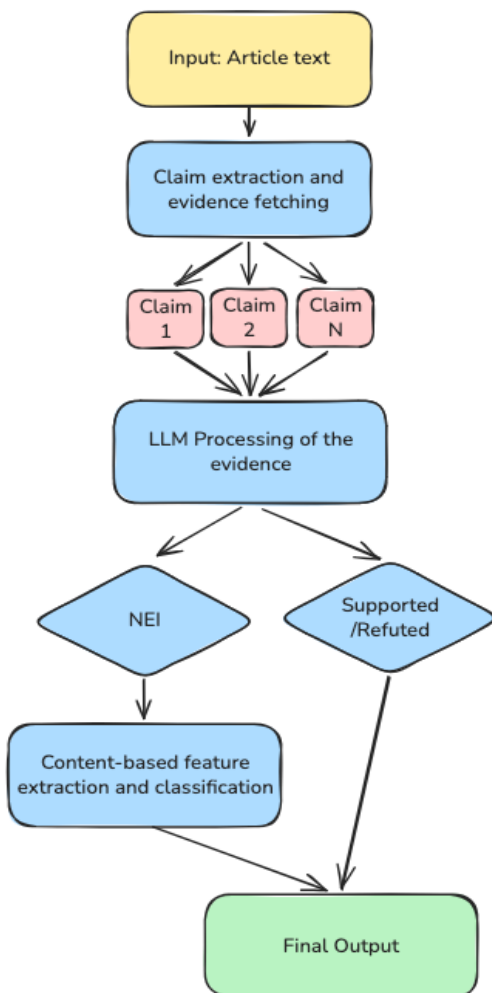
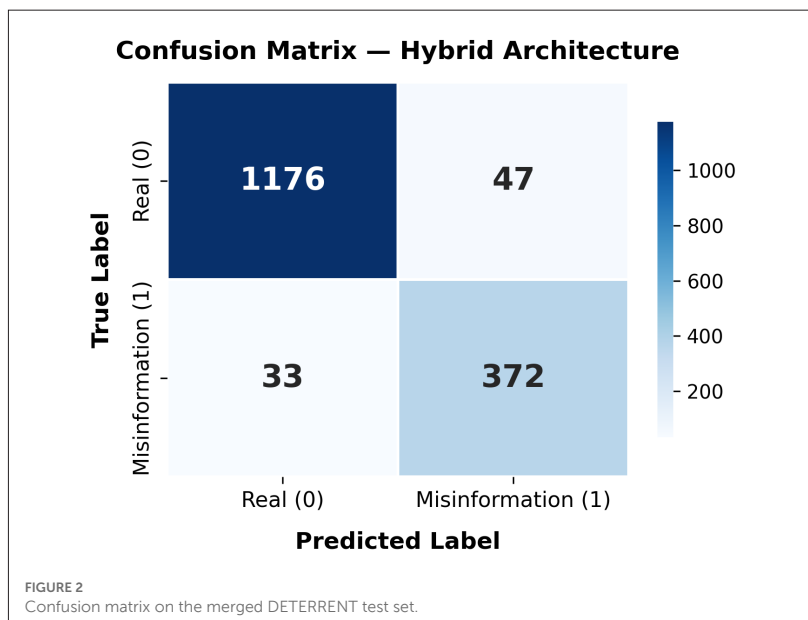


FIGURE 1
Hybrid architecture for medical misinformation detection.

4. Results

The whole pipeline achieves 95.09% accuracy, 93.50% macro F1, and 95.11% weighted F1. This is across 1,628 test articles (1,223 real, 405 misinformation). Per-class precision/recall: 0.97/0.96 for real articles and 0.89/0.92 for misinformation articles. This is also illustrated in the confusion matrix in Fig. 2. The DETERRENT baseline evaluation was done on cancer (96.52% accuracy) and diabetes (92.06% accuracy). If the average of those results is taken, the approach proposed in this study exceeds it.

The KB stage was able to provide an evidence-grounded decision for 24.3%, while 75.7% were classified by the 2nd stage (Random Forest classifier). It is worth noting that in the KB stage, no misinformation article was misclassified as real, achieving a macro F1 of 0.917.



5. Conclusion

The narrative survey done for this study revealed several limitations of existing approaches. The approach proposed in this study tries to address those limitations by providing evidence-based ground where it is possible and classifying using semantic analysis if no evidence ground was found. The main limitation of the conducted experiment is the constructed knowledge base, which contains only information from the training split of the dataset, thus yielding low coverage for evidence-grounded decisions. Future work should incorporate external knowledge bases, which would significantly improve the coverage and reliability.

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Digital interventions for menopause care: A systematic review and landscape analysis of mobile applications

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Citation

Moretó, S., Planas-Guich, E., Cabutto, D., Fuster-Casanovas, A., Carrion, C., Pérez-Navarro, A. Digital interventions for menopause care: A systematic review and landscape analysis of mobile applications.

Aims

The current work has two main goals: 1) to examine the menopause app marketplace and determine the quality and evidence-base of existing tools;

2) to identify and synthesize existing scientific evidence on digital health interventions for menopause, focusing on their effectiveness and usability.

Background

While menopause is a natural part of life and not a disease, it frequently involves a variety of clinical signs that affect health and overall wellbeing, such as vasomotor, genitourinary, musculoskeletal, and psychological symptoms. During this stage, individuals may experience a reduction in quality of life across occupational, personal, and social domains (1,2).

In recent years, there has been a significant rise in the adoption of Digital Health (1) —including mobile applications, telemedicine, virtual reality and wearables (3–6) —to manage menopause and enhance well-being. While existing reviews have provided additional knowledge about digital tools for menopause (7,8), further research is still required to produce more comprehensive evidence regarding their specific functionalities and clinical effectiveness.

Methods

To provide a comprehensive overview of digital health tools for menopause, this research was designed in two stages: an initial mapping of available mobile apps and a subsequent systematic review of scientific evidence.

For the mobile app review, we explored the Android and iOS marketplaces (April 1–20, 2025) using search terms in both English and Spanish. From the initial results and following previously agreed inclusion criteria, we selected free apps that had been updated in the previous 24 months and demonstrated documented user interaction (at least 1,000 downloads on Android or 10 ratings on iOS). Once the apps were organized in a database, we screened all titles and descriptions. Three reviewers assessed the selected apps using the Mobile App Rating Scale (MARS) (9). Additionally, a keyword search was completed in PubMed to identify whether the apps had scientific evidence supporting their usability or effectiveness.

For the systematic review, a literature search was conducted on November 7, 2025 in PubMed, PsycINFO, Scopus, and Web of Science, complemented by searches in literature reviews and trial registries for Randomized Controlled Trials (RCT). Inclusion criteria focused on participants in any stage of menopause or health professionals using digital health tools (including mixed interventions). Randomized and non-randomized trials, as well as pre-post studies with or without control groups, were included. The search results were imported into Covidence®. Study selection was performed independently by three reviewers, disagreements were resolved by two additional reviewers at both the screening and full-text stages. Data from all included studies were independently extracted by two reviewers and verified by two additional reviewers to ensure consistency. Risk of bias is being assessed using Joana Briggs Institute (JBI) tools (10,11). The systematic review is being conducted following PRISMA guidelines (12) and is registered in PROSPERO (CRD420251167206).

Results

Mobile app review: After removing duplicates, 225 apps were screened, out of which only 20 met the eligibility criteria. The majority of the apps (90%) were designed for menopausal individuals, while 10% were support tools for healthcare practitioners. Regarding language, 35% were available in Spanish. Half of the apps were free, while the remaining 50% followed a 'freemium' model. Health professionals were the main source of app information, and a 35% of the apps provided links to scientific resources. The average MARS quality score was 3.28 (SD=0.52). Three apps had published evidence regarding their effectiveness, including one RCT (3), a prepost study (4) and a retrospective analysis (13). These studies assessed the efficacy of the apps on the reduction of menopausal symptoms, specifically targeting vasomotor and psychological symptoms.

Systematic review (preliminary results): A total of 5,472 records were identified, and 2,816 duplicates were removed. Sixty-six full-text articles were assessed, and 33 studies were included. Of the included studies, 48.48% were RCTs and 51.52% used a quasi-experimental design. Digital health interventions, including apps, websites, and platforms, targeted sleep

disturbances, vasomotor and psychological symptoms, and lifestyle habits. Preliminary results from the quality assessment indicate that the included studies are generally of moderate quality. However, significant heterogeneity was observed across methodologies and outcome measures: most studies used validated questionnaires, although there was no consensus on the specific tools to measure the same outcome. The majority of studies reported significant improvements in symptom reduction or lifestyle changes. Common limitations identified were small sample sizes, lack of blinding, reliance on self-reported measures, and limited generalizability of the results. The majority of these studies demonstrated effectiveness in symptom reduction.

Conclusions

There is a wide variety of digital health interventions for menopause, primarily focused on symptom management and lifestyle habits. Nonetheless, significant heterogeneity exists in the available evidence. Furthermore, commercial apps often lack scientific validation and face significant barriers regarding language accessibility and cost. This gap represents a public health concern, as users lack access to reliable information and the necessary tools to discriminate evidence-based content from non-validated sources. Providing evidence-based tools that consider social determinants of health and prioritize the user is necessary to achieve better health outcomes and improve the overall experience for users during menopause.

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Effectiveness of health belief model (HBM)-based educational intervention on breast self-examination: A cluster randomized controlled trial in Lahore, Pakistan

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Abstract

Background

Breast cancer has been reported by GLOBOCAN 2022 as the second leading cancer globally. ¹ In South Asian countries, 76000 women lost their lives to breast cancer per year. ² Pakistan is at the top with the highest incidence of female breast cancer in the WHO Eastern Mediterranean region (EMRO) ³ and this rate is likely to accelerate. One in nine women is at risk of developing breast cancer during her lifetime in Pakistan. ⁴ The incidence of female breast cancer in 20-59-year-olds is at its peak in Asia, i.e., 50.5%. ³ Approximately 40,000 women die from breast cancer annually in Pakistan, reflecting a dire need for early detection and intervention. ⁵ The majority of cases in Pakistan are diagnosed at advanced stages due to a lack of awareness, cultural

barriers, and limited access to early detection services.⁶ Early detection through screening methods such as breast self-examination (BSE), clinical breast examination (CBE), and mammography significantly improves survival outcomes.⁷ However, these practices are poorly adopted in Pakistan, and studies report that awareness about screening methods is less than 30%, and adherence to regular screening is even lower.⁸ Only 10% of women perform breast self-examination occasionally.⁶ Breast self-examination is an easy, effective, and efficient method suitable for women of all ages for early detection of breast cancer.⁹ This study seeks to assess the baseline knowledge, health beliefs, and practices related to breast cancer and breast self-examination in women aged 18 to 39 years by adapting the modified version of MT-CHBMS and determining the differences between the intervention and control groups.

Methods

A single-blinded, cluster-randomized controlled trial (cRCT) was conducted in Lahore District, Union Councils (UCs) were used as clusters. After matching on key baseline characteristics, one UC was randomly assigned to the intervention arm (HBM-based

BSE education) and another to the control arm (general reproductive health education). Women aged 18–39 years residing in the selected areas for more than six months were eligible. Using systematic random sampling from every *n*th household, 410 participants were enrolled (205 per group). Sample size was calculated using G*Power version 3.1 (effect size 0.3, power 0.80, $\alpha=0.05$). The questionnaire included four sections: Section I comprises socioeconomic characteristics. Section II consists of questions based on risk factors of breast cancer, symptoms of breast cancer, BCS methods, and BSE. Responses were measured by utilizing the nominal scale of “True”, “False”, and “I do not know”. Section III: Health Belief Model Scale. All HBM scales are positively associated with breast cancer screening behaviours, except for the barrier scale, which is negatively linked. Section IV: Breast self-examination practice and frequency per month were also asked of the participants.

TABLE 1: Health Belief Model Construct Scores (N=410)

HBM Construct	Mean Score	Interpretation
Perceived Susceptibility	2.6	Moderate
Perceived Severity	4.2	High
Perceived Benefits of BSE	3.8	High
Perceived Barriers to BSE	3.7	High
Self-Efficacy for BSE	2.3	Low
Health Motivation	3.6	Moderate

Results

The total of 410 female participants had a mean age of 32.3 years. 60% were married, 39% single, and 0.97% were divorced or widowed. Knowledge of breast cancer risk factors is very poor among the participants. For most risk factors, 65-90% of females answered "I don't know" or incorrectly. 72% females know a breast lump as a sign of breast cancer. Only 34% recognize nipple retraction as a sign of breast cancer. Similarly, only 39% knew skin changes, and 38% knew armpit lumps as a sign of breast cancer. Knowledge of the correct BSE technique is very poor. Only 38% knew to cover the entire breast, and only 21% knew that BSE should be done monthly. 79% have never performed breast self-examination. Only 3% perform BSE monthly as recommended.

For Perceived Barriers, a higher score means more barriers (worse outcome). For all other constructs, a higher score means a more positive belief.

Conclusion

Breast self-examination is a cost-effective method of screening, but still, females do not know how to perform it. Their knowledge of breast cancer risk factors and breast cancer screening is also low. Baseline findings confirmed critically low levels of awareness and

practice. The intervention group showed improvements across all domains compared with the control group, which remained unchanged. Widespread implementation through community health programs could help reduce late-stage diagnoses and improve survival outcomes in Pakistan.

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Enhancing health education and regional patient engagement through a hybrid metaverse model: The case for facilitated public health infrastructure

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Hölzer, S., Schmidt, L., Erfurth, C. Enhancing health education and regional patient engagement through a hybrid metaverse model: The case for facilitated public health infrastructure.

Background

Digital Public Health (DPH) increasingly relies on technology-mediated communication and services, but benefits depend on people's ability to access, understand, and use them. Digital health interventions can reduce inequalities when supported by digital health literacy and inclusive implementation strategies.(1)

Yet, digital health literacy remains unevenly distributed (2,3). Technology-centred public health approaches may unintentionally widen inequalities (4–6). Meanwhile, digital and analogue dissemination channels are increasingly affected by information overload, reduced attention, and

communication fatigue (7–9). Against this backdrop, interest in immersive digital environments, including web-based 3D, metaverse approaches, and Extended Reality (XR), has grown. These interactive 3D spaces can shift health communication from passive, text-heavy formats toward experiential learning and interactive storytelling (10), supporting healthcare delivery, education, patient engagement (11–13), health literacy, and community engagement (14). However, immersive systems may create navigation and self-efficacy barriers, especially as standalone tools. Evidence on hybrid models combining immersion with facilitation, and their role as regional digital public health infrastructure, remains limited.

Aims

We explore the working hypothesis that these environments achieve their full public health potential not as isolated tools, but as hybrid interventions where human facilitators (key users) bridge the gap between complex immersive technology and the diverse needs of the population.

Methods

To explore the potential of the metaverse for health education and patient engagement, this study utilized a design-oriented mixed-methods approach.

Interventions: Two metaverse-based rooms were developed:

1. **Showroom Digital Health:** Guided, narrative-based introduction to digital health concepts, including telemedicine, electronic health records, digital health applications, video-based therapy, and local/regional initiatives.
2. **SMART LIVING 360°:** Curated solutions for aging and living in place, including assistive technologies, smart-home/health applications, and links to resources on nutrition, addiction prevention, and care support.



FIGURE 1
Exemplary view of the two metaverse-based virtual rooms: SMART LIVING 360° (left) and Digital Health Showroom (right).

Access modalities: To evaluate the impact of social support on technology acceptance, technical access was provided through three distinct channels, while the level of guidance was systematically varied:

- **Web-based access** (desktop/mobile browser; point-and-click navigation) with optional QR-code–based augmented reality (AR) to project selected technologies into the physical environment.

- **Virtual Reality (VR) access** using *Meta Quest 3* head-mounted displays and controllers.
- **Video-only access**, non-interactive walkthrough

The experience was mirrored on a 47-inch screen to increase bystander visibility. Participants used the rooms independently or with facilitation by a Key User providing technical support and contextual explanation. This setup enabled analysis of how social facilitation affects engagement quality, dwell time, disorientation, perceived utility, and knowledge recall.

Settings and participants: The evaluation was conducted in “Third Spaces”, defined as neutral, publicly accessible community locations outside healthcare or home settings. These were selected to reach diverse groups and enable spontaneous and intentional engagement. The study took place in health/science promotion events, neighborhood workshops, and adult education centers.

Data collection and analysis: We triangulated systematic observation, including navigation, assistance, dwell time, and engagement/frustration cues; flash interviews on perceived value, facilitator role, and recall; and interaction analytics, including hotspot engagement and in-platform signals. Qualitative data were coded using Thematic Analysis(15) and integrated with quantitative indicators to examine effective engagement, following Perski et al. (2017) (16).

Results

Across all settings, N = 84 participants engaged with the virtual environments (Showroom Digital Health n = 68, SMART LIVING 360° n = 16). Participants ranged from approximately 10 to 80 years. Approximately 85% reported limited prior experience with immersive or XR/VR-based systems.

Participants described the immersive setup as an **“attention anchor”** that drew them in and sustained engagement beyond typical booth encounters.

“It’s an attention anchor [...“Hingucker”...].

I would have walked past a flyer.” (Participant, Health Fair)

Particularly in high-traffic environments (health fairs), participants frequently approached out of curiosity. Participants often described the environment as an “entry point” into health topics:

“That looked interesting, so I came closer and then I actually learned something.” (Participant, Health Fair)

Systematic observation showed that even digitally skeptical participants were drawn to visual storytelling. Flash interviews indicated that the playful environment lowered barriers to serious health topics. However, in VR-saturated venues, such as the Long Night of Science, participants reported fatigue and preferred passive watching.

“I’ve tried so many VR things today…”

I’d rather just watch.” (Participant, Long-Night-of-Science) Clear differences emerged between standalone and facilitated use.

AQ2

TABLE 1: Comparative Observation Matrix

Indicator	Standalone Use	Facilitated Use
Mean dwell time	4.5 min	11.7 min
Participants showing navigation difficulty	~85%	~20%
Content-related questions asked	0.2 per participant	2.4 per participant
Correct recall of ≥ 1 concept (e.g. what is telemedicine)	~14%	~83%
Reported “useful experience”	~52%	~91%

Mean overall dwell time was 11.7 minutes, exceeding neighboring informational booths (<1 minute for flyers; 2–4 minutes for brief conversations). Most participants asked at least one content-related follow-up question. Without guidance, many hesitated or disengaged early: *"I didn't really know where to go next. I was afraid of clicking something wrong."* (Participant, Health Fair). Facilitated use turned the experience into social learning: *"When she explained it, it suddenly made sense...I wouldn't have explored this alone."* (Participant, Workshop)

Qualitatively, facilitation shifted engagement from interface struggle to meaning making and action orientation. Questions moved from *"How does this work?"* to *"How does this help me?"*. VR-specific barriers were salient: most VR users required assistance within the first two minutes. Bystander engagement showed weaker memorability, suggesting active navigation and facilitated interaction may support retention.

Setting differences: Health fairs maximized reach and first contact; workshops and adult education enabled deeper reflection and collective discussion. The science-fair setting showed high curiosity-driven uptake with more time for reflection.

Two additional observations emerged during field deployment (described as **exploratory signals** rather than formal findings): (i) in several sessions, women appeared more likely to approach and remain engaged when the facilitator was female; and (ii) no clear age-related pattern in openness or interest was observed.

Conclusion

Metaverse-based virtual rooms can contribute to DPH education and regional knowledge dissemination by acting as **"attention anchors"** that attract citizens and initiate dialogue around complex health topics. However, our findings demonstrate that immersive environments alone do not constitute inclusive public health infrastructures. Standalone use involved substantial navigation barriers and low knowledge recall, highlighting a risk of exclusion. Accordingly, **the Metaverse is not (yet) a public-health "self-service" solution:** across

settings, the rooms functioned most effectively as **facilitated socio-technical interventions**. Human facilitators were essential for supporting navigation, contextualizing content, and translating immersive experiences into personally meaningful understanding and locally actionable pathways.

Future work should move beyond a one-size-fits-all approach toward **target-group sensitive deployment**. Effects may differ for people with greater prior immersive experience, and content design warrants more experimentation (e.g., separate rooms for novices' vs experienced users; varying levels of narrative guidance; stronger links to regional services; interactive demonstrations).

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Evaluating the environmental impact of generative AI in healthcare: A systematic review of current practice

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Overview

Generative artificial intelligence (GenAI) has transformative potential for healthcare, however the substantial computational resources required raise concerns regarding environmental impact. Yet it is unclear how the environmental impact of these models is evaluated in clinical settings. This study assessed current practices for evaluating the environmental impact of GenAI applications within healthcare settings.

Background

GenAI models can generate original content including text and images. As healthcare systems face growing pressures from ageing populations, rising costs, and staff shortages (B. Page et al., 2024), have been promoted as solutions for increasing productivity and efficiency (Abramoff et al., 2023). However, their substantial computational requirements raise concerns regarding their environmental footprint, including energy consumption, carbon emissions and water usage (de Vries, 2023; Samuel & Lucassen, 2022). Conversely, efficiency savings may offset some environmental costs, as their environmental impact may be less than human labour (Ren et al., 2024). Despite these concerns, it remains unclear how often environmental impacts of GenAI are considered within healthcare settings (Doo et al., 2024; Samuel & Lucassen, 2022). Given that healthcare systems face the dual burden of managing climate-related health impacts whilst being major greenhouse gas emitters themselves, there is a responsibility to ensure implemented technologies are environmentally sustainable (Braithwaite et al., 2023; Lekadir et al., 2025). However, to facilitate sustainable decision-making, environmental impacts must be transparently measured and reported (Dhar, 2020). This review's primary objective was to assess how environmental impact is measured and reported in studies evaluating GenAI in healthcare.

Secondary objectives included summarising the reported environmental impact of GenAI and comparing the environmental and clinical performance trade-offs of GenAI versus non-GenAI solutions.

Methods

Adhering to PRISMA guidelines, a systematic search was executed on 10 July 2025 across MEDLINE, Embase, preprint servers (ArXiv, BioRxiv, MedRxiv) and major AI conference proceedings (NeurIPS, ICML, ICLR, IEEE) for primary research studies describing GenAI use - defined as any AI programme capable of generating original content (e.g. large language models, diffusion models, etc) - within a healthcare context, published from June 2018 onwards in English. Studies meeting these criteria underwent full-text

screening and were included if they provided a quantitative measurement or qualitative discussion of GenAI's environmental impacts. Given our aim to assess reporting practices, any metric used to measure environmental impact was included. This encompassed metrics such as carbon dioxide emissions, direct and indirect energy usage (e.g. measured electricity consumption vs computational time), water or mineral usage, and waste generation.

Quantitative reporting methods and qualitative discussions were then summarised.

Where GenAI was compared to a non-GenAI solution quantitatively, relative environmental impact was evaluated. Given high heterogeneity and low study numbers, meta-analysis was infeasible; SWiM guidelines for vote counting were followed (McKenzie & Brennan, 2019). Where multiple comparisons were made for a similar task, Mann-Whitney U tests assessed for statistically significant differences between the mean environmental impact ($p < 0.05$).

Screening and data extraction were performed independently by two reviewers, with conflicts resolved by a third.

Results

Of the 1,513 records screened, only 15 (0.99%) met inclusion criteria. Eight (53.33%) reported quantitative outcomes and all 15 (100%) included a qualitative discussion (Table 1). The most commonly reported quantitative measure was training and/or inference time (4 studies), followed by measures of model size or computational complexity (3 studies), direct energy usage (2 studies) and carbon dioxide emissions (2 studies). Qualitative discussions most commonly addressed high energy or water consumption, though many provided only general commentary rather than specific evaluation of the models studied. Few studies considered the downstream environmental impacts of GenAI implementation (e.g. saving human resource), and none performed a formal analysis.

Direct quantitative comparison of GenAI and non-GenAI solutions were performed in only 3 studies, each assessing different clinical tasks, limiting generalisable conclusions (Table 2). Across all three, GenAI was associated with greater environmental burden despite comparable or inferior clinical performance. Stammers et al. (2025) reported that GenAI models consumed substantially more energy (45.65 vs. 0.55 kWh per run; $p=0.003$) and required longer inference times (5699 vs. 62 minutes; $p=0.002$); $p=0.002$. than non-generative models for cohort identification. Similar increases in inference time were observed for GenAI models by Tait et al. (2025) and Ojemann et al. (2024) relative to BERT-based comparators (Table 2). Discriminative models frequently outperformed their generative counterparts on the clinical task: Tait et al. (2025) reported 98.67% vs. 83.33% accuracy for clinical feature extraction ($p<0.001$), and Stammers et al. (2025) reported an F1 score of 94.06% vs. 91.58% for cohort identification ($p=0.010$).

Conclusion

Despite concern regarding the environmental impacts of GenAI, few studies report these outcomes or consider them in the discussion. Where reported, there is a lack of standardisation, with a range of metrics used. The small number of head-to-head comparisons between GenAI and non-GenAI approaches restricts generalisable conclusions; nevertheless, it is noteworthy that non-GenAI solutions demonstrated similar or superior performance with substantially lower resource requirements, motivating exploration of non-generative alternatives where appropriate. Ultimately, it remains unclear how the environmental impact of GenAI in healthcare should be evaluated and reported, and by who - underscoring the need for standardised reporting guidelines and clearer expectations from regulators and commissioning bodies on the required level of environmental impact assessment.

TABLE 1: Summary of included studies and their environmental impact reporting

Study	Setting	Healthcare Domain	GenAI Type	Clinical Tasks	Methods	Comparator	Quantitative GenAI vs Non-GenAI Comparison?	Environmental Impact: Quantitative Metric(s) Reported / Qualitative Discussion Focus	Summary of Environmental Impact Finding/Discussion
Quantitative Metrics Reported									
Doo et al. (2024)	USA	Radiology	Open-source GPT LLMs (LLaMA2 variants, Vicuna 1.5 variants; 5 model sizes)	Disease label classification from chest radiograph reports	Investigated the trade-offs between classification accuracy and energy consumption when utilising different LLM models and sizes to identify disease labels within a publicly available chest radiographic data set.	Comparison within GPTs.	No	GPU energy consumption (kWh); Energy Efficiency Ratio (accuracy/kWh).	Vicuna 1.5 7B was the most energy efficient (0.13 kWh) and had the highest efficiency ratio (737.28). Smaller fine-tuned models achieved superior accuracy and sustainability.
Gao et al. (2023)	USA	Clinical Medicine	T5 Seq2Seq Transformers (T5-Base, T5-Large)	Clinical NLP benchmark tasks: medical knowledge, diagnosis generation, summarisation.	Created DR-BENCH: a benchmark consisting of a series of NLP tasks which collectively are required for testing the clinical reasoning ability of GenAI models. Tested the ability of vanilla and fine-tuned versions of the T5 models on this benchmark.	Comparison within model sizes, and with/without fine-tuning.	No	CO2 emissions generated over all experiments (kg)	Total 106kg CO2 and 1 128 hours across all experiments. Qualitative discussion mentions that the cost and memory requirements for T5-Large were greater than T5-Base, but no comparison in quantitative metrics is provided.
Ojemann et al. (2025)	USA	Neurology	Open-source GPT LLMs (LLaMA2 13B)	Seizure freedom classification from unstructured clinical notes	Tested the performance of prompt-engineered Llama2 13B in extracting seizure freedom from epilepsy clinic notes and compared it against zero-shot and fine-tuned Bio+ClinicalBERT models.	Encoder-Only LLM: Fine-tuned Bio+Clinical BERT	Yes	Inference time (seconds)	LLaMA2 13B required 866-6924s depending on prompting strategy vs 23s for Bio+ClinicalBERT.
Quidwai et al. (2024)	USA	Oncology	RAG-enhanced GPT (Mistral Instruct 7B)	Clinical question answering and personalised treatment recommendations	Tested an AI intervention which works by retrieving relevant information from Multiple Myeloma literature using semantic search (the queries are embedding using an embedding model). Then the GPT large language model utilises the retrieved passages to generate responses to clinical queries.	Closed-source GPT LLMs without RAG: GPT-3.5-turbo-16K & GPT-4-32k	No	Model parameter count (indirect measure of energy use).	RAG-enhanced GPT achieved comparable accuracy to GPT-3.5/4 at 1/10 th parameter count.
Restrepo et al. (2024)	Brazil, Columbia	Ophthalmology, Dermatology, Public Health	GPT LLM (LLaMA2 7B), VLM (CLIP) + Foundation Vision Model (DINOv2)	Medical image classification (diabetic retinopathy, skin lesions), and outbreak prediction (dengue)	Compared three approaches to adapting multimodal models for biomedical tasks in low-resource settings, using three medical	1. Conventional fine-tuning using raw images and text with Vision Transformer (ViT) and BERT encoders.	No	Memory consumption; training & inference times	Embedding-based approach (DINOv2 + LLaMA2) used less memory (0.50 MB vs 747.94 MB model size) and was substantially faster (e.g. BRSET dataset training length 0.95s vs 538s) than raw

					image-text datasets.	2. Using pre-trained foundation models - either DINO v2 for images plus LLaMA2 for text, or CLIP for both - to extract embeddings, then trained a lightweight classifier on embeddings alone.			data approach.
Stammers et al. (2025)	UK	Gastroenterology	Open-source GPT LLMs (5x open-source models including LLaMA, DeepSeek, Qwen and Mistral variants)	IBD patient cohort identification from free-text clinical records.	Evaluated 15 open-source NLP algorithms for identifying patients with IBD from free-text clinical documents (gastroenterology letters, endoscopy and histopathology reports).	Various non-GenAI NLP algorithms ranging from simple regex models to encoder-only LLMs (e.g. BERT-based models).	Yes	Inference +/- training time (minutes); Energy consumption (kWh); CO2 emissions (grams).	GPT models had greater computational time, energy consumption and CO2 emissions than non-GenAI comparators.
Song et al. (2025)	South Korea	Otolaryngology	Cooperative Observation Network (CON)	Anomaly detection for middle ear disease (tympanic membrane)	Trained a CON model – based on GAN - which uses clinical audiometry data and tympanic membrane endoscopic images to identify normal vs abnormal scans.	Two alternative GANs: a standard GAN, and GANomaly – a previously published GAN variant designed for anomaly detection.	No	Computational load (FLOPs) at inference	CON achieved 0.21G FLOPs – approximately 10x lower than GAN (2.98G) and GANomaly (3.04G).
Tait et al. (2025)	UK	Oncology	Open-source GPT LLMs (LLaMA series variants, 8x models total)	Extraction and classification of oncological features from unstructured clinical notes.	Trained and evaluated a series of discriminative and generative NLP models on clinical notes from oncology patient to extract oncological features.	Models built on encoder-only LLM backbones.	Yes	Inference time (seconds)	Generative models took longer than deterministic models (988.3195s vs 94.120s). Larger models and fine-tuning both increased runtimes.
Qualitative Discussion Only									
Brown et al. (2024)	USA	Healthcare Planning	Closed-source GPT LLMs (GPT-3.5, GPT-4)	Clinical outcome prediction (hospital discharge & ICU transfer)	Evaluated the ability of traditional ML models and GPT LLMs for clinical outcome prediction tasks, when supplied with features derived from EHR patient data.	Gradient-boosting tree algorithm.	N/A	Qualitative discussion of the environmental and financial cost of implementation GPT LLMs in healthcare.	Greater environmental cost, including energy cost, of implementing and maintaining GPT LLMs compared with traditional ML models explicitly acknowledged but not evaluated. Study notes operational costs may prohibit use in lower resourced centres.
Keloth et al. (2024)	USA	Public Health	Open-source GPT LLM (LLaMA2 7B)	Social determinants of health extraction from clinical notes	Compared a finetuned GPT LLM to non-generative algorithms for the task of extracting social determinants of health from clinical notes from four US institutions.	Traditional ML models (XGBoost) & non-generative deep learning models (TextCNN, Sentence-BERT).	N/A	Qualitative discussion of high computational requirements of GenAI models.	Acknowledges that GPT LLM training and inference demands significant computational memory and processing power, leading to high energy consumption.
Kim et al. (2025)	South Korea	Emergency Medicine / Anaesthetics	GPT LLM (Y-KNOT 8B, built upon LLaMA3 8B)	Automated clinical documentation drafting (ED discharge summaries, pre-anesthetic	Retrospective evaluation of Y-KNOT – a bilingual biomedical GPT LLM. Evaluated through scores on	None	N/A	Qualitative discussion of the large computational and energy requirements of	Authors mention that they consciously selected an 8B parameter model for minimal latency and environmental

				assessments)	bilingual medical exam questions, and the quality of ED discharge summaries and pre-anaesthetic assessments (scored by physician review).			GPT LLMs.	sustainability.
Naseer et al. (2025)	Pakistan, Saudi Arabia	Elderly Care	Generative Adversarial Network (GAN) used to generate synthetic health data to train algorithms for the subsequent algorithms.	Prediction and classification of falls, health anomalies and environmental hazards in smart home settings	After training algorithm using the synthetic data generated by the GAN, the authors conducted a prospective evaluation in smart home environment with 50 elderly users over 6 months. Data came from IoT sensors incl. heart rate monitors, accelerometers and environmental sensors.	None	N/A	Qualitative discussion of the challenges of integrating models with large energy consumption in IoT systems.	The authors identified the energy consumption of continuous IoT monitoring as a key challenge in implementing these systems. Mitigation strategies discussed include adaptive sampling, low-power protocols, and edge-computing.
Nowrozy et al. (2025)	Australia	Digital Health & Data Security	Closed-source GPT LLM (GPT-4)	EHR access control & cybersecurity classification	Evaluated a GPT-4 framework, integrating GPT-4, on using 120+ use case scenarios derived from Australian legislation (Privacy Act 1988, My Health Records Act 2012, Health	None	N/A	Qualitative discussion of energy and resource consumption required by the proposed GPT-4 framework.	Study identifies high energy consumption and substantial resource requirements (high-performance computing, energy consumption) as sustainability concern for implementation.
Slendorio et al. (2024)	Italy	Oncology	Closed-source GPT LLMs (9 models incl. GPT-4, Claude 3, Gemini)	Clinical decision support (question answering)	Records Act 2001). GPT LLMs tested on questions based on a single clinical case in clinical nursing oncology. Answers graded by a multidisciplinary expert panel using a 7-point Likert scale.	None	N/A	The authors propose a criterion for determining the suitability of GenAI models for clinical practice should be a qualitative assessment of the eco-sustainability of model outputs.	All 9 models scored ≥ 6/7 on eco-sustainability Likert scale. The assessment evaluated whether models provided advice to reduce health system environmental impact, not the models' own footprint.
Villena et al. (2025)	Chile	Public Health	Open-source GPT LLMs (LLaMA series variants, 2x models total)	Clinical NLP tasks: referral triage, specialty classification, named entity recognition (NER)	Trained and evaluated a series of discriminative and generative NLP models on a dataset of Chilean national waiting list referrals, to test the performance of different fine-tuning strategies and algorithms.	Encoder-Only LLMs	No	Qualitative discussion of the large computational requirements of GPT LLMs.	The authors make a comment on the challenge that GPT LLMs like LLaMA have for healthcare uses due to their scale.

Abbreviations

BERT = Bidirectional Encoder Representations from Transformers; CNN = Convolutional Neural Network; CON = Cooperative Observation Network; CO₂ = Carbon Dioxide; EHR = Electronic Health Record; FLOPs = Floating Point Operations; GAN = Generative Adversarial Network; GPT = Generative Pretrained Transformer; IBD = Inflammatory Bowel Disease; IoT = Internet of

Things; kWh = Kilowatt-hour; LLM = Large Language Model; ML = Machine Learning; NER = Named Entity Recognition; NLP = Natural Language Processing; RAG = Retrieval-Augmented Generation; ResNet = Residual Neural Network; VLM = Vision Language Model.

TABLE 2: Quantitative Outcome Data for Studies Comparing Generative AI Models with Non-Generative AI Alternatives

Study	Clinical task	GenAI Model(s)	Non-GenAI Comparator	Environmental Impact Metric	GenAI Result	Non-GenAI Result	Effect Direction	Performance in Clinical Task
Ojermann et al. (2025)	Extracting epilepsy outcomes from unstructured free text notes.	LLaMA2 13B (GPT-based) with 5x different prompts to generate different responses (e.g. one-word response, reasoning, etc)	Encoder-Only LLM: Fine-tuned Bio-Clinical BERT	Inference time to complete 300 test clinical notes (seconds) Fine-tuning training time not included in outcome measure.	Range of median times: 866s (IQR 865-868) (one-word) to 6924s (instruction fine-tuning with 8 examples)	Median time: 23s (IQR 22-29)	Harmful LLaMA2 13B required 37.7 -301x longer inference time than the non-generative comparator.	Best GenAI classification accuracy: 74% (dates in context prompt). Best non-GenAI classification accuracy: 84% (Fine-tuned Bio-ClinicalBERT).
Stammers et al. (2025)	Identifying patients with inflammatory bowel disease from free-text clinical notes.	Open-source GPT-based models: 1. Mistral 0.3-7B 2. MA2-LLaMA3-8B 3. Deepseek-R1-Qwen2.5-32B 4. Deepseek-R1-Qwen3-32B 5. Deepseek-R1-LLaMA70B	Rule-Based Models: 1. Regex 2. Spacy Traditional Machine Learning: 3. BOW 4. TF-IDF 5. Word2Vec Encoder-Only LLMs: 6. SBERT 7. BioClinicalBERT 8. RoBERTa	Inference +/- training time to process 2752 test clinical notes (minutes) GenAI - Inference time only Rule-Based & Traditional ML - Training + Inference time Encoder-Only LLMs - Fine-Tuning + Inference time.	1. 1075 mins 2. 1080 mins 3. 4142 mins 4. 4146 mins 5. 18050 mins	1. 1.56 mins 2. 12.48 mins 3. 1.46 mins 4. 1.48 mins 5. 33.65 mins 6. 5.61 mins 7. 213.55 mins 8. 228.29 mins	Harmful GPT models were substantially slower than all non-generative comparators. Mean computational time required = 5698.6 vs. 62.26 minutes; p<0.002.	Best GenAI F1 score: 91.58% (Deepseek-R1-LLaMA70B) Best non-GenAI F1 score: 94.06% (Fine-tuned BioClinicalBERT)
Stammers et al. (2025)	As above	As above	As above	Energy consumption required to process 2752 clinical notes (kWh) Inference +/- training analysis as above.	1. 4.54 kWh 2. 4.55 kWh 3. 27.75 kWh 4. 27.78kWh 5. 163.65 kWh	1. 1.56 mins 2. 12.48 mins 3. 1.46 mins 4. 1.48 mins 5. 33.65 mins	Harmful GPT models used more energy than all non-generative comparators.	As above
						5. 6.61 mins 7. 213.55 mins 8. 228.29 mins	Mean kWh = 45.65 vs. 0.55 kWh per run; p=0.003	
Stammers et al. (2025)	As above	As above	As above	CO2 emissions generated to process 2752 clinical notes (grams per kWh) Inference +/- training analysis as above.	1. 938.50g 2. 942.90g 3. 5745.90g 4. 5751.50g 5. 33884.40g	1. 1.04g 2. 6.21g 3. 1.63g 4. 1.64g 5. 37.36g 7. 416.73g 8. 441.75g	Harmful GPT models generated more CO2 emissions than all non-generative comparators. Mean CO2 emissions = 28025.94 vs 114.45g; p=0.003	As above
Tait et al. (2025)	Extracting oncological features from unstructured free text notes.	1. LLaMA-3 8B (fine-tuned) 2. LLaMA-3 8B (fine-tuned) 3. LLaMA-3.1 8B (fine-tuned) 4. LLaMA-3.1 8B (fine-tuned) 5. LLaMA-3.2 1B (fine-tuned) 6. LLaMA-3.2 1B (fine-tuned) 7. LLaMA-3.2 3B (fine-tuned) 8. LLaMA-3.2 3B (fine-tuned)	Models built on encoder-only LLM backbones: 1. ArcTEX 2. RoBERTa 3. RoBERTa (fine-tuned) 4. RoBERTa (fine-tuned) + SetFit 5. BioBERT 6. BioBERT (fine-tuned)	Inference time required to process 500 test cases (seconds)	1. 2581s 2. 3195s 3. 2559s 4. 2612s 5. 988s 6. 1025s 7. 1480s 8. 1912s	1. 114s 2. 108s 3. 107s 4. 94s 5. 120s 6. 115s	Harmful GenAI models took longer at inference than non-generative models. Mean inference time required = 2044 vs. 110 seconds; p=0.002.	Best GenAI F1 score: 98.67% (ArcTEX) Best non-GenAI F1 score: 83.33% (LLaMA-3.1 8B (fine-tuned))

Abbreviations

BERT = Bidirectional Encoder Representations from Transformers; BOW = Bag of Words; CO₂ = Carbon Dioxide; GPT = Generative Pretrained Transformer; IQR = Interquartile Range; kWh = Kilowatt-hour; LLM = Large Language Model; RoBERTa = Robustly Optimised BERT; SBERT = Sentence-BERT; TF-IDF = Term Frequency-Inverse Document Frequency.

Effect Direction

‘Harmful’ = GenAI had a greater environmental impact than Non-GenAI comparator; ‘Beneficial’ = reverse. Reflects environmental outcome only, not clinical accuracy.

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Evidence quality & methodological gaps in ChatGPT – healthcare literature

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Abstract

Large language models (LLMs) such as ChatGPT are increasingly becoming transformative tools in healthcare. However, despite the rapid growth of scientific publications on the role of ChatGPT in healthcare, the reliability of the evidence supporting its clinical and educational use remains uncertain. This umbrella review aimed to evaluate the methodological quality of existing systematic reviews and meta-analyses examining ChatGPT in healthcare

conducted from inception till February 2024. Seventeen eligible studies were assessed using the AMSTAR-2 tool to evaluate the risk of bias. Of the 17 included reviews, 70.6% were rated as low quality and only 29.4% as moderate quality. The most frequent methodological weaknesses included lack of justification for study design selection, inadequate reporting of funding sources, insufficient consideration of risk of bias when interpreting findings, and poor explanation of heterogeneity. Despite widespread notion about the potential of ChatGPT, the current evidence base lacks the methodological robustness required for confident clinical translation. There is a need for higher-quality evidence synthesis in the rapidly expanding field of artificial intelligence (AI) in healthcare.

Background

The rapid emergence of artificial intelligence (AI) in healthcare has been accompanied by growing enthusiasm of LLMs, particularly ChatGPT, as tools capable of transforming patient care, clinical decision-making, medical education, and research. However, the reliability of the evidence supporting its integration into healthcare remains uncertain.

This umbrella review was therefore conducted to assess the methodological quality of existing systematic reviews and meta-analyses investigating ChatGPT in healthcare, to identify common methodological shortcomings, and to highlight gaps that must be addressed before safe clinical translation can occur.

Methods

This study was designed as umbrella review synthesizing evidence from existing systematic reviews and systematic reviews with meta-analysis. It was conducted according to PRISMA guidelines, and the protocol was prospectively registered in PROSPERO (CRD42024510926). A comprehensive search was performed across five databases: Cochrane Library, PubMed, Scopus, Web of Science, and Google Scholar, covering all publications from inception until February 3, 2024. No restrictions were placed on publication year or language during the search stage.

Inclusion criteria were peer-reviewed systematic reviews and systematic reviews with meta-analysis focusing on ChatGPT within consumer health and healthcare contexts. Reviews that were not systematic in nature, lacked a clear focus on ChatGPT, were not fully available in English, or were only available as abstracts were excluded.

Two reviewers independently screened titles, abstracts, and full texts, with disagreements resolved by consensus.

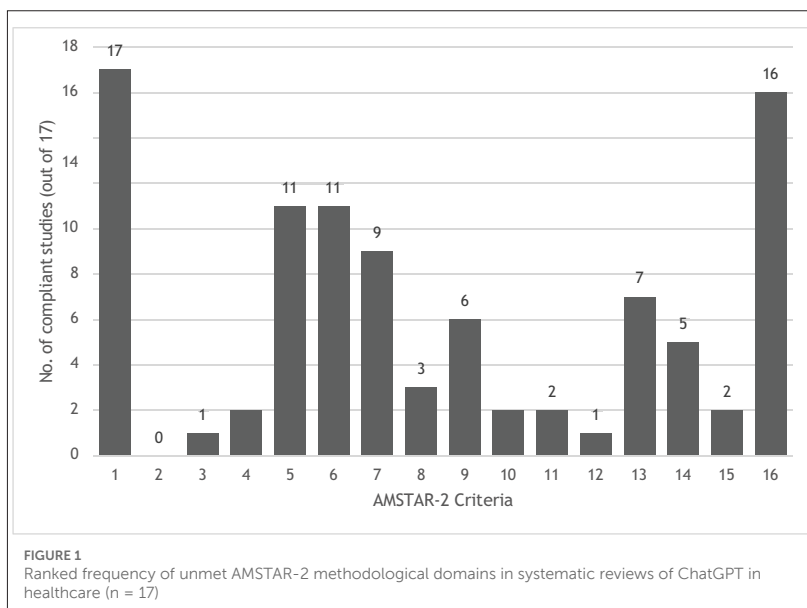
The methodological quality of the included reviews was assessed using the AMSTAR-2 (A MeaSurement Tool to Assess systematic Reviews) checklist (1), which evaluates 16 domains related to review rigor, including protocol registration, study selection processes, justification of study designs, risk of bias assessment, handling of heterogeneity, reporting of funding sources, and conflict of interest disclosures. As per the guidelines, reviews were categorized as high, moderate, low, or critically low quality based on the presence of critical flaws and non-critical weaknesses. The AMSTAR-2 appraisal was initially performed by a single reviewer (ARR) and then verified by another (AT), with both agreeing on the evaluation outcomes without any disagreement.

Results

A total of 17 reviews met the inclusion criteria (15 systematic reviews and 2 meta-analyses), all of which were published in 2023 and early 2024.

The AMSTAR-2 quality assessment revealed substantial methodological limitations across the included reviews. Only 5 out of 17 reviews (29.4%) were rated as moderate quality, while the remaining 12/17 (70.6%) were rated as low quality. None achieved a high-quality rating.

The most frequently missing or inadequately addressed AMSTAR-2 domains included: a. Lack of justification for the selection of study designs included in the reviews; b. Poor reporting of funding sources for included studies; c. Inadequate consideration of risk of bias when interpreting review findings; and d. Insufficient explanation or discussion of observed heterogeneity.



Many reviews included heterogeneous study types such as editorials, opinion pieces, commentaries, and descriptive articles alongside empirical studies. These methodological weaknesses are particularly concerning given that many reviews made strong claims about ChatGPT's transformative potential in healthcare without being supported by robust, high-quality evidence synthesis. Without robust evidence synthesis, premature clinical adoption risks being guided by enthusiasm rather than evidence. The most frequently unmet AMSTAR-2 domains are summarized in Figure 1. Overall rating of articles included is indicated in Table 1:

A quarter (23.5%) of the papers did not perform a risk of bias assessment in their included studies, which is one of the criteria to determine the quality of systematic review.

TABLE 1: AMSTAR-2 rating of the included articles (n=17)

Author, year	AMSTAR-2 rating
Garg et al, 2023 (2)	Low
Schopow et al, 2023 (3)	Low
Sallam, 2023 (4)	Moderate
Li et al, 2024 (5)	Low
Younis et al, 2024 (6)	Low
Temperley et al, 2024 (7)	Low
Badge et al 2023 (8)	Moderate
Bečulić et al, 2024 (9)	Moderate
Levin et al, 2024 (10)	Low
Muftić et al, 2023 (11)	Moderate
Klang et al, 2023 (12)	Low
Tiwari et al, 2023 (13)	Low
Roman et al, 2023 (14)	Low
Bera et al 2024 (15)	Moderate
Bugaj et al 2023 (16)	Low
Wong et al 2023 (17)	Low
Ruksakulpiwat et al 2023 (18)	Low

Conclusion

This umbrella review demonstrates that the current body of systematic reviews examining ChatGPT in healthcare is predominantly of low methodological quality. Common deficiencies in study design justification, bias assessment, heterogeneity reporting, and funding transparency limit the reliability of their conclusions.

Before ChatGPT can be confidently integrated into healthcare practice, stronger evidence synthesis grounded in rigorous systematic review methodology is essential.

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FHIR implementation for electronic medical records in indonesia: A National-Scale interoperability and digital public health governance through SATUSEHAT platform

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Citation

Aisyah, D.N., Fitria, S., Mayadewi, C.A., Rahman, F.M., Afifah, A., Faradiba, D. FHIR implementation for electronic medical records in indonesia: A national-scale interoperability and digital public health governance through SATUSEHAT platform.

Aim

The objective of this paper is to provide a comprehensive analysis of the data standardization and integration process within Indonesia's national health data ecosystem, the SATUSEHAT Platform. This study details the implementation of Fast Healthcare Interoperability Resources (FHIR) within the Indonesian healthcare context and evaluates its initial public health implications concerning coverage, equity, and system capabilities. By operationalizing FHIR as a national standard, this paper highlights the strategic transformation of fragmented health information systems into a cohesive network, demonstrating its relevance and potential scalability for other digital health initiatives.

Background

Digital health transformation is frequently constrained not by the absence of digital systems, but by fragmentation and lack of interoperability [1-3]. In Indonesia, prior to national standardization, healthcare facilities widely adopted heterogeneous Health Information Systems (HIS) that operated in silos, resulting in duplicated data entry, administrative inefficiencies, and limited data reuse for public health decision-making [4]. A 2020 survey indicated that over 60% of healthcare workers experienced redundant documentation, adding an estimated 30-40 minutes per patient encounter and reducing time available for clinical care [4]. In response, Indonesia launched Ministerial Regulation No. 24 of 2022 that mandates the use of Electronic Medical Records (EMR) and integration with the SATUSEHAT platform for all healthcare facilities [5-7]. The adoption of HL7 FHIR R4 as the national interoperability standard positioned Indonesia among the first LMICs to pursue interoperability at population scale using global standards [8].

Methods

This study employs a descriptive implementation analysis of Indonesia's national interoperability rollout, drawing on program documentation, platform metrics, and administrative data from the SATUSEHAT implementation between 2022 and 2024. The analysis focuses on three core components; master data governance, FHIR-based data standardization and integration pathway, and monitoring and evaluation process.

Master Data Governance

To establish a national single source of truth, four foundational master indices were developed and governed centrally [9].

1. **Master Patient Index (MPI):** a validated patient identity framework linked to national population registries.
2. **Master Healthcare Facility Index (MSI):** standardized identification and metadata for healthcare facilities nationwide.
3. **Pharmaceuticals and Medical Devices Dictionary (KFA):** a harmonized registry integrating regulatory datasets to standardize medicines and devices.
4. **Master Data of Healthcare Workforce:** a national index of health workforce credentials and distribution.

These master datasets provide the structural backbone for consistent data exchange across disparate local systems.

FHIR-Based Data Standardization and Integration Pathway

Clinical data elements were mapped to HL7 FHIR R4 resources. Semantic interoperability was ensured through alignment with international terminologies, including ICD-10 and ICD-9-CM for diagnoses and procedures, SNOMED-CT for clinical concepts, and LOINC for laboratory and radiology observations. The integration includes a six-step workflow: (1) prioritization by clinical use case, (2) FHIR resource mapping, (3) terminology

alignment, (4) interoperability guide and API development, (5) quality assurance testing, and (6) national rollout via a Training-of-Trainers model. Facilities were required to progress through three distinct operational phases; registered (formal enrollment), connected (sandbox testing and API access), and integrated (full terminology mapping and live data transmission).

Monitoring and Evaluation

A mixed method approach was implemented to monitor and evaluate the integration process. A national dashboard was developed by the Ministry of Health to monitor data transmission processes across facilities and regions. Desk review meetings were carried out by the MoH digital transformation team with IT vendors across the nation to obtain information related to challenges for the integration process.

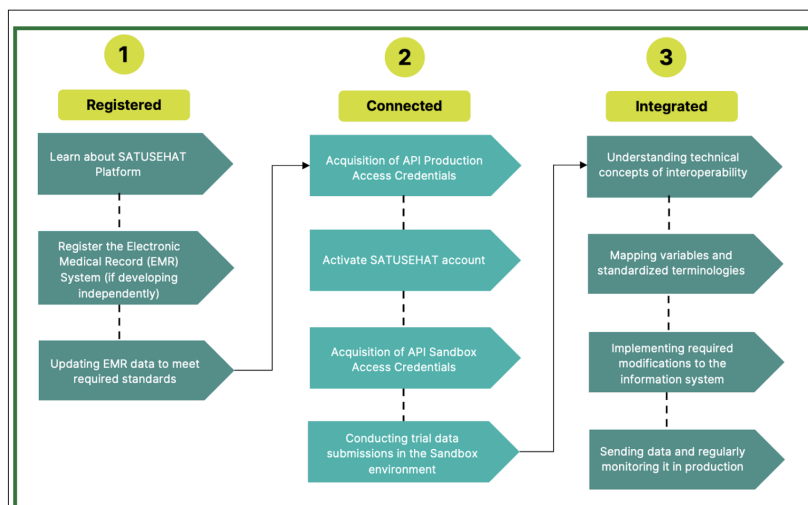
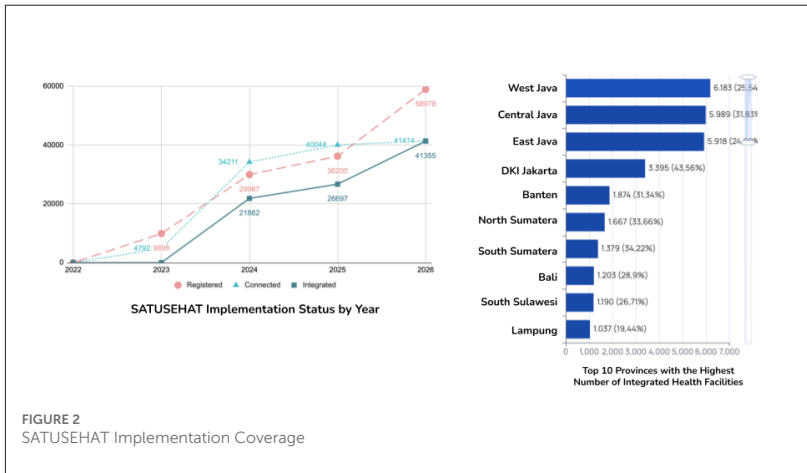


FIGURE 1
Stages of SATUSEHAT Implementation

Results

Implementation monitoring from October 2023 to March 2026 demonstrates significant progress in national adoption, alongside distinct geographic and facility-level variations. FHIR-based interoperability enabled the deployment of multiple national use-case modules, including antenatal care, immunization, tuberculosis surveillance, and registries for major noncommunicable diseases such as cardiovascular disease, cancer, and stroke. By 2024, the MPI covered 261.7 million validated patients. While MSI expanded from approximately 35,000 healthcare facilities in 2022 to over 138,000 facilities nationwide in January 2026. The pharmaceutical and medical device registries standardised more than 66,000 entries, reducing local variation in coding practices.

Out of a national target of 144,872 healthcare facilities, 58,969 facilities (40.7%) have successfully registered with the system. Progressing through the integration funnel, 57,141 facilities (39.44%) have secured production API access, while 41,385 facilities (28.57%) are fully integrated and actively transmitting standardized data to SATUSEHAT. Longitudinal tracking indicates a steady, incremental increase in integrations from late 2023 through early 2026.



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The largest volume of targeted facilities consists of smaller, decentralized entities, Independent Practices (57,050) and Pharmacies (48,193), which vastly outnumber larger institutions like Clinics (24,317), Primary Care Centers (10,348), and Hospitals (3,320). Integration success is heavily concentrated in the most populous regions. This was led by West Java (6,183), Central Java (5,989), and East Java (5,918). While regions like DKI Jakarta show strong proportional integration (43.56%), broader mapping indicates lower integration densities in eastern and more remote archipelagic provinces. Desk review meetings with 152 IT vendors identified four primary technical challenges during integration: non-compliant payload formatting (Bad Request), failure to generate authentication tokens (Unauthorized), attempts to bypass role-based access restrictions (Forbidden), and an inability to manage data queuing, which resulted in frequent rate-limit violations (Too Many Requests). This highlights the ongoing challenge of technical equity across regions and developers.

Conclusions and Public Health Implications

SATUSEHAT represents a critical achievement in addressing fragmented health infrastructure through FHIR and international terminologies. Beyond a technical implementation, the platform provides actionable public health data for national disease surveillance and healthcare delivery optimization. However, differences in adoption across facility types and geographic contexts indicate that technical interoperability alone may not be sufficient to ensure uniform public health benefits. Sustaining these gains will require continued investment in digital infrastructure, local IT capacity building, and robust privacy monitoring and data governance to ensure that technical interoperability translates into durable and equitable digital public health impact, offering valuable lessons for other LMICs.

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From burnout to resilience: Using transformative AI to optimize health workforce sustainability

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Mønter, L. From burnout to resilience: Using transformative AI to optimize health workforce sustainability.

Aims / Overview

Burnout among health and care workers is one of the most pressing threats to health system sustainability worldwide.^[1,4] While individual-level interventions remain common, they often fail to address the structural determinants of workforce distress. This paper explores how transformative AI - defined as AI that shifts health systems from reactive crisis management to anticipatory, system-level governance - can become a strategic enabler of workforce resilience. Drawing on work conducted under the WHO Europe Strategic Partners' Initiative on Data and Digital Health (SPI-DDH), it argues that the strategic value of AI in this domain depends on a foundational reorientation: from monitoring individual workers approaching crisis, toward detecting the organizational conditions that generate that crisis. This distinction - from tracking the frog to controlling the temperature of the water - defined the analytical lens of the SPI-DDH Working Group 4 process.

Background

The scale of the problem is brought into sharp relief by the WHO/Europe MeND survey (2025)^[1] - the largest survey ever conducted on the mental health of health workers in Europe, drawing on over 90,000 responses across 29 countries. One in three doctors and one in four nurses are experiencing symptoms of depression or anxiety, at five times the rate of the general population. One in ten report thoughts of being better off dead. The survey attributes this not to individual fragility but to workload and organizational cultures that have normalized overload. As WHO Regional Director Hans Kluge stated, the field must move decisively to end the culture of 'work till exhaustion' and hold health leaders accountable - not as a soft aspiration, but as a measurable responsibility.

Despite growing awareness, organizational responses remain reactive and fragmented, relying on retrospective surveys or individual coping interventions insufficient to capture dynamic workforce strain.^[4,5] Meanwhile, health systems generate vast volumes of underused data - from staffing rosters and incident reports to service utilization - that could inform proactive workforce planning. AI offers new opportunities to integrate and analyze these complex, multi-source datasets to uncover hidden patterns of stress and organizational bottlenecks.^[5,6]

Methods

This paper draws on Working Group 4 of the WHO Europe SPI-DDH. The SPI-DDH framework mandates a multivocal evidence synthesis methodology, integrating peer-reviewed literature and implementation-facing grey literature across diverse European health system contexts - a deliberate design choice reflecting the initiative's cross-national, policy-oriented nature. A comprehensive review covered 2014–2024. Over 1,200 sources were initially identified; following screening for relevance, methodological rigor, and applicability, 16 scientific articles and 4 grey literature sources were selected for in-depth analysis. Findings were synthesized to identify recurring patterns, practical use cases, and lessons for health system leaders, with particular attention to approaches addressing structural determinants of burnout.

Results

The analysis revealed three interrelated AI-enabled strategies. Before describing each, the conceptual frame is essential.

A central insight from the working group was the need to distinguish two fundamentally different intervention logics. If health workers are the proverbial frog in a gradually heating pot, most existing interventions focus on the frog – building tolerance, extending endurance, or rotating workers through recovery periods. Conventional burnout responses have implicitly followed one of three frog-focused paths: seeking workers with higher tolerance for adverse conditions (risking indirect victim-blaming); recruiting for reduced emotional sensitivity (antithetical to the empathic core of health professions); or implementing rotation schemes that extend individual endurance without altering underlying load.^[4] None addresses the organizational conditions generating burnout. Applied at the individual level, AI risks replicating this same logic. The working group's reorientation – toward systems-level data identifying the temperature, rather than individual data identifying how close to crisis each worker was – shaped both the literature review scope and the typology of promising AI applications.

First, predictive analytics support early identification of burnout hotspots by combining operational data (staffing levels, patient acuity, overtime) with workforce indicators (absenteeism, turnover, incident reporting).^[5] Oriented toward departmental aggregates rather than individual profiles, these models function as temperature sensors – estimating systemic risk and enabling proactive intervention before stress escalates.

Second, AI-powered red-zone detection and visualization tools enhance situational awareness for leaders, translating complex datasets into dashboards highlighting high-risk units and workload spikes.^[5] These provide organizational leadership with a thermometer, not a list of ailing frogs.

Third, adaptive workload redistribution systems dynamically adjust staffing and scheduling based on real-time strain indicators, directly targeting structural drivers of heat accumulation.^[5,6] Across reviewed studies, successful

implementation depended on co-creation with health workers, transparent communication about AI's purpose, and safeguards against surveillance or punitive data use – conditions with direct implications for data architecture and governance design.^[2,3]

Conclusions

AI has transformative potential in addressing burnout – but only when oriented toward the right problem. The three strategies identified are most impactful when deployed as systems-level temperature monitors, not instruments for tracking individual workers approaching crisis. The MeND survey findings reinforce this: the driver of Europe's health workforce mental health emergency lies in organizational structure and culture, not individual vulnerability.^[1]

Embedding AI within ethical governance frameworks is a prerequisite, not an afterthought. Governance mechanisms must include data minimization and purpose limitation, transparent audit processes, co-design workflows engaging frontline workers, and clear accountability structures – consistent with the EU AI Act's classification of workforce management AI as high-risk^[3] and with WHO's ETHICs principles for AI in health.^[2] Without these safeguards, AI risks exacerbating the surveillance concerns and power imbalances that erode workforce trust.

The resilience of health systems depends on the people inside them. How we treat that workforce is not a secondary concern. It is the primary one.

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From play to prevention: Digital gamification methods to combat AMR in African and European schools

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Citation

Münter, L., Sønderberg, N., Dantas, C. From play to prevention: Digital gamification methods to combat AMR in African and European schools.

Aims / Overview

Antimicrobial resistance (AMR) remains a critical global health challenge, yet the most accessible preventive measure - everyday hygiene - continues to suffer from weak engagement and limited behavioural uptake, particularly among children and adolescents. Gamified digital education has shown promise in translating awareness into action, but the field lacks robust evaluation frameworks capable of assessing not only learning outcomes, but also behavioural change, contextual relevance, equity, and scalability across regions.

This extended abstract presents an inter-regional initiative to strengthen AMR prevention in African and European school contexts by embedding evaluation frameworks at the core of a digital developer hub for gamified hygiene education. The initiative integrates participatory workshop design, co-creation of game-based tools, and theory-driven evaluation to ensure that innovation is accompanied by evidence generation, adaptive learning, and policy relevance. The overarching aim is to demonstrate how evaluation-

informed gamification can support sustainable, accountable, and scalable AMR education strategies.

Background

Despite sustained global policy attention, AMR continues to rise, driven in part by persistent gaps in hygiene literacy and preventive behaviours(1). While gamification has been widely promoted as an engaging educational approach, many digital hygiene and AMR interventions remain under-evaluated or assessed through narrow outcome measures, such as short-term knowledge gains. This limits their credibility, transferability, and integration into national or regional strategies. Game mechanics and behaviour-change targets must also be developmentally appropriate, accounting for normative exploratory behaviours across the target age groups.

Evaluation frameworks are essential to addressing this gap(2,3). Hygiene literacy research, implementation science, and behavioural evaluation literature consistently highlight the need for theory-informed approaches that link inputs, mechanisms, and outcomes across multiple levels— from individual behaviour to institutional uptake and system-level impact. In both African and European contexts, the absence of shared evaluation standards has contributed to fragmented evidence, weak feedback loops, and limited learning across initiatives.

Building on prior work conducted through the European Health Futures Forum and subsequent engagements at DPH2025 in Madeira, Digital Health Africa in Berlin, and the International e-Health Forum in Casablanca, this initiative responds to a recurring insight: innovation in gamified AMR education must be accompanied by embedded evaluation capacity. Without this, promising tools risk remaining pilots rather than becoming scalable, trusted components of AMR prevention strategies.

Methods

The proposed approach combines participatory design, digital innovation, and evaluation-informed implementation. Central to the model is the

establishment of an inter-regional digital developer hub that integrates evaluation frameworks from the earliest stages of tool development.

The methodological approach includes:

- Theory of Change development(4): Co-creation of explicit theories of change linking gamified mechanics to hygiene behaviours, learning outcomes, and AMR-relevant practices.
- Evaluation framework design: Development of a shared evaluation framework incorporating mixed methods, including quantitative indicators and qualitative methods.
- Pilot testing and Learning Loops: Iterative piloting of gamified tools in diverse African and European school settings, with embedded monitoring and feedback mechanisms to inform continuous improvement.
- Equity and context sensitivity: Integration of context-specific indicators to assess inclusivity, accessibility, digital infrastructure and device access, and differential effects across settings and populations.

This approach ensures that evaluation is not an afterthought but a core design feature guiding development, adaptation, and scale-up.

Results

The evaluation-centered approach is designed to generate several interlinked outcomes. First, the initiative will produce a minimum of five gamified hygiene education concepts that are explicitly aligned with evaluation frameworks, enabling systematic assessment of both learning and behavioural outcomes. Building on published evidence that gamified interventions can support health behaviour change in school-age children, the hub situates its design within an established evidence base rather than starting from zero (5,6).

Second, the digital developer hub will establish a shared evaluative language among designers, educators, public health actors, and funders, facilitating comparability across tools and contexts. This addresses the current absence of a pipeline for evidence-informed gamified AMR education.

Third, capacity for evaluation-informed innovation will be strengthened among youth developers and educators, enabling them not only to create digital tools but also to generate credible evidence on what works, for whom, and under what conditions.

Finally, the integration of evaluation findings into funding pathway discussions will support more sustainable investment models by aligning innovation outputs with accountability, learning, and policy needs.

Conclusions

By placing evaluation frameworks at the center of gamified AMR education, this initiative advances a more rigorous and sustainable model of digital health innovation. The proposed inter-regional developer hub demonstrates how co-creation, gamification, and theory-driven evaluation can be combined to move AMR prevention beyond isolated pilots toward scalable, evidence-informed practice.

This approach strengthens the credibility of gamified hygiene education for policymakers, funders, and education systems, while preserving creativity and local relevance. Embedding evaluation from design to deployment enables continuous learning, equity-sensitive adaptation, and strategic scale-up - positioning gamification as a legitimate and accountable component of long-term AMR prevention strategies in both African and European contexts.

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Harnessing open-source digital surveillance for vector-borne diseases within the one health framework

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Citation

Hernandez, P., Janigan, E. Harnessing open-source digital surveillance for vector-borne diseases within the One Health framework.

Abstract

Populations around the world are increasingly exposed to climate change-driven emergencies which spread climate-sensitive infectious diseases, mainly vector-borne and water-borne. Additional factors including migration, global trade, and international travel are expanding vector-borne disease (VBD) like malaria, dengue, West Nile virus, Zika and Chikungunya to non-endemic areas.

For effective control and response, the World Health Organization (WHO) promotes integrated infectious disease surveillance following the One Health approach. Yet, digital surveillance often remains fragmented, with human, animal, and environmental health data collected and managed in different systems.

SORMAS (Surveillance Outbreak Response Management & Analysis System) is an open-source digital platform for routine disease surveillance, early detection, and outbreak management. In line with the One Health framework, an environmental surveillance module was developed in SORMAS

following an agile and user-centered approach. This addition facilitates data collection from environmental samples, including vectors.

By integrating human and environment information, SORMAS enables data triangulation and accurate risk assessment, which is the basis for effective decision-making in the control of climate-sensitive diseases, including VBD. The ongoing work to integrate climate data will further strengthen the early warning capabilities of the system.

This innovative development lays the groundwork for the full integration of One Health in digital disease surveillance and outbreak response worldwide.

Background

A growing number of countries and populations are at risk of climate change-related emergencies such as floods, heat waves, droughts, wildfires and hurricanes, which contribute to the spread of climate-sensitive infectious diseases, mainly vector-borne and water-borne (1). Additional factors such as migration, global trade and international travel exacerbate the expansion and re-emergence of vector-borne diseases (VBD) in non-endemic areas (2). Mosquitos and ticks are the main responsible for the transmission of VBD such as malaria, dengue, West Nile virus, Zika, Chikungunya and Lyme disease (3).

To address these challenges, the WHO advocates, through the Quadripartite¹, for enhanced disease surveillance following the One Health approach (4, 5). However, the digital surveillance landscape remains fragmented (6), with isolated digital tools collecting human, animal and environmental health data (7, 8).

SORMAS (Surveillance Outbreak Response Management & Analysis System) is an open-source digital platform developed to support routine disease surveillance, early detection, and outbreak management. Started in 2015, it is now recognized as a Digital Public Good (9) and mature Digital Health Global Good (10), and has been implemented across diverse settings, including low-, middle-, and high-income countries (11, 12). The system enables real-time,

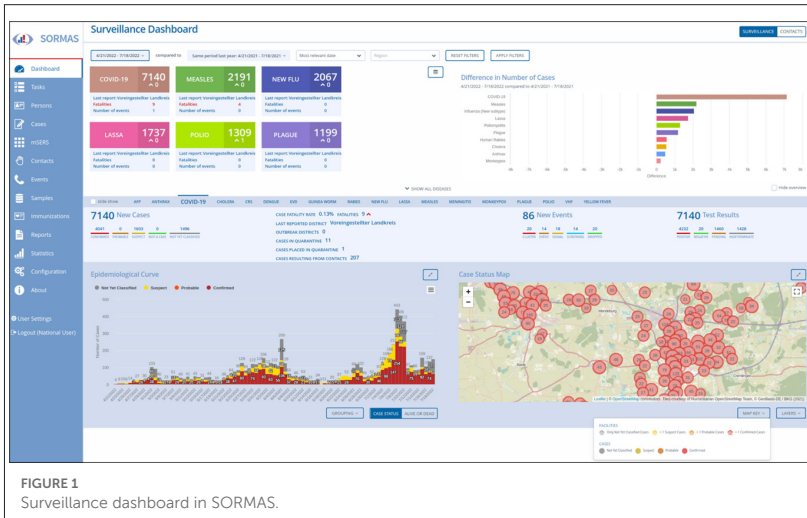


FIGURE 1
Surveillance dashboard in SORMAS.

multidirectional data exchange among stakeholders and offers disease-specific modules covering more than 40 infectious diseases. SORMAS can be used on both computers and mobile devices, with functionality available online and offline. Its users span all levels of epidemic response, from community-based personnel to national ministries of health, including staff at Emergency Operations Centers (EOCs), healthcare facilities, points of entry, laboratories, and community health services, as well as surveillance officers, case managers, and contact tracers.

Methods

An environmental surveillance module was developed in SORMAS through agile epidemiological and system requirements analysis, in an iterative, user-centered approach. Ideation and design took place through several rounds of expert consultation and feedback. GitHub issues were then prepared for software development, which are published on the SORMAS Foundation GitHub page (<https://github.com/SORMAS-Foundation>). Programming also followed an agile methodology, incorporating continuous cycles of user and

expert testing, feedback, and integration. All documentation related to the new features is publicly available on the SORMAS Foundation Wiki (<https://wiki.sorm.as/>), in line with open-source principles.

¹ Quadripartite members: Food and Agriculture Organization (FAO), the United Nations Environment Programme (UNEP), the World Health Organization (WHO), and the World Organisation for Animal Health (WOAH)

Results

The SORMAS environmental surveillance module supports planning and management of sampling of water, vectors, soil or air. It enables documentation of samples by time and GPS location, tracks the transportation of samples to the laboratory, and records diagnostic procedures and its results. Additionally, the module integrates findings with data from other sources, such as notifiable disease reporting, event-based surveillance and clinical management data. It also supports the management of response measures based on these integrated results.

As an example, in a scenario where mosquitoes are collected from a nearby forest and test positive for dengue, health authorities may want to monitor the local population closely for human dengue cases, send out public warnings, and test patients with suspicious symptoms. The environmental surveillance module would enable SORMAS users to track the dengue-positive mosquito samples and generate a risk map by combining this information with human dengue cases in the area. The system could also link relevant local events (e.g. outbreak among group who hiked in nearby forest), ultimately providing timely and accurate information to support public health decision-making.

The module will be enhanced further with additional environmental features, including the expansion of diseases and vectors available for pathogen testing and integration of climate data, which will strengthen the early warning capabilities of the system.

SORMAS

Mosquito breeding ground (U73J7W)

ENVIRONMENTS LIST | ENVIRONMENT

ENVIRONMENT ID: U73J7W-GDGP7-4QW3B-T10B06M

EXTERNAL ID:

REPORT DATE:

REPORTING USER:

INVESTIGATION STATUS:

ENVIRONMENT MEDIA:

VECTOR TYPE:

ENVIRONMENT NAME:

DESCRIPTION:

Location

COUNTRY:

REGION:

DISTRICT:

COMMUNITY:

FACILITY CATEGORY:

FACILITY TYPE:

FACILITY:

Number of characters: 0 / 1000000

Tasks
There are no tasks for this environment
[NEW TASK](#)

Environment samples
No samples have been created for this environment
[NEW SAMPLE](#)

Events
No event linked to this environment
[LINK EVENT](#)

FIGURE 2
Environmental surveillance form example within SORMAS.

Conclusions

By integrating human and environment information, SORMAS facilitates data triangulation and reliable risk assessment, to improve responses to climate-sensitive diseases, including VBDs. The ongoing work to integrate climate data will also enhance the system's early warning capabilities.

This innovative development lays the groundwork for the full integration of One Health in digital disease surveillance and outbreak response worldwide. In addition, user uptake is facilitated by the fact that SORMAS is a mature software, whose features have been developed in a user-centered, iterative approach, and is already in use at scale in LMIC and HIC settings.

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Highly personalized healing stories - the impact of generative AI on interactive narrative therapy

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Citation

Koenitz, H., Eladhari, M.P. Highly personalized healing stories - the impact of generative AI on interactive narrative therapy.

Abstract

Individual adaptation is a key element when it comes to successful mental health therapy. In terms of digital tools, many strategies for personalization have been developed throughout the last six decades which can be applied for interactive narrative therapy. Generative AI tools make many of these strategies accessible to a wider range of digital works creators and therapy practitioners. In this paper, we discuss such strategies and consider their impact in interactive narrative therapy.

Introduction

The connection between interactive narrative (Koenitz, 2023; Koenitz et al., 2015; Montfort, 2003; Murray, 1997) and narrative therapy (White & Epston, 1990) has been drawn previously (Brown & Chu, 2021; Eladhari & Koenitz, 2023; Georgieva & Georgiev, 2022; Harmon et al., 2021), yet the implications of recent developments in generative AI (GenAI) for have been less explored. A recent article questions the ability of interactive narratives to personalize the experience sufficiently to be useful for therapy (Chang

& Alvaro, 2025). These authors take the assumed lack of adaptability as an argument to instead resort to metaphors, where interactors can learn from situations which are abstract generalized fantasy versions of mental health issues such as depression. While we do see merit in the use of metaphors, we feel interactive narrative provides plenty options for personal adaptation with thoughtful design. Adding GenAI into the mix elevates these options to a new level of adaptability which offer a promising route for therapeutic applications. In the following short paper, we will provide an overview of these options and then put them in relation to steps in narrative theory.

Adaptation In Interactive Digital Narratives

Eliza, an early AI experiment of a therapy chatbot (Weizenbaum, 1966) is credited as the first digital work showing the expressive potential of interactive digital narrative (IDN) in Janet Murray's seminal book (Murray, 1997) on the topic. Eliza provided a chatbot interface and used a text parser which applied common phrases to prompt the interactor for replies, identified keywords and re-used text input by the user/interactor to produce dialogs which can be surprisingly convincing for a short time. For example, the system related the words *father* and *mother* to the category *family*. Here is an example conversation with Eliza (system replies capitalized):

Men are all alike. IN WHAT WAY
They're always bugging us about something or other.
CAN YOU THINK OF A SPECIFIC EXAMPLE
Well, my boyfriend made me come here.
YOUR BOYFRIEND MADE YOU COME HERE
He says I'm depressed much of the time.
I AM SORRY TO HEAR YOU ARE DEPRESSED
It's true. I am unhappy.
DO YOU THINK COMING HERE WILL HELP YOU NOT TO BE UNHAPPY
I need some help, that much seems certain.
WHAT WOULD IT MEAN TO YOU IF YOU GOT SOME HELP
Perhaps I could learn to get along with my mother. TELL ME MORE ABOUT
YOU FAMILY
My mother takes care of me.

WHO ELSE IN YOUR FAMILY TAKES CARE OF YOU?

My father. [...]

(Weizenbaum, 1966)

An important aspect of Eliza's successful conversations is convincing narrative design as Janet Murray points out (1997), since the software scripts the interactor to accept their role as a patients in a Rogerian therapy.

This level of adaptation – combining computational functions and thoughtful design – was possible already six decades ago. Ever since, adaptation in many forms has become standard in video games and other types of software products. Established personalization strategies run the gamut from simple text variables – e.g. letting the interactor input a name and then using it throughout, to pre-configured text snippets allowing for a flexible presentation (e.g. describing different progressions and endings), difficulty settings in games, changeable color and graphics for example for colorblind interactors, personalized music playlists in workout programs, and fully customizable avatars in video games, including personality traits and affect (Eladhari, 2014).

The Impact of GenAI

What has changed with generative AI is the ability to produce a variety of content on demand, without setting up specialized systems beforehand, including text, images, video, sound effects and music. Large language models (LLMs) can provide contextual information, for example that London, UK is in the global north, while Bogota, Colombia is in the global south. In addition, more specific information, e.g. to understand the societal circumstances under which a child grew up in Sweden in the 1980s can be provided by current leading LLMs such as Claude Opus 4.7 (Figure 1) or ChatGPT 5.5 applying a generic non-personalized prompt.

What are the significant events for a woman growing up in Sweden in the 1980s?

Synthesized Swedish 1980s history across political, cultural, and social domains >

A few clusters of events shaped that experience.

The Palme assassination (February 1986) was the defining national shock of the decade. Prime Minister Olof Palme was shot on Sveavägen in central Stockholm walking home from the cinema with his wife. For a country that had thought of itself as peaceful and exceptional, it ruptured something. The case remained officially unsolved until 2020. Anyone in Sweden at the time remembers where they were.

Chernobyl (April 1986) had unusual resonance in Sweden because it was Swedish monitoring stations at Forsmark that first alerted the world to the disaster. Fallout drifted across northern Sweden, contaminating reindeer pastures in Sápmi, mushrooms, and berries. Children of the 80s grew up being told which forest foods were safe. This came on the heels of the 1980 referendum on nuclear power, which had been a major civic mobilization.

Cold War proximity was a constant background. The 1981 grounding of Soviet submarine U-137 (the "Whisky on the rocks" incident) inside Swedish territorial waters near Karlskrona, and recurring submarine incursions through the decade, kept neutrality and defense in the news. The Berlin Wall came down in November 1989.

Gender equality politics were unusually formative in Sweden during this period. The Equal Opportunities Act and the Equality Ombudsman (JämO) were established in 1980. Parental leave continued to expand, with active policy pressure for fathers to take their share. Public childcare expansion made it normal for mothers to work. Women like Anita Gradin, Mona Sahlin, and Margareta Winberg became prominent political figures. Astrid Lindgren remained a moral authority — her 1985 essay "Min ko vill ha roligt" helped trigger animal welfare legislation.

Pop culture was rich and specific. ABBA had dissolved in 1982 but saturated everything; Roxette and Europe ("The Final Countdown," 1986) carried the late-decade torch internationally. Bergman released *Fanny and Alexander* in 1982. Children grew up with *Bamse*, *Pippi*, *Emil*, *Vi på Saltkråkan*, and *Lilla Aktuellt*. Lucia in December and Midsummer were the year's emotional poles.

FIGURE 1
Prompt and output from Claude Opus 4.7.

Image and video generation can turn textual descriptions into images and running videos on platforms like Google Veo 3 while Suno can create whole songs from a prompt. Taken together, GenAI tools provide a new level of personalization through adaptation of text, images, videos, and music.

Interactive Narrative Therapy

We take an existing adaptation (Figure 2) of narrative therapy (White & Epston, 1990) into the interactive realm as a starting point to consider the potential of personalization possible with GenAI. This framework (Eladhari & Koenitz, 2023) already acknowledges GenAI in the form of text generation and image generation helping clients to co-author and visualize their initial narrative experience. Given current technical capabilities, video and music generation should be added here. Prompts and survey questions to identify positive times in patient's lives can be individualized and positive outcomes at later stages can be projected into the future as generated narratives. Finally, GenAI analysis can help identify significant actions which helped create the new self-narrative, these can then be shared in digital form with other patients to learn from them.

Narrative Therapy	Interactive Narrative Therapy
Adopt a collaborative co-authoring consultative position.	The client as interactor co-authors their narrative experience with an IDN or IAN work. INTP help clients convey their own narratives with technical means (NLP, writing prompts/cards, drawing prompts/cards, text generation and image generation)
Help clients view themselves as separate from their problems by externalizing the problem.	The interactor position externalizes the problem.
Help clients pinpoint times in their lives when they were not oppressed by their problems by finding unique outcomes.	Prompts and survey questions help clients to pinpoint times in their lives when they were not oppressed by their problems and help in finding unique outcomes.

FIGURE 2
Excerpt of mapping narrative therapy to interactive narrative therapy (Eladhari & Koenitz, 2023). IAN = Interactive analogue narrative, INTP = Interactive narrative for therapeutic purposes, NPL = Natural Language Processing.

Conclusion

In this short overview we discuss personalization options for interactive narrative therapy by applying GenAI tools. We demonstrate that these tools can work without transferring critical personal patient data. In future work we will explore concrete applications.

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Large language models for health literacy: A comparative trustworthiness evaluation with Retrieval-Augmented generation

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Citation

Aleksandrenko, H., Komenda, M. Large language models for health literacy: A comparative trustworthiness evaluation with retrieval-augmented generation.

Background

Generative artificial intelligence (GenAI) is transforming health information delivery, offering unprecedented accessibility while introducing challenges related to confirmation bias (1) and the spread of misinformation (2). Large language model (LLM)-based GenAI chatbots are increasingly serving as sources of health information (3). However, their reliance on pre-trained knowledge without verifying the information source poses risks in health literacy contexts, where reliable and trustworthy information is essential (4).

Health information portals offer an alternative foundation for LLM-based GenAI chatbots, featuring expert-verified, continuously updated content repositories overseen by public health authorities. For instance, the Czech Republic's National Health Information Portal (NZIP) (5, 6), launched in 2020 under multi-stakeholder governance, serves multiple audiences, including citizens seeking health awareness and literacy, healthcare professionals, researchers, and policymakers requiring evidence-based health information for data-driven

decision-making (7). NZIP receives an average of 530,946 monthly visits (January–October 2025) and contains 2,151 verified articles and recommended links, 7,534 health terms, 441 games, and 401 health data outcomes.

Retrieval-Augmented Generation (RAG) enables models to generate more reliable content by leveraging external knowledge retrieval, showing promise in healthcare applications (8). However, empirical evidence comparing the trustworthiness of health information delivery between RAG-enhanced and general-purpose LLM-based GenAI chatbots remains limited (9).

Aims

This study aims to evaluate and compare the trustworthiness of in-house purpose-built

RAG-enhanced versus general-purpose LLM-based GenAI chatbots in health information delivery.

Methods

We conducted an expert evaluation study to assess the trustworthiness of four LLM-based GenAI chatbots. The NZIP AI Assistant is built on the Claude Sonnet 4 API, utilizing RAG architecture with natural language search across 3,672 verified NZIP website pages, and declines to respond when verified information is unavailable. For comparison, three general-purpose LLMs were evaluated: ChatGPT (GPT-4o), Microsoft Copilot (GPT-4), and Google Gemini (Gemini 1.5 Pro), all accessed via free-tier versions relying on broad internet training data.

Eleven expert evaluators with medical, public health, and academic backgrounds who were involved in the design, development, and maintenance of the NZIP independently assessed AI-generated responses to health-related queries in a blinded manner. Each evaluator assessed approximately six questions across all four chatbots within their area of expertise.

The evaluation framework consisted of 67 unique health questions. Each evaluator rated the responses of LLM-based GenAI chatbots on

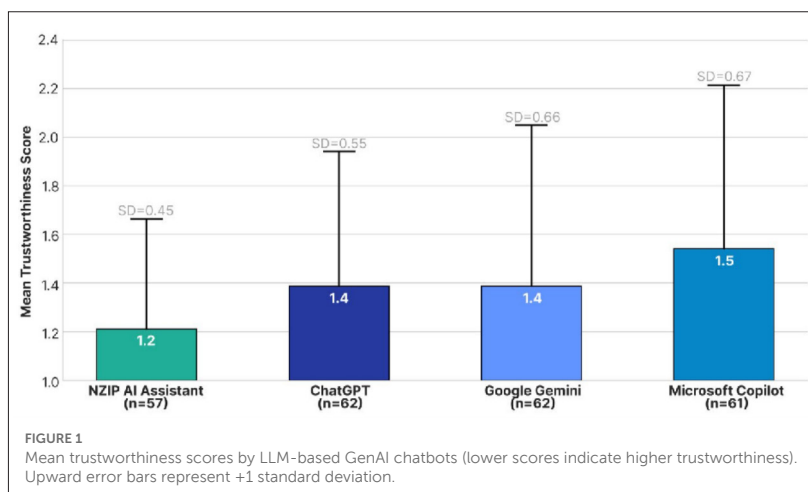
trustworthiness using a 5-point Likert scale (1 = highest trustworthiness, 5 = lowest trustworthiness), with an option to indicate an inability to assess. In total, 242 valid assessments were obtained (90.3%); 26 were excluded due to insufficient or missing responses. Trustworthiness criteria encompassed: reliability and accuracy of information, transparency of knowledge limitations, grounding in credible sources, logical reasoning, and absence of misleading or speculative claims.

To capture the multidimensional nature of health literacy, a multi-label category classification system was implemented, allowing questions to belong to multiple categories simultaneously. Three primary categories were defined: Healthcare System Navigation (25 questions), covering insurance, provider access, and healthcare procedures; Prevention & Lifestyle (28 questions), addressing health promotion, disease prevention, and wellness behaviors; and Disease & Symptoms (48 questions), encompassing clinical information, diagnosis, and treatment. As questions could belong to multiple categories, 30 questions (45%) spanned two or three domains.

Statistical analysis included descriptive statistics (mean, median, standard deviation) for overall and category-specific performance. We employed non-parametric tests, specifically the Kruskal-Wallis test, for overall comparison across all four LLM-based GenAI chatbots and category-specific Kruskal-Wallis tests to examine performance patterns within each health information domain. Statistical significance was set at $\alpha=0.05$. All analyses were conducted using R statistical software, along with the tidyverse packages.

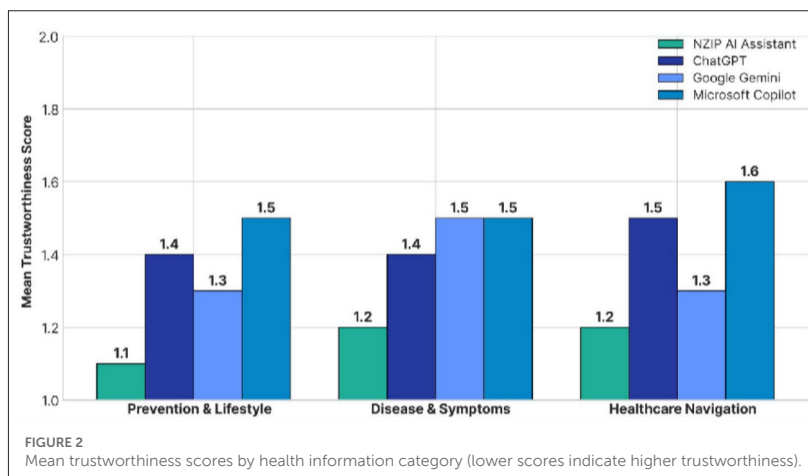
Results

NZIP AI Assistant achieved the lowest mean score of 1.2 (indicating highest trustworthiness), compared to 1.4-1.5 for other LLM-based GenAI chatbots (Fig. 1). The Kruskal-Wallis test revealed statistically significant differences in trustworthiness across the four chatbots ($\chi^2 = 9.6$, $df = 3$, $p = 0.02$). The mean performance gap between NZIP AI Assistant and other chatbots ranged from 0.2 to 0.3 points. The standard deviation for the NZIP AI Assistant (0.5) was narrower compared to other chatbots (0.6-0.7).



Distribution analysis revealed that 80.7% of NZIP AI Assistant evaluations received a rating of 1 (highest trustworthiness), compared to other chatbots: ChatGPT (64.5%), Google Gemini (69.4%), and Copilot (54.1%). Conversely, ratings of 3 or higher were observed in 1.8% of NZIP AI Assistant evaluations, compared to 3.2% to 6.5% for other chatbots. No chatbots received a rating of 5 (lowest trustworthiness).

Category-specific analysis showed consistent patterns across all three health information domains, with varying performance gaps (Fig. 2). In Prevention & Lifestyle queries, the NZIP AI Assistant achieved its highest-rated trustworthiness across categories, with a mean score of 1.1, compared to 1.3-1.5 for other chatbots. For disease and symptom information, the NZIP AI Assistant scored 1.2, compared to 1.4-1.5 for competitors, demonstrating the most consistent gap across chatbots. Healthcare System Navigation queries showed that the NZIP AI Assistant scored 1.2, compared to 1.3-1.6 for other chatbots, with the largest gap ranging from 0.1 to 0.4 points.



Multicategory questions spanning 2-3 domains simultaneously exhibited similar patterns, with the NZIP AI Assistant achieving a mean score of 1.1, compared to 1.3-1.5 for other chatbots. Within-category Kruskal-Wallis tests yielded p-values of 0.08-0.14, which did not reach individual statistical significance due to the smaller sample sizes.

Conclusions

This study provides empirical evidence that RAG-enhanced purpose-built LLM-based GenAI chatbots demonstrate significant differences in health information trustworthiness compared to general-purpose ones, offering distinct advantages in supporting health literacy goals. Consistent performance patterns across categories suggest that these advantages are generalizable, though their magnitude may vary across health literacy domains.

From a national health systems perspective, these results inform strategic considerations for implementing trustworthy AI, aligning with the World Health Organization's guidelines on Ethics and Governance of AI for Health (10) and the EU AI Act's requirements (11). Furthermore, these results suggest that national health information portals serve as a practical and reliable data

source for GenAI solutions, aligning with the European Health Data Space Regulation (12).

Declaration of AI and Ai-Assisted Technologies In The Writing Process

During the preparation of this work, the authors used Grammarly for English grammar checks. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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Large-scale adaptation and tailored implementation of evidence-based primary cancer prevention programmes: The PIECES project.

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Citation

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Aims/Abstract/Overview

The PIECES project has created a framework and a web-based tool, the Primary Cancer Prevention Implementation Toolkit (PCP-IT), aimed to support the implementation process of evidence-based interventions. The PIECES study is ongoing to evaluate the performance, working mechanisms and usability of PCP-IT in 20 implementation sites in 11 countries.

Background

Many effective interventions for primary prevention of cancer (PCP) have robust supportive evidence demonstrating that reducing exposure to preventable risk factors can reduce cancer risk. However, an effectiveness gap with respect to expected results is generally observed when these interventions are applied in practice. Reasons include individual and behavioural factors among target populations, organizational barriers, or political factors^{1,6}. Therefore, the implementation of evidence-based PCP interventions is often inadequate, compromising their effectiveness.

Tailored implementation aims to prospectively adapt PCP interventions to improve contextual fit, identify barriers in the local context, match those to evidence-informed strategies, and adapt them into workable action plans in collaboration with relevant stakeholders, as well as to apply and evaluate them. A tailored process provides a theoretical basis, planning, strategies, and processes that account for heterogeneity in the settings where the intervention will be applied. These include healthcare organization, socioeconomic characteristics of the target population, as well as the social and political circumstances, budget, and regulatory constraints, amongst others.

To this end, an online stepped approach can support the use of scientific evidence about barriers and strategies and provide methods for implementers to identify objectives, barriers to those objectives, match them to known effective implementation strategies and develop them into actionable work plans, accompanied by monitoring and evaluation plans.

Methods

The PIECES project ⁷ has developed a framework providing a systematic process that facilitates the evidence-informed identification of individual, organizational or health policy level PCP interventions and programmes and their reported effectiveness. The process supports selection and adaptation of activities according to their intended implementation setting and context, with development of tailored implementation strategies to effectively implement and scale-up PCP programmes into practice.

The PCP-IT tool has been developed based on ItFits-toolkit originally created in the ImpleMentAll project ⁸, evolved and adapted to the PCP setting; all are accessible through www.itfits-toolkit.com.

A knowledge repository of scientific evidence of effective intervention programmes in PCP, determinants of practice and implementation strategies, created by a thorough review of scientific and grey literature, informs the system on evidence-based interventions for tobacco control, second-hand smoking, alcohol consumption, physical activity, HPV vaccination, and sun exposure. Detailed information on the components of programmes with the highest level of evidence has been mapped and tailored to the target settings and populations of interest. The repository includes modules for stakeholder consultation, an online community of implementation practitioners, and structured stepped process flows, instructions, examples and worksheets for working with the materials.

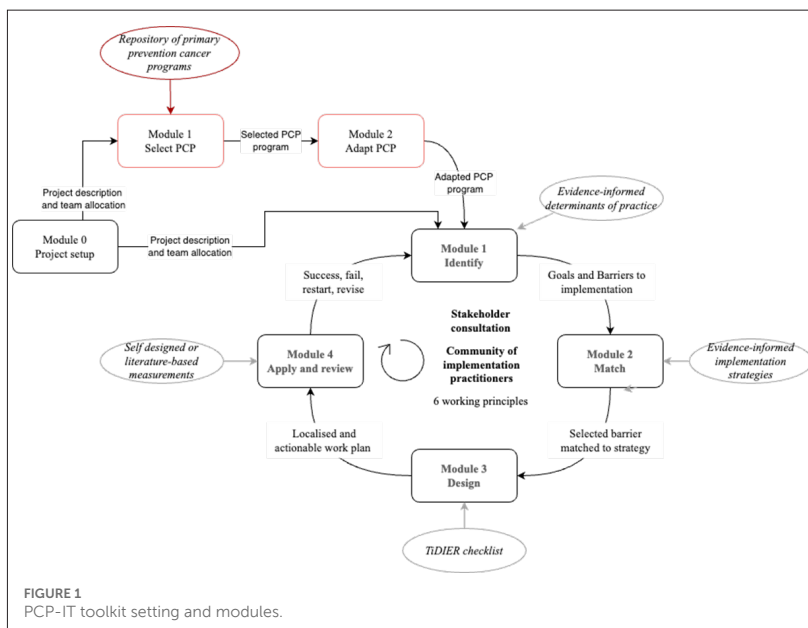
The toolkit consists of six modules designed to guide implementers through the process of selecting and adapting PCP programmes, identifying and addressing barriers, and developing and applying tailored implementation strategies.

Implementers are guided through the implementation process in 6 steps (see Figure 1):

- (1) **Select** amongst over 150 evidence-based primary cancer prevention programmes provided by the repository a program for PCP to be implemented, checking information about objectives, working principles, and the strength of proven effects.
- (2) **Adapt** the program to fit the intended target group and environment where it will be implemented; decide what can be adapted and what remains essential for intervention effectiveness.
- (3) **Identify** implementation goals, the range of barriers to address to achieve those goals, and how to prioritise these goals and barriers.

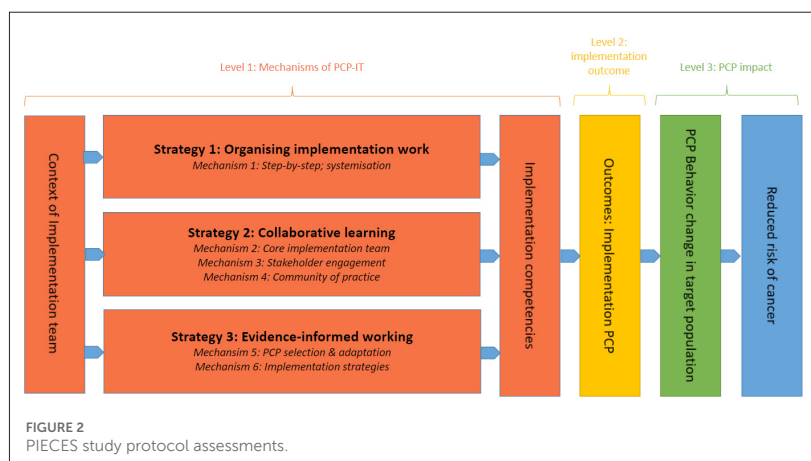
- (4) **Match** barriers with potential solutions through implementation strategies previously used to help overcome barriers in a range of situations.
- (5) **Design** an actionable implementation plan that is aligned with the needs and possibilities of the team and organisation, detailing a work plan and who does what and when, and a plan for evaluating implementation success.
- (6) **Review & apply** the implementation strategies, keep track of progress, and review the impact of implementation work.

Implementation sites can translate and run local instances of the toolkit, enabling local teams to work with the toolkit in their own language.



To evaluate the performance and working mechanisms of the PCP-IT, the online toolkit is currently being tested in practice by 20 implementation teams in Spain, the Netherlands, Albania, the UK, Italy, Ukraine, Germany, and Ireland spanning public health, clinical, occupational, educational, and community environments.

An international multi-site case comparison study (PIECES study) is ongoing to 1) generate in-depth knowledge of the performance and working mechanisms of PCP-IT according to the UK MRC guidelines for process evaluations, Proctor outcomes, and realist evaluation frameworks¹⁰⁻¹² 2) assess the usability of the PCP-IT, whether implementation teams increase their organizational readiness for implementing change, user satisfaction and determine what works for whom in using the PCP-IT to select, adapt and implement PCP programmes and 3) whether the PCP-IT is feasible and effective in improving the uptake, delivery, fidelity and sustainability of PCP programmes. Finally, some information on actual impact of the interventions can be gathered (figure 2). Full completion is expected by 2027; updated study status will be shared at the meeting.



Conclusions

A web-based tool has been created to facilitate and support implementation activity. By easing access to exhaustive evidence-based information and splitting the process of implementation into manageable tasks, the toolkit provides teams with the latest evidence-based knowledge of PCP programmes, barriers, and strategies in an accessible format for use, facilitates interaction with stakeholders outside the core implementation team, provides instructional support on how to work through the different implementation steps, and provides a forum for sharing experiences with other sites who have gone through similar processes.

The toolkit is currently being used by 20 implementation sites in 11 European countries for a range of primary cancer prevention interventions. The multi-site case comparison PIECES study will provide by the end of 2027 in-depth knowledge of the performance and working mechanisms of PCP-IT and assess whether the PCP-IT is usable and whether implementation teams are satisfied with the tool.

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Low vision assistive devices: From traditional to digital assistive technology

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Senjam, S.S., Bansal, G. Low vision assistive devices: From traditional to digital assistive technology.

Background

Vision impairment is a significant public problem due to its high prevalence and its negative impact on affected individuals. The Vision Loss Expert Group estimated in 2019 that approximately 43.2 million people were living with blindness and 295.3 million with low vision worldwide.[1,2] In India, approximately 5 million and 15 million people are blind (less than 3/60 in the better eye) and visually impaired (<6/18 to 3/60 in the better eye), respectively.[3] Historically, individuals in these groups have relied on various traditional assistive devices, such as optical and non-optical vision assistive devices, to perform activities of daily living and their routine solutions. These devices pose various limitations, for example, limitation of field of vision, focus only a fixed distance, absence of contrast enhancement features, and non-wearable. However, in the recent past the advance in the Human-Computer-Interaction make a paradigm shift in vision assistive technology, increasing significant accessibility among persons with visual impairments. Wearable AI powered smart glasses and Head-Mounted-Display vision devices based on Augmented or Virtual Reality are some of the advanced

vision devices for persons with visual impairment.[4] Such advances in technology are transforming the use of traditional visual assistive device to a truly visual free interaction using alternative body senses, such as sound, haptic, gesture. Among these devices, smartphone is one of such advanced devices that everyone of us possess, including users from low middle income countries.[5] [6]Therefore, it is worth to conduct a study of use among visually challenged individuals in low resources country rather than focusing on high end devices. There are many built-in accessible features and third-party accessible applications that enable people with visual impairment to perform daily activities, independent functioning, movement, social inclusion and participation, education, etc. Smartphone are universally designed, so they are unlikely to induce any social stigma or negative reactions from public while used by persons with visual disabilities.

Aim

To explore patterns of use and factors influencing the adoption of accessible smartphone applications among adult individuals with low vision and blindness attending a tertiary eye care vision rehabilitation clinic in north India.

Methods

In this cross-sectional study, participants with low vision and blindness aged 18 years and above were consecutively recruited from a low vision rehabilitation clinic in 2023.

Data on smartphone ownership and application use were collected through a semi-structured interview. The questionnaire was developed using a nominal technical group in which experts from relevant field were invited. Applications were classified as either specialised (accessible) or mainstream. Distance and near visual acuity were also recorded using a Snellen full lines vision chart for distance and N-notation for near vision. Data was analysed using STATA 15.1 version. Patients were categorised into two groups based on best-corrected vision acuity (binocular vision): those with $<6/18$ up to $1/60$ and those with less than $1/60$. A multivariable regression analysis was used to identify predictors of special app use. A P-value at <0.05 was considered statistically significant.

Results

In total, 115 persons with low vision and blindness, with a mean age of 26.3 years (SD+12.8), were interviewed. Among them, 28 respondents (24.6%) were dependent on family members for their daily solutions. Of the 98 mobile users, 87 (88.8%) used smartphones, and 35(40.2%) reported using one or more special apps. While InstaReader (20%), and KiBo (16%) were the most used apps by participants with vision acuity < 1/60 blindness for reading purposes, Visor app for magnification and Be My Eyes as sighted help was used by 14.7% and 6.7% participants with vision acuity < 6/18 to $\geq$ 1/60 respectively (Figure 1). Listening to music (91.5%), communication through messaging and entertainment (88.5% each), reading (80.0%), and setting reminders were among daily activities which were facilitated by specialized applications (Figure 2). Multivariable analysis showed that participants with a higher education (OR=15.57; 15.1 for graduate and above, $p \leq 0.05$), and being unmarried (OR=9.92, $p=0.01$) were predictors for the use of special apps. Higher education may provide an opportunity for digital literacy among participants, whereby use of apps is higher. Another potential reason can be students with graduate and above, digital devices are primarily used for their learning either in college or workplaces. Further, to the best of our experiences, activities such as entertainment, messaging, and listening to music are commonly undertaken by unmarried individuals. Therefore, the use of such applications may be higher among this group.

Conclusion

Many individuals with visual impairment own smartphones but do not utilize specialized applications for daily tasks, despite strong evidence of their benefits. These applications can support activities such as communication, reading, navigation, entertainment, and task management, thereby improving independence and quality of life. However, potential barriers like lack of awareness, limited training, usability challenges may hinder their adoption. To rule out this, a study is warranted in the future. Promoting the use of accessible applications through structured training, community-based interventions, and integration into vision rehabilitation services is essential. Strengthening digital literacy and ensuring user-friendly design can further

enhance uptake, enabling individuals with visual impairment to achieve greater autonomy and functional efficiency in daily living.

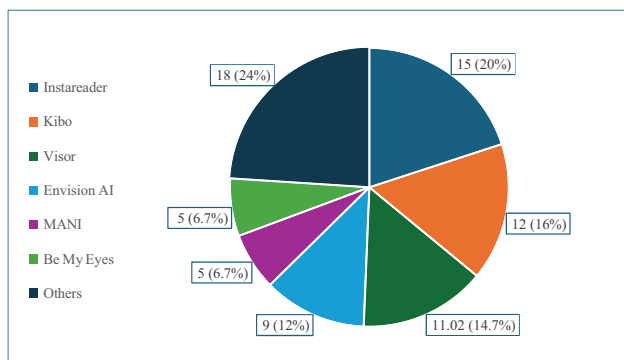


FIGURE 1
Special Applications for people with visual impairment.

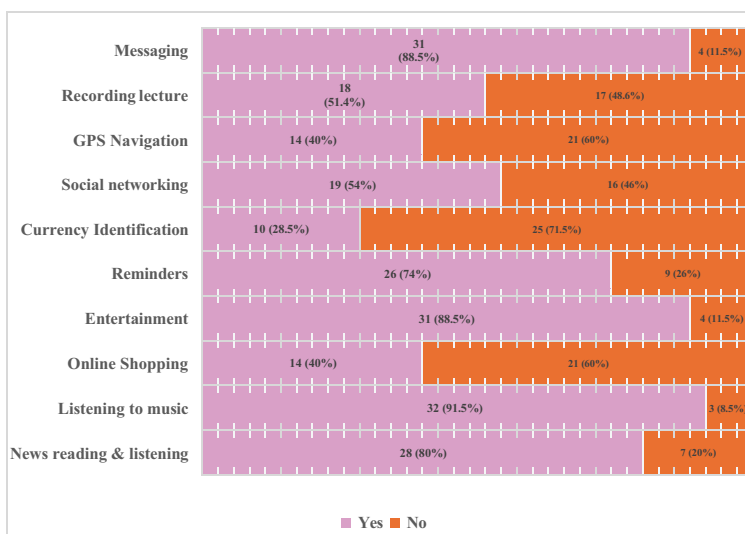


FIGURE 2
Activities facilitated by specialized applications.

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Mapping digital tools for the COVID-19 vaccination roll-out in poland: Scoping review of interventions and implementation focus

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Overview

COVID-19 vaccine uptake in Poland lagged behind the EU average, reflecting misinformation, hesitancy, and barriers to access. Evidence on how digital tools supported the roll-out has remained dispersed. This study mapped digital health interventions supporting COVID-19 vaccination in Poland. We conducted a scoping review of peer-reviewed and grey sources in Polish or English published January 2021–July 2024. Searches covered MEDLINE, Embase, Scopus, Web of Science, Google Scholar, and institutional websites. Two reviewers screened sources and charted interventions by technology, target group, and function. Interventions were categorized using the classification derived from Cataldi et al.'s framework. We identified 28

interventions. Telemedicine and e-health solutions were most common (9 solutions each), followed by online platforms and mobile apps (4 solutions each), e-referrals (3 solutions), chat support (3 solutions), and hotlines and digital certificates (2 solutions each). Process-oriented initiatives predominated (14 solutions). Patient-focused tools (8 solutions) were mainly informational; only one intervention explicitly targeted Ukrainian refugees. Digital tools mainly supported vaccination logistics, while interactive demand-side and anti-disinformation components were rare. Future strategies should add patient-centered, culturally tailored, two-way digital communication, especially for vulnerable groups.

Background

Despite the availability of safe and effective COVID-19 vaccines, vaccination uptake in Poland remained below the European Union average for substantial parts of the pandemic (1,2). In Poland, vaccine hesitancy was a major driver of the coverage gap, shaped by both individual willingness and broader system factors. It was amplified by rapidly circulating misinformation and inconsistent risk communication (3). Digital solutions, therefore, gained attention not only as a way to facilitate access and organize services, but also as scalable channels for education and infodemic management (4). A range of digital tools already embedded in the health system and expanded during the pandemic, including the Internet Patient portal and e-referrals, were used to facilitate vaccine information provision, registration, documentation, certification, and follow-up (5). Although such tools can reduce friction in the vaccination pathway and enable targeted outreach, their implementation and reach are shaped by socio-cultural factors, media consumption patterns, and structural features of the health system. However, evidence on how these tools were applied across the Polish context remained dispersed and had not been systematically mapped.

The aim of this study was to identify digital health interventions that supported the COVID-19 vaccination roll-out in Poland, with a particular focus on tools enabling communication, outreach, registration, service delivery, and follow-up.

Methods

We performed a scoping review within the EUVABECO project (6), combining peer-reviewed literature and Internet-based grey sources. We included Polish- or English-language materials published between January 2021 and July 2024. Searches covered MEDLINE, Embase, Scopus, Web of Science, and Google Scholar, complemented by systematic screening of national and international institutional websites for official reports, public announcements, and legal/regulatory documents. Search strings combined terms related to COVID-19 vaccination, communication, and digital/electronic tools in the Polish setting.

Two independent, blinded two-person review teams screened titles/abstracts and then assessed full texts. Data extraction was conducted in parallel using a pre-specified coding scheme. Disagreements were resolved through within-team discussion followed by cross-team reconciliation to reach consensus; if needed, a senior reviewer adjudicated. For each included source, full-text intervention descriptions were systematically reviewed and coded to capture (a) the technology/tool category and (b) the intended target population. Based on these coded narratives, standardized labels were identified to summarize recurring intervention patterns. In parallel, the same blinded teams independently assigned each activity to the Cataldi framework categories; any discrepancies in these assignments were resolved using the same consensus procedure (7). As a scoping review, this study mapped the availability and characteristics of interventions but did not assess their effectiveness on vaccination outcomes.

Results

We identified 28 interventions incorporating digital tools. Telemedicine and broader e-health solutions were most common (9 interventions each). Online platforms and mobile applications appeared in 4 interventions each; e-referrals in 3; chat-based support in 3; hotlines in 2; digital certificates in 2; and chatbots and standalone information systems in 1 intervention each.

Public health process-oriented interventions predominated (14 initiatives) and focused on system functions, including centralized registration/

scheduling, electronic referrals, documentation/reporting, and certification. Key examples included the Internet Patient Account (mojelKP mobile app), e-referrals, the EU COVID Certificate, and the national vaccination registration platform. Other system-supportive solutions included a WhatsApp-based virtual assistant and consolidated telephone hotlines providing COVID-19 information and navigation support.

Parent/patient-focused interventions comprised 8 initiatives and primarily reduced informational and procedural barriers through patient portals, official websites, apps, and digital communication channels. Examples included online pre-vaccination questionnaires, family-record functions within mojelKP, chat-based sign-language interpretation, and multilingual telephone information services. The keyword “information” was assigned to 5 interventions, while terms indicating motivation, persuasion, or behavior change were not observed.

Population-tailored approaches were scarce. One intervention explicitly addressed Ukrainian refugees by providing online multilingual resources, particularly for families with children.

Policy-oriented interventions (6 initiatives) focused on implementation and oversight, including digital certification and regulatory mechanisms governing eligibility and access, as well as e-referrals for foreign nationals. Most mapped tools primarily streamlined vaccination logistics and organization rather than enabling personalized communication or behavior-change support, and were centrally designed and deployed nationally.

Conclusions

Digital health tools in Poland were used mainly to strengthen operational delivery of COVID-19 vaccination, likely improving coordination and reducing procedural barriers. However, the mapped portfolio suggests an imbalanced digital response, with supply-side functions dominating over demand-side needs related to hesitancy, trust, and misinformation. Interactive, educational, and anti-disinformation components were uncommon, indicating underuse of digital channels for targeted confidence building.

Integrating Ukrainian refugees into national e-health structures was an important step toward inclusive digital public health, yet the small number of refugee-tailored initiatives suggests that heterogeneous linguistic, cultural, and access needs may have remained insufficiently addressed. Future preparedness and routine immunization strategies should complement logistics with patient-centered and culturally sensitive digital features, including tailored messaging, two-way communication, and evidence-informed approaches to counter misinformation, especially for vulnerable and hard-to-reach populations. These findings can inform future public health strategies by highlighting areas where digital tools can be optimized, particularly in addressing vaccine hesitancy and misinformation alongside logistical support.

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Mobile application for digital mosquito surveillance: Three-year deployment results and network of networks expansion in Brazil and Portugal

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Abstract

Vector-borne diseases, including mosquito-borne diseases such as dengue, Zika and chikungunya, account for more than 17% of infectious diseases and cause over 700,000 deaths annually [1]. In many endemic settings, mosquito surveillance remains paper-based, creating transcription burden and estimated 2–4 week delays between field collection and actionable data [6,7]. We present deployment results from a co-designed mobile surveillance application developed with environmental agents in Madeira, Portugal, and Cabedelo, Brazil. The Madeira deployment has supported continuous ovitrap monitoring since 2022. In Cabedelo, the system achieved a 23-second median form completion time, recorded 11,576 property inspections across 4,557 properties, and identified 357 breeding sites across 250 properties. These deployments catalysed the Vector-Borne Disease Digital Surveillance Network of Networks (NoN), launched in March 2025 to support interoperable surveillance infrastructure for scale-up to 500+ agents across four municipalities and three Brazilian states. We describe the system architecture, participatory design process, deployment metrics and NoN framework for national-scale integration with Brazil's Ministry of Health systems.

Background

Mosquito-borne diseases are endemic across Brazil and other tropical regions, with growing risk linked to climate change. Since the 2015 epidemic in northeast Brazil, our team has developed a mobile surveillance app [2,3], IoT sensors for ovitrap monitoring [4], and a dashboard for managers and policymakers [5]. The current system brings these advances into routine government use, with future deployment planned for gamification and predictive analytics modules.

Existing tools include CDC Epi Info Vector Surveillance [11], WHO/DHIS2 entomology modules [12], and integrated dengue platforms such as VITEC [13]. Recent scoping reviews show that few platforms integrate field collection, laboratory processing and management oversight within a single system [9,10]. While existing systems demonstrate the value of digital vector surveillance, they did not fully match the project requirement for a maintained, role-specific and municipality-configurable platform spanning field collection, management oversight and future laboratory integration. We therefore developed a bespoke modular system while preserving interoperability with national health systems.

The operational challenge is substantial. Madeira maintains 204 ovitraps and Cabedelo 125, each requiring routine inspection. Paper-based data collection introduces transcription errors, delays public health response, and prevents managers from viewing field activity in real time. The mobile app was designed to make surveillance data available the same day and support faster, evidence-based decisions. This evaluation assesses operational feasibility, speed and adoption; it does not claim measured reductions in mosquito abundance or disease incidence.

Methods

We used an iterative participatory design methodology consistent with human-centred design principles for public health applications [8]. First, semi-structured interviews with environmental agents in Madeira and Cabedelo documented paper-based workflows, pain points and field constraints, including device availability and network coverage. Second, requirements were translated into interactive Figma prototypes, reviewed remotely with stakeholders and refined through feedback. Third, in-person workshops in Madeira (2022–2023) and Cabedelo (2024–2025) allowed agents to test designs on target devices. Their input shaped role-based access controls, contextual help, offline functionality and manual GPS coordinate entry [2]. Co-design continued after deployment through iterative workshops in Brazil.

The mobile app uses Ionic-React with TypeScript for cross-platform deployment. The backend uses Node.js/ExpressJS, MongoDB Atlas, JWT authentication, geolocation functionality and cloud hosting. The system supports three user roles: managers, agents and laboratory users. Managers oversee assignments and analytics; agents conduct field visits and submit forms; laboratory users process samples and record egg or larvae counts. A separate Next.js manager dashboard uses the same backend.

Madeira deployment began in 2022 and provides over 36 months of operational experience. Cabedelo deployment began in October 2024, with 15+ months of data available by December 2025. This abstract focuses on Cabedelo performance metrics.

Results

Between October 2024 and December 2025, 15+ active agents in Cabedelo recorded 11,576 property inspections across 4,557 visited properties. The app identified 357 breeding sites across 250 properties, corresponding to 5.5% property-level breeding-site prevalence. Across 9,610 measurable submissions, median form completion time was 23 seconds and mean completion time was 34 seconds. Most submissions were rapid: 89.8% were completed in under one minute and 97.5% within two minutes. Forms with elapsed time exceeding 10 minutes were excluded as artefacts of agents leaving forms open between sessions.

Use increased steadily over time, peaking during mosquito season from August to November 2025 at more than 1,500 monthly inspections. Agent proficiency also improved during deployment: median completion time fell from 31 seconds in the first quarter to 19 seconds in the final quarter, while sub-one-minute submissions increased from 84% to 92%. Agents reported high satisfaction after training. Implementation challenges included poor connectivity, device compatibility, offline data caching, and managing parallel paper and digital workflows during transition.

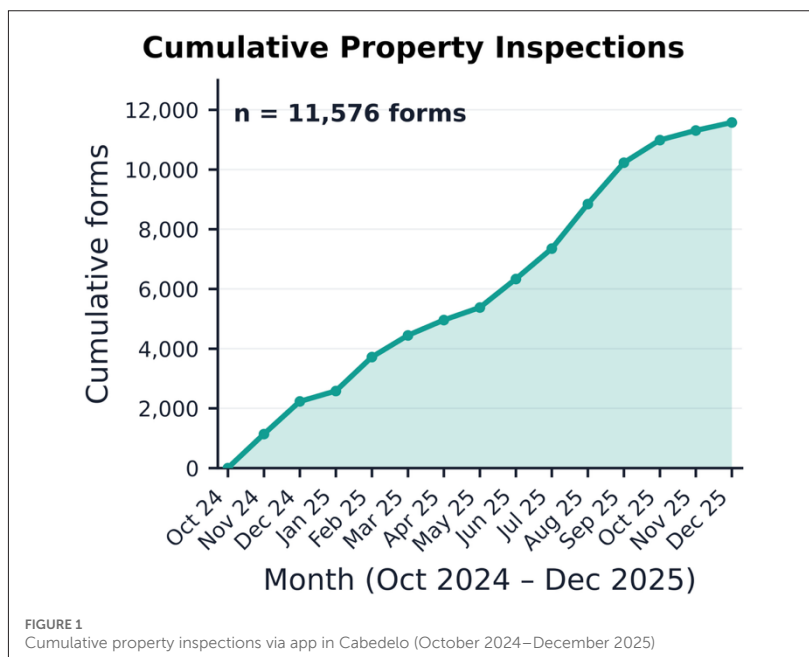
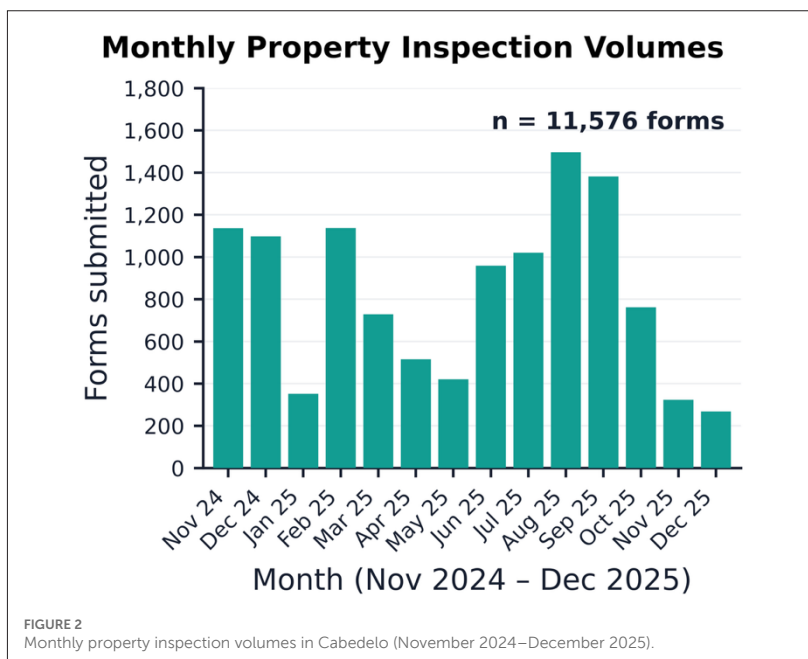


Figure 1 shows cumulative submissions increasing steadily, indicating sustained adoption. Figure 2 shows monthly inspection volumes and temporal variation, including a January 2025 holiday dip, recovery to baseline by February, lower volumes from March to May uring peak dengue incidence months, and peaks above 1,500 submissions during August–November 2025.

Future Work

The Cabedelo results contributed to the launch of the Vector-Borne Disease Digital Surveillance Network of Networks on 17–18 March 2025 in Recife, with participation from WHO and Brazil's Ministry of Health [5]. The NoN connects separate digital tools into shared infrastructure for coordinated surveillance and response. It supports protocol standardisation, technical interoperability, sustainable funding and agent training.



In 2026, we aim to scale the app to more than 500 agents across four municipalities in Paraíba, Pernambuco and Rio de Janeiro. Standardised data formats will allow aggregation across municipalities while preserving local flexibility. Preliminary dashboards show real-time maps, surveillance trends, agent activity and automated weekly reports. Planned machine learning work will combine 11,576+ field records with sensor, weather and ecological data to predict breeding hotspots 2–4 weeks ahead [4,10], enabling health teams to act before outbreaks develop. Downstream effects on intervention quality, breeding-site reduction and disease surveillance outcomes will require a separate outcome evaluation during the 2026 scale-up.

Conclusion

Three years of field use show that mobile apps built with environmental agents can shift mosquito surveillance from delayed paper reporting to

same-day operational visibility. In Cabedelo, agents completed forms in a median of 23 seconds while recording 11,576 inspections and identifying breeding sites in 5.5% of visited properties. The NoN marks a shift from isolated tools to connected infrastructure for regional coordination, with planned scale-up from 15 to 500+ agents across multiple Brazilian states. Future integration with predictive modelling may support more proactive public health interventions.

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Moderation analyses of illness perceptions and ehealth literacy on effectiveness of a digital inhaler program in asthma patients: A post-hoc analysis of the ACCEPTANCE trial

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Introduction

Digital inhalers, which combine inhalers, electronic monitoring device (EMD) and mobile applications, are used to capture inhaler use data and provide insights into medication adherence for asthma patients. Our previous ACCEPTANCE study showed a digital inhaler program improved medication adherence for 6 months and asthma control for 12 months [1]. However, the effectiveness might vary across patients due to different individual characteristics. Illness perceptions and eHealth literacy (eHL) are modifiable factors that may influence the effectiveness of digital inhalers, yet their impacts remain unknown.

Objectives

This study aims to investigate the moderation effects of illness perceptions and eHL on the effectiveness of a digital inhaler program, and associations of illness perceptions and eHL with demographics.

Methods

This is a post-hoc analysis of a 12-month cluster randomized controlled trial in Dutch primary care settings. Patients with suboptimally controlled asthma were recruited from general practices, pharmacies and social media from November 2019 to September 2022 (T-1), and entered a 6-week run-in period to get familiar with the EMD Turbu+ device for monitoring inhaler use data. After final eligibility confirmation, non-adherent patients were divided into different groups based on a 1:1 cluster level randomization (T0). The intervention group used a digital inhaler program (EMD Turbu+ device and Turbu+ Insights application) that enabled self-monitoring and provided detailed information (e.g. visualized inhaler use, reminders, missed dose and overuse messages, motivational messages, and a symptoms and triggers tracker). The control group used a passively monitored inhaler ((EMD Turbu+ device) without the Insights application). The study flowchart was shown in Figure 1. Main outcomes include: 1) medication adherence monitored continuously by the EMD, 2) asthma control measured with the Asthma Control Questionnaire (ACQ) at 3 (T3), 6 (T6), 9 (T9), 12 month (T12). Moderators were measured at baseline: 1) illness perceptions measured by the Brief Illness Questionnaire (B-IPQ) and 2) eHL measured by the eHealth Literacy Questionnaire (eHLQ). Multivariable linear regression analyses were performed using data within 6 months, including main effects (model 1) and moderation effects with interaction terms (model 2). Sensitivity analysis was conducted by including ACQ data within 12 months.

Results

A total of 164 patients were included in this study. Both the intervention and control group showed relatively positive perceptions of asthma (mean B-IPQ scores: intervention vs. control = 34.5 ± 8.3 vs. 35.6 ± 8.1) and a medium level of eHL (mean eHLQ scores: intervention vs. control = 98.6 ± 12.0 , 102.0 ± 14.9). No moderation effects of illness perception and eHL were found on medication adherence and asthma control ($p > 0.05$) (Table 1), indicating the digital inhaler program demonstrated equivalent effectiveness across different patients. Sensitivity analysis confirmed the result. Younger age was associated with higher eHL ($b = -0.27$, $p < 0.01$).

Conclusion

The digital inhaler program improved medication adherence and asthma control regardless of illness perceptions and eHealth literacy levels of asthma patients. Younger patients reported higher eHL. Given that the digital inhaler program requires minimal digital skills to navigate and features a user-friendly design, future studies should consider other potential moderators to identify who may benefit most from digital inhalers or need additional support. Meanwhile, accessibility, usability, acceptability, and inclusiveness should be prioritized during the intervention design.

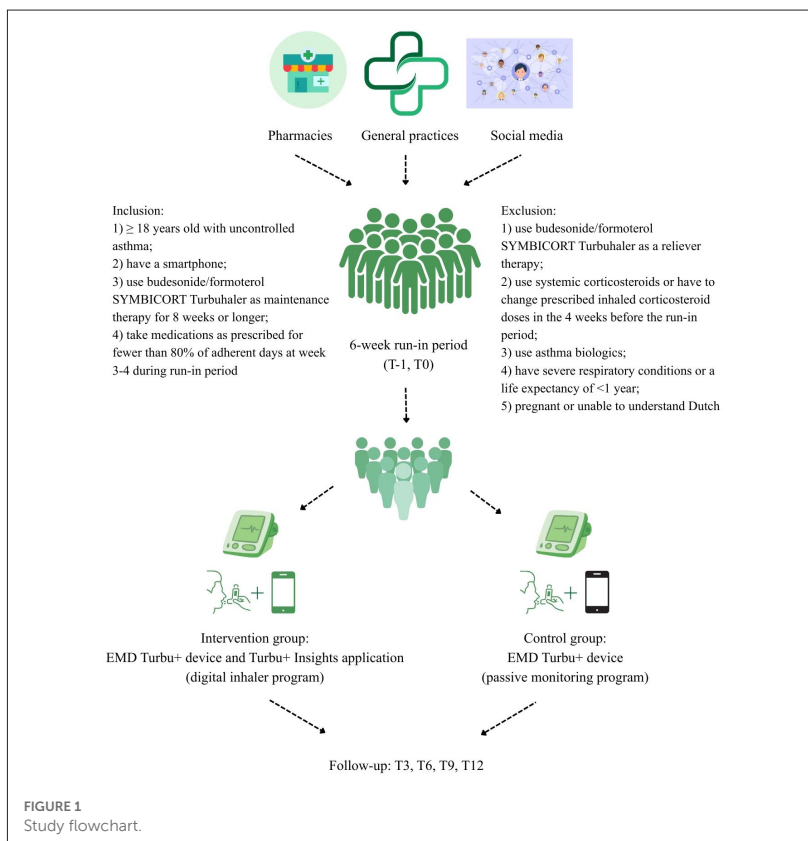


TABLE 1: Moderation analyses of illness perceptions and eHealth literacy on medication adherence and asthma control

Predictors	Medication adherence (week 2-26)				Asthma control (T0-T6)			
	Illness perceptions as moderator		eHL as moderator		Illness perceptions as moderator		eHL as moderator	
	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2
Study arm (intervention vs. control)	0.08	0.08	0.06	0.06	-0.17	-0.17	-0.22*	-0.22*
B-IPQ ^a	-0.00	0.00			0.03*	0.04*		
eHLQ ^b			-0.01*	-0.01*			-0.00	-0.00
Time (in weeks)	-0.00*	-0.00*	-0.00*	-0.00*				
Time (T3 vs. T0)					-0.17*	-0.17*	-0.17*	-0.17*
Time (T6 vs. T0)					-0.28*	-0.28*	-0.28*	-0.28*
Age	0.00*	0.00*	0.00	0.00	0.00	0.00	-0.00	0.00
Gender (male vs. female)	0.07	0.07	0.08	0.08	-0.21*	-0.21*	-0.22*	-0.22
Educational level (middle vs. low)	0.09	0.09	0.11	0.11	-0.15	-0.13	-0.26	-0.26
Educational level (high vs. low)	0.12	0.13*	0.15*	0.16*	-0.02	-0.01	-0.11	-0.11
Study arm * B-IPQ		-0.01		0.00		-0.01		
Study arm * eHLQ				0.00				-0.00

Note: numbers in the table are *bs*.

a: centered value.

*: $p < 0.05$.

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The menopause mission - digital empowerment for women's healthy ageing nina

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Aims / Overview

Menopause affects half of the global population yet remains one of the least understood and least systematically addressed phases of the life course within public health and health system design. Despite growing attention in media and policy discourse, menopause continues to be characterized by fragmented evidence, persistent stigma, and uneven access to effective care. These gaps reflect broader, structural failures in public health governance related to women's health.

This paper examines how digital health platforms can serve as catalysts for menopause awareness, empowerment, and data generation and as an entry point for broader transformation in women's health research, policy, and practice.

Background

Although menopause is a universal biological transition, its social, clinical, and public health dimensions remain poorly addressed. Many women experience significant physical, psychological, and cognitive symptoms, yet report limited understanding of menopause, difficulty accessing reliable information, and feelings of being dismissed or unheard within healthcare settings. At the same time, healthcare providers often lack formal training in menopause management, as this topic has historically received minimal attention in medical education and research. As a result, public health strategies, workplace policies, and health system planning frequently fail to account for menopausal needs, exacerbating gender-based inequities in health and well-being.

Recent advances in digital health offer new opportunities to address these challenges. Digital platforms can support awareness-raising, peer connection, symptom tracking, and access to evidence-based interventions. Importantly, they also generate new forms of real-world data that can complement traditional clinical research, helping to fill longstanding evidence gaps. However, without intentional design and governance, digital solutions risk reproducing existing biases or generating datasets that are unrepresentative and inequitable.

Methods

The approach presented in this abstract is grounded in participatory, interdisciplinary collaboration led by the EHFF Women's Health Working Group and work in the HERA project. Drawing on the HERA project framework (herawomen.eu), the design integrated co-creation and iterative learning to capture menopausal experiences across personal, social, and institutional contexts.

Key methodological elements include:

- Mapping of societal and systemic gaps: Identification of cultural, clinical, and policy-level challenges arising from incomplete understanding of menopause.

- Review of current digital solutions and the access and use of them: Exploration of existing digital platforms supporting menopause awareness, symptom tracking, CBT, pharmaceutical guidance, dietary advice, and lifestyle interventions.
- Participatory dialogues: Structured discussions that integrate lived experience with clinical and public health perspectives to surface unmet needs and priority areas.
- Equity analysis: Examination of how digital tools can contribute to more accurate, inclusive, and ethically governed datasets on menopause and women's health.
- Systems implications: Reflection on how improved menopause data and awareness can inform health system planning, workforce education, workplace policies, and public health strategies.

Results

Our review of existing digital empowerment platforms examined their support for symptom awareness, informed decision-making, and access to intervention pathways including CBT, pharmaceutical, dietary, and lifestyle options. The existing evidence landscape shows inconsistent use of clinical guidelines, little integration of health literacy or professional education, and low integration of workplace adaptation to menopausal needs (1,2).

The narrative review of published platform assessments reveals this fragmentation also (3,4). Symptoms-tracking applications and awareness platforms are well represented, but are rarely multilingual, adapted to users with lower health literacy, or drawing upon evidence-based CBT delivery. Tools are also rarely designed from an equity or data governance perspective – nor integrated into existing health systems – raising concerns of siloed and biased data.

This equity issue points to a significant opportunity also. This is also supported by the core of the vision-led architecture for women's health that

was published in The Lancet by Persson et al.(5), that converged on data and data platforms as a non-negotiable foundation for women's health system transformation and an untapped potential.

Our synthesis firmly places menopause as a public health issue, highlighting how incomplete understanding, persistent stigma, and data gaps continue to shape women's experiences and health system responses across Europe and beyond.

This has significant implications of improved menopause-related data for public health and health system planning. This includes the potential to inform clinical guidelines, strengthen professional education, improve workplace policies, and support more equitable resource allocation.

TABLE 1: Digital Menopause Platform Landscape - Gaps and Governance Implications

Platform Type	Evidence Base	Equity Reach	Data Governance Maturity
Symptom tracking apps	Moderate	Narrow (digital-literate, English-speaking)	Low - largely ungoverned
Awareness & information platforms	Low-Moderate	Moderate	Low
CBT delivery tools	Emerging	Narrow (high-literacy, high-income)	Variable
Peer community platforms	Low	Moderate-broad	Very low - privacy risks
Pharmaceutical guidance tools	Low	Narrow (HCP-mediated)	Low
Integrated life-style platforms	Low	Narrow	Low

AQ2

Evidence base = volume and quality of published efficacy evidence.

Equity reach = accessibility across literacy, language, and socioeconomic strata.

Data governance maturity = presence of ethical data frameworks, consent architecture, and equity-by-design principles. Sources: (1,3,5)

Conclusions

This paper underscores menopause as a neglected yet pivotal public health issue with far-reaching implications for health equity, system performance, and societal well-being. By centering menopause within a digital health and systems transformation framework, the EHFF Women's Health Working Group demonstrates how digital health platforms offer powerful tools for empowerment, awareness, and data generation, but their impact depends on inclusive design, ethical governance, and integration into public health strategies.

Treating menopause not as a niche issue but as a public health priority creates opportunities to improve evidence, challenge stigma, and design health systems that support women across the life course; beginning with menopause as a catalyst for change.

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Utilising digital tools to ensure high coverage of mass drug campaigns amongst never treated and hard to reach populations, critical groups to achieve disease elimination goals

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Aims

To utilise various digital innovations to support planning, monitoring and evaluation of mass drug administration campaigns, to ensure high coverage in hard-to-reach and never treated populations.

Background

There are a number of neglected tropical diseases (NTDs) outlined by WHO as targeted to reach elimination by 2030 (1). A key intervention

for preventative chemotherapy NTDs (trachoma, lymphatic filariasis, onchocerciasis, soil-transmitted helminths and schistosomiasis) include mass drug administration (MDA) often distributed house-to-house by community drug distributors (CDDs) or school-based through teachers. Many of these NTDs are primarily found in rural, low-resource areas. As programmes move to the last mile challenges, the importance of reaching never treated populations is paramount to treat any remaining reservoirs of infection (2). Often, never treated populations overlap with rural, hard-to-reach locations, areas MDA campaigns can struggle to cover adequately. Digital solutions have been critical to help identify these populations and monitor and ensure the drugs reach those that need them. These abstract outlines the various successful methodologies applied by Sightsavers and partners. Sightsavers is an international NGO supporting various Ministry of Health (MoH).

Methods

Digital technologies were designed to address specific challenges identified in MDA.

1. Improvements to real - time monitoring of MDA campaigns, ensuring supervisor's issues can be escalated and responded to as they arise and monitoring drug coverage to ensure the campaign is on track to meet minimum thresholds. This system was designed using CommCare (3) for data collection (a form digitising the supervisors checklist and a form to enter MDA treatments given by sub-district, age and gender) and a Metabase (4) visual dashboard to support management, highlighting issues arising and tracking actions and showing disaggregated progress towards coverage targets.
2. Real-time tracking of CDD locations to ensure optimal geographic coverage. Base maps were developed using satellite imagery and pre-identified resident structures aided through GRID3 (5) and visual validation. Community drug distributor GPS were tracked through ArcGIS Quick Capture and real-time visualisation of their movements mapped onto the pre-populated satellite imagery using ArcGIS Field Maps, both mobile applications developed by ESRI (6).

3. Enhanced MDA microplanning using geospatial web maps to:
 - Identify supervision gaps, based on geographical review of past supervision visits
 - Hard-to-reach areas, categorised by a walking travel time of more than four hours, based on distance from towns, topography and access
 - Review of supervision areas to ensure they are realistic and fair for all, taking into account population density, travel time and CDD numbers.

This was supported through the freely available ESPEN geospatial microplanner tool (7).

4. Monitoring supervision visits in real-time and ensuring supervisors can be directed to areas missed. This was achieved through webmaps (both ArcGIS and ESPEN microplanner) that integrated with CommCare, through an application programming interface (API). Data related to supervisor locations submitted from CommCare were extracted on a daily basis, automatically updating the maps with communities that had been supervised or were within one hour of a supervisor visit (marked as green) and highlighting those that had not yet been supervised (marked as red). Priority areas (hard-to-reach) identified during the micro-planning process were also demarcated on the map, to track if they were visited by a supervisor.

Results

These approaches have been deployed completely or in part across a number of sub-Saharan African countries, including Kenya, Guinea-Bissau, Nigeria and Liberia. This has shown improved results in MDA coverage, especially in previous populations with poor coverage, increased geographic reach, including hard-to-reach areas and stronger supervision allowing for improved course corrective actions.

In Kenya, there were challenges of persistence of trachoma infection amongst the nomadic Maasai community in rural Kajiado County. Improved real-time monitoring and deployment of the CDD tracking system (method 1 and method 2) resulted in an increase in MDA coverage in Kajiado West sub-county from 56.9% (42.9–69.9) in 2021 to 88.2% (78.9–93.7) in 2023, as evidenced by coverage evaluation surveys. With an even larger improvement of 85% in coverage amongst children under five, who were previously inadequately reached, and a drop in who were never treated from 29.1% to 10.6%. Figure 1 below shows the improvement in geographic coverage in a ward in Kajiado South sub-county over time with areas previously missed identified (black outline) and reached more efficiently in future campaigns. Improved accountability and planning also resulted in deployment of more optimal CDD numbers and more effective movement plans, ensuring better reach of previously missed areas.

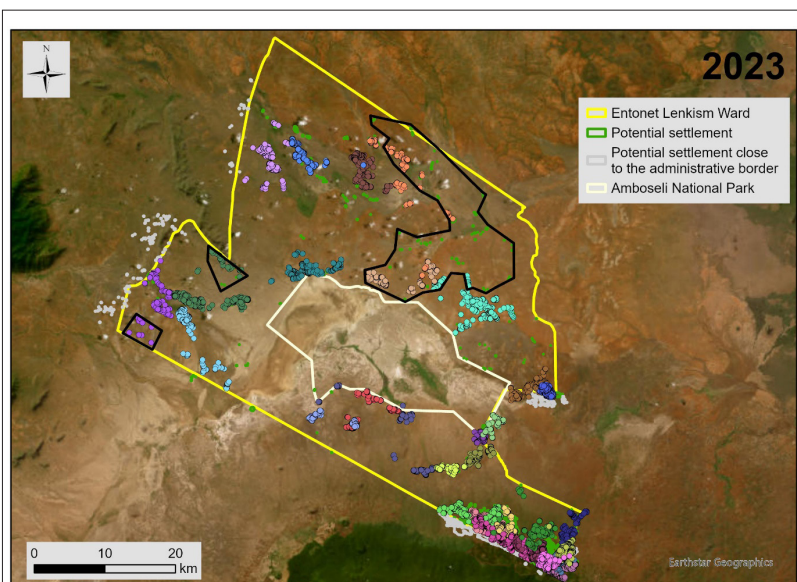


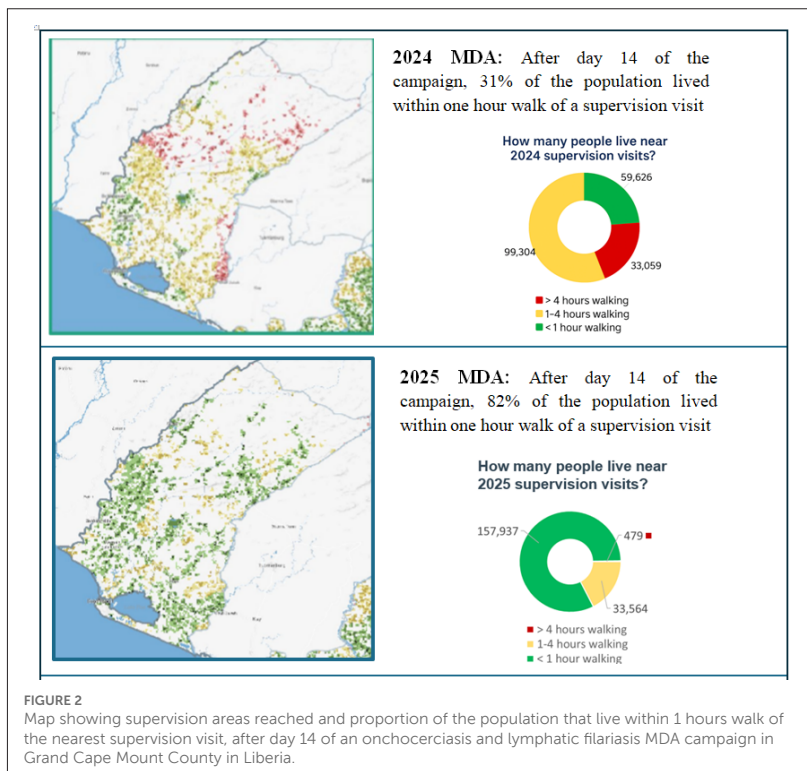
FIGURE 1
Distribution of the GPS data collected by the CDDs during the 2023 MDA campaign in Entonet-Lenkism Ward. Black areas are where the MDA campaign did not reach in 2022

In Gabu and Bafata health districts in Guinea Bissau, enhanced MDA microplanning using the ESPEN Geospatial Microplanner and online ArcGIS webmaps (methods 3 and 4) integrated with the real-time monitoring of supervisors was implemented. The supervision gaps and hard-to-reach areas were identified by MoH supervisors for priority supervision, with 64 communities visited for the first time during the most recent MDA, as evidenced on the bespoke ArcGIS online webmaps. Supervision areas assigned to supervisors was also realigned to make them fairer in terms of work-load and a feasible geographic location to cover.

Initial results in Liberia, where a similar approach was employed, indicate that daily review of the updated supervision gap analysis can improve supervision coverage, especially in rural areas. In Grand Cape Mount County in Liberia, the proportion of the population living within one hour of a supervision visit grew from 31% to 84% after the tool was introduced, and the proportion of the population living more than four hours away from the nearest site shrank from 17% to less than 0.5%, figure 2.

Conclusions

Millions of treatments are provided globally every year through mass drug campaigns, yet the planning and real-time oversight, is often inadequate. Important populations are often missed as they are harder to reach and persons go never treated, who are possible sources of continued infection critical to reach in elimination campaigns or vaccination programmes. Digital solutions as outlined in this abstract are essential, often low-cost tools, that need to be embedded in all MDA campaigns, to ensure effective coverage and reach to these special populations. It further ensures quality assurance and allows timely corrective actions to be achieved if and when issues do arise.



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Posters & demos track

AI preparedness in Low-Resource systems and why digital health governance matters: A case study of ghana's lightwave health information management system

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Ama-Gyaa, A.M., Joachim, A., Akosua, F. AI preparedness in low-resource systems and why digital health governance matters: A case study of Ghana's lightwave health information management system.

Background

As artificial intelligence (AI) is increasingly integrated into digital health systems in low- and middle-income countries (LMICs), governance has emerged as a critical determinant of whether AI strengthens equitable, people-centred public health or amplifies existing structural weaknesses. The World Health Organization (WHO) Global Strategy on Digital Health 2020-2025 identifies governance, accountability, transparency, and trust as foundational enablers of digital transformation. Yet in many LMICs, AI initiatives risk being layered

onto systems with unresolved deficits in data quality, institutional oversight, and legal accountability. Emerging governance maturity frameworks underscore that health AI readiness requires structured, measurable governance foundations before deployment (Hussein et al., 2026).

Ghana's Lightwave Health Information Management System (LHIMS) was introduced to strengthen health data management across public facilities. However, its rollout exposed critical structural weaknesses in four domains: interoperability standards, vendor dependence and subsequent lock-in, data stewardship and sovereignty, and institutional capacity. These deficits, if unaddressed, create a fragile and inequitable foundation for AI deployment in public health settings.

Aims and Objectives

This study examines how digital health governance informs strategies for AI preparedness in low-resource health systems, using Ghana's LHIMS as a case study. Specifically, the study:

- Examines existing governance structures against WHO digital health principles
- Assesses data management practices and interoperability readiness
- Analyses documented implementation evidence from LHIMS rollout to infer systemic governance conditions
- Identifies structural factors that facilitate or hinder responsible AI integration
- Derives actionable, inference-grounded recommendations for governance reform based on documented outcomes

Methods

This study employs a qualitative documentary and policy analysis design. No primary data collection or human subjects research was conducted. The analysis draws exclusively on publicly available secondary sources,

including Ghana's national digital health policy documents, official LHIMS implementation and rollout reports, WHO digital health governance frameworks, and peer-reviewed

literature on AI governance in health systems. Governance gaps and systemic risks are inferred from documented implementation outcomes, policy text, and contextual evidence within these sources. Thematic analysis was applied to organise findings across key governance domains, mapped against WHO digital health governance principles with particular attention to accountability, data stewardship and sovereignty, vendor dependence, and institutional capacity. This is a systems-level documentary analysis. It does not evaluate individual performance, attribute fault to specific actors, or represent the views of any institution or government body.

Results

Analysis of LHIMS implementation documentation revealed inferred governance gaps with direct implications for AI preparedness across six domains (Table 1).

TABLE 1: Governance gap assessment - LHIMS mapped against WHO digital health governance principles and AI preparedness risk

WHO Governance Domain	LHIMS Status	AI Preparedness Risk
Data stewardship and sovereignty	Weak	Ambiguous authority over secondary data use for AI model development
Vendor accountability and transparency	Absent	Proprietary architecture limits external validation and national oversight of AI functions
Interoperability standards	Weak	Fragmented, inconsistent datasets increase risk of biased AI outputs
Accountability pathways	Absent	No clear responsibility when AI-informed decisions affect health outcomes
Ethical oversight	Absent	No mechanism to review or audit AI-enabled clinical decision support
Workforce AI capacity	Partial	Insufficient competency to govern, audit, or critically evaluate AI tools

Available implementation documentation suggests that ambiguities in data stewardship created uncertainty over legitimate authority for secondary data use, including AI model development, with inferred risks to patient autonomy, privacy, and informed consent. Evidence of vendor lock-in under LHIMS, indicated by proprietary system architecture and restrictive contractual arrangements documented in rollout reports, points to reduced transparency, limited external validation, and weakened public sector autonomy over digital health priorities. Vendor lock-in in digital health systems represents a particularly acute risk in LMIC contexts, where negotiating leverage is limited, procurement oversight is variable, and technical capacity to independently audit or migrate away from proprietary systems is constrained. When a vendor controls system architecture, data formats, and access protocols, the public sector loses practical authority over its own health data infrastructure. This dependence extends beyond cost implications: it determines who can interrogate AI-enabled functions, who can validate algorithmic outputs, and who can intervene when system performance diverges from public health priorities. These structural conditions are inferred to have undermined accountability mechanisms and performance monitoring capacity at both facility and national levels, creating a governance environment in which AI integration would amplify, rather than resolve, existing institutional vulnerabilities.

Documented gaps in oversight structures raise fundamental questions about accountability when AI-informed decisions influence clinical or public health outcomes. Without clearly defined accountability pathways, as inferred from the LHIMS governance architecture, patient protection and public trust are structurally at risk. Critically, AI systems trained on the incomplete and inconsistent datasets characteristic of LHIMS-type implementations may reproduce or amplify existing inequalities in health access and outcomes. In low-resource settings where data gaps often reflect structural disparities, poorly governed AI integration risks embedding bias into clinical decision-support tools and predictive systems, disproportionately affecting already marginalised populations and undermining the equity commitments central to Sustainable Development Goal 3 (SDG-3).

Conclusion

The LHIMS experience demonstrates that digital health systems without robust governance serve as a fragile and high-risk foundation for AI in public health. Governance gaps do not merely slow AI adoption; they actively increase the risk of biased, inequitable, and unaccountable AI outputs. For LMICs globally, governance readiness is a prerequisite, not a parallel track, for ensuring that AI strengthens health system resilience, equity, and public trust. Ghana's experience offers transferable lessons for global policy, underscoring the need to align digital transformation with people-centred, ethically grounded AI governance frameworks.

Key message: Governance readiness is not a parallel track to AI adoption — it is the prerequisite. Without it, AI integration in low-resource health systems risks amplifying the very inequities it is designed to address.

Recommendations

Seven recommendations emerge directly from the inferred governance findings of this documentary analysis (Table 2). Each recommendation is grounded in a specific structural gap identified through the LHIMS experience and is intended to offer transferable, actionable guidance for health ministries, digital health programme managers, and policy makers in comparable LMIC settings preparing to integrate AI into public health systems.

TABLE 2: Recommendations grounded in LHIMS governance findings

#	Recommendation	Rationale from LHIMS Experience
1	Institutionalize digital health governance	LHIMS lacked a central governance body with defined authority over procurement, data, and vendor relations
2	Strengthen data management and interoperability	Fragmented data infrastructure across roll-out facilities produced inconsistent datasets unsuitable for AI training
3	Implement continuous monitoring	Governance gaps in LHIMS went unaddressed because no

#	Recommendation	Rationale from LHIMS Experience
	and adaptive policy feedback	systematic performance review mechanism existed
4	Develop a legal and regulatory framework for AI in health	Absence of AI-specific law left patient data vulnerable to secondary use without informed consent
5	Integrate AI readiness into national health strategy	AI integration was reactive rather than planned; readiness must be embedded at policy design stage
6	Build workforce AI competency	Facility staff lacked capacity to interrogate vendor outputs or identify algorithmic bias in LHIMS functions
7	Promote public-private partnerships with governance guardrails	Vendor lock-in under LHIMS eroded public sector autonomy; future partnerships require enforceable oversight clauses

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Algorithmic exposure to body image content on TikTok: A Mixed-Methods international study with implications for mental health and platform accountability

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Citation

Branley-Bell, D., Murphy-Morgan, C., Moore, L. Algorithmic exposure to body image content on TikTok: A mixed-methods international study with implications for mental health and platform accountability.

Overview

Social media platforms, such as TikTok, represent a significant and rapidly evolving frontier for public health. With billions of users globally, TikTok's recommendation algorithm curates personalised feeds that can expose individuals to a spectrum of body image content - from body acceptance material to content promoting extreme thinness, fitness ideals, and pro-eating disorder (pro-ED) messaging. This study investigates the extent and nature of algorithmic exposure to body image content on TikTok, its psychological impacts, and the coping strategies users employ, aiming to identify actionable implications for digital public health interventions, platform accountability, and policy.

Background

TikTok is a dominant platform for body image, diet, and fitness content, leveraging short-form video to engage billions of users worldwide. The platform's algorithm creates personalised feeds that can expose users to idealised body imagery and weight-related messaging [3,4,6]. Body image content on TikTok encompasses a spectrum from harmful to health-promoting material, reflecting four primary body ideals: the *thin ideal* (striving for a slender physique, often emphasising weight loss and calorie restriction), the *fit ideal* (pursuing a muscular or toned body, often through intense exercise), *body purity* (focusing on clean eating, detoxification and lifestyle perfectionism), and *body acceptance* (promoting self-compassion).

Whilst body acceptance content can foster supportive communities and challenge unrealistic standards [1], exposure to thin and fit ideals has been linked to internalisation of appearance standards, social comparison, body dissatisfaction, and disordered eating [5]. The body purity movement, though framed as wellness-oriented, can promote orthorexia and moralise food choices [2]. Most concerning is the documented presence of pro-eating disorder (pro-ED) content that actively encourages restrictive eating, excessive exercise, and extreme weight loss, which TikTok's algorithm may inadvertently amplify to vulnerable users, raising critical questions about platform accountability and user wellbeing.

This study examines: (i) the degree to which users encounter content promoting body ideals; (ii) whether users actively seek this content or encounter it through algorithmic curation; (iii) impacts on psychological wellbeing; and (iv) coping strategies users employ to avoid distressing content.

Methods

A mixed-methods online survey recruited 335 international TikTok users via Prolific (99 male, 231 female, 5 non-binary; aged 18–66 years, $M=28.34$ yrs, $SD=9.32$ yrs). Participants averaged 8.4 hours per week on TikTok ($SD=9.02$ hrs), and the majority (95.2%) had used the platform for more than one year. Pressure to conform to societal appearance ideals was measured

using the Sociocultural Attitudes Towards Appearance Questionnaire for Social Media (SATAQ-SM). The survey assessed body image-related content exposure, reasons for exposure, psychological impacts, and coping strategies. Quantitative data were analysed using multiple regression and independent samples t-tests; qualitative open-ended responses were thematically analysed to capture user experiences, algorithmic awareness, and coping mechanisms.

Results

Nearly all participants (>99%) identified TikTok as reinforcing specific body ideals: thinness (20%), muscularity (31.6%), body purity (26.9%), and body acceptance (34%). Over one-third (36%) reported that exposure had directly affected their own body perception. Regression analyses revealed that exposure to thin ($\beta=.141, p=.02$) or fit ideals ($\beta=.198, p=.002$) significantly predicted greater pressure to conform, whereas exposure to body acceptance content was associated with reduced pressure ($\beta=-.118, p=.036$).

Participants were almost evenly divided between those who actively searched for body image, diet, or fitness content (49.3%), and those who encountered it passively through algorithmic curation (51.7%). Motivations for active searching ranged from health-orientated ("to get healthier") to comparison-driven ("to compare my body with those of models and see where I can improve").

Most concerning, one-third of all users (33%) had been exposed to pro-ED content, despite only 6% intentionally seeking such material. Those exposed scored significantly higher for pressure to conform ($p=.007, d=.29$) and reported a stronger impact on their own body perception ($p<.001, d=.37$). Participants expressed frustration that the algorithm surfaced this content following engagement with benign material (e.g., recipes), underscoring the involuntary nature of much harmful exposure. Amongst the small subset (6.3%) who actively sought pro-ED content, motivations included reinforcing existing eating disorders, but also curiosity, educational interest, and "pushing back" by reporting videos.

Participants demonstrated awareness of algorithmic influence but felt limited agency. Coping strategies included engaging with alternative content to “retrain” the algorithm and “mindful scrolling” – preparing mentally before opening the app to rapidly scroll past unwanted content without engaging, to avoid algorithmic reinforcement. Notably, very few participants reported using platform-provided moderation tools, and none mentioned the “reset algorithm” function. Users focused on modifying their own behaviour rather than utilising technological safeguards, suggesting potential gaps in awareness, trust, usability, or fear of unintended consequences (e.g., losing desired content, followers). This pattern warrants further investigation in the authors’ planned follow-up study.

Conclusions

These findings have direct relevance for digital public health strategy. TikTok simultaneously facilitates body-positive communities and exposes users (frequently without their intent) to harmful content, including pro-ED material. The low uptake of existing platform moderation tools points to a critical gap between available technological safeguards and real-world user behaviour, highlighting both a design challenge and a public health opportunity.

For digital health interventions, the findings support the development of evidence-based digital literacy programmes that explicitly address algorithmic awareness and equip users (including adolescents and those with existing body dissatisfaction or eating disorder histories) with practical skills to navigate and reshape their feeds. For platform governance, results provide an empirical basis for policy recommendations including: mandatory algorithm transparency, co-designed and intuitive content filtering tools, and accessible user empowerment features that mitigate harm without degrading positive platform experiences.

For clinicians and mental health professionals, these findings reinforce the importance of routinely assessing social media exposure in body image and eating disorder assessments. The data further highlight that passive algorithmic exposure – not just intentional content-seeking – constitutes a

meaningful pathway to psychological harm, and should be considered in clinical frameworks.

A 24-month longitudinal follow-up study is currently underway with the same cohort, examining changes in exposure to body image content, body image outcomes and shifts in usage patterns (including any reduction in engagement), evolution in users' coping strategies, and uptake and real-world impact of platform content moderation tools.

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An offline-first health information system for public health reporting in low-connectivity settings

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Mure, S., Salim, A., Antabi, N., Ashworth, H. An offline-first health information system for public health reporting in low-connectivity settings.

Introduction

Reliable public health surveillance depends on timely and complete reporting of routine clinical data. However, in many low-resource and humanitarian settings, health facilities experience intermittent electricity, unstable internet connectivity, and limited technical infrastructure. Clinical encounters are frequently recorded on paper, and health data are delayed, incomplete, or entirely absent from national reporting systems (1,2).

Most digital health platforms assume continuous connectivity and centralized cloud infrastructure, making them difficult to implement in environments where infrastructure is unreliable (3). Few electronic health record (EHR) systems designed for low-resource settings have demonstrated scalability across diverse humanitarian contexts (3). This creates surveillance gaps in which high-risk populations are underrepresented in routine health data.

Strengthening routine health information systems is a recognized priority for improving public health surveillance and health system planning (4).

This study evaluates whether an offline-first clinical information system can function as a practical public health reporting infrastructure in settings where continuous connectivity is not available, enabling routine clinical documentation and population-level reporting. Offline-capable digital systems have been proposed as a strategy to improve data availability in infrastructure-constrained health systems (5).

The Hikma Health EHR is an established free and open-source offline-first clinical information system that has been implemented across multiple low-resource and humanitarian settings globally. This study presents findings from a high-volume deployment in conflict-affected clinical settings in Syria to evaluate operational feasibility in severe connectivity constraints.

Methods

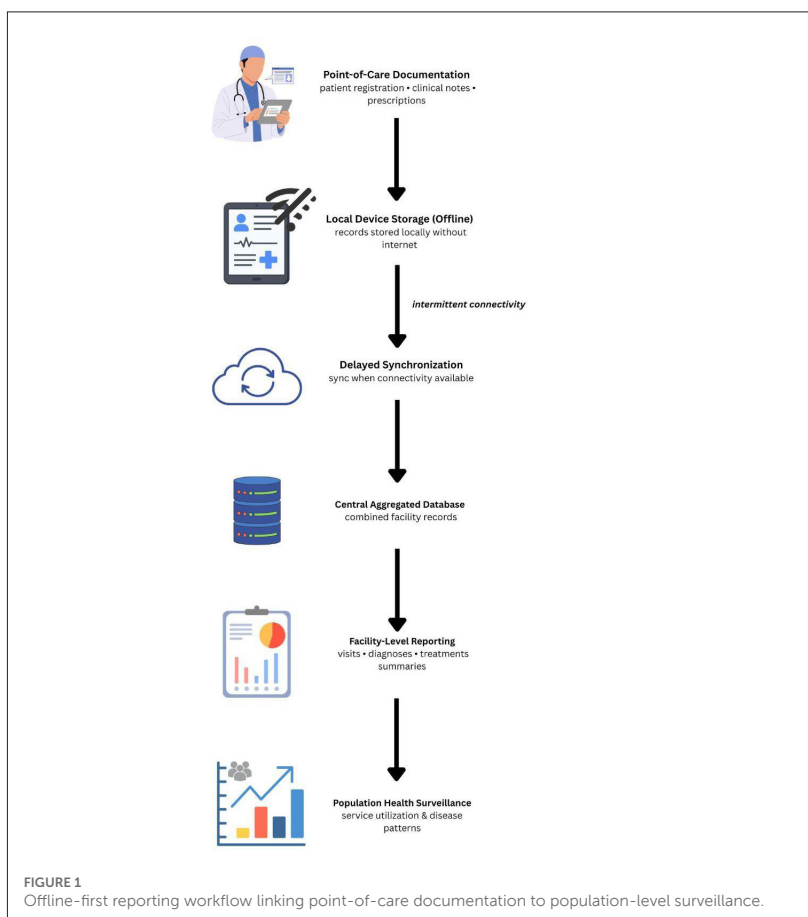
An offline-first EHR system was designed to function without continuous internet access by storing patient data locally at the point of care and synchronizing records once connectivity becomes available. The architecture supports patient registration, clinical documentation, medication dispensing, and longitudinal follow-up tracking. An overview of the offline-first clinical documentation and synchronization workflow is shown in **Figure 1**.

The platform is a React Native mobile application backed by a local SQLite database. Clinical workflows, including encounter documentation, vitals, ICD-coded diagnoses, prescriptions, and pharmacy dispensing, execute locally without network access. The application operates on standard Android and iOS mobile devices and uses incremental timestamp-based synchronization to exchange only modified records, minimizing bandwidth requirements after prolonged offline periods.

The Hikma Health EHR supports 3 modes of operation: (a) offline with cloud synchronization over HTTPS & encrypted RPC channels, (b) offline with peer-

to-peer synchronization to a local hub over an encrypted RPC channel, and (c) a fully online mode for facilities with stable connectivity.

Synchronization supports reconciliation of delayed or concurrent updates across devices.



Data security & Access control

Patient data are stored locally in sandboxed SQLite databases inaccessible to other device applications. Sensitive credentials and tokens are stored using platform-managed secure storage rather than general device storage.

Network communications, including data synchronization, occur over TLS over HTTPS to the cloud server and the local sync hub uses an AES-GCM-encrypted RPC channel with public-key identity verification which ensures synchronization traffic is protected even on untrusted networks. Each peer carries an explicit trust status (active, revoked, or untrusted) and renovation immediately detaches a device from the network.

Access within the platform is governed by a capability-based permission model scoped per clinic. Capabilities are independently grantable per user per clinic to manage the amount and type of access users have to patient data.

Device loss risks are mitigated through device authentication, restricted local data access, account deactivation, and device revocation from synchronization networks.

Servers are deployed and controlled by the implementing organization, ensuring complete data ownership and guardianship and further complying with jurisdictional compliance requirements where applicable.

This evaluation focused on a multi-site deployment conducted with the Atlantic Humanitarian Relief (AHR) medical mission in conflict-affected clinical settings in Syria with intermittent electricity and limited internet connectivity. Clinicians used the platform across mobile and facility-based clinics, with records captured offline and synchronized once connectivity was restored. Structured workflows enabled generation of aggregated encounter summaries and longitudinal patient records. Evaluation assessed operational feasibility through system usage metrics and synchronization success in settings where cloud-dependent systems are not practical.

Results

The system enabled offline digital documentation in settings previously dependent on paper records. Facilities were able to record clinical visits without internet connectivity and synchronize records once connections were restored.

During a five-day high-volume humanitarian medical deployment conducted as part of broader ongoing system implementation efforts, clinicians documented over 5,200 patients across 10,895 clinical visits while operating without continuous connectivity. A total of 3,856 prescriptions were recorded and dispensed within the system. Records synchronized successfully once connectivity was restored, supporting offline documentation and delayed reporting.

The system recorded patient histories, repeat encounters, and medication dispensing, maintaining patient-level records across visits where follow-up occurred.

Synchronized records were aggregated into routine summaries of visits, diagnoses, and treatments. These reports reduced reliance on retrospective paper reporting and enabled more timely monitoring of service utilization and disease patterns.

Conclusion

Offline-first health information systems can enable routine clinical documentation and public health reporting in settings where connectivity-dependent platforms are not practical. The Hikma Health EHR offers a cloud-based, free, and open-source offline-first option that is scalable in lower-resource settings. Compared to other solutions such as Sijilli, there is no required hard-ware and the EHR is customizable without significant software engineering (3). In this implementation, facilities without reliable internet access were able to capture structured clinical data, synchronize records, and generate aggregated reporting from routine encounters. From a prior mixed-methods feasibility study of the Hikma Health EHR system in both mobile (Lebanon) and stationary clinics (Nicaragua), quantitative data showed

that after about three hours of training and three weeks of use, participants were comfortable using the HH EHR and the HH EHR decreased patient interview times by three minutes (6). Qualitative data showed that these findings were due to the ease and simplicity of the modular workflows (6). By enabling facilities operating under infrastructure constraints to contribute standardized clinical data, offline-first systems address surveillance blind spots and support inclusion of underserved populations in routine public health monitoring. This approach provides a pathway for incorporating facilities currently excluded from digital reporting networks into surveillance systems.

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Beyond technology: Barriers to VR integration in healthcare

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Introduction

In the field of health sciences, implementation science has become one of the most prominent approaches for studying the barriers to introducing new technologies. This is typically achieved through evidence-based methodologies employing both quantitative and qualitative research methods (Albers et al. 2020). One of the challenges of this field is how virtual reality can be implemented in healthcare settings and what are the possible barriers that hinders its adoption. Although implementation may be hindered by a variety of factors, geopolitical aspects should also be taken into consideration. Additionally, the complexity and usability of a technology can significantly influence the pace of implementation.

Applications designed for VR technologies explore a range of different purposes within healthcare: e.g. therapeutic treatment, education, training, simulation and planning. Because virtual reality is an immersive medium delivered via head-mounted displays (HMDs), where three-dimensional rendered environments update in real time according to the user's head movements, it has the potential to elicit stronger perceptual and affective responses, therefore, VR represents a promising tool for psychological interventions in healthcare settings too (Kouijzer et al. 2023). Despite its growing potential, real-world hospital implementation remains challenging and context-dependent, as studies show that successful VR integration in clinical environments depends on institutional readiness, staff training, and patient acceptability (Riches et al. 2023).

These barriers can be described by the TASK framework (Yang et al. 2022) which deals with the relationships between a Task and the task implementer's Affect, Skills, and Knowledge. The framework contains the following type of barriers: emotion barriers, logic barriers, knowledge barriers, resource barriers. This study reports preliminary results from a qualitative interview study from Budapest, Hungary with multiple stakeholder groups, examining attitudes, barriers, and implementation possibilities regarding the use of VR in hospital psychological care.

Methodology

A group of healthcare professionals and patients was recruited over a six-month period (June to December 2025) from hospitals in Semmelweis University and Mazsihiszi Hospital that were collaborating with professionals working with VR technology in healthcare (see Fig. 1). The study was reviewed and approved by the Medical Research Council of Hungary. The questions were uncovering the personal attitudes of health care professionals as well as of the patients towards introducing new technology into their daily healthcare research practice and how they see the barriers of this implementation. Data analysis was done based on thematic analysis (Braun and Clarke 2006).

Gender	Role	Young /Mid-career/Senior/Elderly adult (Career level or Elderly adult age group)	Experience with VR (none/some/expert)
Female	Physiotherapist	Young	Expert
Female	Clinical psychologist	Young	Some
Female	Department head psychologist	Mid-career	Some
Male	Medical doctor	Senior	None
Male	Patient 1	Elderly adult	None
Male	Patient 2	Elderly adult	None
Female	Health psychologist	Mid-career	Expert
Female	Nurse	Mid-career	None
Female	Mental health specialist	Senior	None
Female	Patient 3	Elderly adult	None
Female	Patient 4	Elderly Adult	None
Male	Speech therapist	Young	Some

FIGURE 1
Demographics and the participants' level of experience with VR

Results and Discussion

In the following we describe the themes identified in the interviews.

1. **Training and Competency Requirements.** Participants consistently emphasized that VR integration requires structured, practice-based training rather than minimal technical instruction. Experiential learning and troubleshooting competence were considered essential, particularly in geriatric and rehabilitation contexts. As one clinician stated, “Not just a ‘how-to’ document, but practical training — more like a workshop than reading material” (psychologist). Similarly, it was highlighted that “The use of VR would require short training on settings, program initiation, and prior assessment of the patient’s condition” (logopedist). Brief or superficial instruction was regarded as insufficient for responsible clinical application.
2. **Time Burden, Workflow Integration, and Structural Preconditions.** VR implementation was described as time-intensive and logistically demanding. The active usage time underestimated the overall workload, which includes device setup, calibration, supervision, hygiene, and coordination with other therapies. One professional summarized this as: “If someone plays for 40 minutes, the session will definitely be 60 minutes because it has to be set up” (physiotherapist). Limited device numbers, charging needs, and disinfection requirements further constrained feasibility. Sustainable integration was therefore linked to institutional embedding and clearly defined roles. As one respondent emphasized, “There should be a person dedicated to this” (psychologist), while another noted the necessity of “a communication chain and protocol — who informs whom, and when” (clinical psychologist). Without such structural support, VR remained difficult to embed within already overloaded clinical routines.
3. **Professional Motivation and Institutional Attitudes.** Adoption appeared to depend strongly on individual enthusiasm and team dynamics. Some professionals expressed openness toward innovation: “I really like trying out every new thing” (mental health specialist). At the same time,

institutional skepticism, particularly among physicians, was reported: "Patients respond well, but doctors are very, very skeptical about it" (department head psychologist). Research engagement and cohesive teamwork facilitated implementation efforts.

4. **Perceived Patient Benefits.** VR was widely described as technology with potential in mood-enhancing, stimulating, and capable of mitigating the effects of hospitalization. It provides sensory engagement, distraction from symptoms, and opportunities for reminiscence. For example, "In a spinal cord-injured patient, the VR experience noticeably improved mood" (speech therapist), and it was seen as something that "it can counterbalance hospitalization by giving a sense of escape" (clinical psychologist). Mastering a new technological tool was also perceived as strengthening self-confidence.
5. **Risks and Adverse Reactions.** Some patients reported vertigo, fear, or balance disturbances during immersive experiences. One patient described: "When I looked at the edge of the image, it felt frightening, like standing at the edge of a cliff" (patient 1). Clinicians similarly observed that patients may "not feel the ground under their feet" (physiotherapist). Fatigue and cognitive status were considered important screening factors prior to VR use.
6. **Customization and Design Considerations.** Most institutions relied on off-the-shelf applications, yet professionals expressed interest in simplified, rehabilitation-oriented or reminiscence-based content. Financial constraints limited custom development: "A specialized application would cost a lot of money" (physiotherapist). Patients also expressed preferences for emotionally safe content: "I would choose something calming, not frightening" (patient). Simplicity was emphasized: "It should not be too complex — rather simplified" (physiotherapist). Naturalistic environments were consistently preferred over artificial or overly gamified imagery.

TASK-Based Barrier Matrix				
Personal motivation	✓			
Negative patient reactions	✓			✓
Patient satisfaction / preferences		✓		
Training & competency requirements			✓	
Design understanding		✓		
Procurement / device purchase				✓
Institutional constraints				✓
Wi-Fi availability				✓
	Emotion	Logic	Knowledge	Resource

FIGURE 2
TASK matrix.

These categories can be structured based on the TASK framework (see fig. 2). Emotion barriers involve personal motivation and negative patient reaction (however, these can also signalize resource barriers), while patient satisfaction is based on personal preferences. Training and competency requirements are related to knowledge related barriers, as it requires one to understand how VR can work within the psychological therapy. Personal preferences could be interpreted as logical barriers: these require proactive thinking, as well as time and dedication. In the same way, design understanding translates to the therapist who aims to find the best application available for VR platforms, but often this requires thinking out of the box. Resources could be understood as referring to how the technological device and the application can be purchased (e.g., via public procurement) in a complex socio-technological environment, while institutional constraints and, for example, the availability of a reliable Wi-Fi network are also often barriers.

Conclusion

The findings suggest that while VR holds substantial promise for psychological care in hospital settings, its successful implementation depends less on technological potential and more on addressing interrelated emotion, logic, knowledge, and resource barriers as conceptualized within the TASK framework.

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Bridging risk management and public health response among multiple sectors: The role of oman's national center for emergency and health risk management

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Citation

Al Busafi, M.S., Al Nabhani, A.A., Al Zakwani, J.A. Bridging risk management and public health response among multiple sectors: The role of Oman's national center for emergency and health risk management.

Introduction

Public health emergencies—infectious disease outbreaks, environmental hazards, and natural disasters—are complex, evolving threats demanding coordinated multi-sectoral responses. Effective preparedness requires timely resource mobilisation alongside clear governance, structured risk management, and unified communication across sectors. Fragmented coordination, ambiguous roles, and weak information-sharing undermine emergency management, requiring integrated frameworks that embed public health expertise within national governance.

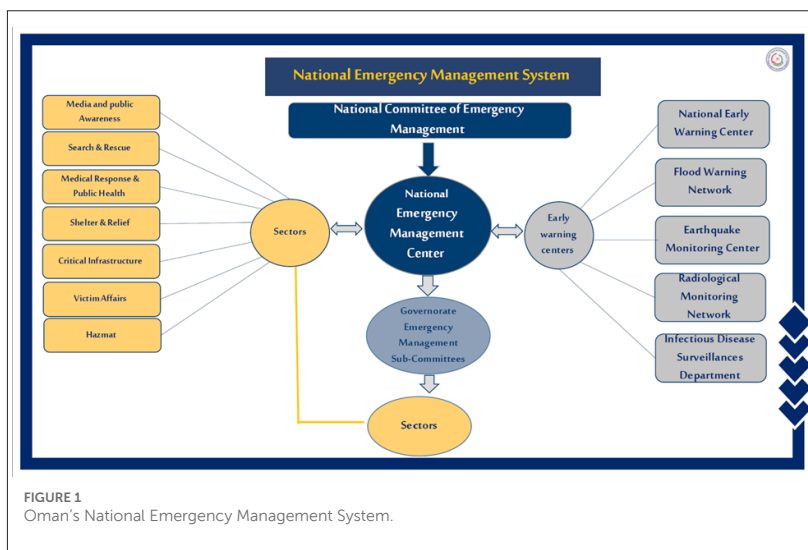
In Oman, the National Center for Emergency and Health Risk Management (NCEHRM), under the Ministry of Health, fulfils this role. It coordinates the

Medical and Public Health Response Sector within the National Emergency Management System led by the National Center for Emergency Management (NCEM, n.d.), ensuring operations are strategically aligned, evidence-informed, and efficiently implemented for both immediate crises and longer-term risks. NCEHRM also oversees all public health risks recorded in the National Risk Register and its publicly available edition.

NCEHRM's Structure and Roles

NCEHRM comprises three complementary departments. The Risk Management Department identifies and manages risks at ministerial and national levels and oversees public health entries in the National Risk Register. The Emergency Operations Department coordinates communications across all phases of emergency management—during Center activation and in routine operations—and coordinates with the recently established Omani National Emergency Medical Team (NEMT). The Referral Department is the primary liaison with healthcare institutions nationally, and occasionally internationally, facilitating patient transfers when institutions face communication breakdowns, critical bed shortages, or other unforeseen challenges.

Within NCEM's eight-sector framework, NCEHRM ensures medical and public health operations are coordinated across all phases—mitigation, preparedness, response, and recovery. The other seven sectors are depicted below as shown in Figure 1. Integrating specialised public health expertise across these sectors enhances response efficiency and effectiveness at national and sub-national levels. In parallel, The National Risk Register is a dynamic, structured instrument for identifying, monitoring, and prioritising health threats, equipping decision-makers with actionable intelligence for mitigation and response. A new public edition is disseminated to the population, enabling communities to understand potential threats and adopt preventive measures (Al Roya, 2025).



Approach

This poster presents a descriptive, practice-focused analysis of Oman's national emergency management system, emphasising NCEHRM's role in public health preparedness and risk management. The analysis drew on official policy documents, organisational frameworks, sectoral coordination mechanisms and public health threats, focusing on how NCEHRM manages its routine workload while directing the Medical and Public Health Response Sector, integrating risk intelligence through the National Risk Register, and coordinating with the Omani National Emergency Medical Team (NEMT).

Key Features and Impact

NCEHRM has emerged as a transformative institution within Oman's public health emergency architecture, with several features that enhance national preparedness and response.

Operational Efficiency through the Omani NEMT

NCEHRM's coordination of the NEMT has reduced operational costs by consolidating specialised medical response under unified command, optimising resources, and limiting reliance on costly emergency arrangements. The model's value was demonstrated during

Cyclone Tej (October 2023), when the NEMT was first activated—unlike Cyclone Mekunu (2018), despite comparable severity. Mekunu resulted in six fatalities in Oman and caused ~US\$540,000 in damages to the Ministry of Health, whereas Tej resulted in no reported deaths and only ~US\$26,000 in damages. Early NEMT activation averted more than US\$500,000 in damages and economic losses, achieved a 95% reduction in lives lost relative to prior events, and enabled timely evacuation of in-hospital patients from at-risk areas.

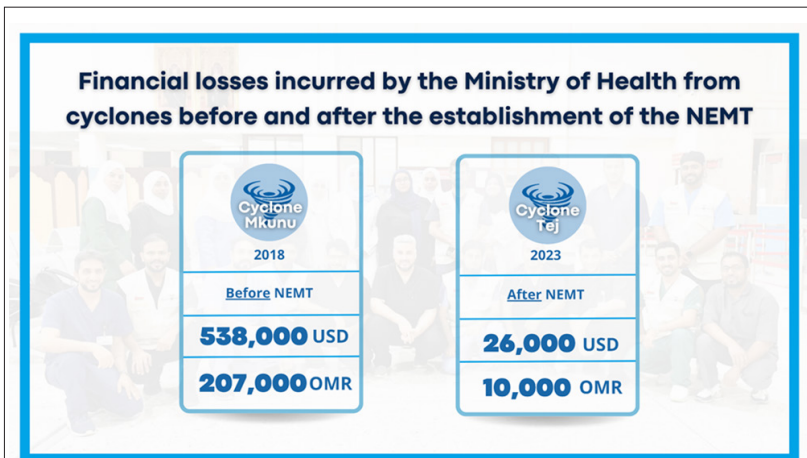


FIGURE 2

Comparative financial losses incurred by the Ministry of Health from cyclones before and after the establishment of the NEMT.

AQ1

Dengue Fever Outbreak

WHO reported Oman among countries with the highest confirmed dengue cases in the Eastern Mediterranean Region in 2023, alongside Pakistan and Saudi Arabia (WHO, 2023). The Medical & Public Health Response Sector proved effective during the recent dengue fever surge across several Omani governorates, accompanied by visible spread of *Aedes aegypti*. NCEHRM activated and convened with relevant entities to implement mitigation plans covering environmental sanitation, biological control, and provision of essential diagnostic and therapeutic supplies. This multi-sectoral approach interventions flattened the infection curve within three months, with sustained control measures throughout the year (Al Balushi et al., 2023).

Institutional Support

NCEHRM has strengthened healthcare institution resilience by deploying personnel to hospitals facing staff shortages or operational strain. Support has reached major institutions including Khoula Hospital, the Royal Hospital, Al-Nahdha Hospital and Sultan Qaboos Hospital in Salalah during the Khareef season—a period of annual surges in tourism, infectious disease outbreaks, and road traffic accidents. This reinforcement has safeguarded service continuity in critical specialties during peak demand.

Coordination of Complex Patient Referrals

The Referral Department oversaw more than 220 emergency referral cases in 2025, each involving patients whose clinical needs exceeded the referring institution's capacity and required NCEHRM-coordinated transfer by land or air. This volume reflects the Department's role in maintaining seamless inter-facility coordination when communication or bed shortages threaten timely care.

Omani NEMT Accreditation

NCEHRM has formally initiated the process of accrediting the Omani NEMT as a WHO Emergency Medical Team Type 1 (Fixed) —the team has now reached the trials and simulations stage of the WHO classification pathway, in which operational readiness is tested through structured exercises that validate clinical, logistical, and administrative capabilities against the core

and technical standards set out in the WHO Classification and Minimum Standards for Emergency Medical Teams. Successful classification will position Oman among a select group of nations with WHO-verified national EMTs, strengthening domestic surge capacity and enabling regional deployment when required (WHO, 2021).

Conclusion

Oman's experience illustrates how a specialised public health institution can act as the central integrator of emergency preparedness, multi-real coordination, and population-focused risk management. By coordinating the Medical and Public Health Response Sector and managing the National Risk Register—with active population engagement— NCEHRM ensures timely, evidence-informed responses to health threats, strengthening governance, enhancing efficiency, and promoting community preparedness and resilience.

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Co-design and usability testing of a digital tool to support autistic adults with disclosure decision-making

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Citation

Emeline Han, K.L., Crane, L., Woodcock, K. Co-design and usability testing of a digital tool to support autistic adults with disclosure decision-making.

Abstract

Overview

Autistic adults face significant decisional conflict, stress, and anxiety when navigating disclosure of their autism diagnosis or autistic identity, yet evidence-based support for making these decisions remains limited. This project aimed to co-design and test a digital disclosure decision-making guide with and for autistic adults. Seven autistic adults shaped and refined the digital prototype through three participatory workshops, where they identified priorities around accessibility, privacy, flexibility, and authentic representation, and provided detailed feedback on language, content, design, and functionality. A mixed-methods usability evaluation grounded in the Technology Acceptance Model (TAM) is currently underway. A combination of think-aloud interviews and post-use surveys is being used to assess TAM constructs including perceived usefulness, perceived ease of use, user attitudes, and behavioural intentions. Complete usability findings

will be presented at the conference, with recommendations for inclusive development and implementation of digital decision support tools.

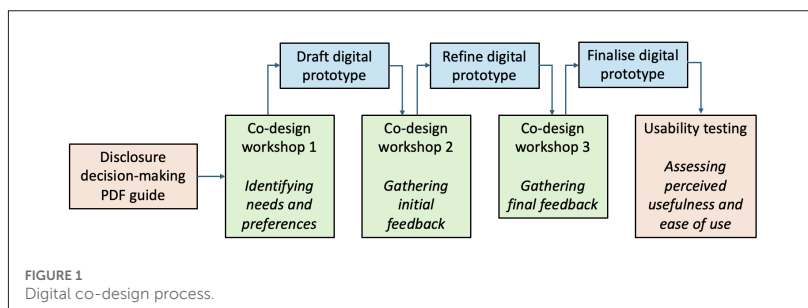
Background

Deciding whether, when, and how to disclose an autism diagnosis or autistic identity is a complex and ongoing process. Research shows that autistic adults' disclosure decisions are influenced by internal factors such as how strongly they identify as autistic and how positively they perceive this identity, as well as external factors such as how they anticipate others will react and how safe they feel in a specific environment [1-3]. Disclosure carries potential benefits (e.g., increased understanding, acceptance, and inclusion) and risks (e.g., stigma, discrimination, and exclusion) that have significant implications for autistic adults' social and psychological wellbeing [4-5]. While both concealment and disclosure have been associated with higher distress and poorer wellbeing among autistic adults at a single point in time [6], disclosure has been associated with better mental health outcomes over time [7]. Despite the significant decisional conflict, stress, and anxiety autistic adults report when navigating disclosure dilemmas, there is limited structured, evidence-based support to help autistic adults make informed and empowered disclosure decisions.

The author previously co-produced a self-guided disclosure support intervention with autistic collaborators based on systematic review evidence and community consultation. A pilot study suggested that this intervention was feasible and acceptable, while participants recommended adapting the self-help guide from a paper-based format into a web-based format to improve accessibility and engagement. This poster will present work-in-progress from a follow-on project that aims to: (1) co-design a digital version of the disclosure decision-making guide *with* and *for* autistic adults, and (2) evaluate the usability of the resulting tool.

Methods

Following a user-centred design process (Figure 1), the author held a series of three workshops with seven autistic adults and an inclusive digital designer to iteratively develop and refine the digital prototype based on user needs and preferences.

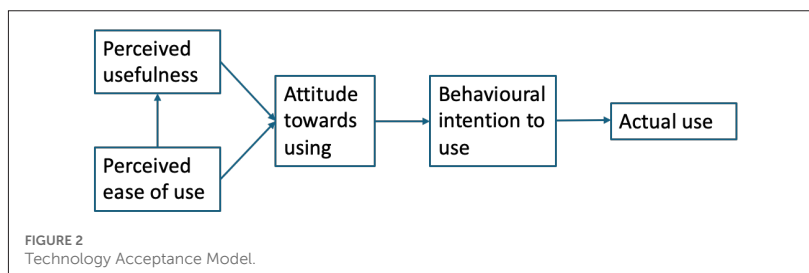


After finalising the digital prototype, a small-scale mixed-methods study is currently being conducted to test its usability using a combination of interviews and surveys.

- 1) **Think-aloud interviews:** 10 autistic adults have taken part in think-aloud interviews, meeting the target sample size for studies using this method [8]. Participants were asked to verbalise their actions, initial impressions, and thoughts while navigating through parts of the prototype, giving real-time insights into user interactions.
- 2) **Post-use surveys:** 12-14 autistic adults are being recruited to complete the full guide and a post-use survey, following the recommended sample size for questionnaires assessing website usability [9]. These surveys will assess the usability of the digital guide using the Technology Acceptance Model (TAM; Figure 2) [10].

Applying TAM, the post-use survey will collect:

- 1) Quantitative data on *perceived usefulness*, *perceived ease of use*, and *actual use* of the guide.
- 2) Qualitative data on *attitudes* and *behavioural intentions* towards using the guide, barriers/enablers to usability, and suggestions for improvement.



Data collection is ongoing. Quantitative data will be analysed descriptively using SPSS and qualitative data will be analysed thematically using NVivo.

Results

Initial workshop findings indicated strong support for a web-based format that enables private, self-paced, and flexible engagement – particularly valued when navigating sensitive and personal decisions around disclosure. Participants identified key priorities including accessibility features accommodating different information processing styles (e.g., multimodal content, ‘save-and-return’ functionality); minimalist design to reduce cognitive load and sensory overwhelm; lived experience examples representing diverse contexts and identities; as well as robust data security and confidentiality assurances. Subsequent prototype review workshops generated detailed feedback on language, content, navigation, clarity, functionality, and credibility, which was incorporated prior to usability testing. Full results from usability testing will be presented at the conference, including quantitative data on TAM constructs and more in-depth qualitative insights into user experiences.

Conclusion

This project presents the first co-produced digital guide specifically designed to support autistic adults with disclosure decision-making. Findings will advance evidence-based and community-led approaches to supporting autistic mental health and wellbeing. The study will also provide broader actionable recommendations for inclusive design and implementation of

digital decision support tools with and for other marginalised or underserved populations.

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Consolidation of medical organizations in CIS healthcare systems: A strategic institutional framework for scaling AI in patient-centered care

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Citation

Imandossova, A., Abishev, O., Issatayeva, N. Consolidation of medical organizations in cis healthcare systems: A strategic institutional framework for scaling AI in patient-centered care.

Background

Despite growing investments in AI, most CIS healthcare systems remain structurally incapable of scaling it. Healthcare systems across the Commonwealth of Independent States (CIS) face rising noncommunicable disease burdens, fiscal constraints, and accelerating digitalization. Artificial intelligence (AI) spanning clinical decision support, predictive analytics, and resource optimization is increasingly recognized as a strategic tool for improving care efficiency and access. Effective AI deployment, however, requires large volumes of high-quality, interoperable data and coherent

governance structures. In most CIS countries, care delivery remains fragmented: providers operate with heterogeneous health information systems and weak data-sharing mechanisms, constituting a fundamental barrier to scalable AI implementation.

Interoperability enablers such as HL7 FHIR and data governance frameworks are central to digital health scaling, yet low- and middle-income countries commonly face legacy systems and governance fragmentation [1]. In Russia, imperfect regulatory frameworks and insufficient funding remain key barriers to digital maturity [2]. Kazakhstan has advanced considerably, moving from paper-based records to a national eHealth strategy since 2013 and approving a National AI Development Concept through 2029 [3,4]. Uzbekistan is actively consolidating fragmented facilities as part of its National Health Compact 2030 [5]. These divergent trajectories highlight the critical role of institutional preconditions for AI adoption (Table 1).

TABLE 1: Digital Health and AI Readiness Across Selected CIS Countries

Country	National EHR	Consolidation Level	AI / Digital Strategy	Interoperability	Key Barriers
Kazakhstan	National EHR since 2013; interoperability platform 2017	High - PHC networks integrated; hospital clustering active	National AI Concept 2024–2029	National platform; partial HL7 FHIR	Digital literacy gaps; urban–rural divide
Russia	EGISZ operational; mandatory EHR from 2024	Moderate - regional clusters; federal fragmentation	National Digital Health Program; AI pilots in radiology	HL7 v2 prevalent; limited FHIR R4	Legacy systems; regulatory gaps; vendor lock-in

Uzbekistan	EMR scaling; WHO-supported HMIS in development	Active - merging silo facilities into unified polyclinics (2025)	National Health Compact 2030; digital pillar	Interoperable EMR in progress; standards pending	Fragmented facilities; governance gaps; funding
Azerbaijan	Partial EHR	Moderate -	Digital agenda	Nascent; HL7	Low public

Country	National EHR	Consolidation Level	AI / Digital Strategy	Interoperability	Key Barriers
	rollout; recentralized MoH since 2019	recentralization created unified governance	developing; no formal AI strategy	not mandated	investment; weak data privacy framework
Belarus	Centralized system; EHR coverage expanding	High - centralized MoH model; unified hospital network	State digital program; AI limited to pilots	Centralized infrastructure; low international standards	Limited AI incentives; regulatory clarity lacking

Sources: [2–6].

Objective

This study examines organizational consolidation as an institutional mechanism to enable large-scale, patient-centered AI implementation in CIS healthcare systems, analyzing how consolidation supports data integration, governance alignment, and standardized clinical practice. It introduces consolidation as a previously underexplored institutional precondition for scalable AI in healthcare systems.

Methods

A qualitative comparative institutional analysis was conducted alongside a structured policy review of five CIS countries (Table 1). Three dimensions were examined: (1) governance and ownership models; (2) data integration and interoperability frameworks; and (3) regulatory and ethical mechanisms for cybersecurity and patient trust. Sources included national policy documents, WHO/Europe country reports [6], and peer-reviewed literature. Interoperability was assessed against HL7 FHIR R4, IHE profiles, SNOMED CT, LOINC, and ICD standards that enable semantic interoperability and safeguard patient autonomy [7,8].

Results

Table 1 reveals substantial variation across the CIS region. Kazakhstan and Belarus demonstrate the highest consolidation and electronic health record (EHR) coverage; Uzbekistan is actively integrating fragmented facilities; Russia maintains broad national infrastructure but faces legacy and governance barriers; Azerbaijan lags in standards adoption. These differences correlate directly with AI readiness. In the absence of comparable CIS-level data on AI adoption rates itself a reflection of the measurement and governance gaps described above evidence from the United States provides the closest available benchmark for this relationship. Hospitals in multi-hospital systems adopt predictive AI far more often (81–86%) than independent facilities (31–37%) [9], confirming that consolidation creates the organizational and data infrastructure necessary for AI deployment. Adoption of predictive AI in hospital networks rose from 66% to 71% between 2023 and 2024, driven primarily by system-affiliated institutions [9]. Consolidated systems further enable standardization of clinical pathways via SNOMED CT, LOINC, and ICD, reducing practice variability and improving AI-driven decision support reliability. Consolidation also strengthens data governance, clarifying accountability and ensuring ethical oversight essential for patient trust. Notably, in Kazakhstan, consolidation of PHC networks has already enabled early-stage integration of AI-driven decision support tools at scale. These findings suggest a structural relationship: without consolidation, AI adoption remains episodic and pilot-driven; with consolidation, it becomes systemic and scalable.

Discussion

Consolidation should be understood as a critical enabler of digital transformation, not merely an administrative reform. The CIS region serves as a natural laboratory for studying the relationship between institutional structure and AI scalability in healthcare, with lessons applicable to health systems globally. As Table 1 illustrates, CIS countries with higher consolidation levels demonstrate greater digital and AI readiness. However, consolidation alone is insufficient: dedicated investments in interoperability infrastructure, workforce capacity, and regulatory clarity are required [1]. AI clinical applications depend directly on data governance quality [10]. Poorly designed consolidation risks marginalizing patient voices or prioritizing efficiency over equity. In the absence of consolidation, AI risks reinforcing fragmentation rather than solving it. Strategic alignment with digital health policies anchored in recognized standards is therefore essential.

Conclusion

Consolidation of medical organizations in CIS countries provides a key institutional foundation for patient-centered AI integration. The comparative analysis reveals a clear pattern: higher consolidation correlates with greater AI readiness. When grounded in established standards (HL7 FHIR R4, IHE, SNOMED CT, LOINC [7,8]) and embedded in national digital health agendas, consolidation enhances capacity to deploy AI that improves care quality, coordination, and equitable access. Policymakers should treat consolidation as an integral, evidence-informed component of national AI and digital health strategy. Without institutional consolidation, large-scale AI in healthcare will remain unattainable.

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Continuous neurodegenerative care through a computer vision home-based rehab tool

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Citation

Barnés, A., Ortiz, J., Izquierdo, A., Campuzano, R., Gallego, G. Continuous neurodegenerative care through a computer vision home-based rehab tool.

Introduction

Non-communicable diseases are the leading cause of global mortality, accounting for approximately 71% of deaths worldwide [1]. Within this broader context, neurological disorders, including stroke, are emerging as a significant public health challenge [2].

By 2050, it is projected that there will be 21.43 million incident stroke cases and overall, 224.86 million disability-adjusted life years attributable to stroke globally. As a consequence, the number of individuals living with post-stroke impairments will increase rapidly, leading to persistent functional limitations, reduced quality of life, and a growing burden on caregivers. These conditions require ongoing management, with exercise and rehabilitation playing essential roles across three stages of prevention: primary (risk reduction), secondary (disease modification), and tertiary (symptom management).

As healthcare systems already face significant pressure to provide sustained care, the increasing demand for rehabilitation services and tertiary

prevention further exacerbates this burden. This growing demand highlights a misalignment between continuous disease progression and intermittent clinical evaluations. As a result, the delivery of timely, data-driven, and individualized care is hindered. Furthermore, access to rehabilitation services is often limited by structural barriers, such as shortages of specialized professionals and restricted clinical services, particularly in isolated areas.

Addressing this gap raises key questions: Can home-based systems provide continuous monitoring of functional performance? Can biomechanical metrics derived from markerless motion capture reliably detect changes in functional performance over time in home-based settings? Additionally, can continuous monitoring enable earlier therapeutic adjustments and more proactive care? Could it improve outcomes while reducing clinical and socioeconomic burdens?

Methods

This study focuses on upper-limb rehabilitation in chronic post-stroke patients, using stroke as a model for secondary neurodegeneration. Exercises are based on the Fugl-Meyer Assessment for Upper Extremity, targeting movements such as joint flexion, extension, and hand-to-face actions [5]. Thereby, evaluating motor domains, including range of motion, coordination, and control.

Exercise selection is limited by tracking reliability, excluding movements with inconsistent landmark detection. Table 1 (participant characteristics) is not included, as the clinical protocol will begin in May 2026 pending ethical approval. Patient eligibility, however, will consider mobility, paralysis, and cognitive status.

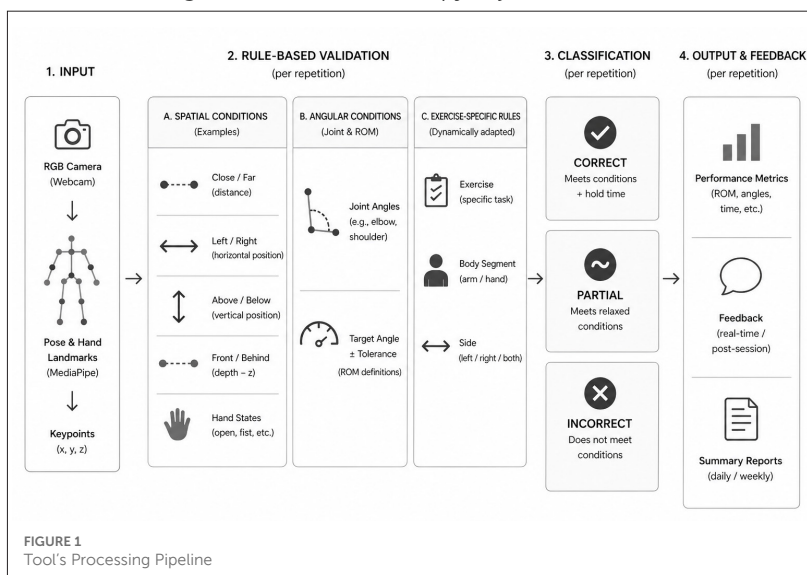
A standard RGB camera is used for markerless motion capture, enabling the collection of objective kinematic data without wearable sensors. The system estimates full-body and hand key points in real time using MediaPipe [3], [4] while reconstructing the user's pose and computing biomechanical metrics such as joint angles, position, and alignment. It currently monitors exercise performance alongside self-reported pain.

AQ6

The platform is web-based, built with Vue.js for the frontend and Node.js for the backend, and is accessible on any device with a camera. Data is stored and processed in Google BigQuery, supporting large-scale and longitudinal analysis.

Exercise performance is evaluated using a rule-based validation system in which each repetition is assessed against predefined spatial and angular conditions between anatomical landmarks (e.g., close/far, left/right, above/below). These rules adapt to each exercise and body segment, allowing classification of movements as correct or incorrect. Importantly, incorrect repetitions are categorized (e.g., insufficient range of motion), providing detailed feedback on performance.

To support adherence, the system provides real-time feedback and tracks engagement metrics such as session completion, repetitions, and execution time. Clinicians receive periodic summaries (daily or weekly), enabling remote monitoring and data-driven therapy adjustments.



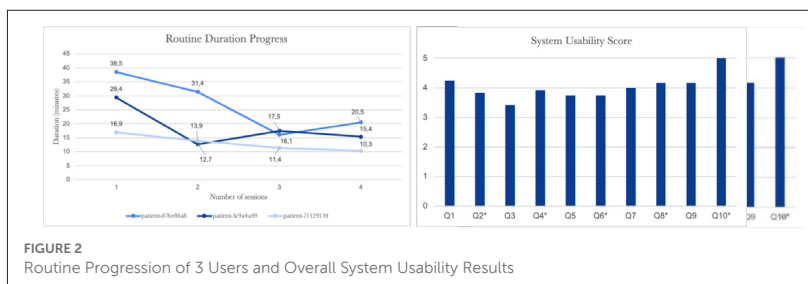
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Preliminary Results

At the time of writing, formal clinical outcomes are not available, as the study is currently awaiting the hospital's ethics committee approval. Therefore, this section reports preliminary findings based on in-house evaluations conducted with 10 participants outside the target clinical population, including healthy individuals and users with unrelated conditions.

It is observed that the system enables time-efficient and scalable home-based rehabilitation, with session duration emerging as a key indicator of usability and adaptation. Initial sessions were longer and more variable (overall mean of 21.4 minutes), reflecting onboarding processes such as system familiarization and setup. In contrast, repeated sessions stabilized between 10 and 15 minutes, indicating rapid user adaptation and reduced cognitive and operational demands.

Usability, assessed using the System Usability Scale (SUS), yielded a mean score of 40.25/50 (80.5/100), indicating good overall usability and strong learnability. Participants reported high confidence, low perceived complexity, and minimal need for technical support, with many expressing willingness to use the system regularly. However, variability in responses and qualitative feedback highlighted limitations, particularly in motion tracking accuracy, especially for hand movements, as well as environmental and device-related constraints. Overall, these findings support the system's feasibility while identifying key areas for improvement prior to clinical validation.



AQ1

Preliminary Conclusion

Preliminary internal findings indicate that the proposed system is feasible for home-based rehabilitation and monitoring of functional performance. The reduction in session duration and a SUS score of 80.5/100 suggest it is user-friendly and can support sustained engagement in non-clinical settings. Moreover, the system can generate objective, longitudinal biomechanical data, offering a deeper understanding of motor performance compared to traditional assessments, potentially facilitating earlier identification of functional changes.

Despite these promising results, current findings must be interpreted with caution. Evaluations were conducted in a non-target population, and key limitations, particularly in motion tracking accuracy (notably for hand movements), environmental constraints, and device variability, may affect reliability in real-world conditions. These factors highlight the need for further technical refinement and validation.

While impacts on outcomes and costs remain to be confirmed, the system shows potential to reduce clinical and socioeconomic burdens, particularly within OECD healthcare models focused on prevention and data-driven care [6]. Building on these initial findings, additional pilots are underway in Spain, expanding the evaluation to adults over 65 to validate implementation strategies and assess the tool's scalability within the target population.

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Developing a digital perinatal sexual health and wellbeing programme for deployment in the squeezezy app

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Citation

Mapp, F., Robson, C. Developing a digital perinatal sexual health and wellbeing programme for deployment in the Squeezezy App.

Introduction

Sexual behaviour and sexual function fluctuate significantly during the perinatal period (pregnancy and the year after giving birth). Sexual dysfunction is experienced by up to 83% of postnatal women¹ with huge variation in symptom type, severity and impact on wellbeing. Many women do not seek help from healthcare services and common responses are to ignore, normalise and/or self-diagnose and self-treat issues. There is a clear need for high-quality, tailored information for women and their partners. The Squeezezy App is the most downloaded pelvic health app with over 80,000 monthly active users and an Organisation for the Review of Care and Health Applications (ORCHA) score of 83%². Within the Squeezezy platform, there are digital products to support pregnancy pain, pelvic organ prolapse and maternity care but these are focussed on clinical management of symptoms and do not take a holistic approach to improving sexual health and wellbeing. Squeezezy is therefore well-positioned to host perinatal health information.

Through a funded collaboration by UCL Innovation & Enterprise and the National Institute for Health and Care Research (NIHR) with the Squeezezy App,

we aim to develop an evidence-based digital sexual health and wellbeing programme for deployment in the Squeezy App for women who are pregnant or have given birth within the past year.

Methods

We used the GUIDED reporting guideline³ to inform the programme development process which comprised eight main stages:

- i) Develop shared understanding of the Squeezy App context, purpose, target population and usual product development cycle.
- ii) Conduct a scoping review to map evidence about existing perinatal sexual health and wellbeing programmes. FM searched MEDLINE and SCOPUS for academic literature and a Google search sorted by relevance to find relevant unpublished work. Stakeholders also provided resources e.g. NHS information sheets, internal reports and data dashboards. Results were analysed for key content themes and charted by resource type. Themes were synthesised and grouped to create a thematic framework to underpin the new perinatal sexual health and wellbeing programme.
- iii) Develop programme theory to map out exactly how and why a digital programme is expected to improve sexual health and wellbeing for perinatal women using the Squeezy App.
- iv) Identify existing components from other digital products and programmes within the Squeezy App family that could be adopted into this new programme.
- v) Agree guiding principles for the programme.
- vi) Stakeholder engagement – FM contacted healthcare professionals, some who were using the Squeezy App for maternity care and some who weren't. Pregnant and postnatal women and their partners were also asked for input. Informal interviews were arranged with stakeholders

to discuss a) the sexual health and wellbeing needs of perinatal women, b) existing care pathways within perinatal healthcare and c) to generate specific ideas for content. Building on the initial thematic framework from the scoping review, key themes were identified from stakeholder interviews and charted by stakeholder type. A Patient and Public Involvement (PPI) group were formed to provide ongoing feedback to help shape content wording, tone, format, interactiveness and overall usefulness of the programme.

- vii) Programme iterations and tailoring – information was collated from the above developmental stages, categorised thematically and sorted into programme modules for further iteration and editing. Additional tailoring of the information presented was undertaken to account for delivery method (vaginal birth, assisted/instrumental delivery and caesarean and previous sexual experience).
- viii) Remaining uncertainties – FM and CR will map what remains unknown at the end of the programme development process using feedback from Squeezy users who have accessed the perinatal sexual health and wellbeing programme.

This is an ongoing programme of work and final numbers of stakeholders, literature included, content themes and nature of PPI input may change.

Results

Stakeholders included ten healthcare professionals (specialist pelvic health physiotherapists (n=5), perinatal pelvic health midwife (n=1), public health midwife (n=1), community midwife (n=1), GP (n=1) and Obstetrician (n=1)) from eight different NHS perinatal pelvic health services and related healthcare services all directly involved in antenatal and/or postnatal care. Five women and one male partner of different ages, ethnicities, socio-demographic backgrounds, geographic locations and stages of pregnancy and postnatal were also involved.

The scoping review confirmed the lack of sexual health, function and wellbeing information tailored specifically for women and their partners in the perinatal period and a clear need for high quality information. Three overarching themes were identified which became our guiding principles for developing the programme: 1) using trauma-informed approaches; 2) digital inequalities and the potential to widen health inequalities; 3) foregrounding pleasure principles in relation to sexual activity.

Through discussion between FM and the Squeazy App development team, emerging evidence from the scoping review and ongoing input from all stakeholder groups, we narrowed the scope of this work to focus on postnatal sexual health and wellbeing as there is the greatest need and biggest potential for impact of a new digital programme. We then developed an initial programme theory based on a logic model format detailing the necessary inputs, activities, outputs, outcome and impact as well as key assumptions and contextual factors to further guide development of programme content.

From 16 themes identified in the literature and stakeholder interviews, we developed 11 programme information modules: introduction and sexual norms; vulval anatomy; perinatal

physiology; libido; having sex after birth; birth trauma; managing fear of sex; pelvic floor health and common symptoms; sexual wellbeing; feeling pressure to have sex; FAQs and resources. The PPI group were integral to finalising the set of modules and deciding the essential content for each.

Lessons learned

- Align work with commercial interests, current policy and research priorities
- Be clear about the boundaries of collaborators' knowledge and expertise
- Develop aim and scope for the project early and collaboratively

- Engage a range of stakeholders throughout development and deployment
- Agree on Intellectual Property at the start

Conclusion

This collaborative, multi-stage development process between researchers, healthcare professionals, perinatal women and their partners and digital product experts has enabled the creation of an evidence-based, user-centred perinatal sexual health programme designed specifically for the Squeezy app. The programme addresses clear gaps in information pertaining to resuming pleasurable sex after birth, signs and symptoms of sexual dysfunction and effective self-care strategies as identified from the literature and stakeholder input. Ongoing refinement and evaluation of this programme will determine the acceptability, longterm feasibility and potential to support improved sexual health and wellbeing during the perinatal period.

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Developing digital biorisk management laboratory assessment tool (LAT) for indonesia one health university network

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Citation

Setiawan, A.H., Iskandriati, D., Naroeni, A., Pamungkas, J. Developing digital biorisk management laboratory assessment tool (LAT) for Indonesia one health university network.

Overview

Indonesia introduced its laboratory biorisk management standard in 2017 and updated it in 2019. The Indonesia One Health University Network (INDOHUN) subsequently developed the Laboratory Assessment Tool (LAT) to be used as for its laboratory network. This paper describes the LAT development process based on the national standard.

AQ4

Background

Indonesia hosts thousands of life science laboratories across government and private sectors that handle and store microorganisms for research, education, and diagnostics [1]. To ensure safe practices, all laboratories are required to implement biorisk management in line with national and international standards. In 2017, the National Standardization Agency of Indonesia introduced SNI 8340:2016, adopting the CEN workshop Agreement (CWA) 15793, and updated it in 2019 to align with ISO 35001 [2,3]. To complement existing biosafety and biosecurity tools, INDOHUN developed a LAT to evaluate readiness for implementing these standards.

Methods

The early version of LAT is a systematic and semi-quantitative, and encrypted Microsoft Excel 2013 format. As the standard was updated into SNI ISO 35001:2019, INDOHUN initiated to update and improve the previous LAT, referring to the new standard by migrating the Excel-based version into a web-based version in 2020. The LAT development and migration consisted of several steps, which are 1) Step I: Need and scope identification, 2) Step II: Questionnaire development (including migration to web-based), 3). Step III: LAT Configuration, 4) Step IV: LAT trial-assessment, 5). Step V: LAT finalization and dissemination, and 6) Step VI: LAT training.

Results

The first version of LAT consisted of 16 elements and comprised 55 questions. The new version consisted of 11 elements and 54 questions (table 1). Each of those questions has four scoring options, with the highest level of compliance receiving a score of 4 and the most basic level receiving a score of 1. The question was developed based on both SNI 8340:2016 SNI ISO 35001:2019 standard through expert consensus and pilot validation.

TABLE 1: List of elements of LAT questionnaire

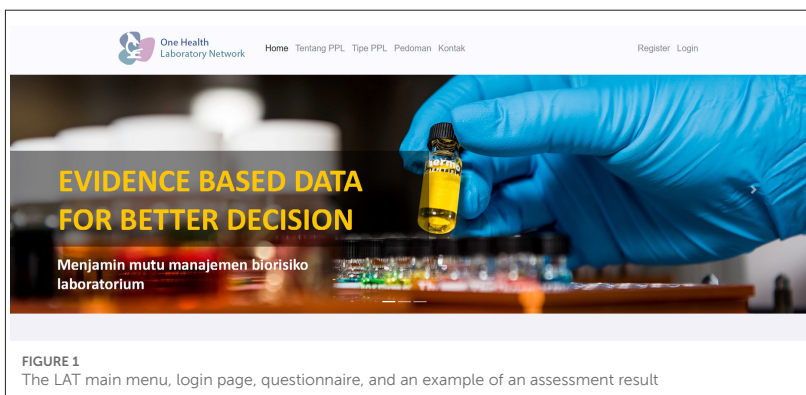
No	Element	Number of Questions
The first standard: CWA 15793 – SNI 8340:2016		
1	Biorisk management	13
2	Risk assessment	5
3	Information and inventory of biological hazards and toxins	5
4	General safety	1
5	Personnel and competency	2
6	Good microbiological technique	3
7	Personnel protective equipment	3
8	Human resource background	1
9	Occupational health program	3
10	Emergency response and contingency plan	4
11	Physical facility requirements	1
12	Equipment and maintenance	2
13	Decontamination, disinfection, and sterilization	5
14	Transportation of biological hazard and toxin	3
15	Biosecurity	1
16	Accident and incident investigation	3
Total		55

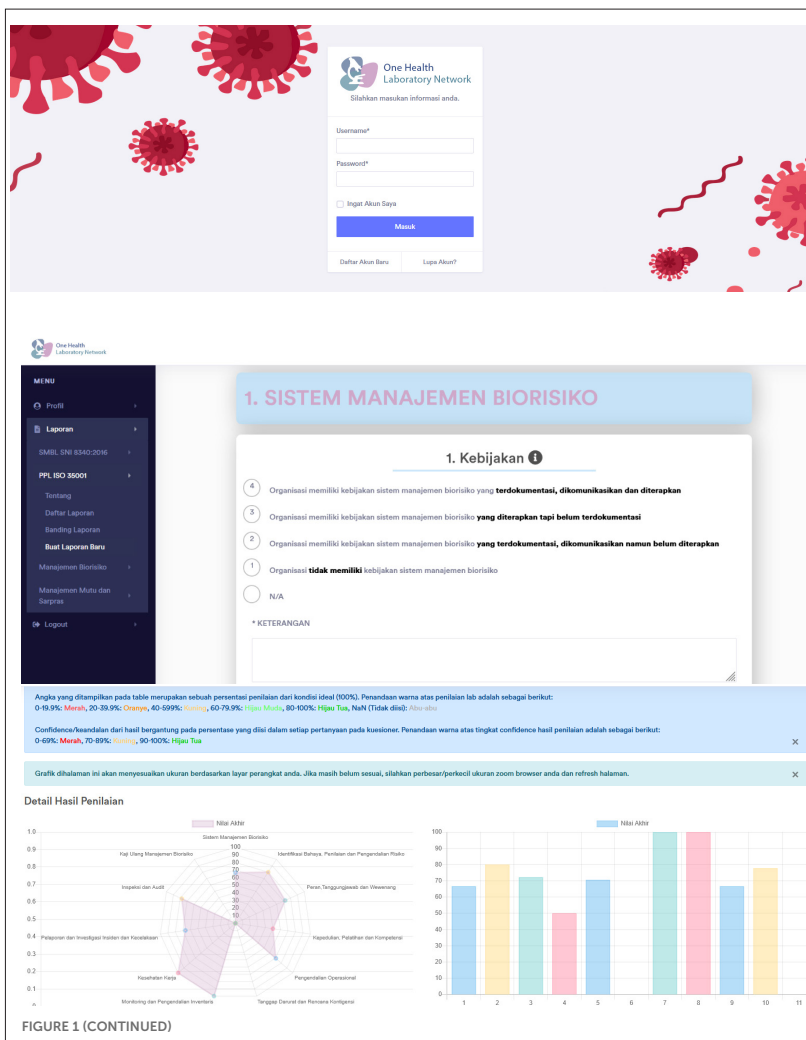
The updated standard: SNI ISO 35001:2019		
1	Biorisk management system	5
2	Hazard identification, risk assessment and control	5
3	Roles and responsibility	6
4	Training and competency	4
5	Operation control	17
6	Emergency response and contingency plan	4
7	Monitoring and inventory control	2
8	Occupation health and safety	3
9	Accident and incident investigation	3
10	Inspection and audit	3
11	Checking and corrective action	2
Total		54

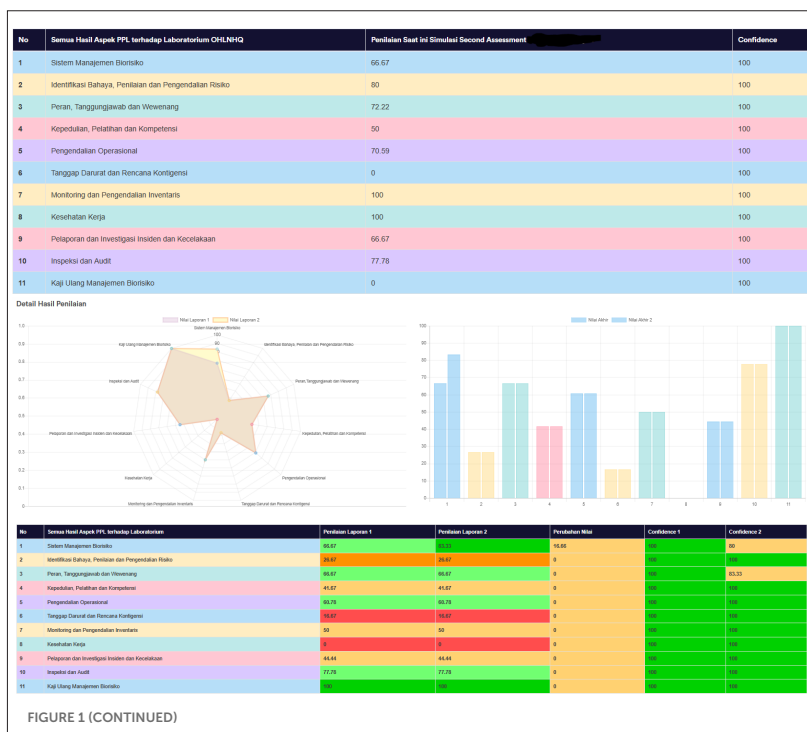
The web-based features (<http://perangkatpenilailab.id/>) include a main menu, login homepage, and a questionnaire (figure 1). The FAO Laboratory Mapping (LMT) was used as a reference to score responses to each question, using a scale ranging from 1 to 4 [4]. As a comparison, FAO LMT is focusing on animal health laboratory safety and biosecurity, while WHO Risk Assessment Tool (RAST) mobile app provides a structured, logic-based framework to evaluate laboratory biosafety and biosecurity risks to help identify hazards and assess mitigation strategies based on the WHO Laboratory Biosafety Manual 4th edition [5]. However, this LAT focuses more on laboratory biorisk management implementation readiness to comply with the Indonesian standard. During the development phase, a pilot assessment was conducted across five BSL-2 university laboratories involved in human and animal research, as well as diagnostic activities. Following this pilot, the LAT was reviewed to assess the relevance of each aspect and element, and revisions were made to improve the questionnaire.

The LAT employs a weighted scoring system adapted from the validated FAO LMT methodology [4]. Element scores are calculated as the sum of individual item scores normalized to a maximum of 100. Results are presented using bar and spider charts adapted from the FAO LMT and modified for general scientific laboratory use, with emphasis on biorisk management in accordance with SNI ISO 35001:2019 (figure 1)

Assessment scores were benchmarked against an optimal value of 100%, corresponding to an ideal score of 4 per question, and expressed as percentages (0–100%). Site confidence reflects LAT completeness, calculated from the proportion of unanswered and N/A items, ranging from 0% to 69% as low confidence, 70% to 89% as medium confidence, and 90% to 100% as high confidence (figure 1). The assessment comprises two stages: an initial baseline and a follow-up to evaluate improvement. Comparative results, applied to repeated assessments at the same site, are visualized using the same bar and spider chart (Figure 1). This approach enables laboratory management to monitor performance, assess intervention outcomes, and support continuous improvement planning.







Conclusions

The LAT was established to assess laboratory achievement on biorisk management implementation. The tool is projected not only for human and animal health laboratories but also for other relevant life science laboratories. It was recognized that trials in various laboratory settings provide a comprehensive evaluation of this tool to ensure that it is effective across life science laboratories. It allows us to describe the achievement of a standard by giving the percentage of conformity of each element shown by a bar, a spider chart, and a percentage. The immediate assessment results provided by the automatic analysis indeed make the tool effective and helpful for laboratories willing to obtain SNI ISO 35001:2019 certification.

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Digital addiction and mental health in older adults: A meta-analysis of correlations with depression and sleep quality

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A meta-analysis of correlations with depression and sleep quality.

Abstract**Introduction**

Smartphone and internet use among older adults has expanded rapidly, reshaping communication, social engagement, and access to health-related information [1]. While digital technologies may reduce social isolation and improve connectivity, increasing evidence suggests that problematic or addictive smartphone use may be associated with adverse mental health and sleep outcomes in this population [2]. Aging is intrinsically linked to physiological changes in sleep architecture, a higher prevalence of insomnia, and increased vulnerability to mood disorders, particularly depression, which may amplify the potential harms of excessive or dysregulated digital engagement [3,4]. Although observational studies [5-9] have reported associations between smartphone addiction, impaired sleep quality, and depressive symptoms in older adults, findings remain heterogeneous, and

a quantitative synthesis focusing specifically on this population has been lacking.

Materials and Methods

A systematic review and meta-analysis were conducted in accordance with Cochrane Collaboration guidelines and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [10]. PubMed, Embase, and the Cochrane Library were searched from inception to January 2026. Eligible studies were cross-sectional or longitudinal observational studies including adults aged ≥ 60 years that assessed smartphone addiction or problematic smartphone/mobile phone use using validated instruments and reported Pearson correlation coefficients between smartphone addiction and depressive symptoms and/or sleep quality. Sleep quality was assessed using the Pittsburgh Sleep Quality Index (PSQI) across all included studies, while depressive symptoms were measured using validated scales such as the Geriatric Depression Scale (GDS-15 or GDS-30) or the depression subscale of the DASS-21. Random-effects models were applied, with correlation coefficients transformed using Fisher's z and back-transformed for interpretation. Heterogeneity was assessed using Cochran's Q and Higgins' I^2 statistics. Leave-one-out sensitivity analyses were conducted when substantial heterogeneity was observed. Study quality was evaluated using the Newcastle-Ottawa Scale [11], and certainty of evidence was assessed using the GRADE framework [12].

Results

Three [5,6,7] observational studies met the inclusion criteria, comprising 1,023 participants from China, Turkey, and Brazil, with a mean age of 66.2 years. All studies employed cross-sectional designs. The pooled analysis demonstrated a statistically significant, moderate positive correlation between smartphone addiction and poor sleep quality ($r = 0.34$; 95% confidence interval [CI]: 0.06 to 0.57), indicating that higher levels of problematic smartphone use were associated with worse sleep quality among older adults. In contrast, the pooled correlation between smartphone addiction and depressive symptoms was positive but not statistically significant ($r = 0.26$; 95% CI: -0.15 to 0.60). Considerable heterogeneity was observed for both outcomes ($I^2 > 95\%$).

Sensitivity analyses suggested that one study exerted a disproportionate influence on pooled estimates and heterogeneity; exclusion of this study attenuated effect sizes and substantially reduced heterogeneity, particularly for the sleep quality outcome. Visual inspection of funnel plots suggested asymmetry, consistent with possible publication bias or small-study effects, although interpretation was limited by the small number of included studies.

Discussion

The findings indicate that smartphone addiction is moderately and significantly associated with impaired sleep quality in older adults, supporting the clinical relevance of problematic digital engagement as a contributor to geriatric sleep disturbances. Proposed mechanisms include excessive nighttime device use, cognitive-emotional hyperarousal, exposure to blue light, and circadian rhythm disruption, which may be particularly consequential in older adults with reduced sleep resilience [7,13]. In contrast, the association between smartphone addiction and depressive symptoms was less consistent and did not reach statistical significance, possibly reflecting methodological heterogeneity, differences in measurement instruments, and the multifactorial nature of late-life depression. Bertocchi et al. (2022) [5] observed that in low-income settings, the importance of smartphones correlated with reduced depressive symptoms, as the device may serve as a bridge for social support rather than a source of addictive behavior. This highlights that technology's impact on geriatric mental health is shaped by social context and the nature of its use. Although problematic digital engagement appears to be a relevant and potentially modifiable risk factor for sleep disturbances in aging populations, current evidence is limited by heterogeneity and a small number of studies [16,17]. Identifying addictive smartphone use as a modifiable factor is important for developing interventions aimed at preserving sleep quality and mental health.

Conclusion

This meta-analysis shows that smartphone addiction is linked to poorer sleep quality in older adults, while its association with depressive symptoms remains unclear. Future studies should use longitudinal designs and

standardized, age-appropriate tools to better understand causality and guide interventions promoting healthy digital use and mental well-being.

Key-words: Depression; Meta-Analysis; Older adults; Sleep quality; Smartphone addiction.

Abbreviations

CI, Confidence Interval; DASS-21-D, Depression, Anxiety and Stress Scale - 21 items (Depression subscale); GDS, Geriatric Depression Scale; GDS-15, Geriatric Depression Scale - 15 items; GDS-30, Geriatric Depression Scale - 30 items; GRADE, Grading of Recommendations, Assessment and Development and Evaluations; IAT, Internet Addiction Test; MPAS-SV, Mobile Phone Addiction Scale - Short Version; NOS, Newcastle-Ottawa Scale; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; PSQI, Pittsburgh Sleep Quality Index; r , Pearson correlation coefficient; SAS, Smartphone Addiction Scale; SAS-SV, Smartphone Addiction Scale-Short Version; SD, Standard Deviation.

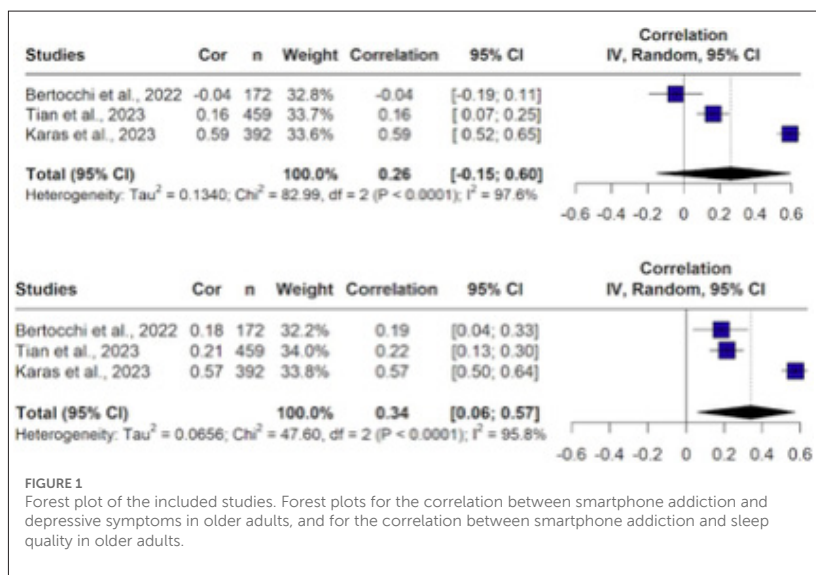
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TABLE 1: Characteristics of Studies Included in the Meta-Analysis

Character- istic	Bertocchi (2022)	Tian (2023)	Kara, s (2023)
Country	Brazil	China	Turkey
Study design	Population-based cross-sectional	Cross-sectional	Cross-sectional
Sample size (N)	668	459	392
Mean age (SD)	69.50 (7.47)	63.44 (3.96)	65.62 (0.94)
Female, %	50.1%	56.9%	22.7%
Addiction measure	IAT	MPAS-SV	SAS
Depression measure	DASS-21-D	GDS-15	GDS-30

Abbreviations: DASS-21-D, Depression, Anxiety and Stress Scale - 21 items (Depression subscale); GDS-15, Geriatric Depression Scale - 15 items; GDS-30, Geriatric Depression Scale - 30 items; IAT, Internet Addiction Test; MPAS-SV, Mobile Phone Addiction Scale - Short Version; SAS, Smartphone Addiction Scale; SD, Standard Deviation.

* Total sample (N = 668) included smartphone users, basic cell phone users, and non-users. Data for correlation analysis were derived from the smartphone user subset (n = 172).



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Digital and psychological interventions to prevent the onset of depression among perinatal women: Preliminary results of a meta-analysis with implications for public health

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Introduction

The perinatal period can be a challenging time for women, often associated with mental health issues such as depression (Howard & Khalifeh, 2020). Current studies place the prevalence of perinatal depression (PND) at around 20% (Woody et al., 2017). This condition not only have a negative impact on the woman, but her partner and baby can also be affected (Slomian et al., 2019). At a societal level, the costs associated with PND are around £74,000 per mother-child pair in the United Kingdom (Luca et al., 2020).

These issues point to the maternal PND as a public health problem and highlight prevention as an essential action for addressing it (Ormel et al., 2020).

Although previous studies have shown a low but significant preventive effectiveness of psychological interventions for maternal PND, they still fail to specifically distinguish between effectiveness associated with prevention and treatment (Buntrock et al., 2024). This is because the studies frequently focus on women at risk of PND and do not exclude those already experiencing depression (Motrico et al., 2022). Accordingly, clinical practice guidelines have emphasized the need to develop care models that allow for the early identification of this condition to promote tiered care models within the public health system. The benefits of implementing such models include resource efficiency and personalized care (World Health Organization, 2022).

Although there is currently insufficient evidence, digital tools represent an appropriate format for implementing these interventions from public health services, as they are scalable, have a lower cost, and allow for the personalization of content, thus favoring aspects such as adherence and

motivation towards intervention programs (Progga & Rubya, 2025; Torous et al., 2025).

For these interventions to be successfully integrated into public health services, they must be developed from a patient and public involvement (PPI) perspective. This methodology could foster the development of patient-centered, sustainable models that are tailored to the needs of women in the perinatal period (Zhou et al., 2024).

In this context, we conducted a systematic review and meta-analysis to assess the effectiveness of digital and psychological interventions to prevent perinatal depression among women without diagnosed depression.

Methods

The PRISMA statement (Page et al., 2021) and the Guidance for Reporting Involvement of Patients and the Public 2 checklist were followed (Staniszewska et al., 2017). A comprehensive search was performed from databases inception to 4 July 2025 across 13 electronic databases. Eligibility criteria were defined using the PICO(S) framework (Higgins & Green, 2011). Eligible studies were randomised controlled trials (RCTs) including women without depression at baseline, evaluating digital or blended (digital and face-to-face format) psychological interventions compared with usual care, no intervention, attention control, or waitlist controls, and reporting outcomes related to the prevention of PND, assessed as prevention of depression onset or the reduction of symptoms of depression.

The most relevant data extracted included participant characteristics (e.g. pregnancy or postpartum), intervention features (e.g. orientation, guidance), outcomes (e.g. instrument used), and follow-up assessments. Risk of bias was assessed using the Cochrane Risk of Bias 2 tool (Sterne et al., 2019). Cohen's *d* was used to calculate the standardized mean difference (SMD) (Cohen, 1989). Random effects models were used for pooling effect sizes. Heterogeneity was evaluated using the *Q* statistic and *I*² index (Higgins et al., 2019). Publication bias was assessed through Egger's test (Stuck et al., 1998), the Begg test and Mazumdar rank correlation analysis (Begg & Mazumdar,

1994), and the Duval & Tweedie trim-and-fill procedure (Duval & Tweedie, 2000).

Sensitivity analyses explore variations of the SMD through a fixed-effect model; calculating Hedges's g ; excluding the study that had the strongest influence on the heterogeneity; calculating the average SMD across all follow-up assessments; and including or excluding studies with low and high risk of bias, respectively.

Meta-regression analyses were performed using the random-effects model. Risk of bias and sample size were included as covariates (the latter if publication bias was observed). The Knapp and Hartung adjustment was implemented (Knapp and Hartung, 2003). Permutation tests were used to correct for multiple testing (Higgins & Thompson, 2004). The quality of evidence was appraised using the GRADE approach (Balshem et al., 2011).

PPI was incorporated through a webinar with perinatal mental health professionals. The webinar served to present the main results and to gather the professionals' interpretations and opinions on these results. The data was analyzed using qualitative content analysis (Krippendorff, 2018).

Results

Of the 12,603 records identified, 27 RCTs were included in the qualitative analysis, and 24 RCTs were incorporated into the meta-analysis. Most of the studies recruited women during pregnancy (17 RCTs) and included both primiparous and multiparous women. The mean age of women was around 27 years old. Thirteen RCTs implemented universal preventive strategies. Psychological interventions were developed exclusively during pregnancy (10 RCTs) or the postpartum period (10 RCTs). Psychoeducation (15 RCTs) was the most common therapeutic approach. Most interventions were delivered online (22 RCTs), with five RCTs combined online and face-to-face formats. Follow-up assessments ranged from mid-pregnancy to one year postpartum.

Twenty RCTs were rated as high-risk studies. Meta-analytic results demonstrated a small and significant preventive effect in favor of the

intervention group compared to the control group (SMD = -0.192 ; 95% CI -0.288 to -0.097 ; $p < 0.001$), with substantial and significant heterogeneity ($I^2 = 64.9\%$; $Q_{25} = 71.25$, $p < 0.001$) (see Figure 1). Evidence of publication bias was detected, and the pooled SMD ceased to be significant when studies with a high risk of bias were excluded.

Meta-regression analyses indicated that interventions delivered from pregnancy to the postpartum period or only during the postpartum period were associated with greater effectiveness (see Figure 2). According to GRADE, the quality of evidence was low.

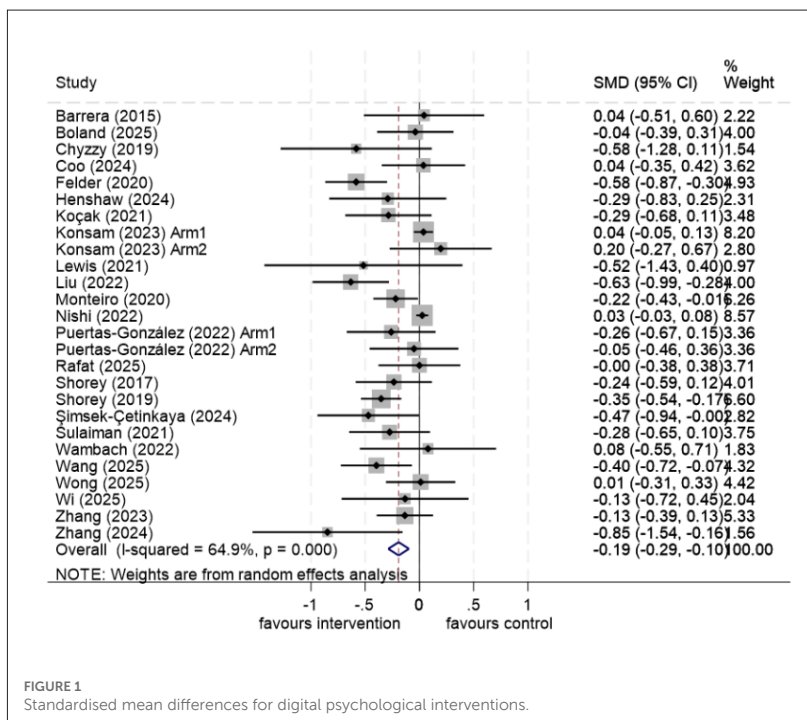


FIGURE 1
Standardised mean differences for digital psychological interventions.

Independent variables	Unadjusted Coefficient ^{a,b}	p	p (95% CI) ^{c,e}	Adjusted Coefficient for risk of bias and total sample ^{b,d}	p	p (95% CI) ^{c,e}
Parity (primiparous only)	-0.115 (-0.272 to 0.042)	0.143	0.867 (0.862 to 0.871)	-0.027 (-0.250 to 0.195)	0.800	0.992 (0.991 to 0.993)
Previous history of depression (Women without history of depression)	0.060 (-0.149 to 0.269)	0.556	0.999 (0.998 to 0.999)	0.092 (-0.116 to 0.300)	0.363	0.990 (0.989 to 0.992)
Exclusion criteria for PND at baseline (diagnostic interview)	-0.105 (-0.271 to 0.060)	0.198	0.732 (0.726 to 0.738)	-0.148 (-0.369 to 0.073)	0.177	0.749 (0.743 to 0.755)
Type of prevention (universal)	-0.147 (-0.365 to 0.071)	0.174	0.440 (0.434 to 0.447)	-0.222 (-0.471 to 0.026)	0.076	0.331 (0.324 to 0.337)
Antenatal-only delivery	0.167 (0.015 to 0.318)	0.033	0.180 (0.175 to 0.186)	0.236 (0.059 to 0.413)	0.012	0.120 (0.116 to 0.125)
Comorbidity of anxiety assessment (no)	-0.061 (-0.269 to 0.147)	0.546	0.998 (0.997 to 0.999)	-0.123 (-0.342 to 0.096)	0.252	0.975 (0.973 to 0.977)

Note.

^a F (6.19) = 7.82; p = 0.0002; I² residual = 0%; Adjusted R² = 100%.

^b Knapp and Hartung method used for estimation of standard errors and 95% confidence intervals (CIs)

^c Higgins and Thompson permutation test applied to calculate p values with multiplicity adjustment (Monte Carlo approach with 20 000 permutations)

^d F (8.17) = 6.82; p = 0.0005; I² residual = 0.00%; Adjusted R² = 100%.

^e Calculated using the *metareg* command in Stata (version 14.5) based on the F-test.

FIGURE 2
Meta-regression analysis.

Six themes emerged through PPI analysis, including measurement approaches, social context, the co-occurrence of perinatal depression and anxiety, parity status, prior experiences of perinatal loss and education about perinatal health.

Conclusions

After isolating the preventive effect of online and psychological interventions on PND and incorporating PPI, this study suggested that preventive effectiveness may be greater when these interventions are delivered across the perinatal period.

However, the small effect of SMD, heterogeneity between studies and publication bias limit confidence in the observed effects. The frequent reliance on self-report measures and limited use of diagnostic interviews further constrain interpretation.

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Digital citizenship for health: Policy brief and youth-authored statement from DTH-Lab

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Citation

Goeschl, S., Gray, W. Digital citizenship for health: Policy brief and youth-authored statement from DTH-Lab.

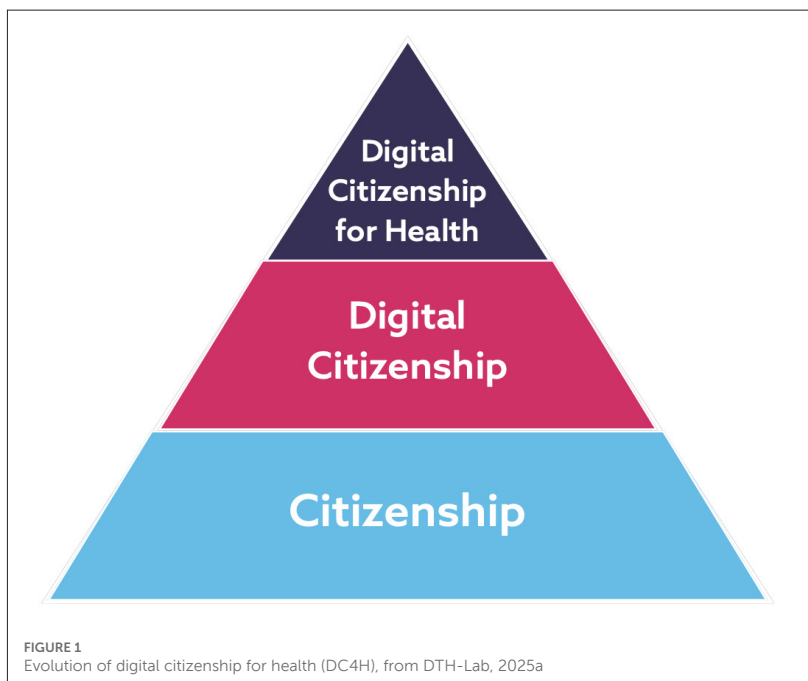
Introduction

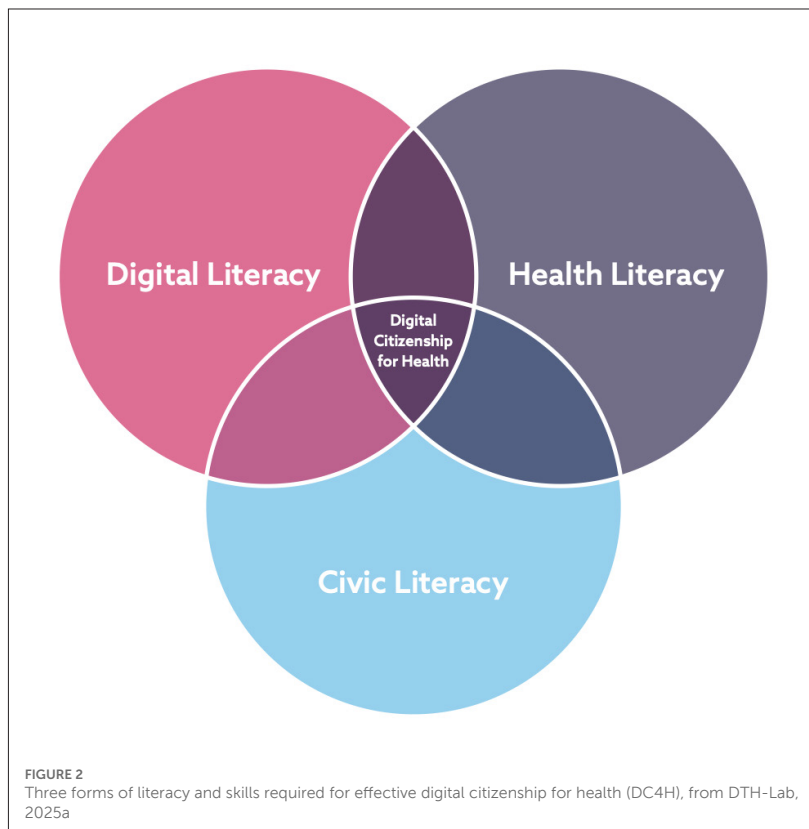
The Lancet and Financial Times Commission on Governing Health Futures 2030's (GHFutures) report highlights the interconnections between digital, health and civic literacy, claiming that none can be expressed effectively without the others in a digital age (Kickbusch et al., 2021). Yet these components are often treated as separate, limiting young people's ability to engage confidently as digital citizens, particularly those in marginalized groups.

Digital citizenship for health (DC4H) provides a framework to pursue governance approaches through digital citizenship education that increases health equity, enfranchises communities and advances public participation in health and well-being, particularly for youth health and well-being (DTH-Lab, 2025a). DC4H extends the notions of citizenship and digital citizenship to the health domain, with each level representing an evolution of the previous one as digital innovations become increasingly integrated into health systems and

everyday life (Figure 1). It builds on digital citizenship while accounting for the profound effects of digitalization on societal health and well-being, as digital technologies create new spaces for health to be promoted or harmed and new ways for healthcare to be delivered.

Effective DC4H requires combined digital, health and civic literacy and skills (Figure 2). Its promotion enables individuals and communities to engage in healthcare choices, advocate for their rights, and hold healthcare providers, policymakers and technology developers accountable for equitable access to quality services. DC4H also supports participation in the design and governance of digital transformations for health and the co-creation of equitable digital environments, while emphasizing collective responsibility for ethical digital, data and AI governance.





Methods/Data

This work builds on the conceptual foundations from the GHFutures2030 Commission report (Kickbusch et al., 2021) as well as the Digital Transformations for Health Lab's (DTH-Lab) policy brief on DC4H and draws on a participatory Study Session on DC4H (Digital citizenship for health: Empowering young people to co-create tools and innovate solutions), organized by DTH-Lab in partnership with the Youth Health Organization

and the Council of Europe Youth Department (DTH-Lab, 2025a; DTH-Lab, 2025b). The study session was held from 17-21 February 2025 at the European Youth Centre in Budapest and brought together young people to reflect on digital transformations and health, contributing to a youth-authored statement that informed the results and calls to action presented here.

Results

The statement outlines four key concerns identified by youth in 2025 for advancing DC4H. Key concern 1: Accessibility and inclusivity

Digital exclusion persists due to unequal access to infrastructure, devices, usability and affordability. Inequalities across countries remain stark, alongside lack of diverse health data, culturally sensitive solutions, algorithmic bias, and restricted access caused by complex platforms and paywalls. While digital spaces are central to participation, representation in decision-making remains limited. A value-based approach to digital health transformations should be taken, integrating shared values, cultural considerations and diverse perspectives from design to implementation to ensure accessibility and inclusivity.

Key concern 2: Digital well-being and online communities

Digital environments affect mental, physical, social and environmental health, including through digital overload, sedentary lifestyles, social isolation and the growing environmental footprint of digital technologies. Rather than blanket restrictions, preventative measures are required through literacy, regulation and accountability. At the same time, online spaces offer opportunities for connection and civic engagement, highlighting the importance of balanced online-offline life and recognition of young people as rights holders.

Key concern 3: Literacy and education

Combined digital, health and civic literacy is essential for meaningful participation, yet gaps persist in education systems, alongside regional disparities in digital skills and curricula insufficiently adapted for inclusive digital learning. In health and STEM fields, preparation for digital-first systems remains inadequate, with currently limited training in digital health. Representation in research and policymaking is still lacking, and there is a need to build literacy among youth through both formal and non-formal education to enable co-creation and policy engagement.

Key concern 4: Trust in a digital age

Trust remains fragile due to limited transparency, weak consent mechanisms, poor explainability of AI, cybersecurity vulnerabilities and shifting power toward private actors. Mis- and disinformation further erode confidence, while effective fact-checking practices are lacking. Strong digital and data governance is required to ensure safe data handling, reliable health information and accountable use of emerging technologies, forming the foundation for rebuilding trust and sustaining engagement.

Conclusion

To enable all young people to become digital citizens for health, multisectoral and intergenerational collaboration between governments, international organizations and other actors is needed. Together, they must adopt a human and child rights approach to digital and data governance, invest in combined digital, health and civic literacy, meaningfully engage diverse youth in health and digital governance processes, and prioritize DC4H in education and capacity-building activities. Youth must be recognized as active partners in shaping digital transformations in health and innovation to create inclusive, equitable and transparent digital ecosystems that prioritize privacy, accessibility and well-being. Through collaboration between governments, organizations, and young people, digital spaces can better reflect diverse needs and promote health and well-being. Empowering youth with the tools, knowledge and opportunities to engage in digital health governance and citizenship will support a healthier, more inclusive society and an innovative digital future.

The authors of the youth-authored statement (DTH-Lab, 2025b) call on intergovernmental organizations, national governments, health systems and institutions, private sector, academic publishers and journals, education institutions and fellow young people to:

1. Recognize young people as key partners in promoting DC4H with an active role in shaping their own health futures.
2. Promote open access to trusted health information and resources that mitigate risks associated with mis/disinformation.
3. Increase efforts for underprivileged youth to participate in digital, health and civic spaces.
4. Encourage and provide offline interactions with young people that promote physical, mental health and well-being.
5. Improve youth leadership skills across sectors to participate in decision making processes in governance and civic institutions.
6. Develop government-endorsed digital platforms that provide accurate, fact-checked and trustworthy health information.
7. Mandate rigorous verification protocols and promote the use of youth-friendly language and visuals to enhance understanding and trust.

Disclosures

This submission draws on two previously published documents from the Digital Transformations for Health Lab (DTH-Lab). ChatGPT 5.2 (chatgpt.com, OpenAI 2026) was used for initial reformulation of certain paragraphs from these source documents. All content was human-edited and source-verified.

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E-health literacy as a correlate of usability in digital vaccination decision support: Evidence from a Polish adaptation study

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Citation

Sikorska, M., Koeck, J.-L., Stropalova, O., Kaag, F., Jakubowski, S., Smola, P., Maj, A., Duplaga, M. E-health literacy as a correlate of usability in digital vaccination decision support: evidence from a Polish adaptation study.

Overview

Digital public health increasingly relies on usable decision-support tools to translate vaccination policy into actionable guidance. Within the EUVABECO project, the Mentor®, a vaccination guidance app, was adapted for Poland to provide adult immunization recommendations to lay users and is freely available online. We conducted a prospective, single-group usability evaluation ($N = 31$) using task-based persona scenarios and free exploration, with System Usability Scale (SUS) as the primary endpoint. Mean SUS was 71.50 (SD 3.63), and 64.5% scored ≥ 68.00 , indicating good usability. SUS correlated with e-health literacy (eHL) ($\rho = 0.44$, $p = 0.01$) but not with age or vaccination hesitancy; married participants rated usability higher ($p = 0.02$). Overall, the adapted tool achieved good usability, and the results suggest that eHL may be an important determinant of perceived usability in citizen-facing vaccination decision support.

Background

Digital public health programs increasingly rely on software tools that translate vaccination policies into actionable guidance for the public and health professionals. Their population impact depends not only on clinical accuracy but also on usability, because usability affects whether users can complete tasks efficiently, understand outputs, and trust recommendations. The EUVABECO project (European Vaccination Beyond COVID-19), funded under the European Commission's EU4Health program, aims to help European Union Member States adopt and scale effective vaccination practices by transferring and adapting solutions across countries, supported by structured implementation planning and pilot validation in new settings (1). Within EUVABECO, this transfer strategy is operationalized through complementary types of digital solutions supporting vaccination services, including decision support systems, tools enabling targeted outreach by social screening, electronic information leaflets, modeling and forecasting tools to estimate intervention impact, and an electronic vaccination card supporting continuity of vaccination history across care settings (1).

Mentor® was originally developed and deployed in France by a consortium partner (Syadem) (2, 3) and was subsequently localized for Poland.

The Polish implementation was narrowed to provide adult vaccination recommendations for lay users and aligned with national immunization requirements while preserving core decision support functionality. This study evaluated the usability of the Polish version of the application using the System Usability Scale (SUS) (4) and examined whether SUS ratings differed by sociodemographic characteristics, vaccination attitudes, and eHL.

Methods

We conducted a prospective, single-group usability evaluation with $N = 31$ adults recruited to reflect, as far as possible, heterogeneous profiles (sex, age, and place of residence). Data collection was implemented online using two survey modules administered immediately before and after structured application use. The pre-use module captured sociodemographic characteristics, patterns of digital tool use (Internet intensity), sources of vaccination information, vaccination attitudes (including a vaccination hesitancy score), and eHL assessed with the eHEALS instrument (5).

Application testing followed a standardized, task-based protocol combining persona scenarios with free exploration. Participants completed three scenario walkthroughs representing common adult vaccination decision situations. Each scenario required entry of profile information and navigation through recommendations and explanatory content, designed to activate distinct recommendation pathways and interface elements. After completing the scenarios, participants explored the application freely to approximate real-world use and to identify additional friction points not covered by scripted tasks.

In the post-use module, perceived usability was measured with SUS as the primary endpoint (4). Additional outcomes included Net Promoter Score, the User Experience Questionnaire, and open-ended questions on difficulties, strengths, and suggested changes. Analyses focused on SUS. We calculated SUS total scores using standard scoring rules. Associations between SUS and continuous predictors (eHL, age, vaccination hesitancy score, and the number of vaccination types taken in adulthood) were assessed using Spearman correlations. Group comparisons across sociodemographic strata

were evaluated using rank-based tests (Mann–Whitney U or Kruskal–Wallis), given the small sample size and expected non-normality of usability ratings.

Results

Participants had a mean age of 42.5 years (SD 12.4), and 58.1% were women. Most lived in cities with at least 500,000 residents (54.8%). The sample was highly educated (74.2% held a master's degree or equivalent), and 54.8% were married. Digital readiness was high: 54.8% reported proficient internet skills, and 51.6% reported more than 10 hours of Internet use per week.

All participants completed the post-use evaluation. Mean SUS was 71.50 (SD 3.63), indicating good perceived usability. The proportion scoring at or above the acceptability threshold ($SUS \geq 68.00$) was 64.5% ($n = 20$). SUS showed modest variation across strata. Married participants rated usability higher than participants who were single or divorced (mean 78.30 vs 59.30; Mann–Whitney test, $p = 0.02$). SUS did not differ by education (master's vs lower education; $p = 0.53$), place of residence (Kruskal–Wallis, $p = 0.16$), or income (Kruskal–Wallis, $p = 0.56$). No differences were observed by self-assessed digital skills ($p = 0.68$) or weekly internet intensity ($p = 0.36$). eHL correlated positively with SUS (Spearman $\rho = 0.44$; $p = 0.01$). SUS was not associated with age ($\rho = 0.08$; $p = 0.65$) or vaccination hesitancy ($\rho = -0.17$; $p = 0.35$). Open-ended feedback confirmed the feasibility of completing scenario tasks and provided actionable suggestions to improve the clarity of explanations and navigation.

Conclusions

The Polish localized Mentor application achieved good perceived usability following scenario testing and free exploration. These findings support EUVABECO's transfer approach by indicating that a decision-support tool developed in one national context can be adapted for use in another Member State while retaining acceptable usability. The positive association between eHL and usability is plausible because eHL reflects skills needed to interpret and appraise health information, follow multi-step guidance, and resolve uncertainty when recommendations depend on user-entered data. These competencies may reduce cognitive load during task completion.

In contrast, weak effects for other variables may reflect limited variability in this pilot sample (higher education and digital readiness), ceiling effects on basic internet skills, and the global nature of SUS, which may be less sensitive to differences in vaccination attitudes when users focus primarily on navigation and comprehension. Larger evaluations should confirm subgroup patterns, include a more diverse set of users, and assess whether usability improvements translate into better understanding of recommendations and outcomes relevant to adult immunization.

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How did state premiers frame COVID-19 risk during the first 100 days of the pandemic? A comparative discourse analysis

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Citation

Back, E., Kostkova, P.P., Roberts, S. How did state premiers frame COVID-19 risk during the first 100 days of the pandemic? A comparative discourse analysis.

Introduction

During the past two decades, the World Health Organization (WHO) has declared several Public Health Emergencies of International Concern (PHEIC) – including H1N1 influenza, two Ebola outbreaks, Zika, two Monkeypox strains, and COVID-19 – and modelling predicts increased pandemic frequency and intensity in the years ahead (Marani *et al.*, 2021). The global health threat this presents is well recognised by state institutions.

This poster shares early insights from an ongoing doctoral research project, which explores public perceptions of pandemic risk in the wake of COVID-19 through a comparative study. The study covers the United Kingdom (UK) and two other countries with contrasting experiences of COVID-19: South Africa (SA) and New Zealand (NZ). Research questions include whether and how

people are continuing to use certain health protection measures or engaging in new health-seeking behaviours; how they rate the likelihood of another infectious disease pandemic in the near term; and what, if anything, they believe should be done to reduce this risk.

The project uses an interdisciplinary mixed-methods approach, drawing from fields including global health, and risk and disaster reduction, and from social and cultural theory. As such, it is designed to explore people's risk perceptions from different angles, including by considering how these perceptions may be shaped by dominant risk narratives and other state discourse.

The project will collect and analyse primary data. Quantitative data will be generated via a survey conducted within a short timescale across all three countries and subjected to statistical analysis. Qualitative data will be gathered through free text fields in the survey, as well as through focus groups with young people, as evidence suggests exposure to epidemics during the "impressionable years" (ages 18 to 25) has a negative and lasting effect on confidence in political leadership and institutions as well as public health systems (Aksoy *et al.*, 2020). All qualitative data will be organised using MAXQDA software then inductively coded and analysed thematically.

The first phase of the project includes a scoping review and a discourse analysis, which provide a contextual baseline for the study and will ultimately inform the development of survey questions and focus group topic guides.

The poster shares initial findings from the analysis of pandemic-related state discourse in NZ, SA and the UK. This work examines the statements on COVID-19 that were delivered by national premiers – namely Prime Minister Johnson in the UK, Prime Minister Ardern in NZ, and President Ramaphosa in SA – during the first 100 days of the pandemic, which informed public perceptions of COVID-19 risk during that period and beyond. The first 100 days were selected as international partners are now focused on ensuring medical countermeasures – rapid diagnostics, therapeutics and vaccines – will be available within 100 days of a future pandemic being declared. As they

acknowledge, “non-medical public health interventions like social distancing, isolation, contact tracing and personal protective equipment” will continue to be “essential” during the first 100 days (Pandemic Preparedness Partnership, 2021, p.7). The way states frame and communicate risk to the public during this early phase is therefore vital in shaping perceptions, informing behaviours, and limiting health and other impacts.

Methods

The table below captures the statements by national premiers that were identified as being in scope for the discourse analysis. These were delivered in the period through to 19 June 2020 (100 days after WHO declared COVID-19 a pandemic on 11 March).

Transcripts of these statements were downloaded from the relevant government web archives. The discourse analysis incorporated three dimensions: content analysis, characterising each statement and documenting key statistics, such as word counts; narrative analysis, capturing key messages and mapping a narrative arc for each premier over the 100-day period; and thematic analysis, based on inductive coding of each statement using MAXQDA software. The latter involved two rounds of coding, identification of overarching ‘risk frames’ and other parent codes, followed by a third round of coding. A Foucauldian lens was applied, paying particular attention to how state premiers use language to frame COVID-related risk and request observance of the ‘rules’ and ‘guidance’ developed to ‘manage’ such risk.

National premier	Statements to 19/6/2020	Total words
Arden (NZ)	10 addresses or releases 18/3-8/6	Range 304-1207; average 500
Ramaphosa (SA)	8 messages, addresses or statements 24/3-17/6	Range 1429-3470; average 2759
Johnson (UK)	20 statements 3/3-16/6	Range 421-2083; average 1030

Results

Five thematic risk frames were identified: securitisation – e.g., existential threat, military references, state power and authority, state protection of citizens, emphasis on rules and requirements; moralisation – e.g., unequal risk exposure and impacts, right and wrong behaviour, emphasis on care, compassion, need to look after each other; technification – e.g., emphasis on data, science, tools, technologies, state expertise and resources; individualisation – e.g., threats to freedoms, emphasis on personal responsibility, protecting oneself and family/friends; and socio-economic – e.g., risks to economy, business, livelihoods, way of life, alongside potential for recovery. Securitisation framings were dominant, alongside moralisation (Arden, Ramaphosa) and technification (Johnson). Socioeconomic framings also featured strongly in Ramaphosa's statements.

Three situational framings were also evident (heightened threat, diminished threat, risk reduction). The language used to communicate COVID-related risk reduction measures ranged from direct instructions (commanding, telling) and obligations (must, should) to softer persuasion (urge, encourage) and solicitation (ask, appeal), with Johnson more direct, Ramaphosa less direct, and Arden combining strong commands with softer asks.

Conclusions

The poster shares work in progress from an ongoing research project, which has potential for both instrumental and conceptual impact. Once complete, the study will provide insights into how different approaches to COVID-19 risk communication and management have influenced perceptions of pandemic risk over the longer term, which may inform future pandemic preparedness and response.

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Implications of compound climatic extremes and human mobility for predicting dengue dynamics in Vietnam

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Citation

Fletcher, C., Belman, S., Gibb, R., Do, K.Q., Pham, Q.D., Thao Nguyen, T.T., Nam, V.S., Lan, P.T., Colón-González, F.J., Palmer, J., Lowe, R. Implications of compound climatic extremes and human mobility for predicting dengue dynamics in Vietnam.

Introduction

Dengue fever represents a major public health challenge across the globe, with Southeast Asia among the most affected regions (WHO 2025). Vietnam has previously reported the highest number of cases in the region (Wartel et al. 2017; ECDC 2025; WHO 2025). Despite this, variation in transmission dynamics exists between the subtropical north, where outbreaks are intermittent, and the tropical south, which experiences a more seasonal endemic pattern (Colón-González et al. 2021; Quyen et al. 2018). Modelling dengue outbreaks in emerging settings requires different model specifications compared to regions with regular seasonal outbreaks (Gibb et al. 2023; Lee et al. 2021). Incorporating human mobility data into forecasting frameworks previously demonstrated improved predictive model performance (Kiang et al. 2021; Khan et al. 2025). Here, we present a district-level dengue modeling approach to test a joint climate-human mobility framework. We explore the implications of including different structures of spatial connectivity across the regions of Vietnam for incorporation into a district-level early warning system framework for dengue outbreak risk prediction. In brief, we (i) explore additive and interacting climatic, oceanic, and socioeconomic drivers of dengue at the district level across 8 subregions (Fletcher et al., 2025), and then (ii) compare different methods accounting for spatial structure and connectivity, including mobility covariates and matrices.

Methods

We apply our framework across Vietnam including 670 districts (note: these district boundaries changed in 2025) using monthly dengue case data spanning 2001–2021.

We developed a Bayesian hierarchical mixed modelling framework incorporating seasonal, interannual, and spatial random effects to account for unmeasured space-time variations in dengue risk. We then included climatic, oceanic and socioeconomic indicators together with the space-time random effects. We favoured covariates that improved goodness-of-fit across a range of metrics and with parameter distributions that excluded zero. We identified two best models to explore whether spatial connectivity improved model performance. These models included a 3-way interaction

between the 6-month average temperature anomaly (lagged 0 months), and a 6-month precipitation index (either total precipitation or a drought index; lagged 4-5 months and lagged 0 months), with additive effects for the 6-month averaged temperature (lagged 0 months), Dipole Mode Index (lagged 4 months), and proportional access to hygienic toilets. For all covariates, the effects were stratified by 8 subregions using random slopes.

We next incorporated multiple formulations of spatial connectivity to determine whether they improve model performance. We included neighborhood adjacency matrices, gravity models, travel time estimates, and human mobility probability matrices derived from Meta Data for Good movement distribution maps.

We also developed three refined human mobility indicators from the movement distribution maps to better capture transmission dynamics.

(i) Mobility Inflow

The first covariate, normalized inflow ($\mathcal{I}_j$), quantifies the relative volume of individuals arriving into the district j from all other origins. We first calculate the Expected Raw Inflow (F_j) as the sum of individuals moving from all origin districts i to destination j , weighted by the origin population (P_i):

$$F_j = \sum_{i \neq j} M_{ij} P_i$$

We then normalized (F_j) by the maximum inflow observed across all districts to produce the final covariate:

$$\text{Inflow}_j = \frac{F_j}{\max_k(F_k)}$$

Where M_{ij} represents the probability of movement from district i to district j in the asymmetric origin-destination matrix (M). By setting $i \neq j$, we ensure the metric reflects only mobility rather than local residence.

(ii) Stay Probability

The staying probability (S_i) represents the proportion of individuals who remain within their home district during the observed period. This is captured by the diagonal elements of the mobility matrix:

$$S_i = M_{ii}$$

(iii) Conditional Outgoing Entropy

To measure the diversity of travel destinations, we calculate the Conditional Shannon Mobility Entropy (H_i^*). This metric specifically describes the geographical spread of people leaving a district excluding those who are staying M_{ii} .

We first calculate the conditional transition probability (M_{ij}^*), which re-scales the probabilities so that movement to all other districts sums to 1:

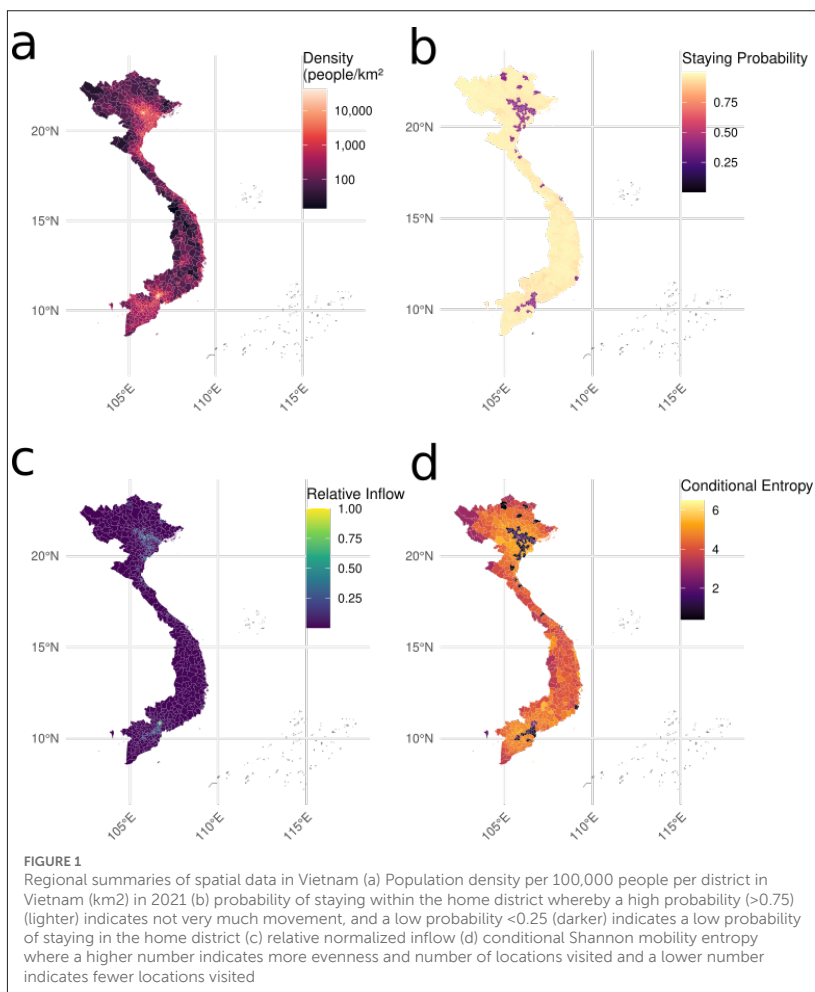
$$M_{ij}^* = \frac{M_{ij}}{\sum_{k \neq i} M_{ik}}, \quad \text{for } j \neq i$$

We then calculate the entropy based on this conditional distribution:

$$H_i^* = - \sum_{j \neq i} M_{ij}^* \ln(M_{ij}^*)$$

A high H_i^* indicates that travelers from district i disperse to many different locations, whereas a low value suggests movement is concentrated toward a few specific destinations.

Mobility indicators were incorporated both as fixed covariates and as spatially structured random effects within a modified Besag-York-Mollie (BYM2) framework with structured and unstructured spatial effects.



AQ1

Results

We consistently found a negative association between both staying probability and conditional mobility entropy and with dengue in the

Northwest (83% non-urban) and Central Highlands (72% non-urban) sub-regions. This indicated that when people stayed in their home district, or moved evenly to the surrounding districts, dengue risk decreased. Mobility inflow was not a useful covariate whereby all parameter estimate distributions included zero. When we incorporated spatial connectivity matrices rather than an adjacency matrix to account for residual spatial confounding, the goodness-of-fit consistently improved in a model which incorporated total precipitation, but only the travel time matrix showed improvement in a model which incorporated a drought index.

Conclusions

While it is clear that climate contributes to dengue dynamics in endemic settings, our study demonstrates that combining human mobility and socioeconomic drivers with climatic factors may improve estimates of dengue incidence in non-endemic settings. Specifically, human mobility covariates have a stronger effect on dengue risk in high elevation, rural regions such as the Central Highlands and Northwest sub-regions; the mechanism of this effect needs further exploration. This builds on existing research to further support that human mobility covariates may be valuable for dengue prediction in emerging regions. This can inform design for early warning systems, provide actionable insights for public health decision-making, as well as further elucidate the role of human mobility in dengue risk.

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Integrated epidemic intelligence for early warning of arboviruses: A digital public health pilot in a vulnerable territory of Brasília, Brazil

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Abstract

Arboviral diseases represent public health challenges in Brazil, particularly in socially vulnerable urban territories such as Estrutural, Brasília/BR, where conventional surveillance detects outbreaks with delays due to fragmented data flow. To overcome these limitations, this pilot study presents an integrated Epidemiological Intelligence system grounded in the principles of Digital Public Health. The methodology combines indicator-based, syndromic, event-based, laboratory, and genomic surveillance streams, utilizing geocoding of residential addresses and spatiotemporal cluster detection models. The results demonstrate that this multidimensional approach can generate early warnings before the surge in routine notifications, employing laboratory data to validate signals and reduce false positives. The combination of interoperable data systems with community-reported signals enhances micro territorial situational awareness. This integration facilitates targeted vector control, field investigations,

and equitable local health decision-making in resource-limited urban environments.

Background

Arboviral diseases, including dengue, chikungunya, and Zika, represent critical public health challenges in Brazil. In socially vulnerable urban territories, such as the Estrutural region in Brasília, the convergence of environmental risks, climate variability, and uneven healthcare access significantly intensifies transmission. Conventional surveillance systems frequently detect outbreaks too late due to reporting lags, fragmented data flows, and inadequate integration of epidemiological, clinical, and laboratory sources. In peripheral areas, micro-spatial transmission dynamics emerge long before becoming visible in routine notification systems, restricting the ability of health services to anticipate and control outbreaks. To overcome these structural limitations, moving toward an integrated, territory-sensitive surveillance model is urgent. Aligned with World Health Organization guidelines, this proposal advocates for an “Epidemic Intelligence” framework that amalgamates indicator-based and event-based surveillance. Grounded in Digital Public Health principles, the system integrates interoperable clinical, laboratory, genomic, and community-informed signals with residential geocoding. This multidimensional approach overcomes informational latency, significantly improving early warning sensitivity, empowering local primary care teams to guide targeted territorial responses, and reinforcing health equity in vulnerable urban settings.

Methods

The methodological framework for the integrated Epidemic Intelligence system, designed under a Digital Public Health approach, provides a robust structure for monitoring arboviruses in socially vulnerable urban areas, such as the Estrutural region. The system integrates multiple surveillance streams: indicator-based, syndromic, event-based, laboratory, genomic, and community surveillance. Indicator-based data incorporate arbovirus notifications from the Notifiable Diseases Information System (SINAN), laboratory confirmations from the Laboratory Environment Management System (GAL), and near realtime syndromic information from primary health

care consultations. Concurrently, event-based surveillance encompasses rumor monitoring via natural language processing of social media and informal reports, alongside structured observations from community health workers during household visits. Genomic surveillance data are further integrated to identify viral lineages and validate introduction events. To ensure interoperability, these heterogeneous databases are integrated through standardized data pipelines utilizing unique identifiers, temporal harmonization, and residential geocoding. The core spatial analysis relies on Kulldorff's spatial scan statistic, implemented in the SaTScan software, applying a discrete Poisson model to compare observed and expected event counts under the null hypothesis of spatially uniform risk. The statistical significance of identified space-time clusters is rigorously assessed by calculating the log-likelihood ratio (LLR) and performing Monte Carlo simulations. To ensure response specificity, early warning alerts are exclusively generated when statistically significant space-time clusters converge with concordant signals from at least two independent surveillance sources. Ethically, the study strictly complies with data protection regulations; personal identifiers are restricted solely to linkage and geocoding procedures, and all subsequent analyses are conducted on fully anonymized datasets.

Results

The integrated Epidemic Intelligence system enhanced early detection capabilities and territorial situational awareness compared to routine surveillance practices. By amalgamating routine notifications with near-real-time clinical, laboratory, and community-based signals, the system disseminated alerts prior to formal increases in SINAN registries, providing a temporal window for preemptive interventions. The incorporation of laboratory and genomic data proved essential for event validation; it mitigated false positives arising from syndromic or rumor-based signals while offering insights into viral lineages, cluster formation, and local transmission dynamics. Community-based monitoring identified nascent transmission signals in vulnerable micro-territories with constrained healthcare access, addressing structural deficiencies inherent in traditional surveillance. Address-level geocoding enabled the precise identification of high-resolution

space-time clusters, facilitating targeted vector control, epidemiological field investigations, and optimized primary care responses. Operational feedback demonstrated that primary care teams were motivated to assimilate this syndromic and community intelligence into their regular workflows. This multidimensional integration translates heterogeneous public health data into actionable priorities for localized community-level interventions.

Conclusion

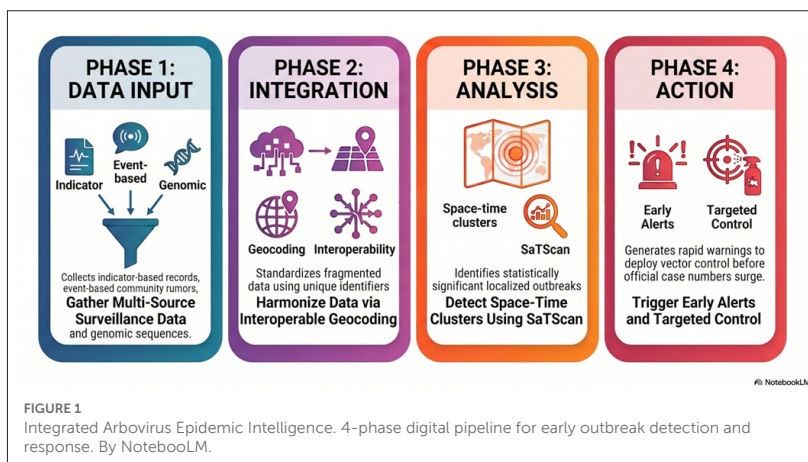
This pilot study demonstrates the operational feasibility of a WHO-aligned integrated Epidemic Intelligence system for arbovirus early warning in primary healthcare settings. The proposed model operationalizes Digital Public Health principles by combining indicator-based and event-based surveillance with laboratory data, genomic information, residential geocoding, and space-time cluster detection. The synthesis of interoperable systems, advanced analytics, and community-informed signals provides surveillance sensitivity at a micro-territorial scale. This integrated approach supports equitable, timely, and actionable decision-making, addressing the informational latency observed in traditional surveillance within socially vulnerable urban areas. The expansion of this model offers a basis for climate-sensitive arbovirus surveillance and epidemic preparedness in comparable urban contexts

Ethics and data protection

All data processed is anonymized and the location scale is not compatible with territory identification

Graphical Abstract

This mini-review (Figure 1) summarizes research efforts on unifying health data and proposes a holistic data visualization strategy, highlighting the contribution of data science engineering strategies to evidence-based health decision-making



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Routine primary-care encounters as early warning signals for climate-sensitive illness: Integrating environmental exposure data into clinical records

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Introduction

Climate change is increasing the frequency and severity of extreme heat and air pollution events, both of which are associated with exacerbations of respiratory and cardiovascular disease (1,2). Public health surveillance systems typically rely on emergency department visits, hospitalizations, or mortality data which detect impacts only after clinical deterioration has occurred and often underrepresent socially vulnerable populations who primarily access primary care.

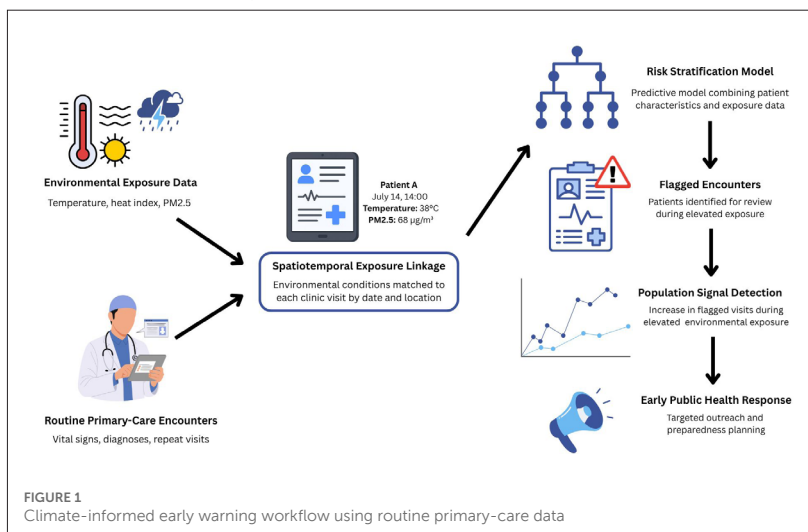
Electronic health records rarely incorporate environmental exposure variables despite their importance as determinants of health in vulnerable

populations. Routine primary-care encounters may provide earlier signals of climate-related health deterioration, and have long been recognized as a potential data source for syndromic surveillance (3). Linking environmental exposure data to routine clinical encounters allows health systems to detect health impacts during routine care rather than after acute deterioration.

This study evaluates whether integrating environmental exposure data with routine clinical encounters can identify elevated health risk and support proactive intervention, enabling primary-care records to function as an early-warning resource for climate-sensitive illness.

Methods

A Health and Environmental Record System (HERS) was developed to integrate environmental exposure indicators with patient clinical records. The system integrates environmental exposure data with clinical records to enable interpretation and action at both the individual clinical level and the population health level. An overview of the climate-informed early-warning workflow is shown in **Figure 1**.



Electronic health record data from a primary-care network serving approximately 6,000 unhoused patients annually were used for analysis. Environmental exposure data included daily temperature and particulate matter (PM_{2.5}) measurements obtained from publicly available monitoring datasets and locally deployed air-quality sensors. Exposure values were assigned to each clinical encounter based on the date and geographic location of the visit, allowing alignment of environmental conditions with individual patient encounters.

Using de-identified clinical data, a random forest model generated risk scores for worsening health based on changes in vital signs and return clinic visits. Environmental exposure variables were included as model features alongside demographic and clinical indicators. Patients with chronic respiratory disease (asthma and COPD) and cardiovascular disease were prioritized.

During periods of elevated environmental exposure, the system flagged patients within the clinical record and generated lists for provider review and follow-up. The system was evaluated within an existing primary-care electronic health record environment. Adaptation to offline-first health information systems is available to support use in low-connectivity settings.

Results

Over a three-month implementation period, 903 unique patients were screened across 3,310 clinical encounters. The system identified 235 patients at elevated risk across 702 visits and flagged 2,706 medications for review.

Flagged patient lists prompted provider outreach, medication review, and counseling related to heat and air pollution exposure. Flagged encounters increased during periods of higher environmental exposure, suggesting routine care data may function as an early indicator of climate-related health burden.

Providers reported improved situational awareness of environmental risk and prioritized patients who might otherwise have been identified only after acute presentation.

Conclusion

Integrating environmental exposure data with routine clinical encounters can transform primary care records into a digital public health early-warning resource. Compared with traditional surveillance based on hospitalizations or mortality, primary-care data may enable earlier detection of climate-sensitive illness and enable proactive care.

This approach demonstrates how clinical information systems can contribute to population-level monitoring and health system adaptation to climate-related risks. Future work will evaluate scalability and deployment within low-connectivity health systems where traditional surveillance infrastructure is limited.

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Standalone applications for major depressive disorders: A systematic review

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Introduction

Standalone mobile health (mHealth) applications have been increasingly adopted as accessible digital health interventions (DHIs) to support early detection and the self-management of depressive disorders¹⁼³. However, the extent to which the components of these DHIs translate into consistent clinical benefits remains uncertain⁴⁻⁵. This study aimed to evaluate the effectiveness of automated app-based interventions in reducing depressive symptom severity among adults and to analyze their therapeutic and technological components.

Methods

A systematic review and aggregated analysis of randomized clinical trials (RCTs) were conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines⁶. Searches were performed in PubMed, Embase, and the Cochrane Library. Eligible studies were those that included app-based digital interventions for adults with

depressive symptoms, assessed using the Patient Health Questionnaire-9 (PHQ-9) according to Diagnostic and Statistical Manual of Mental Disorders (DSM-IV).⁷ All included trials were fully automated self-help interventions with minimal or no human support⁸. Each intervention was characterized according to the presence of key design features, including the use of cognitive-behavioral therapy (CBT) approaches⁹, gamification strategies¹⁰, conversational elements, and artificial intelligence (AI) functionalities. Discrepancies were resolved by arriving at a consensus between authors.

Results

Seventeen RCTs¹¹⁻²⁷ met the inclusion criteria. Studies were conducted across multiple geographic regions and exhibited substantial variability in sample size, participant characteristics, population specificity, and scale-based measures, as detailed in Table 1: Participants generally presented with mild to moderate depressive symptom severity, and samples were predominantly female. Most studies were conducted in high-income countries. Four^{15,20,21,25} did not report sufficient data (mean and standard deviation values) to be included in the quantitative synthesis, as shown in Figure 1. Substantial statistical heterogeneity ($I^2 = 84.3\%$) was observed. The aggregated analysis demonstrated a statistically significant reduction in depressive symptom severity, favoring standalone mHealth interventions, with a pooled effect estimate of -1.83 (95% confidence interval -2.71 to -0.94 ; $p = 0.0007$). Individual effect sizes varied considerably, ranging from large reductions in PHQ-9 scores to modest or non-significant effects, reflecting differences in intervention design, duration, and population characteristics. Among DHIs, CBT was the most frequently adopted therapeutic framework. Gamification elements were the second most common feature, followed by conversational components and AI-based functionalities.

Discussion

mHealth interventions represent one of the most prevalent categories among DHIs, largely due to their adaptability across diverse populations and settings²⁸. The findings suggest a potential positive impact on the treatment

of depressive disorders. However, interpretation of these results is influenced by methodological constraints and heterogeneity in intervention design.

Methodologically, small sample sizes, predominantly female samples, and reliance on passive comparators are important factors contributing to heterogeneity across studies, substantially compromising external validity²⁹. Additionally, the exclusion of severe cases is clinically appropriate and ethically justified for safety reasons²⁸; however, it restricts the applicability of findings primarily to mild-to-moderate presentations.

Regarding theoretical foundations, interventions were grounded primarily in CBT frameworks and delivered through diverse mobile formats, including chatbots, mini-apps, and messaging systems. Effect sizes varied considerably, with gamified or AI-based apps reporting substantially greater effects than minimally guided interventions, suggesting that the degree of therapeutic structure and interactivity may influence clinical outcomes. This is further supported by a large-scale trial¹² showing that behavioral activation outperformed cognitive restructuring, indicating that not all CBT components contribute equally to symptom reduction.

From an implementation perspective, Brazil's Digital Health Strategy 2020–2028 emphasizes the integration of mobile platforms into the Public Health System³⁰. However, adapting digital tools for vulnerable populations remains challenging, as demonstrated by a Brazilian trial in which a messaging-based intervention for older adults in economically disadvantaged areas did not significantly reduce depressive symptoms¹⁹.

Future studies should prioritize active comparators, longer follow-up periods, and more targeted populations to better identify the intervention components most strongly associated with clinical benefits.

Conclusion

Standalone mHealth interventions showed a statistically significant reduction in depressive symptom severity among adults with mild-to-moderate presentations. However, methodological heterogeneity limits the

generalizability of these findings. Further robust multicentric studies with stricter methodological standards are needed to establish evidence-based guidelines for mHealth development.

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Strengthening digital health capacity in low-resource settings: A capacity-building proposal from Pakistan

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Citation

Amna Arslan, S.T., Tanweer, A. Strengthening digital health capacity in low-resource settings: A capacity-building proposal from Pakistan.

Background

Digital health technologies are transforming healthcare delivery globally, yet their adoption in low-resource settings remains uneven (1). Pakistan, with its diverse healthcare infrastructure and emerging digital initiatives, faces significant challenges in health informatics capacity, workforce training, and system interoperability (2). Addressing these gaps is critical for achieving equitable digital health transformation and improving public health outcomes.

Objective

This project proposes to assess the current state of health and nursing informatics capacity in Pakistan, identify barriers and opportunities for digital health integration, and propose actionable strategies for capacity building. The work seeks to generate potential evidence to inform future

policy and practice while fostering grassroots engagement in digital health development.

Methods

We used preliminary findings from semi-structured interviews with nutrition professionals (n=10) on their experiences and perceptions regarding digitalization and workforce capacity in Pakistan. Based on those findings, we plan to employ a mixed-methods approach for capacity building, combining quantitative design, qualitative synthesis and stakeholder engagement. First, a quantitative mapping of existing health information systems and digital health education programs among nursing students and educators in selected academic and health care institutions will be conducted. Second, qualitative data will be collected through semi-structured interviews and focus groups with healthcare professionals, educators, and policymakers to explore perceptions, challenges, and readiness for digital transformation. Third, stakeholder engagement workshops will be organized to co-design recommendations aligned with local needs and global standards.

Quantitative data will be analyzed using descriptive statistics, while qualitative data will be examined through thematic analysis. Ethical approval will be obtained from relevant institutional review boards.

Preliminary Findings

Preliminary findings revealed several recurring barriers to the implementation of digital health in Pakistan. Participants highlighted limited digital literacy, weak data interpretation skills, fragmented digital infrastructure, and insufficient interoperability between systems as major challenges. Participants also emphasized that healthcare practices remain highly dependent on contextual understanding, communication, and human judgment, raising concerns regarding over-reliance on digital systems and potential loss of human interaction in care processes. Participants further emphasized the need for locally adaptable training frameworks, interdisciplinary collaboration, and sustained institutional support to strengthen workforce readiness for digital transformation.

Expected Outcomes

The project is expected to generate empirical evidence on a capacity map of health informatics infrastructure and training in Pakistan, to identify key gaps in digital health literacy, interoperability, and governance, to define a set of context-sensitive recommendations for strengthening digital health systems and workforce capacity, and to develop framework for future collaborative initiatives to support ongoing research and training in digital public health.

Significance

By addressing workforce and system readiness, this initiative contributes to equitable digital public health transformation in a low-resource setting. By situating the research within the humanitarian-development nexus, the project will contribute to sustainable digital health strategies in resource-constrained settings and inform global policy dialogues on digital public health.

Keywords: Digital Health, Health Informatics, Capacity Building, Low-Resource Settings, Public Health, Pakistan, Collaborative Research

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Use of artificial intelligence in the pharmaceutical sector: A bibliometric analysis

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Citation

Casajust, P., Peñarroja, V., Ficapal, P. Use of artificial intelligence in the pharmaceutical sector: A bibliometric analysis.

Introduction

The process of developing a new drug is expensive, lengthy and risky for pharmaceutical companies (Deore et al., 2019). Developing a new medicine takes on average 10–15 years of research and costs more than 2 billion dollars before it becomes available for patients' use (Berdigaliyev & Aljofan, 2020). In recent years, the rapid advancement of artificial intelligence (AI) has reshaped the pharmaceutical sector, influencing multiple stages of drug development. These new tools have the potential to help pharmaceutical companies accelerate the overall process and work more efficiently, thereby reducing both time and costs (Paul et al., 2021; Rajesh & Elumalai, 2025). The aim of this research is to perform a bibliometric analysis of the existing literature on the use of AI within pharmaceutical companies.

Methods/Data

A bibliometric analysis was conducted to identify studies addressing the use of AI in pharmaceutical companies. Searches were performed in the Web of Science database. The search strategy combined terms related

to AI, the pharmaceutical industry and use: TS=((("artificial intelligence" OR "generative") AND (Pharma OR "pharmaceutical") AND (company OR industry) AND (use OR adoption))). No restrictions were applied regarding publication year. The final search was conducted on 15 February 2026. All retrieved records were exported and analysed through VOSviewer. Relevant data were summarized to describe the current trends in the use of AI within pharmaceutical companies.

Results

A total of 453 publications were identified in the Web of Science database. Figure 1 presents the distribution of publications by year. Publication dates ranged from 1993 to 2026. An increase in the number of publications was observed from 2018 onward, reaching a peak in 2025, which was the year with the highest number of publications ($n=113$, 24.9%). The United States was the most productive country ($n=103$, 22.7%), followed by India ($n=88$, 19.4%). Co-authorship analysis identified several author clusters. The most productive and most cited author was Basit Abdul, with 10 publications and 711 citations, followed by Simon Gaisford and Moe Eldawabi, both with 8 publications. Keyword co-occurrence analysis identified six main thematic clusters (Table 1).

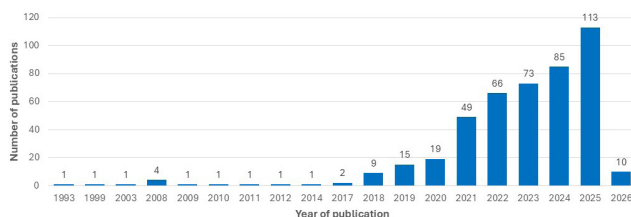


FIGURE 1
Distribution of publications by year (1993–2026)

TABLE 1: Keyword co-occurrence analysis

Cluster	Items
Cluster 1	0, artificial intelligence, artificial neural networks, artificial-in-telligence, automation, big data, challenges, clinical trials, framework, future, generative ai, health, healthcare, impact, industry 4.0, innovation, intelligence, management, opportunities, performance, pharmaceutical industry, pharmacovigilance, risk, simulation, sustainability, technologies, technology, transformation, trials
Cluster 2	chemistry, covid-19, delivery, descriptors, discovery, docking, drug design, in-vitro, inhibitors, models, molecular design, nanoparticles, pharmaceutical-industry, pharmacokinetics, prediction, regression, solubility, strategies, systems
Cluster 3	3d printing, additive manufacturing, artificial neural network, artificial neural-network, computer vision, design, dissolution, drug-delivery, formulation, neural-networks, optimization, pharmaceutical, process analytical technology, quality, release, tablets
Cluster 4	cancer, classification, diagnosis, disease, drug development, drug discovery, expression, generation, identification, medicine, platform, precision medicine, safety, target
Cluster 5	bioinformatics, database, drug repurposing, drugs, information, integration, machine learning, model, modeling, qsar, toxicity, validation
Cluster 6	ai, algorithms, artificial intelligence (ai), deep learning, health care, industry, neural networks, neural-network, personalized medicine, pharmaceuticals, system

The largest cluster (29 items) focused on industry transformation and strategy, highlighting the integration of AI, big data, and Industry 4.0 within pharmaceutical companies and addressing management, sustainability, and organisational impact on healthcare and pharmacovigilance. A second cluster (19 items) centred on molecular design and pharmacokinetics, including molecular docking, solubility, nanoparticles and the study of

inhibitors. Additional themes included advanced manufacturing and formulation technologies (16 items), highlighting the technical side of drug discovery such as 3D printing and process analytical technology; clinical applications (14 items) in cancer diagnosis, disease detection, and precision medicine; computational modelling and bioinformatics approaches (12 items) such as machine learning and Quantitative Structure-Activity Relationship (QSAR) modelling; and fundamental AI architectures (11 items), including deep learning and neural networks. Recent publications increasingly focus on generative AI applications in molecular design and target identification, reflecting the growing role of deep learning in early drug discovery. However, limited attention has been given to the organisational and human impact of AI adoption. When terms related to employee wellbeing were incorporated into the search strategy, only four publications were identified in the Web of Science database, highlighting a significant research gap in this area.

Conclusions

This bibliometric analysis demonstrates a rapid and sustained increase in scientific interest in the application of AI within pharmaceutical companies, particularly since 2018. The literature is predominantly focused on technical and clinical dimensions, including molecular design, computational modelling, advanced manufacturing and precision medicine. The growing evidence of generative AI and deep learning highlights a shift toward more sophisticated and data-driven approaches in early drug development. Although Web of Science provides high-quality indexed literature, future research could benefit from comparing findings against other bibliometric databases, to enhance robustness and generalisability. However, despite this technological expansion, limited attention has been given to the organisational and workforce implications of AI adoption. The limited publications addressing employee wellbeing suggest a significant research gap. Future studies should therefore complement technological innovation with research into the human and organisational impact of AI implementation within pharmaceutical companies. In particular, studies may explore the impact of AI on job displacement and changes in job roles and responsibilities, the potential for overreliance on or dependence on AI systems, and associated psychosocial risks following AI implementation, such

as increased stress or job insecurity. Addressing these dimensions will be essential to maximise the benefits of AI while ensuring sustainable integration within pharmaceutical companies. However, operationalising such research may be challenging due to limited access to organisational data and restrictions related the use of sensitive or proprietary data. Furthermore, future studies should explicitly consider the risks associated with AI-driven approaches, including model failures and brittleness, limited explainability of complex algorithms and privacy concerns.

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When the novelty wears off: Exploring long-term self-tracking patterns

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Citation

Bork, A., Wagener, N. When the novelty wears off: Exploring long-term self-tracking patterns.

1 Introduction & Background

Physical activity is strongly associated with improved physical health [37], mental health [31], and overall quality of life [27]. Nevertheless, since the COVID-19 pandemic, activity levels have declined [34], while sedentary behaviour and screen time have increased [30]. In response, personal informatics (PI) technologies such as activity trackers, smartwatches, smart scales, and smart rings have become widespread tools for collecting and reflecting on data about physical activities.

This focus is also reflected in Human–Computer Interaction (HCI) research: 83% of PI publications address health [6]. Despite extensive work on self-tracking since early Quantified Self research [18], studies overwhelmingly focus on short-term interventions, clinical contexts [11, 19], or brief engagement studies [7, 12, 23, 25, 35], rather than sustained practices.

In contrast, self-tracking is widely conceptualised as an evolving long-term practice shaped by reflection, lapses, and changing goals [8, 16, 24, 29]. Despite this conceptualisation, even longitudinal studies rarely extend beyond a few months. Prior work shows that engagement typically declines after the initial novelty phase [32], with discontinuation and lapses being common [7], highlighting *adherence* as a key challenge also recognised in behaviour change research [4]. While some studies identify recurring usage phases [22] or the influence of preexisting motivation [10], multi-year tracking remains largely unexplored [33].

This gap is critical, as long-term tracking could provide early detection and prevention of chronic conditions [19], particularly when systematically implemented by healthcare providers [21]. However, it remains unclear whether and how such benefits are realised in practice, whether longterm tracking reaches diverse populations beyond already health-conscious groups, and whether it meaningfully contributes to public health [9].

To address this gap, we explicitly shift the focus from short-term use to sustained, multi-year tracking, answering calls in HCI for longitudinal research [20, 33] and moving beyond short-term intervention paradigms [14]. We investigate user profiles of long-term self-trackers who have used at least one self-tracking device continuously for more than two years, drawing on self-donated tracking records, survey data, and in-depth interviews. We are guided by the following research questions:

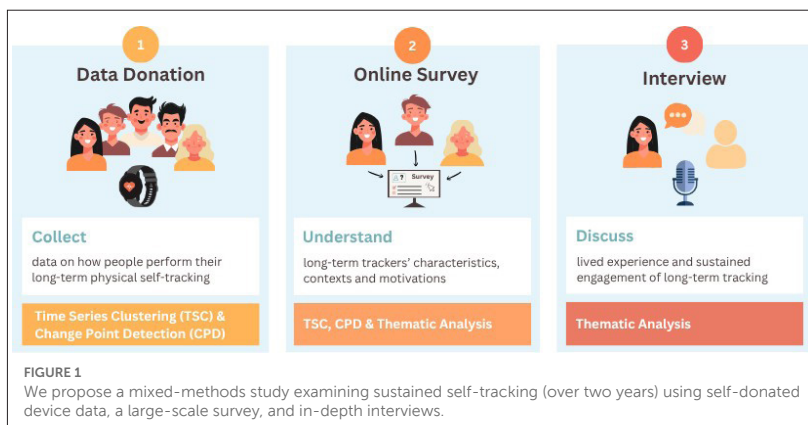
RQ₁: What demographic, motivational, and behavioural profiles characterize long-term self-trackers? RQ₂: What individual-level health and behavioural outcomes are associated with sustained self-tracking beyond the initial novelty phase?

By answering these questions, we explain long-term adherence: we identify user personas, personality traits, and individual life challenges that shape or

impede sustained tracking. These insights provide a foundation for designing technologies that support lasting engagement and for future research on how personal factors influence long-term self-tracking trajectories. We contribute (i) a large-scale dataset of long-term tracker use enabling behavioural pattern identification, (ii) empirical insights into the user profile, use cases, and health behaviour changes associated with sustained tracking, and (iii) design recommendations to support long-term engagement with self-tracking technologies.

2 Study Design & Methods

To understand user profiles and everyday practices associated with long-term tracking, we apply a mixed-method approach combining (1) historical self-tracking records spanning several years obtained via a data donation application, (2) quantitative survey data on long-term trackers' characteristics, both of which enable AI-based pattern recognition, and (3) semi-structured interviews with selected participants to gain in-depth insights into tracking behaviour and specific user needs (see Figure 1). This section describes our intended participant sample and provides our design rationale for each applied method.



2.1 Participant Sample

In contrast to most other longitudinal studies [10, 22], we will recruit at least 100 active long-term self-trackers via Prolific who have continuously tracked with at least one self-tracking device for over two years. Following Meyer et al. [22], we define long-term self-trackers based on inactivity thresholds rather than simple duration of use. Thus, participants are excluded if their tracking history shows an inactivity period exceeding 120% of their previously longest tracking break, with a minimum threshold of 50 days. While we include participants across a wide range of ages, genders, geographical locations, wearable technologies, and health statuses to capture diverse self-tracking practices, our primary focus is on participants without chronic diseases, to focus on everyday self-tracking practices in the general population rather than condition-specific use cases.

2.2 Data Donation

We will deploy a secure data donation application through which consenting long-term self-trackers can share their historical records. The donated datasets will be examined using AI-supported methods, particularly Time Series Clustering [17] and Change Point Detection [1], to identify recurring structures, long-term usage patterns and major change points in usage.

2.3 Online Survey

Prior research shows that surveys effectively capture self-tracking motivations and adoption patterns [6, 7, 26]. Therefore, we use a survey to examine long-term self-tracking, including user profiles, practices, motivations, engagement trajectories, and perceived impacts on behaviour and health. We collect demographic and contextual data, detailed accounts of tracking practices, and reflections on changes over time. Self-reports are complemented by validated measures of physical activity (IPAQ-SF) [5], well-being (WHO-5) [38], motivational regulation (TSRQ) [15], habit strength (SRHI) [36], personality (BFI-10) [28], and mental health (PHQ-9) [13].

2.4 In-Depth Interviews

Prior work has shown that interviews provide rich insights into lived self-tracking experiences, including how engagement evolves over time [7],

which motivates their inclusion alongside our other data sources. We will conduct at least ten in-depth interviews with long-term self-trackers who opt in via the survey by providing contact details separately to ensure data protection. Interviews will be used to pay particular attention to how participants narrate changes in motivation and engagement over time. All interviews will be audio-recorded, transcribed verbatim, and analysed using reflexive thematic analysis [2, 3].

3 Conclusion & Outlook

This work-in-progress addresses the lack of systematic evidence on long-term self-tracking by combining donated behavioural records, survey data, and in-depth interviews with individuals who have tracked for more than two years. It aims to identify who sustains tracking, how motivations and practices evolve, and whether long-term use meaningfully supports healthy behaviour changes. By doing so, the study seeks to inform practical design strategies for improving long-term adherence and highlight life challenges and changing circumstances that current tracking technologies insufficiently address and that require more tailored design approaches than we currently have.

4 Acknowledgements

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An AI language model, Overleaf's spellchecker, and Grammarly were used to correct spelling errors and improve the writing quality. All ideas and content remain the authors'.

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Demos

A Digital public health framework for drug risk assessment: AI-enhanced dashboard + biopsychosocial taxonomy and intersections

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1. Background and Research/Practical Need

Large-scale public health surveys generate rich evidence but are often difficult to use when they include many items and complex constructs. PERCIBET3, conducted in 2022 at Universidad Veracruzana, included 19,151 university students and 278 items assessing drug-related perceptions, risk factors, and consumption patterns. Using the University Students' Drug Use Questionnaire (CODEU-3), it collected data on sociodemographic and academic characteristics, as well as psychosocial and behavioral factors, including family functioning, self-esteem, anxiety and depression, and impulsivity. It also examined attitudes toward drug use, peer and partner pressure, substance use prevalence, risk behaviors, prevention, legalization, and medical use [1].

Although this breadth supports robust analyses, it creates barriers for students, families, clinicians, and researchers who need to locate items, interpret results, and translate findings into prevention strategies. Addiction prevention and public health interventions increasingly rely on biopsychosocial models to understand interactions between biological, psychological, and social determinants [2,3]. However, conventional survey reports rarely allow intuitive exploration through this multidimensional lens.

This demo addresses that gap through an AI-enhanced dashboard combining a biopsychosocial taxonomy, semantic search, cross-domain visual exploration, and dataset-grounded AI summaries. Beyond implementation, it includes initial pilot validation evidence and a planned user evaluation to assess usability, interpretability, and perceived usefulness in stakeholder workflows.

2. Demo Overview and Functionality

Our project is a companion tool to the Adhera Caring Digital Program, transforming PERCIBET3 survey data into actionable biopsychosocial insights. The dashboard goes beyond taxonomy visualization by helping stakeholders translate survey evidence into prevention strategies, program optimization, and evidence-based digital care interventions.

Core capabilities include:**a. Structured Insight Engine (Biopsychosocial Framework)**

Survey items and results are organized under a validated biopsychosocial taxonomy (biological, psychological, social, and cross-dimensional factors), enabling:

- Rapid identification of risk and protective factors
- Structured interpretation of multidimensional determinants
- Alignment between survey evidence and digital intervention modules

This ensures conceptual consistency between research evidence and the Adhera Caring Digital Program content.

b. Intelligent Semantic Search

Interactive visualizations translate complex survey results into accessible insights to support prevention strategies, educational planning, public health communication, and digital program personalization, while maintaining analytical rigor across stakeholder groups.

c. Action-Oriented Visual Analytics

Interactive visualizations convert complex survey results into clear insights to support prevention design, educational planning, public health communication, and digital program personalization, while preserving analytical rigor.

d. Cross-Domain Insight Exploration

Users can analyze intersections between biopsychosocial dimensions (e.g., Psychological x Social), enabling:

- Identification of compound risk patterns

- Hypothesis generation for targeted interventions
- Design of more precise prevention pathways

This supports data-driven refinement of digital care journeys.

e. AI-Assisted Evidence Summarization

The system generates concise summaries at individual item level, selected taxonomy group level and cross-dimensional intersections.

Summaries are constrained strictly to dataset evidence and support faster interpretation, strategic decision-making, and communication with stakeholders.

3. Taxonomy Development, Pilot Validation, AI Workflow and Evaluation Plan

The taxonomy was developed through an iterative process combining biopsychosocial theory, CODEU-3/PERCIBET3 item mapping, and expert review. Items were first classified as biological, psychological, social, or cross-dimensional, with ambiguous items refined through discussion. A pilot participatory workshop with four professionals—health sciences and informatics—assessed conceptual clarity, feasibility, and interdisciplinary interpretability. Feedback informed category labels, classification rules, and dashboard representation using large language models to route user queries and generate dataset-grounded outputs. Natural-language queries requiring data analysis are converted into executable Python code, while metadata- or taxonomy-related queries are answered in natural language using the available structured information. [4]. Reliability safeguards include a graph-based workflow that constrains model actions to predefined operational steps, response-validation layers before user display, simplified reasoning traces for transparency, and user feedback mechanisms to flag errors and support continuous monitoring of failure modes.

Although not yet validated with end users, a future evaluation with at least 20 users using SUS and task-based feedback is planned [5,6].

4. Access

The demo is accessed through the “Adhera Clinical Dashboard” web application. Access requires a user login (credentials provided to approved individuals only)

5. Target Users and Use Cases

The dashboard targets students, families, researchers, clinicians, and public health professionals by enabling accessible exploration of PERCIBET3 survey domains, item-level results, and biopsychosocial patterns. During the demo, users will be able to search items by domain, review summaries and distributions by taxonomy category, explore cross-domain intersections, and generate concise AI-assisted summaries to support rapid interpretation.

6. Data, Privacy, and Ethics

The dashboard uses anonymized results from the PERCIBET3 survey conducted in 2022 at Universidad Veracruzana. No direct identifiers are included. The demo is presented within the context of an ongoing multi-country agreement between Adhera Health and Universidad Veracruzana, supporting responsible use of evidence for public health and addiction prevention. The dashboard operates on anonymized survey results and follows data minimization principles. To mitigate risks of misinformation or overgeneralization, AI-generated summaries are constrained to evidence available in the dataset and are intended to be validated by users and professionals within the research, public health context and comprehensive drug prevention management.

7. Images (Screenshots)

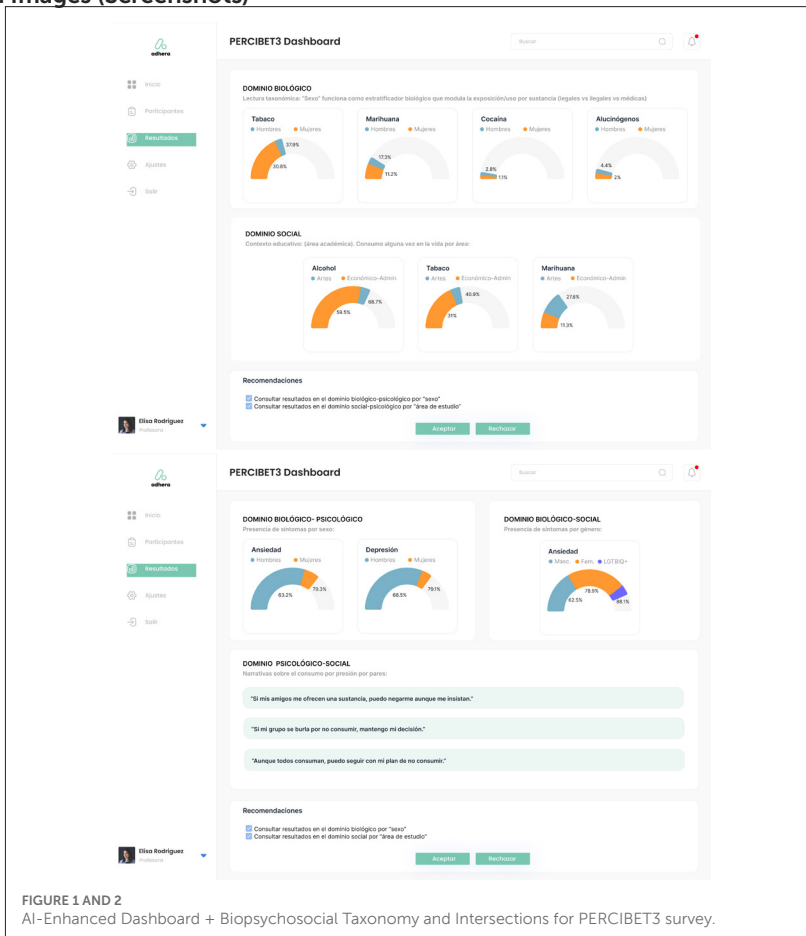


FIGURE 1 AND 2

AI-Enhanced Dashboard + Biopsychosocial Taxonomy and Intersections for PERCIBET3 survey.

8. Generative AI Use Disclosure

Generative AI tools are used within the dashboard to support taxonomy-driven classification, semantic search, and evidence summarization. These functions are implemented with safeguards to promote responsible use,

including transparency, traceability, human oversight, and data minimization [7,8]. AI-generated outputs are constrained to the available dataset and are reviewed within the project workflow,

Final interpretation of the results remains the responsibility of the users and authors, particularly in health and public health contexts where reliable and contextualized interpretation is essential.

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ChatMap to support better public / climate health outcomes

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Citation

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Geospatial data is a key part of the information needed for climate health and public health preparedness and response work but, in many places, is unavailable or inaccessible to those that need it. **Open / participatory mapping is a proven means of generating low cost, high quality and community-validated data at both depth and scale.** Supported by the Wellcome Trust, HOT is developing hypotheses for how open/participatory mapping can be leveraged as a **strategic enabler for public health / climate health work.**

It is an exciting time to be in this position. We are seeing momentous leaps within the open mapping ecosystem, in terms of innovative technology (community AI models, community drone imagery acquisition at scale), and particularly **WhatsApp / chat app-based data collection tools.** We're also seeing an ongoing expansion of the open mapping community, particularly through local and national authorities, grassroots community groups / networks and technical contributors.

Along these lines we would love the opportunity to demo ChatMap: a free and open source tool developed in late 2024 by HOT for collecting geolocated images and other media from WhatsApp, Telegram or Signal groups. It has already been used to support disaster responses (2024 Rio Grande do Sul

floods, Hurricane Melissa, 2025) and by national risk management agencies (Colombia, 2025-ongoing). We believe that it has a lot of potential for supporting public / climate health outcomes as well, with Médecins Sans Frontières (MSF) currently exploring using it to support their operations.

The opportunity for demo participants is to a) Explore potential health-related applications and real-world implementation opportunities that can benefit from the use of ChatMap b) Explore other solutions and similar applications that can be either coupled with HOT's or that can benefit from collaboration c) Get feedback for developments or integrations for HOT tools to better address the needs of practitioners in the Health sector.

The key feature of ChatMap is that it has an incredibly low barrier to entry.

Contributors do not need to either install software, sign up or login and do not need any prior mapping knowledge in order to participate. **All they need is a smartphone with either WhatsApp, Signal or Telegram installed.** In theory this unlocks a pool of over 3.5bn global users that could be called on to support humanitarian/health related mapping outcomes using ChatMap.

Figure 1 below shows the ChatMap process. A smartphone user shares a photo or other media (text, audio, video) from the ground in a messaging app and then shares the associated location. The ChatMap tool parses this information to associate location and media, generating a map (geo-json file) of the geolocated media.

ChatMap was used at scale in **response to Hurricane Melissa in October 2025**. HOT supported the creation of a WhatsApp group with over 2,000 Jamaican residents, who shared over 230 **reports of flooding, damaged buildings, landslides, road blockages and fallen power lines**. The results were added to a public uMap map (see Figure below) and the results were integrated into systems used by the Jamaican government, supporting damage monitoring and household assistance programmes. A representative from a Jamaican Ministry responsible for response and rebuilding stated: 'ChatMap ... has been an invaluable tool in our relief efforts. It has provided critical situational awareness and streamlined the way we monitor damage to buildings across the island.'

From WhatsApp to the map

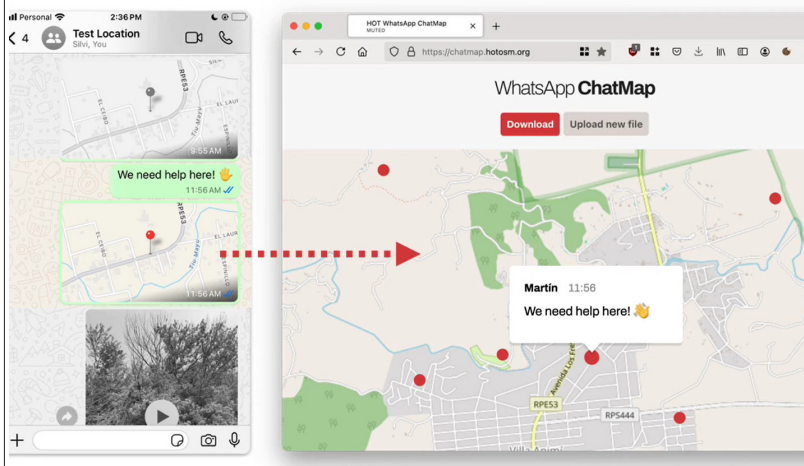


FIGURE 1

Left - WhatsApp group discussion with media and associated point locations. Right - The ChatMap website converting the exported group chat into geolocated data.

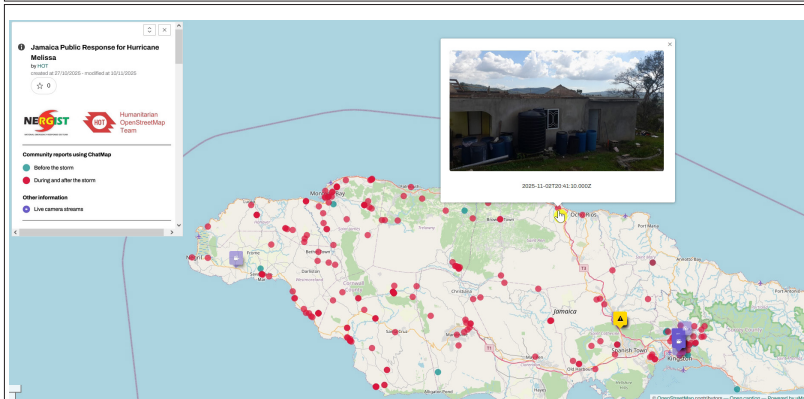


FIGURE 2

ChatMap use in Jamaica in response to hurricane Melissa in 2025 resulted in a publicly accessible interactive map (uMap) with all geolocated media.

AQ1

Although ChatMap was conceived as a tool for disaster response, it was adopted by community and experts for applications at the intersection of health and climate. One of them was Learn, Map & Clean Challenge!. An initiative in partnership with UNICEF - YOMA for mapping trash during the World Cleanup Day, which resulted in the participation of over 2000 youth across more than 30 countries and 2400+ trash points. Alongside, the development of an online course, introduced participants to basic GIS concepts and the tool, enabling them to improve their capacity to apply spatial analysis to real-world community issues.

ChatMap results can either be updated intermittently by using WhatsApp/Signal/Telegram's export function or a 'live' feature can be enabled for WhatsApp, whereby geolocated media is available immediately after the data is shared. The results can be sent to a private repository if needed, and it is also possible to generate ChatMap results with direct messages to a specific number if there is sensitivity around sharing locations/media with a large group of contacts.

ChatMap not only lowers the barrier for crowdsourced data collection, but it does it for data visualization too, as it can be directly integrated with uMap, a free and open source tool that allows people to visualize spatial data in an interactive map. This platform does not require expertise on GIS or data management to create or update the data. This makes the data collected immediately available for analysis and potentially more impactful in cases of disaster response, coordination or health-related emergencies.

At this stage (early 2026) we do not have proven use cases of ChatMap for supporting health-related outcomes, however, HOT has more than 15 years of experience collaborating on health-related projects like: helping the eradication of vector-borne diseases by mapping breeding habitats with Clinton Health Access, mapping mental health-related infrastructure with WHO, mapping healthcare facilities with iLab Liberia during COVID-19 and more. These experiences, as well as consultations with partners like MSF, have helped HOT understand some of the key gaps on health-related projects; one of the most important being timely and interoperable data

that can be collected with minimal resources. ChatMap is precisely what we think that can address this –however the tool is still in its early stages, and would benefit greatly from its exposure to a community of professionals in the health sector that gives the feedback that can help it develop in the right direction.

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Development and implementation of a digital health app for girls and young women with severe menstrual pain in Germany

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Citation

Lysk, S., Wiegmann, S., Pach, D., Mechsner, S., Skupin, L., Witt, C., Paprott, R., Neumayer, H., Konigorski, S. Development and implementation of a digital health app for girls and young women with severe menstrual pain in Germany.

Background

Menstrual pain is one of the most common health problems among women under the age of 25, with an estimated prevalence around 75%, and it has a substantial socioeconomic impact due to school and work absenteeism [1, 2]. Despite its high prevalence, severe menstrual pain remains widely underserved worldwide, even though numerous effective pharmacological and self-care interventions are available.

To this day, the topic is characterized by a lack of knowledge, misinformation, shame, and social taboos worldwide, leading to symptoms that are often not taken seriously over long periods of time [3, 4]. Knowledge about menstruation and evidence-based options for pain reduction is frequently limited among adolescent girls and young women, and existing information and healthcare services often fail to reach very young target groups [5]. In addition, severe menstrual pain is frequently normalized in society, leading to uncertainty about when symptoms warrant seeking professional healthcare. In this context, digital health applications offer new opportunities for improved access to evidence-based information on menstrual pain and for behavior change interventions, particularly for younger users [6, 7].

Period. App

Aim and Concept

Based on this background, the *period.* app was developed within the project MeMaF (“Menstrual Health and Hybrid Care Model for all Young Females”). MeMaF was funded by the German Innovation Committee of the Federal Joint Committee (Grant number: 01NVF21111). The aim of the application is to reduce menstrual pain severity and to provide low-threshold access to valid, target group–adapted medical information.

The app offers multimedia health information on the menstrual cycle and menstrual pain, an in-app menstruation and symptom diary, as well as education on multimodal pain management, including evidence-based complementary medicine options for menstrual pain. After three months of app use, users complete an in-app questionnaire that identifies individuals at high risk for developing endometriosis. This high-risk group is then directly

invited via the app to participate in a new multimodal, hybrid treatment program at the Endometriosis Center of Charité – Universitätsmedizin Berlin. App users can book an appointment for an initial online consultation directly within the app.

App Development And Content

The app was developed using a user-centered, iterative design approach and is available for both Apple (iOS) and Android devices. Prior to development, workshops and surveys with the young and female target group were conducted to assess needs and expectations. Throughout the development process, UI/UX testing was carried out with adolescents and young women. Development took place within a certified quality management system. In 2024, the period. app was released and approved as a medical device in accordance with the European Medical Device Regulation (MDR, Class I). Screenshots of the app are displayed in Figure 1, and a detailed overview of the app content is shown in Table 1:

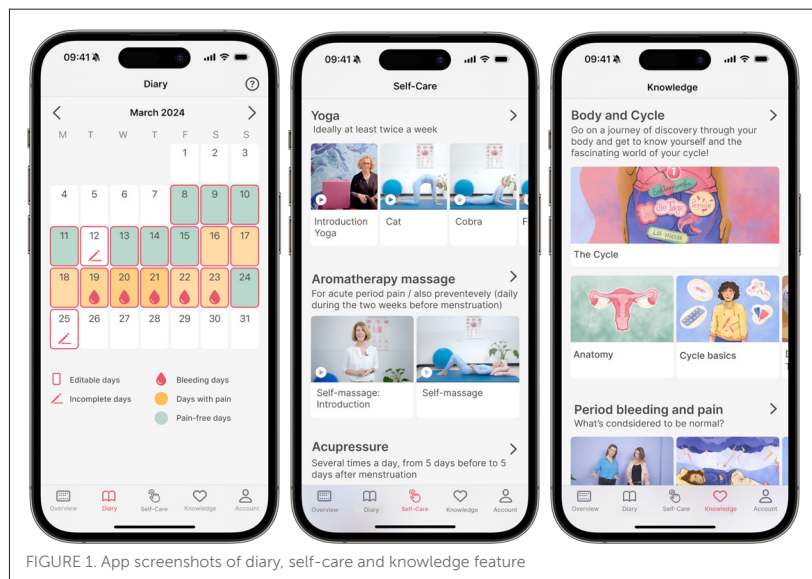


FIGURE 1. App screenshots of diary, self-care and knowledge feature

TABLE 1. Overview of features and content of the period. app

App Feature	Content
Diary (Tracking)	• Daily tracking of menstruation, pain intensity, concomitant symptoms, medication intake, self-care exercises • Monthly overview
Knowledge	• Short texts and videos* (animated and real-life videos with healthcare professionals) • Topics: ○ Menstrual education ("What is normal?") ○ Female menstrual cycle ○ Menstrual hygiene and menstrual products ○ Medication use ○ Self-care strategies for pain relief ○ Common gynecological conditions
Evidence-based self-care	• Yoga videos* • Massage videos* • Acupressur videos*
Dashboard	• Overview graph of pain intensity • Questionnaires (used for app evaluation in study setting, questionnaire-based risk stratification) • Direct appointment booking for affiliated endometriosis center • Suggested knowledge content

* The videos included in the app were specifically developed for girls and young women and are published under Creative Commons licenses, allowing them to be used beyond the app, for example, in schools or educational and training settings.

Evaluation

The app is currently being evaluated in a single-arm pre–post study as part of the MeMaF project and is registered in the German Clinical Trials Register (DRKS00033626). Between March 2024 and February 2025, a total of 3,522 participants were recruited into the pre–post trial, meeting the predefined

recruitment target. Study results are expected to be available in autumn 2026.

Demonstration

The demo will showcase the content of the application, including selected videos, and explain the underlying processes. This includes how users at high risk for endometriosis are identified through the app and how the application functions as a tool to connect users with a specialized endometriosis center, even when it is geographically distant.

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From paper to platform: A co-designed mobile surveillance system for mosquito-borne disease control in Brazil

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Mosquito-borne diseases cause over 700,000 deaths annually [1], yet surveillance in endemic regions remains constrained by paper-based workflows, fragmented data systems, and limited technological infrastructure. Deployed since last year in Brazil, we will demonstrate a multi-profile mobile surveillance application designed to strengthen VBD control through digital transformation of field operations [2].

With the platform now operational across two countries, 11,576 property inspections in Cabedelo and 36 months of continuous ovitrap monitoring in Madeira, this demo invites the DPH community to experience the full field-to-dashboard data pipeline. Last year, the first real-world deployment in Madeira was presented, focusing on the system architecture and early pilot results. This year, the demo will explore shared lessons from field deployment and co-design from our most recent developments in Brazil, the modular design of the app, opportunities for adoption in new regions, and potentialities for integration with predictive modelling workflows [3,4].

The Challenge

This tool is built to directly address critical challenges in current mosquito control efforts in the field. First, using a capacity-conscious design we meet the realities of fieldwork. The application runs on low-cost Android devices already available to government health agencies, functions offline in areas with intermittent connectivity, and achieves 23-second median form completion to minimise disruption to agents' existing workloads. Interfaces were co-designed with field workers of varying technical literacy to ensure accessibility without compromising functionality [5]. Secondly, we have sought to perform harmonization across fragmented systems. The platform provides role-specific interfaces, including granular data entry for environmental agents and laboratory staff, and an aggregated high-level dashboard for managers. It does this while maintaining unified data architecture enabling multi-sectoral coordination across field, laboratory, and management workflows. It was designed with direct input from the agents themselves based on existing paper forms, articulated needs, and current mandates [5]. Thirdly, the tool is built to generate actionable data outputs that support decision-making and translational

research. The data collected aims at informing evidence-based policy, and the architecture was designed with expansion and connection in mind, enabling downstream applications such as forecasting and predictive modelling [3,4].

This demo illustrates how capacity-conscious tools developed through genuine co-design with field-based environmental agents can strengthen VBD surveillance without excluding resource-limited health systems. The role-based architecture offers a replicable model for diverse stakeholders and disease control programmes.

The Design

Participatory design was performed through hands-on workshops with environmental agents. In Cabedelo, training workshops (November 2024) introduced the project, distributed devices, and conducted app testing in workshop settings before accompanying agents into the field to observe paper and digital workflows during the transition period. This approach (researchers working alongside agents during actual property inspections) identified major challenges which would not have otherwise been identified. The November 2024 stakeholder workshop in Recife subsequently validated these design choices with municipal managers and state health secretaries, identifying additional dashboard requirements for cross-jurisdictional reporting.

The App

The application requires Android 8.0+ devices with GPS. It is compatible with project-issued phones deployed to surveillance teams, and with the common types of phones already found in MoH worker's pockets. Authenticated, role-based access controls functionality per profile. Managers access the web dashboard via standard browsers. At the demo, attendees can interact with the agent and lab interfaces on provided mobile devices, submitting forms and observing their data appear in real time on the manager dashboard displayed on a laptop, experiencing the complete field-to-decision pipeline.

The agent and lab interfaces is built with Ionic-React and TypeScript, with Node.js and ExpressJS on the server side and MongoDB Atlas for the

database [2]. The manager dashboard is a separate Next.js web application sharing the same back-end and database. JWT-based authentication secures role-specific access across all three clients.

The system supports three user profiles with bespoke interfaces:

The manager dashboard provides high-level oversight for managers. Features include realtime visualization of agent activity via interactive maps, aggregated statistics on trap status and breeding site prevalence, progress monitoring filtered by user and location, and data export for policy briefings. The agent interface delivers a streamlined mobile experience for field workers. Features include GPS-tagged form submission, one-tap trap selection from pre-loaded municipality lists, offline data caching, and incomplete form tracking. Lab interface enables sample analysis personnel to update agent-submitted data with egg and larvae counts and species identification, maintaining data continuity from field collection through laboratory processing.

Data outputs currently include CSV exports delivered via email, enabling offline analysis and reporting. The manager dashboard, currently under development, will serve as the primary data portal with filtering, visualisation, and direct export functionality.

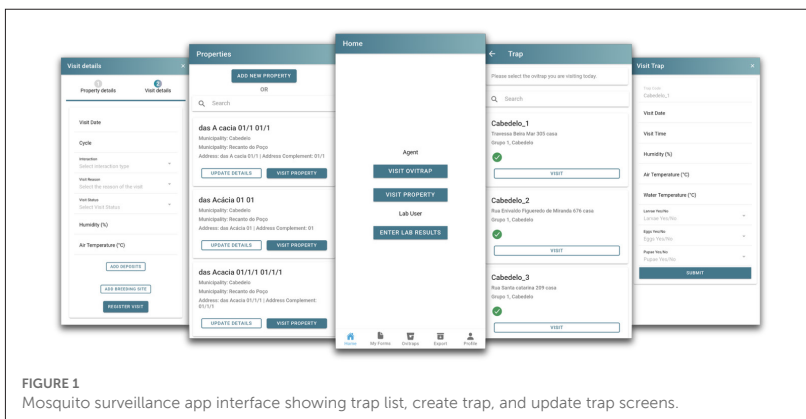


FIGURE 1
Mosquito surveillance app interface showing trap list, create trap, and update trap screens.

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The Deployment

Operational deployment demonstrates contribution to VBD control. In Cabedelo (October 2024–December 2025), 15+ agents recorded 11,576 property inspections identifying 357 breeding sites across 4,557 properties, enabling targeted larvicide application. Same-day data availability (reduced from 2–4 week paper delays [7,8]) allows managers to redirect resources to emerging hotspots. In Madeira, 36+ months of continuous operation across 200+ ovitraps has generated a longitudinal dataset intended to underpin future predictive modelling [3,9].

The Next Steps

The formal NoN launch in Recife (March 2025), with WHO and Brazil Ministry of Health keynotes, established this tool as core infrastructure for national scale-up targeting 500+ agents across multiple Brazilian states, with a transatlantic sister network in Portugal and Southern Europe [10]. The NoN continues to expand, with new municipalities in Brazilian States of Paraíba, Pernambuco, and Rio de Janeiro committed to joining the pilot in 2026. Integration with predictive modelling which combines app-collected surveillance data with climate and ecological variables [3,9] will enable forecasting of breeding hotspots at week granularity for proactive action and intervention at the policy level [4]. Harmonization efforts are underway to align data formats with Brazil's national systems, working toward full digitalization that replaces legacy paper forms. The networking in Portugal and Southern Europe address the expanding *Aedes* range driven by climate change, with potential applicability to other vector-borne disease surveillance contexts, including tick-borne diseases in Southern Europe.

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Maisha ACT: Africa's first large-scale antimalarial consumption tracker

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Overview

Maisha ACT is Africa's first large-scale antimalarial consumption tracker that provides a holistic snapshot of antimalarial consumption in the public and private sector. Previous estimates have substantially underestimated antimalarial consumption in the private sector and focused primarily on WHO-prequalified medicines procured by the public sector. With antimalarial drug resistance (AMDR) on the rise, more accurate estimates that include the private sector—the largest driver of AMDR and largest provider of medicines—are needed to appropriately address this crisis.

Developed by Maisha Meds, this tracker integrates pharmaceutical export data for finished and unfinished pharmaceutical products from S&P Global; donor procurement records; manufacturer shipment data; and retail sales from Maisha Meds' network of ~5,000 private pharmacies, clinics, and drug shops across Kenya, Nigeria, Tanzania, and Uganda. Maisha ACT

covers ~90% of all antimalarial drug consumption across over 40 African countries.

Maisha ACT consists of three groups of dashboards:

1. *Key starting material (KSM) and active pharmaceutical ingredient (API) trade data*

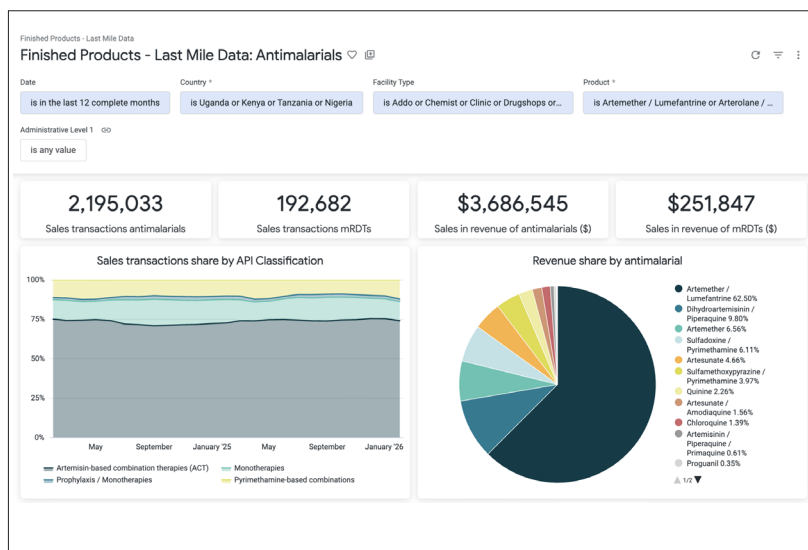
This dashboard tracks the international trade of KSM and API between countries, primarily from China to India. Data comes from S&P Global and monitors the trade of 33 KSM and APIs. Users can filter by year, API/ KSM, exporter country, and importer country. Completed analyses include exported volume in kilograms per year, total value of exports, average price per kilogram, and top exporters and importers.

2. *Finished products trade data*

This dashboard tracks international shipment and donor procurement records of finished pharmaceuticals to more than 40 African countries. Users can filter by exporter, year, data source, importer country, destination sector (public sector, private sector, NGOs), APIs, and brand name. Additional filters include dosage form, strength, and pack size for artemisinin-based combination therapies.

3. *Finished products last-mile data*

This dashboard tracks patient-level sales generated from Maisha Meds' network of ~5,000 private facilities across East Africa and Nigeria. These facilities use Maisha Meds' point-of-sale software: an application with online and offline functionality that allows them to make sales, track patient records, manage inventory, scan barcodes, add SKU numbers, monitor expiration dates, sort products by therapeutic area, track patient and supplier credit, and generate business reports on low-cost tablets.



The majority of participating facilities are in rural or peri-urban regions. Approximately one-quarter of the registered facilities are Accredited Drug Dispensing Outlets (ADDOs) in Tanzania or Patent and Proprietary Medicine Vendors (PPMVs) in Nigeria, which generally serve rural and underserved communities. Data from all facilities is anonymized and aggregated for Maisha ACT, where users can filter by date, country, administrative level 1, facility type, and product to examine data on sales volumes, average price (USD), and revenue. To preserve anonymity, users are unable to see facility-level sales, and administrative level 1 data can only be viewed for regions where at least three facilities have contributed data in the prior month. Furthermore, no patient-level data is stored in the POS software.

Use Cases

Monitoring diversification of antimalarials

Increasing AMDR has led to the development of new approaches to treating malaria, such as the use of multiple first-line therapies (MFTs).¹ Data from

Maisha ACT can support baseline and temporal analyses of antimalarial diversification over time to identify critical geographies for intervention (e.g., regions with an overreliance on a single therapy).

Monitoring antimalarial quality and clinical practices

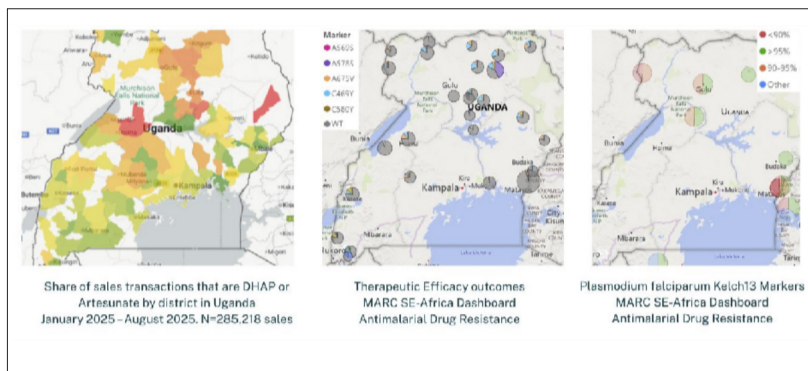
The classification of antimalarials by WHO prequalification status enables users to identify the volume and proportion of antimalarials which are WHO-prequalified and their corresponding prices. For example, ~70-80% of the antimalarials consumed in the private sector in Africa are not WHO-prequalified, and these products are generally cheaper than their WHO-prequalified counterparts. The tracker can also facilitate the identification of practices which are not supported by clinical guidance, such as the use of monotherapies including injectable artesunate to treat uncomplicated malaria.

Triangulating AMDR data

Maisha ACT data can also be used to infer where AMDR might be increasing. Last-mile data from facilities in Maisha Meds' network identified the increased use of dihydroartemisinin-piperaquine (DHAP) and injectable artesunate in some regions of Uganda where markers of resistance have been documented in the MARC SE-Africa antimalarial resistance dashboard.² This could suggest changes in purchasing behavior due to reduced effectiveness of artemether-lumefantrine.

Supporting modeling research

Successfully simulating the infectious disease dynamics of malaria transmission and resistance requires reasonable estimates of the volume and type of antimalarials most often used in a given region. An overreliance on public sector consumption data can produce an incomplete picture and bias inputs. Maisha ACT's inclusion of public and private sector consumption data provides greater insight and accuracy.



Identifying potential markets for new pharmaceuticals

Increasing AMDR has underscored the importance of diversifying antimalarial usage with existing and new antimalarials, including combination therapies that do not rely on artemisinin and/or lumefantrine. Maisha ACT data can support the identification of markets for which patients have greater willingness to pay for newer antimalarial therapies such as DHAP, which can signal a greater appetite for therapies at the same or higher cost. This data can also inform the entry of generic manufacturers by identifying competitive prices and market saturation.

High-level results from Maisha ACT that can support these analyses are updated monthly and published at: <https://dashboards.endmalaria.org/en/maisha-meds>.

Accessing Maisha ACT

Access to Maisha ACT is free for non-profit organizations, with paid subscription options available for commercial users. To learn more and request access to the platform, individuals and organizations can visit <https://maishameds.org/maisha-act/>.

Conclusion

Maisha ACT provides unprecedented insight into public and private-sector consumption of antimalarials in Africa. As case management of malaria becomes increasingly challenging due to AMDR and shortfalls in funding for global health, this platform can facilitate informed decision-making and innovative research for myriad stakeholders.

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MindReset: A scalable eye-tracking-enabled split-second unlearning (SSU) platform for mental health and persistent pain

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Citation

Hudson, M., Kempinski, Y. MindReset: A scalable eye-tracking-enabled split-second unlearning (SSU) platform for mental health and persistent pain.

Aims

MindReset aims to deliver a scalable, mechanism-driven digital intervention for anxiety, sleep-related disturbance, persistent pain, and associated mental health conditions. The platform operationalises the Split-Second Unlearning (SSU) framework within a smartphone application, using software-based eye tracking to detect and interrupt maladaptive Emotional Memory Image (EMI) activation patterns. The objective is to provide an evidence-informed micro-intervention accessible at population scale without specialist hardware or synchronous clinician input.

Background

More than 1 billion people worldwide are currently living with a mental health disorder (World Health Organization, 2025). Anxiety, sleep disturbance, and related conditions represent a substantial share of this burden, contributing to reduced productivity, increased comorbidity, and elevated healthcare utilisation; depression and anxiety alone cost the global economy an estimated US\$1 trillion per year in lost productivity. Access to treatment

remains constrained by cost, clinician shortages, and stigma, and existing digital interventions frequently rely on cognitive reframing or symptom-tracking with modest long-term adherence, particularly among those whose distress arises from automatic, pre-conscious activation patterns that conscious reflection alone cannot reach.

The SSU model proposes that psychological and physiological distress arises from non-conscious sensory representations, termed Emotional Memory Images (EMIs), encoded during overwhelming or threatening experiences (Hudson & Johnson, 2021, 2022; Hudson et al., 2025). When triggered, EMIs initiate rapid fight–flight–freeze responses associated with involuntary eye-movement patterns reflecting automatic memory retrieval below the threshold of conscious awareness. A brief interruption during the reconsolidation window, when the memory trace is temporarily labile, may functionally decouple the emotional memory from its associated stress response, targeting the pre-cognitive activation sequence that cognitive restructuring cannot reach.

Sleep disturbance is a particularly significant EMI-driven manifestation. As circadian rhythms shift toward sleep and the hypothalamic-pituitary-adrenal (HPA) axis begins to stand down, any active EMI functions as an internal alarm signal, triggering hypervigilance and disrupting restorative sleep. This mechanism may underpin insomnia, pre-sleep rumination, and sleep apnoea. MindReset targets the EMI-linked eye-movement patterns that drive this cascade, offering a mechanism-informed route to interrupting sleep-related arousal at its source (Hudson & Johnson, 2021, 2022).

The SSU framework also addresses persistent pain. Hudson and Johnson (2023) introduced the Past Adversity Influencing Now (PAIN) model, proposing that persistent pain is maintained by EMIs formed during first-time adverse past experiences, keeping the nervous system locked in a prior threat state through the same psychophysiological mechanism that underlies anxiety and sleep disturbance. Dysmenorrhea offers a clear example: affecting 50–90% of menstruating women worldwide, it is proposed to arise where an adverse menarche, a first-time event, encodes an EMI that

re-activates the original stress response with each subsequent menstrual cycle, producing recurrent pain in the absence of underlying pathology. A case report documented complete resolution of longstanding dysmenorrhea following a single SSU session, with the patient remaining pain-free at six-month follow-up (Hudson & Johnson, 2021, 2022). The PAIN framework extends MindReset's potential application across the broader landscape of persistent pain conditions that remain unresponsive to biomedical intervention alone.

Methods / System Architecture

MindReset operationalises SSU within a smartphone application using software-based eye tracking via the front-facing camera. No specialised hardware is required beyond a standard consumer smartphone. Sessions last 2–5 minutes, can be self-administered, and can be repeated as needed throughout the day. The platform is designed for asynchronous use, removing the dependency on scheduled clinician contact that constrains conventional digital therapeutic delivery. The four-phase intervention sequence is summarised in Table 1:

TABLE 1: MindReset SSU Intervention Phases

Phase	Description
1. Activation	Audio prompts guide recall of a stressful thought or intrusive memory, opening the reconsolidation window.
2. Detection	Front-camera eye tracking monitors gaze behaviour to detect involuntary movements associated with EMI access.
3. Interruption	A structured visual task disrupts EMI-linked eye-movement patterns while audio maintains neuroplastic receptivity.
4. Verification	The user attempts to re-access the original activation; changes in gaze-response patterns are recorded.

The application records anonymised gaze metrics and session parameters for ongoing analysis and platform refinement. No personally identifiable data are collected, and no-account registration is required, ensuring privacy by design and regulatory simplicity across jurisdictions. The application is publicly available via standard app distribution platforms and has been deployed across multiple countries without adaptation of the core intervention protocol, indicating cross-cultural feasibility.

Results

MindReset has achieved more than 80,000 downloads across multiple countries to date. Exploratory pilot observations across several cohorts indicate promising preliminary signals, summarised in Table 2: Highlights include over 90% of Garmin-monitored users showing decreased stress within the first week, 88% of students reporting reduced depression scores (average –35%), and a 38% reduction in OLBI burnout scores after two weeks. Mental energy monitoring conducted by Prof. Berckmans of KU Leuven found positive impact in 79% of 67 sessions within 15 minutes. A clinical trial addressing PTSD symptoms, led by Sapir College, is currently underway with positive initial results. All quantitative results to date are observational pending completion of controlled trials.

Conclusions

MindReset represents an emerging digital infrastructure approach for addressing stress-related public health challenges at scale. By targeting psychophysiological activation mechanisms underlying anxiety, sleep disturbance, burnout, and persistent pain, united by the PAIN framework's insight that past adverse experiences continue to drive present suffering via EMIs. The platform delivers a mechanistically distinct pathway beyond cognitive appraisal. For sleep, removing the EMI extinguishes the pre-sleep HPA alarm signal; for pain conditions such as dysmenorrhea, clearing the adverse first-event EMI removes the recurring psychosomatic trigger. Its brief session structure and zero-hardware deployment model remove the principal access barriers of conventional therapy, positioning the platform for genuine population-level reach.

TABLE 2: Preliminary Pilot Observations by Cohort

Cohort	Measure	Key Finding	Detail
Wearable stress monitoring (Garmin)	Physiological stress score	>90% showed decreased stress	Reductions observed within first week of use
Mental energy (BioRICS Fitbit)	Energy use & recovery	79% of 67 sessions showed positive impact	Effect within 15 minutes; conducted by Prof. Berckmans, KU Leuven
Student anxiety & depression	PHQ-9	88% experienced decrease in depression	Average reduction of 35%
Workplace burn-out	OLBI	38% decrease in burnout	After 2 weeks of use
PTSD (clinical trial)	PTSD symptom measures	Positive initial results	Ongoing RCT led by Sapir College

The platform is readily integrable within stepped-care frameworks, employee assistance programmes, and university health services. The combination of reconsolidation-informed theory with real-time mobile eye tracking constitutes a novel approach to scalable mental health and pain intervention, with meaningful implications for public health delivery and digital therapeutics research. More broadly, the SSU model and its digital operationalisation via MindReset may signal an emerging shift in therapeutic orientation, from the cognitive era, which has dominated clinical psychology for half a century, toward a post-cognitive paradigm in which pre-conscious psychophysiological activation sequences become the primary target of intervention. If the locus of human suffering lies not in what people think, but in what their nervous systems have learned to anticipate before thought can intervene, then effective treatment must reach further back in the causal chain than cognition alone can access.

Conference attendees will be invited to experience a live SSU session, observe real-time eye-tracking identification phases, and review anonymised gaze-pattern visualisations illustrating EMI activation and interruption.

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PhD & MSc Student Track

Adaptive by design: Adaptive participatory design for sustainable AI-enabled software systems in contexts of variable human capacity

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Overview

Cognitive fatigue is one of the most prevalent and disabling symptoms experienced by people with Multiple Sclerosis (PwMS), affecting up to 80% of individuals and significantly limiting participation in work, education, and daily

life (1-3). Unlike physical fatigue, it is invisible, fluctuates throughout the day, and is highly context-dependent (4-6). Despite the growing use of digital health technologies (DHT); such as AI-enabled mobile health applications and wearable devices; for self-management, most existing tools assume stable cognitive capacity and continuous engagement, limiting their utility when users need support most (7).

This PhD research aims to develop and validate an inclusive participatory design (PD) framework for sustainable, fatigue-aware DHT. Drawing on the *dynamic disability* concept, the project treats fluctuation as a core design constraint rather than an exception (8, 9).

Using a participatory mixed methods approach (10, 11) across embedded case studies in Finland, Spain, and Greece, the research operationalizes this dynamic disability perspective to advance inclusive design methods that enable meaningful participation and reduce digital exclusion among populations with chronic and fluctuating health conditions.

Background

Multiple Sclerosis (MS) is a chronic neurological condition commonly diagnosed in young adulthood, a life stage associated with education, employment, and family formation. PwMS experience a range of physical, cognitive, and psychosocial symptoms, with cognitive fatigue being among the most disabling yet least comprehensively understood. It affects attention and information processing, fluctuates unpredictably and varies by contexts and time, and is inadequately assessed by conventional clinical or digital assessments (1, 4, 5).

Recent research conceptualizes such experiences as *dynamic disability*: impairments that vary depending on internal state, environmental conditions, and task demands (8, 9). From a design perspective, this challenges dominant assumptions in DHT, which typically treat cognition as stable and consistent (10). Rigid interaction patterns fail to accommodate fatigue fluctuations, reducing adherence and leading to disengagement.

Across Europe, standard DHT designs may inadvertently marginalize people with fluctuating abilities, potentially reinforcing health inequities. PD offers a promising response by involving users as expert contributors and translating invisible experiences into design requirements. However, existing PD frameworks provide limited guidance on engaging people with dynamic disabilities without increasing burden (11, 12). A methodological paradox exists: while promoted as inclusive, participatory methods may inadvertently exclude those they aim to serve due to assumptions of consistent participation and cognitive capacity.

This research positions PD not only as a design methodology but as a health intervention that can enhance empowerment and support sustainable, meaningful engagement with DHT. The project will deliver a practical, fatigue-aware PD framework with adaptations across understanding, co-design, and evaluation stages, designed to integrate seamlessly with Agile development processes (11, 13, 14).

Methods

The research adopts a participatory mixed methods approach structured around three embedded case studies conducted over 36 months in PwMS in Finland, Spain, and Greece. This cross-cultural European context explores how cultural values and everyday environments shape experiences of cognitive fatigue and DHT use. The project will follow principles of ethical approval, data management, Open Science, and European citizen science guidelines.

Case Study 1: Understanding Unmet Needs and Digital Health Technology Use

Semi-structured interviews with PwMS across the three countries map unmet needs and current DHT use, with sampling guided by data saturation (10). The study explores: (1) which DHT tools PwMS currently use; (2) how cognitive fatigue affects engagement with and adherence to these DHT tools; and (3) unmet needs regarding self-management on living with MS, physical activity possibilities, possible behaviour change strategies, mental

health support, workplace accommodation, and motivation to engage with DHT. This generates design insights that inform subsequent case studies and framework development.

Case Study 2: Co-Design of an Inclusive Participatory Design Framework

Building on Case Study 1 insights, the second case study develops a structured co-design plan in collaboration with domain experts in PD and cognitive accessibility (10-13). The plan will be refined and validated through workshops with around 5 PwMS. The study will develop practical, fatigue-aware adaptations to commonly used PD methods, including interviews, workshops, questionnaires, etc. This co-design process culminates in the development of an inclusive PD framework tailored to the needs and capabilities of PwMS.

Case Study 3: Validation of Fatigue-Aware Participatory Design Framework

Focus groups with PwMS will evaluate the framework through structured discussion guides assessing its relevance, clarity, feasibility, and perceived inclusiveness across design process stages (10, 11, 15). Participants will reflect on the proposed framework's principles, recommended methods, and fatigue-responsive adaptations, discussing alignment with their lived experiences and fluctuating participation capacity. The focus groups will also explore implementation challenges, ethical considerations, and opportunities for integrating the framework into real-world digital health design processes, including Agile development contexts.

Feedback will be used to refine and finalize the PD framework, ensuring it supports meaningful, sustainable, and empowering participation of people with dynamic disabilities.

Expected Results

As a work-in-progress PhD project, the three interconnected case studies are expected to generate: (1) characterization of how cognitive fatigue influences DHT interaction and identification of unmet needs; (2) a practical PD framework for inclusive research with people experiencing dynamic disabilities; and (3) validation of the framework's relevance and feasibility

through PwMS feedback. Collectively, these outcomes will advance sustainable, inclusive design methods and inform future evaluations of DHT for populations with chronic and fluctuating health conditions.

Conclusions

This research addresses a critical gap in DHT by centering dynamic disability as a core design concern. By integrating the dynamic disability concept with PD, the project reframes inclusion as a temporal and contextual challenge, and DHT as adaptive technologies. The interconnected case studies advance understanding of PwMS digital needs, co-design methods for fluctuating abilities, and usability evaluation for dynamic disabilities. The resulting inclusive, fatigue-aware PD framework offers transferable methodological guidance for researchers and designers developing inclusive DHT for populations at risk of digital exclusion.

Acknowledgments

During the preparation of this work, the author(s) used Grammarly (v1.152.0.0; accessed February 2026) to identify typographical errors and improve grammar. Portions of the text were also drafted with assistance from ChatGPT (OpenAI; vGPT-5.2; accessed February 2026) to enhance readability. All outputs were reviewed and edited by the author(s), who take full responsibility for the final content of the publication.

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Comparing the relationship between high and low tobacco control policies and public engagement with smoking cessation support

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Background

Globally, tobacco use causes over eight million premature deaths per year (World Health Organisation, 2025). Tobacco use and tobacco control measures vary significantly throughout Europe. (WHO, 2025). The United Kingdom (UK) and Germany are culturally and economically similar countries in which tobacco use is a leading cause of death and disability (IHME, 2026). However, approaches to tobacco control differ dramatically, with the UK prioritising a preventative focus and implementing high control, centralised legislation, and Germany championing a self-determined treatment focus,

implementing low control, and decentralised legislation (Cairney, 2019; Tobacco Control Scale, 2021; Zeeb et al., 2025).

As of 2021, the United Kingdom was jointly ranked first in European tobacco control, while Germany ranked 34th out of 37 countries assessed according to the Tobacco Control Scale (TCS) (TCS, 2021), which measures the implementation of policies recommended by the World Bank. This is primarily due to the UK's restrictions on the influence of the tobacco industry and tobacco product advertising, thereby increasing the consistency of anti-smoking rhetoric compared to Germany. Notable differences also exist due to the UK's stricter bans on public smoking, higher tobacco taxes, and greater availability of smoking cessation support.

Sustained smoking cessation is most effectively achieved through combined behavioural support and pharmacotherapy (Stead et al., 2016). The English National Health Service (NHS) has provided and promoted free combined support since 2000 (NHS, 2022). However, a review of the English Smoking Toolkit Study (STS), a nationally representative dataset of public smoking behaviour, found that most quit attempts in England are unaided, with only 2.2% of quits using face to face behavioural support (Jackson et al., 2025). Access to in-person support may be hindered due to well-documented strains on the NHS; said strains are exacerbated in socioeconomically disadvantaged areas, which exhibit the highest risk of smoking related illness (Marmot, 2020). Digital behavioural support may improve the accessibility of support, but usage data still reveal substantial room for improvement (Jackson et al., 2025), highlighting the need for targeted interventions.

German healthcare faces similar strains in more deprived areas, yet primary healthcare is highly accessible through patient-selected publicly funded insurers (Blümel et al., 2022). The German healthcare system has created a novel approach to improve the accessibility of effective behavioural support by prescribing clinically validated Digital Health Applications (DiGAs). With DiGAs targeting smoking cessation, it is the first time that

health insurances are legally obliged to fully cover behavioural support for smoking cessation. However, publicly available smoking cessation support is recent and restricted. Insurers have only been required to provide a one-time reimbursement of combined pharmacological support to individuals with severe tobacco addiction since 2025 (Blümel et al., 2022; IQWiG, 2024). Therefore, whilst the DiGA initiative has increased the availability of behavioural support to the German public, the recency of the program and a decentralised approach to smoking cessation is likely to hinder public awareness of support and effective use cases, limiting engagement.

Consequently, due to the contrasting regional tobacco policy and public health environments, the English and German public are likely to differ in attitudes, knowledge, and biases towards smoking cessation and motivations to seek and engage with support. This novel public-informed insight is expected to inform culturally specific recommendations to promote intervention uptake.

Research Objectives

The aim of this research is to compare the influence of the two contrasting tobacco control environments of England (high control and centralised policies) and Germany (low control and localised policies) on people who smoke, and to identify shared and unique top-down barriers and facilitators to public engagement with smoking cessation support.

Methods

I plan to use a mixed-methods approach to uncover existing discrepancies with smoking cessation engagement, whilst exploring the depth of the subject with bottom-up lived experiences from public and patient involvement (PPI) and top-down stakeholder expertise.

Study 1. Mixed-method survey with people who have smoked or currently smoke in England and Germany.

An online mixed-method survey will be conducted with people who have smoked or are currently smoking in England and Germany, including both

those who have and have not used support to quit to identify barriers and facilitators to support and inform policy interventions. Using the COM-B model (Michie et al., 2011), and PRIME Theory of Motivation (West & Michie, 2021) as frameworks, the questions aim to offer insights to the following areas:

1. Attitudes towards bottom-up and top-down responsibilities for individual healthcare and smoking behaviour.
2. Health-related barriers and motivators to quitting smoking.
3. Internal and situational barriers and motivators influencing the use of smoking cessation support.

Study 2. Qualitative interviews with English and German public health commissioners and smoking cessation policymakers.

Online semi-structured individual interviews will be conducted with between 12-20 stakeholders with diverse experience in professional roles, policy contexts, and geographical settings. Interviews will explore stakeholders' perceptions, experiences, and views of the perceived benefits, feasibility, and acceptability of proposed smoking-related legislation in England and Germany. A reflexive thematic analysis will be performed deductively to understand and code common and pertinent insights. (Braun & Clark, 2008). This understanding will be used to understand and interpret shared and diverging perspectives regarding effective interventions.

Results

Following an initial literature review, and a small pilot study with users of the smoking cessation app Smoke Free and the survey platform Prolific, individual motivators, barriers, and time of engagement with support is likely to differ between English and German populations. For example, English participants may indicate more internal consistency, however, German participants may report barriers reflective of the strength of state-dependant tobacco policies.

Conclusions

This will be the first direct comparison of high- and low-control tobacco policies England and Germany, providing a valuable comparison across two otherwise similar countries. Primarily, this work will incorporate PPI and expert knowledge to establish region-specific and culturally acceptable policies, to promote engagement with more effective smoking cessation and target at-risk groups with tailored support.

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Evaluation of augmented reality applications on smartphones as a tool for enhancing anatomical knowledge among medical students in Pakistan

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Hassnain, S., Hassnain, A. Evaluation of augmented reality applications on smartphones as a tool for enhancing anatomical knowledge among medical students in Pakistan.

Background

Anatomy is the main building block of the medical curriculum and is essential for mastering clinical skills since the human body is central to all examinations. ¹ Traditionally, lectures, tutorials, textbooks, cadaver dissections, and prosected specimen demonstrations have been the primary tools for teaching anatomy. ² Understanding complex anatomical structures is crucial for developing skilled healthcare professionals, so teaching methods should be as effective as possible. ³ Students of the millennial generation lean towards a learning style that is centered on their needs, encourages active participation, and supports independent study. The advent

of smartphone learning has extended educational spaces beyond traditional classrooms, calling for an adjustment in teaching methods. The incorporation of technology in education, particularly through AR applications, aligns with these preferences by providing dynamic and hands-on learning experiences that transcend traditional classroom boundaries.⁵

Augmented reality (AR) has emerged as a transformative educational tool.⁶ These applications can superimpose digital information onto the physical world, offering students interactive and immersive experiences that enhance their understanding of complex anatomical structures.⁷ The accessibility of AR via smartphones makes it a handy and cost-effective tool for students, who can use these applications anywhere and anytime.⁸

Despite AR's promising potential, its use in medical curricula is still limited, necessitating further research to validate its effectiveness and optimal use. This study aimed to determine medical students' familiarity with AR technology, engagement levels, usability challenges, and preferences between AR and traditional learning methods.

Methods

A descriptive cross-sectional study was conducted by using a structured questionnaire to collect face-to-face data from medical (MBBS) students at FMH Medical & Dental College, Lahore, and 22 other medical colleges in Pakistan via Google Forms. The sample was collected through a convenience

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TABLE 1: Distribution of students according to year of MBBS

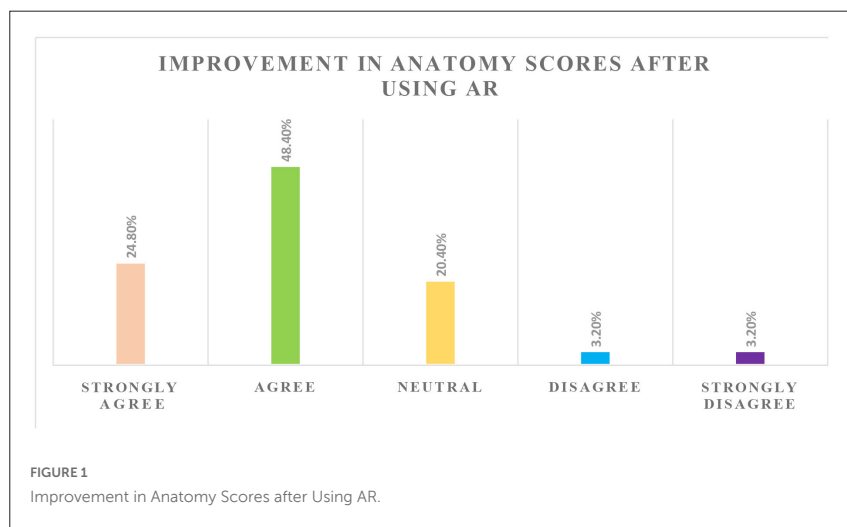
Year of Study	MBBS Students	Percentage (%)
1st Year	255	32.1%
2nd Year	69	8.7%
3rd Year	194	24.4%
4th Year	178	22.4%
5th Year	99	12.5%
Total	795	100%

sampling method. Approval was obtained from the FMH Institutional Review Board (IRB) #FMH-16/08/2024-IRB-1456.

Informed consent was obtained from all participants, ensuring confidentiality and anonymity. After taking the consent, the students were asked to complete the questionnaire. There were a total of six sections, including demographic profile, familiarity with technology, engagement, usability issues, effectiveness, and preferences about these apps. The study utilizes already existing resources of students, such as smartphones and the internet, for online data collection. For in-person filling of the questionnaire, the budget was divided among all the researchers. No external funding was available. Data analysis was performed using SPSS 27.00, employing descriptive statistics and inferential analysis to compare AR applications with traditional methods.

Results

This study surveyed 795 medical students, 453 responses from FMH, and 342 responses from other medical colleges throughout Pakistan. The mean age



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of participants was 21.21 ± 1.91 years. Among the total students, 32% were males and 68% were females. Findings revealed that 66% of students were familiar with AR technology, and 85.1% agreed that AR applications enhanced their learning. The data suggest maximum awareness of AR apps, which indicates that AR is not an entirely new or foreign concept for the students.

70% of respondents reported frequent use of AR, while 30% expressed challenges with its usability. AR was positively perceived by 70% for improving anatomy scores and reducing reliance on traditional methods. Furthermore, 81.7% of students found AR to enhance enjoyment in learning anatomy. A blended learning approach, combining AR and traditional techniques, was preferred by 62.3% of participants, indicating the complementary role of AR in anatomy education.

Conclusions

The study reveals a positive reception of augmented reality (AR) applications on smartphones as an effective tool for enhancing anatomical knowledge among medical students. The findings reveal that AR significantly improves engagement, spatial understanding, and overall learning experience compared to traditional methods. With 85% of students reporting AR as more engaging and 74% preferring its use in future studies, this technology shows promise as a valuable supplement to conventional anatomy education.

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From pilot to policy: Evaluating a hybrid-online PrEP service to inform digital pathways in sexual and reproductive health

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Shanley, A., O’Flaherty, C., McCarthy, N., Lyons, F., Kinchin, I. From pilot to policy: Evaluating a hybrid-online PrEP service to inform digital pathways in sexual and reproductive health.

Background

HIV remains a critical public health issue worldwide. Additional work remains if we are to achieve our commitments at national and international level to ‘end AIDS’ and ‘get to zero new HIV transmissions’ by 2030(1–3). In response to the ongoing HIV epidemic, oral HIV pre-exposure prophylaxis (PrEP) has emerged as a highly effective preventative measure. PrEP is a biomedical

prevention strategy that, when taken as prescribed either daily or around specific periods of risk, is 99% effective at preventing HIV (4–7).

While PrEP provision has steadily increased in recent years, the intervention falls short of comprehensive coverage. The World Health Organisation (WHO) has called for efficient methods of providing PrEP to increase coverage and suggests that achieving a critical mass of people on PrEP and moving closer to eradication of HIV requires services to be tailored to meet the needs of both service users and providers (8).

A digital health approach to PrEP service delivery presents an opportunity to dismantle structural barriers, including geographic disparities, clinic capacity constraints, scheduling challenges, financial impacts and social stigmas, that historically hinder access to this essential HIV preventative tool (9–11).

This study aims to overcome these barriers by co-designing and piloting a hybrid-online service. It adapts an existing digital platform, currently delivering at-home STI testing services for Ireland's national health service, to the PrEP context. Insights from this approach can inform wider application of digital health interventions in sexual and reproductive healthcare that are more flexible, equitable and responsive to changing population needs, demonstrating how digital transformation can complement and strengthen conventional services rather than replace them.

Methods

Following a Patient and Public Involvement (PPI) framework, we engaged PrEP service users and providers to ensure the intervention was inclusive by design. Through a blend of digital and in-person collaborative sessions, diverse users, including migrants, older people, rural residents, and non-native English speakers, provided feedback that informed the study design, patient information materials, and data collection instruments. We conducted a mapping exercise with service providers from two pilot sites to identify end-to-end clinical steps and ensure seamless integration of the online pathway into existing health service standards and infrastructure.

Recruitment commenced at two pilot sites located in urban centres, each with significant peri-urban and rural catchment areas. PrEP users already engaged in services and that meet eligibility criteria are offered the choice to opt-in to receive their care via the hybrid-online pathway. This pathway shifts routine in-person monitoring and testing into a remote, asynchronous user-led process. This model also allows for a return to in-person services if clinically necessary or requested by the service user.

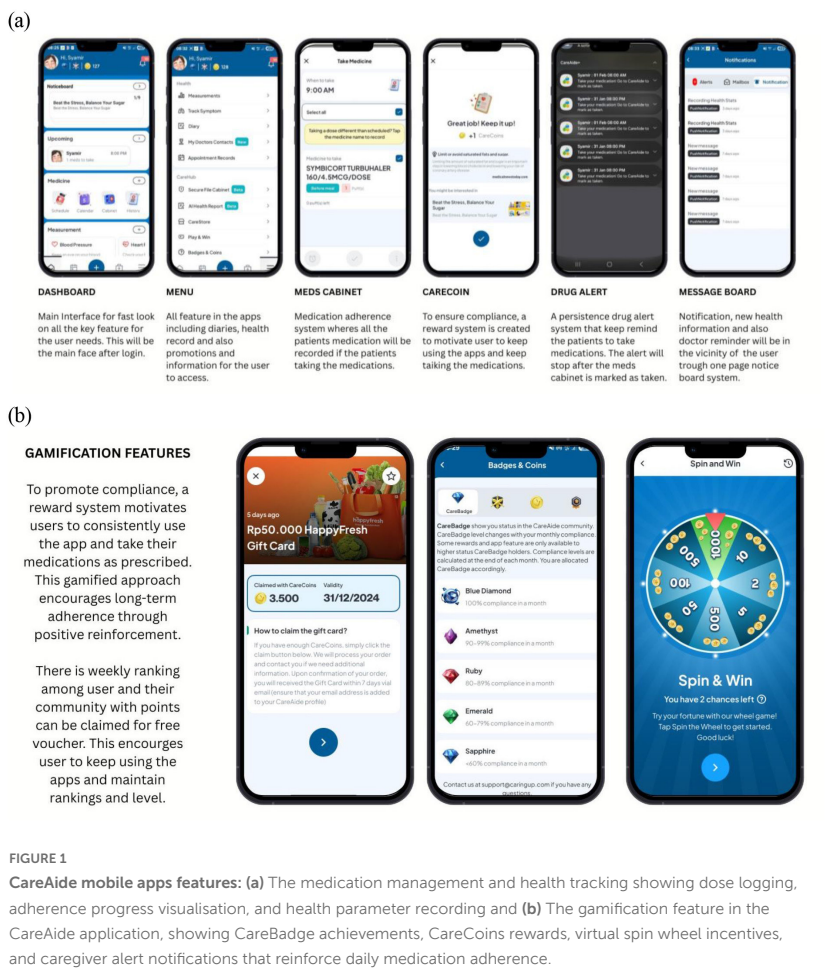
A mixed methods research framework using a pre/post/follow-up study design combines quantitative evaluation of outcomes over time with qualitative, in-depth exploration of experiences. The quantitative method consists of self-administered surveys at baseline, post-intervention, and exit points to measure system usability, acceptability, and patient-borne costs such as travel and work/study disruptions. A secondary survey of those who declined participation captures specific barriers to digital adoption. The qualitative method involves semi-structured interviews with purposively sampled adopters, non-adopters, and healthcare providers to explore lived experiences and the practical workflow implications of a hybrid-online pathway.

By triangulating pilot data with relevant literature and policy documents from within the broader digital healthcare landscape, this research aims to establish a framework for replicability across sexual and reproductive health services and beyond.

Results

Preliminary results from the initial three months of the pilot demonstrate a clear demand for the digital pathway. Recruitment reached 55.2% of the target figures, with a baseline survey completion rate of 73.2%. Demographic analysis confirms the reach of the recruitment strategy, successfully enrolling participants from rural regions, migrant communities, and varied age groups.

Baseline data regarding the traditional in-person service model reveal a significant participant burden. A third of the cohort (33%) travelled over an



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hour to access care and over half (59.6%) spent thirty to sixty minutes in the clinic. This time investment had a significant impact on participant's lives; 86.4% of participants reported changing their work schedules to attend appointments with 43.6% indicating an impact on their work or career progression, while 12.9% noted disruption to social commitments.

Financially, the preliminary data highlight a clear economic burden associated with the in-person model. Participants reported direct out-of-pocket expenditures across several categories, including public transport fares, fuel, and parking fees, with 67.3% of the cohort incurring costs per visit. Beyond direct spending, significant indirect costs are incurred; 12.5% of respondents reported a loss of income or the use of paid (23.7%) or unpaid (17.1%) leave to facilitate clinic attendance.

Conclusion

Data collection is ongoing but early findings suggest a strong interest in this digital intervention across a diverse user base. By shifting low complexity care away from capacity constrained traditional services, the hybrid-online model has the potential to address critical gaps in service efficiency and patient equity while maintaining standards of care. As the pilot continues, acceptability, projected savings and efficiencies among service users and providers will be estimated through post-intervention and exit survey data.

By positioning this research within the broader context of Ireland's digital healthcare strategy, the study will contribute to ongoing policy discussions around the adoption, integration, and sustainability of hybrid-online models of care. The findings can help guide future evidence-based decision-making, ensuring that digital healthcare services align with patient needs while maintaining clinical standards. As such, this study will not only inform best practices for PrEP service delivery but also set the stage for transforming other healthcare services with hybrid-digital solutions, making the healthcare system more responsive to the needs of the modern healthcare service user.

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Level up adherence: gamified digital health app careaide boosts clinical, economic and quality-of-life outcomes in chronic disease management -- A multicentre randomised controlled trial

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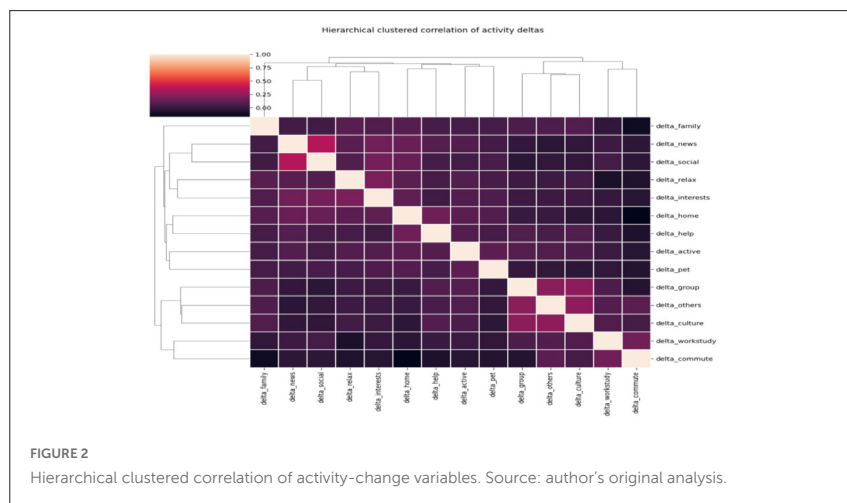
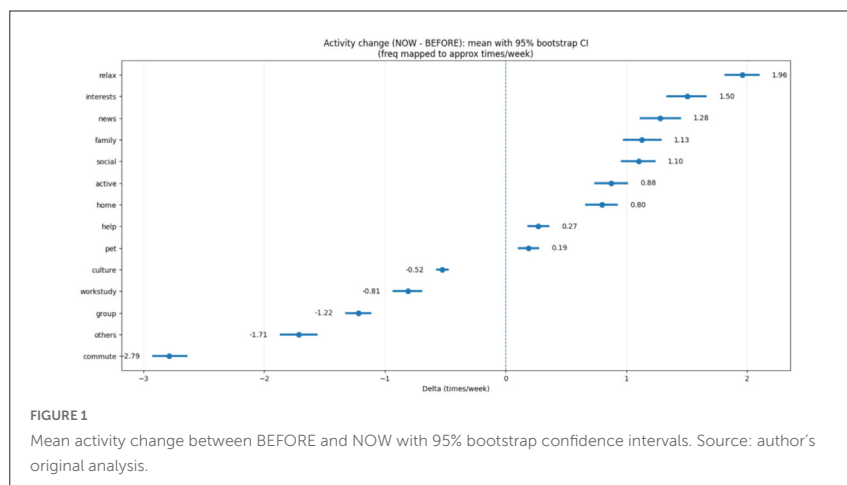
Mohamad Shukeri, M.S., Ibrahim, B., Huri, H.Z., Ong, S.C., Rana, R., Malik, T. Level up adherence: gamified digital health app careaide boosts clinical, economic and quality-of-life outcomes in chronic disease management -- A multicentre randomised controlled trial.

Overview

Medication non-adherence is a key barrier to effective chronic disease management globally. CareAide is a gamified mobile health application that sustains medication adherence through achievement rewards, virtual incentives, and real-time caregiver alerts, turning daily medication taking into a goal-driven experience (Figure 2). The CareAide trial is a multicentre randomised controlled trial (RCT) across three Malaysian public hospitals. Among 1,106 six-month completers, CareAide significantly improved adherence, disease-specific clinical outcomes, and health-related quality of life (HRQoL) across four NCD cohorts, supporting its potential as a scalable and equitable digital health solution.

Background

Non-adherence to chronic disease medications affects over half of patients with hypertension and type 2 diabetes mellitus (DM) in Malaysia, contributing directly to preventable complications and avoidable healthcare costs [1,2,3,4,5]. Conventional reminder and counselling strategies produce modest and short-lived gains because they do not address the motivational drivers of sustained adherence [6]. Systematic reviews confirm that mHealth applications significantly improve adherence and clinical outcomes across chronic disease populations [7,8], yet rigorous RCT evidence from middle-income country settings remains limited. CareAide applies gamification principles grounded in behavioural change theory [9,10]: patients earn CareBadge achievements and CareCoins for each dose taken on time, spin a virtual reward wheel upon streak completion, and trigger automatic caregiver alerts on missed doses (Figure 2). Medication schedules and adherence progress are tracked through an integrated management dashboard (Figure 1). The CareAide Trial is the first multicentre RCT to evaluate this application across four NCD populations simultaneously.



Methods

An open-label, parallel-arm RCT was conducted from June 2022 to October 2024 at University Malaya Medical Centre (UMMC), Hospital Putrajaya (HPJ), and Hospital Pulau Pinang (HPP). Adults aged 18 years and above with hypertension, type 2 DM, asthma, or heart failure, and a baseline Morisky Medication Adherence Scale-8 (MMAS-8) score below six, were randomised 1:1 to CareAide plus standard care or standard care alone. The primary outcome was MMAS-8 [11] assessed at baseline, three months, and six months. Pharmacy-verified Proportion of Days Covered (PDC) [12] was calculated for DM and hypertension cohorts from dispensing records; standard PDC was not applied to asthma or heart failure due to methodological limitations for inhaler-based and complex polypharmacy regimens. Secondary outcomes included EQ-5D-5L [13], AQoL-6D [14], ACQ-7, NYHA functional class, and Mobile Adherence Satisfaction Scale (MASS) [16].

Results

Of 1,320 enrolled adults, 1,106 completed six months of follow-up (intervention $n=556$, control $n=550$; retention 83.8%). Both groups were well matched at baseline. MMAS-8 scores improved significantly in the CareAide arm across all four NCD cohorts ($p<0.001$ for each). High adherence (MMAS-8 at least 6) rose from 0.9% at baseline to 86.3% at six months in the intervention group, versus 1.8% to 40.9% in controls. PDC at or above 80% was achieved by 99.5% of CareAide participants versus 34.3% of controls in the DM cohort ($p<0.001$), and 77.2% versus 57.3% in the hypertension cohort ($p=0.009$). Outcomes by cohort are summarised in Table 1.

Clinical outcomes mirrored adherence gains across all cohorts. Systolic blood pressure was significantly lower in the hypertension intervention group (136.1 vs 147.3 mmHg, $p<0.001$) and LDL was 2.46 versus 3.31 mmol/L ($p=0.04$) [16,17]. DM participants achieved greater HbA1c reduction (-0.80% vs -0.40%, $p=0.004$). Asthma control improved significantly (ACQ-7 adjusted difference -0.414, 95% CI -0.783 to -0.045, $p=0.028$) [15]. Heart failure participants showed a directional improvement in NYHA functional class.

TABLE 1: Medication adherence and key clinical outcomes by chronic disease cohort at six months

Disease	n (I/C)	MMAS-8 Median 6 Months (I vs C)	PDC ≥80% (I vs C)	Key Clinical Outcome at 6 Months	p
Hypertension	289/296	6.50 vs 5.50	77.2% vs 57.3%	SBP 136.1 vs 147.3 mmHg; LDL 2.46 vs 3.31 mmol/L	<0.001
Diabetes Mellitus	176/171	7.75 vs 5.75	99.5% vs 34.3%	HbA1c change: -0.80% vs -0.40% (p=0.004)	<0.001
Asthma	143/133	6.75 vs 5.75	N/A *	ACQ-7 adjusted diff: -0.414 (95% CI -0.783 to -0.045)	0.028
Heart Failure	148/150	6.75 vs 5.75	N/A *	NYHA class: directional improvement observed	<0.001

I = Intervention; C = Control; MMAS-8 = Morisky Medication Adherence Scale-8; PDC = Proportion of Days Covered; SBP = Systolic Blood Pressure; LDL = Low-Density Lipoprotein; HbA1c = Glycated Haemoglobin; ACQ-7 = Asthma Control Questionnaire-7; NYHA = New York Heart Association. * PDC not calculated for asthma or heart failure due to methodological limitations; MMAS-8 is the primary adherence measure for these cohorts.

HRQoL improved significantly on both instruments. EQ-5D-5L utility scores in the asthma and heart failure cohort rose from 0.828 to 0.919 in the CareAide group versus 0.796 to 0.855 in controls (p=0.0003) [13]. AQoL-6D scores in the DM cohort showed a significant between-group difference (d=0.353, p<0.001) [14]. Patient satisfaction was high (mean MASS 3.65, SD 0.57), with lowest-income participants reporting comparable satisfaction to higher-income groups, indicating equitable digital engagement [16].

Conclusions

The CareAide Trial provides multicentre RCT evidence that a gamified mHealth application can produce sustained, clinically meaningful improvements in medication adherence, clinical outcomes, and HRQoL across four chronic NCDs in a middle-income country setting. Convergent

findings from MMAS-8 and pharmacy-verified PDC strengthen confidence in these results. The consistency across disease categories and income groups highlights real-world scalability, with direct implications for national mHealth policy in Malaysia and comparable healthcare systems [18].

Ethics, Registration, and Declarations

Ethical approval was obtained from the Medical Research and Ethics Committee, Ministry of Health Malaysia (MREC ID: 2022311-11069). All procedures adhered to the Helsinki Declaration, Malaysian Good Clinical Practice guidelines (ICH GCP E6(R3)), and the Personal Data Protection Act 2010. The trial was prospectively registered with NMRR (ID: 22-01108-COI (IIR)) and ClinicalTrials.gov (NCT06068309; registered 19 September 2023). Written informed consent was obtained from all participants prior to enrolment. Data are available from the corresponding author on reasonable request. The authors declare no competing interests. This work was funded by Caring Up Malaysia Pte. Ltd., Kuala Lumpur, Malaysia (Grant: UM.0000532/HPV.PV). The authors thank the Director General of Health Malaysia for permission to publish, and acknowledge financial support from the Universiti Malaya Scholarship Scheme (UMSS) to the corresponding author; the scholarship did not fund the research. Use of the MMAS-8 is protected by US copyright laws and was used under licence from Donald E. Morisky, ScD (Certificate No: 2744-4598-7689-9843-1300).

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Lifestyle reconfiguration under public health restrictions: Structural evidence from early COVID-19 survey data

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Su, Q. Lifestyle reconfiguration under public health restrictions: Structural evidence from early COVID-19 survey data.

Introduction

During COVID-19, digital platforms became one of the main places where people received public health guidance, compared risks and made sense of restrictions. Digital public health is therefore not only about the speed of information delivery, but also about how information environments shape everyday behaviour. Earlier work has shown how online misinformation and infodemics can damage trust and risk understanding (Vosoughi et al., 2018; Eysenbach, 2020; WHO, 2020; Cinelli et al., 2020). At the same time, digital health research also points to a wider issue: people live, communicate and adapt within platform-based systems, not outside them (Kostkova, 2015; Kickbusch et al., 2021). Building on this context, this study examines how lifestyle patterns were reorganised during the early COVID-19 lockdown. The aim is not to prove one causal pathway, but to describe what changed across multiple everyday activities and what this says about digital public health monitoring.

Methods

The study is based on the author's original analysis of an online survey collected during the early lockdown period. The dataset records self-reported activity frequencies before restrictions and during lockdown, including commuting, work or study, exercise, family time, relaxation, personal interests, news consumption, social media use, cultural activities and group activities. Variables were cleaned so that naming, scale direction and missing values were handled consistently. Activity change was calculated as NOW minus BEFORE, and the ordered frequency categories were mapped to approximate weekly occurrences so that changes could be compared more clearly.

The analysis was deliberately descriptive. Following exploratory data analysis principles, the purpose was to identify visible structure in the data rather than build a predictive or causal model (Tukey, 1977). Mean activity changes were estimated with 95% bootstrap confidence intervals. Spearman correlations between activity-change variables were then reordered using hierarchical clustering to examine whether activities changed in modules. Age, gender and response timing were used for interpretation only, because the sample was not designed to represent the whole population.

Results

The clearest result is that lockdown did not simply reduce all activities. Instead, it moved activity from public and mobile settings towards home-based and digitally mediated routines. Commuting showed the largest negative mean change, while group activities, cultural activities and other offline outward-facing behaviours also fell. This fits with evidence that mobility restrictions had wider social and economic consequences, not only direct infection-control effects (Bonaccorsi et al., 2020).

By contrast, relaxation, personal interests, family activities, news use and online socialising increased or remained relatively strong. In other words, the pattern is not just 'less activity'. It is a redistribution of time, attention and social life under public health restrictions.

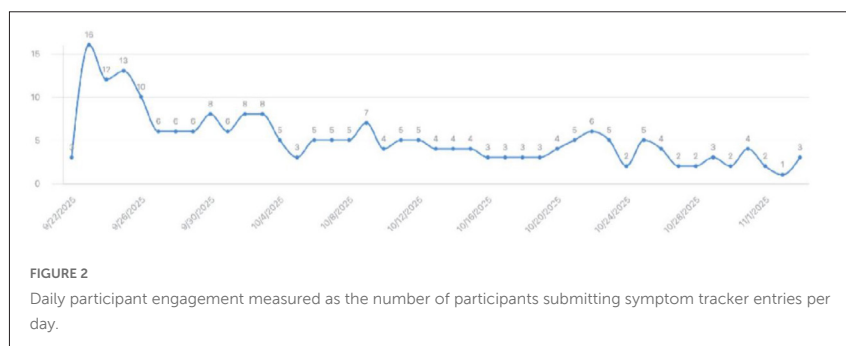
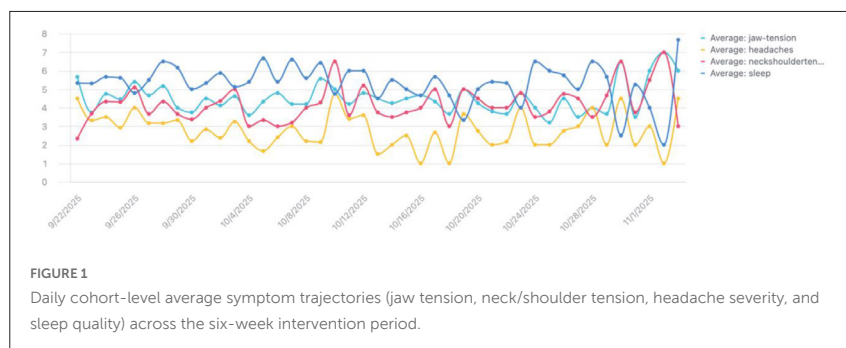


Figure 1 shows this difference more directly. The largest negative changes are concentrated in commuting and offline activities, while the positive side is led by relaxation, interests, news, family and online socialising. The confidence intervals make the contrast clearer: within this sample, these were structured shifts rather than small random movements around zero.

Figure 2 adds a second point. Behavioural change did not appear as one single block. The clustered correlation matrix shows local groups of activity-change variables, suggesting several related but partly independent behavioural modules. Home-based and digital activities tended to sit apart

from mobility- and public-participation activities. This matters because an average result can hide very different individual pathways. For example, the original exercise-transition analysis showed that some respondents maintained regular activity, some reduced it, and a smaller group increased from a lower baseline. This kind of heterogeneity is consistent with social and behavioural evidence from the pandemic, where responses depended on context, resources and daily constraints (Bavel et al., 2020).

Discussion and conclusion

The findings suggest three points for digital public health. First, public health emergencies reshape ordinary routines, not only clinical risk. Data on everyday behaviour can therefore add useful context to emergency monitoring. Second, interventions should avoid treating the public as one behavioural group. People may all be under the same restriction, but their daily routines are not disrupted in the same way. Third, digital and survey data are useful but imperfect. This survey was skewed towards older and female respondents, and all measures were self-reported. Digital traces and online surveys can also reproduce inequalities in who is visible in the data (Gonzalez-Bailon, 2019). The results should therefore be read as exploratory evidence of structure, not as a final population estimate.

Overall, early lockdown produced a modular and uneven lifestyle reconfiguration. Mobility and face-to-face activities declined, while home-based and digitally mediated activities became more central. Recognising this unevenness can support more targeted and fairer digital public health communication, especially when future emergencies again require rapid behavioural adaptation.

Acknowledgement of AI assistance

Generative AI was used only to support language editing, condensation and formatting of the author's original coursework into the required proceedings format. The analysis, interpretation, figures and cited sources are based on the author's original submitted report.

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Personalized sleep bruxism management through phenotyping: A pilot study of whatsapp-delivered behavioral coaching

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Citation

Sengouga, W.N. Personalized sleep bruxism management through phenotyping: A pilot study of whatsapp-delivered behavioral coaching.

Aims / Overview

This study evaluates the feasibility and preliminary validity of a phenotyping-based behavioral intervention for sleep bruxism delivered via WhatsApp. It aims to (1) assess whether a simple phenotyping algorithm can meaningfully stratify participants into clinically relevant subgroups, and (2) explore longitudinal symptom dynamics and engagement patterns using daily self-reported tracking.

Background

Sleep bruxism is a multifactorial behavior involving rhythmic masticatory muscle activity during sleep and is increasingly understood as a manifestation of interacting physiological and psychosocial processes rather than a single disorder (Lavigne et al., 2008; Lobbezoo et al., 2018). It is associated

with sleep fragmentation, autonomic activation, stress-related muscle tension, and insomnia-related hyperarousal (Kato et al., 2001; Manfredini and Lobbezoo, 2010; Riemann et al., 2010). Current consensus definitions emphasize its heterogeneity and the role of arousal-related mechanisms (Lobbezoo et al., 2018).

Despite this complexity, management approaches remain largely generalized and do not account for underlying drivers, which may contribute to inconsistent outcomes. To address this gap, we developed a simplified phenotyping framework categorizing individuals into three archetypes: **AIR** (breathing-related airflow limitation), **EARTH** (stress-related muscle tension and psychological load), and **WATER** (sleep-related hyperarousal and insomnia features). Each archetype is mapped to a targeted behavioral coaching pathway delivered digitally.

This pilot study explores the feasibility of implementing this approach through WhatsApp-based coaching and the utility of high-frequency symptom tracking to capture real-world dynamics.

Methods

Fifty-five participants were recruited and classified into archetypes using a structured questionnaire and rule-based algorithm: AIR (n=4), EARTH (n=34), and WATER (n=17). Participants were enrolled in archetype-specific WhatsApp groups and received daily educational modules, behavioral exercises, and peer-supported interaction over a six-week period.

Intervention content was tailored to each archetype's dominant mechanism. EARTH modules emphasized stress regulation, muscle relaxation, and coping strategies; WATER modules targeted sleep hygiene, cognitive techniques for hyperarousal, and sleep continuity; AIR modules addressed breathing awareness and positional or behavioral strategies related to airflow limitation.

Data collection included:

- Baseline and week-6 surveys assessing bruxism frequency, jaw pain, sleep quality, insomnia symptoms, perceived stress, and breathing-related difficulties
- Daily symptom tracking (jaw tension, fatigue, sleep quality, headache severity)
- Weekly check-ins capturing perceived helpfulness and brief reflections
- Semi-structured interviews at the end of the program

Daily tracker data were aggregated to compute **cohort-level average symptom scores per day**. Engagement was measured as the **number of participants submitting daily entries per day**, providing a proxy for adherence.

Analyses were descriptive and exploratory, focusing on (1) temporal evolution of symptom averages, (2) co-variation between symptom domains, and (3) engagement trajectories over time.

Results

Daily cohort-level averages demonstrated **relatively stable overall symptom levels with pronounced day-to-day variability** across all domains (Figure 1). Rather than following a linear improvement trend, symptoms fluctuated throughout the six-week period, reflecting dynamic patterns of symptom expression.

Jaw tension and neck/shoulder tension showed consistent temporal alignment, frequently rising and falling together, suggesting a shared musculoskeletal or stress-related component at the population level. Headache severity, while generally lower in magnitude, exhibited episodic increases that often coincided with peaks in tension-related symptoms, indicating a potential downstream or threshold-dependent relationship.

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Sleep quality demonstrated a distinct but related pattern, with lower reported sleep quality frequently occurring on days characterized by higher tension and headache scores. This co-variation supports a coupled relationship between sleep disruption and somatic symptom burden.

Although variability was evident, the range of symptom averages remained within a relatively narrow band, suggesting modulation of symptom intensity rather than large shifts in overall burden at the cohort level.

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Engagement patterns are shown in Figure 2. Daily participation was highest at the beginning of the intervention and declined gradually over time before stabilizing at a lower level. This trajectory is consistent with typical digital health engagement patterns, where initial motivation is followed by attrition and retention of a core group of active users.

The decline in reporting is important for interpreting symptom trajectories, as cohort-level averages reflect both symptom burden and participation. However, the persistence of consistent co-variation between symptom domains throughout the study period suggests that observed patterns were not solely driven by changes in the reporting population.

Qualitative Findings (Participant Reflections and semi-structured interviews)

Weekly reflections surveys and semi-structured interviews indicated increased awareness of symptom patterns, particularly links between stress, sleep quality, and physical tension.

Participants reported that archetype-specific content felt relevant to their experiences, with stress-management strategies (EARTH) and sleep-focused interventions (WATER) most frequently highlighted. Many described adopting practical techniques and gaining improved ability to manage symptoms, even when symptoms persisted. Some participants also noted challenges with maintaining daily engagement over time, consistent with the observed decline in tracker submissions.

Conclusions

This pilot study demonstrates the feasibility of delivering phenotype-guided behavioral coaching for sleep bruxism through a widely accessible platform such as WhatsApp. The integration of daily symptom tracking enabled the capture of high-frequency data, revealing dynamic relationships between muscle tension, sleep quality, and headache symptoms.

Findings support the conceptual framework underlying the AIR, EARTH, and WATER archetypes and highlight the importance of considering symptom interdependence and temporal variability in bruxism management. Rather than showing simple linear improvement, symptom trajectories suggest that intervention effects may manifest through modulation of daily fluctuations and interactions between symptom domains.

Sustained, though reduced, engagement over time further supports the practicality of digital delivery for behavioral interventions in real-world settings.

These results provide a foundation for future studies incorporating individual-level analyses, mixed-effects modeling, and integration with objective physiological data to refine phenotyping approaches and optimize personalized digital therapeutics for sleep bruxism.

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Translation and Validation of the Digital Working Alliance Inventory (DWAI) into German using conventional and AI-supported translation procedures

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Citation

Tietjen, A.K., Barth, J., Thomae, A.V., Holl, F., Schläpfer, S., Hirzler, L., Bousraou, Z., Witt, C.M. Translation and Validation of the Digital Working Alliance Inventory (DWAI) into German using conventional and AI-supported translation procedures.

Aims/Abstract/Overview

Users can develop a working alliance with digital health interventions (DHI) which may be associated with the effectiveness of DHIs. Valid measurements in multiple languages are needed to measure the working alliance with DHIs across countries. This study had two aims: (1) to translate the Digital Working Alliance Inventory (DWAI) from English into German using both conventional and AI-supported translation procedures, evaluate the different translation outputs, and establish a final German translation; and (2) to validate the German DWAI (DWAI-G) for measuring working alliance in unguided DHIs.

Background

DHIs have become increasingly important in digital public health due to their flexibility, adaptability, and accessibility for different target groups. These interventions can be categorized as guided (with human support via chat, email, or video) or unguided (fully automated without human involvement). Unguided interventions offer advantages including immediate accessibility, scalability, and lower costs, and have shown effectiveness across various health domains. A possible key factor contributing to the effectiveness of DHIs is the working alliance which reflects the collaborative relationship between users and digital tools. Studies have shown positive associations between the working alliance and favorable health outcomes in unguided formats, though existing alliance measures adapted from face-to-face contexts may not fully capture the unique characteristics of unguided DHIs. To address this gap, Henson et al. (2019) developed the DWAI, a six-item measure specifically designed for unguided DHIs, covering the core domains of the working alliance: goals, tasks, and bond. The DWAI has demonstrated strong psychometric properties in English-speaking populations, with excellent internal consistency (Goldberg et al. 2022). Since there was no German version so far, its use in German-speaking populations was limited and a German translation needed. The emergence of artificial intelligence (AI) technologies, particularly Large Language Models, presents new opportunities for translation procedures that may be more time- or resource-efficient. However, studies evaluating AI-supported translation versus conventional translation procedures and quality are limited.

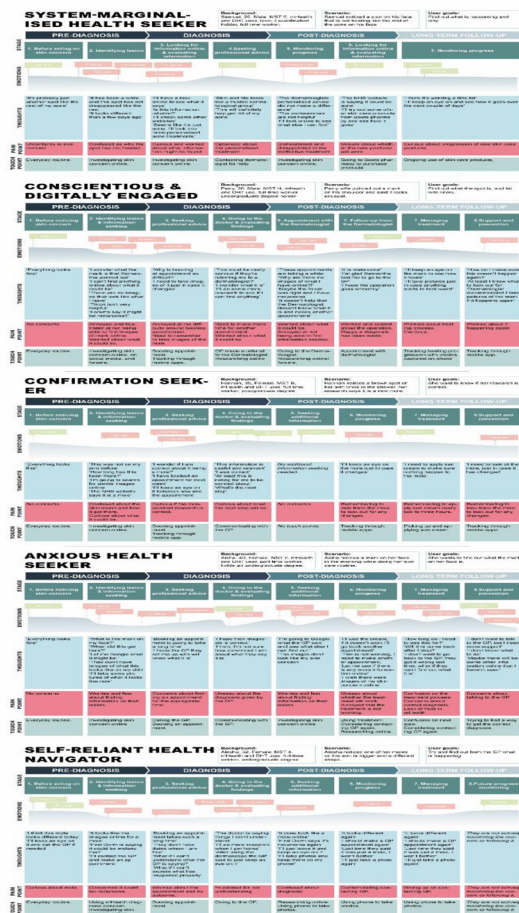


FIGURE 1
User journey maps of personas.

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Methods

The study follows the COSMIN guidelines for patient-reported outcome measures. Four German versions were generated using different translation procedures: (A) Human-Forward-Backward Translation involving independent forward translation by two translators, a synthesis meeting, backward translation by two different translators, and final comparison with the original; (B) AI-Supported Forward Translation; (C) AI-Supported Forward-Backward Translation; and (D) AI-Supported TRAPD Approach (Translation, Review, Adjudication, Pretesting, Documentation), integrating both AI-supported and human translations with comprehensive review, adjudication meetings, pretesting, and thorough documentation. Seven stakeholders evaluated all four translations in a meeting. Each item was rated for quality on a 1 to 10 scale with higher scores indicating better quality. Following ratings, the highest-quality items were discussed and finalized. For validation, data are collected from two ongoing three-arm randomized controlled trials (DRKS00035541, DRKS00035596) involving cancer patients with fatigue or sleep problems. Participants are engaging in a 6-week unguided eLearning course to learn how to perform acupressure or a mindfulness-based technique. The DWAI-G was implemented after the completion of the first module in week 1, with retest in week 2. Psychometric properties are evaluated, including structural validity, internal consistency, test-retest reliability, and construct validity testing through preregistered hypotheses about associations with other constructs (as predicted: #239772).

Results

Four different translations were obtained. The AI-supported TRAPD approach was rated as highest in terms of quality (median = 9, IQR = 4), compared to human-forward-backward (median = 8, IQR = 3.25), AI-forward (median = 7, IQR = 3), and AI-forward-backward (median = 7, IQR = 3) approaches. Six of eight items from the AI-supported TRAPD approach were included in the final German DWAI, along with two from the human-only version and one item from the AI-supported forward-backward version. Notably, AI-supported translations required substantially less time: 0.5 to 1.5 hours for AI-only translation versus 7.3 hours for human-only translation. The most time-intensive AI-supported TRAPD approach needed 13.3 hours, which included in sum 2.5 hours of pretesting.

Conclusions

Our findings concerning the translation of the DWAI showed that AI-supported translation can generate high-quality translations while saving time and resources. Moreover, the results underscored the feasibility and value of AI-supported translation, particularly within an AI-supported TRAPD approach. From our perspective, AI can complement translation procedures of standardized measures efficiently. Human involvement in review and pretesting remains essential to ensure validity and quality. Future work should explore standardized workflows for AI-supported translation, such as optimal prompting, quality criteria of AI-generated translations, and documentation practices, to enhance transparency.

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Understanding participants' skin health journey through a diversity lens: A patient-led study

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Citation

Khatun, N., Spinelli, G., Colecchia, F. Understanding participants' skin health journey through a diversity lens: A patient-led study.

Aim

The study investigates the perception of skin health and the diverse engagement and use of digital health. It identifies pain points, behaviours, and challenges experienced during patient skin health journeys.

Abstract

People of colour often experience difficulties surrounding skin health diagnosis and awareness. Understanding how individuals approach skin health, the tools used to support them, and the barriers faced can help create a patient-centred public health intervention. The user personas and user journey maps have displayed different behaviours linked to pain-points surrounding difficulties with booking appointments and the need to conduct research before and/or after healthcare appointments due to the lack of applicable information, reinforcing the need for inclusive and accessible public health interventions in skin health that respond to diverse user experiences and needs.

Background

Globally, 33% of people conduct self-checks annually, and 11% of these individuals seek professional dermatological checks despite the current recommendation being monthly (La Roche-Posay, 2015). As a result, several cases of skin concerns, such as melanoma, go undiagnosed every year. For people of colour (POC), skin health disparities remain a persistent issue, stemming from multiple factors, including limited awareness and insufficient education across all stakeholders (Croce et al., 2021; Robinson et al., 2011). Barriers affecting interactions between healthcare professionals (HCP) and POC further contribute to delayed diagnoses.

Method

An online survey was conducted to explore participants' skin health journey, their use of mobile health applications (mHealth) and digital health tools (DHT). It was created on JISC, shared online for two months, and participants were recruited via the British Skin Foundation social media platforms. Inclusion criteria: categorising into a Monk Skin Tone (MST) group, 18+, experienced a skin concern, and familiarity with using digital technologies.

Results

A total of 99/100 (n=99) partook. User personas and user journey maps (UJM) (Figure 1), extracted from the survey results, have been synthesised to generate actionable user-centred insights that identify potential intervention areas where health concerns, underlying emotions, and behavioural motivations are understood (Kwon et al., 2025).

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Below are five personas that best illustrate the main contributions. Figure 1 contains background descriptions and aims for each persona.

1. Anxious health-seeker (Aisha): They are heightened by anxiety and fear around the limited resources for their skin concerns, resulting in barriers to booking appointments and delaying access to HCPs.
2. Confirmation seeker (Hannah): This user uses multiple approaches to learn about and monitor their skin. They seek information prior to appointments

and often arrive at conclusions they wish HCP to validate. This stems from trying to reduce anxiety around concerns or wanting health empowerment and autonomy.

3. Conscientious and digitally engaged (Perry): They manage their skin health by tracking concerns, using mHealth, using DHT, and self-education. Their proactive approach enables them to recognise skin changes early, which can later be escalated if needed. Despite occasional negative experiences with HCP, they trust their advice.
4. System-marginalised health-seeker (Samuel): They experience barriers when attempting to engage with their skin health; these include accessing the right treatment, understanding information, or receiving irrelevant information, and sometimes they are left with no choice but to ignore the skin concern.
5. Self-reliant health navigator (Alesha): They conduct all their research independently using DHT; however, they find the information is too vague. They do not contact clinical professionals due to negative experiences, which have eroded their trust in HCPs.

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User journey maps, as shown in Figure 1, have been curated from the survey results and personas.

Synthesising user personas and UJM reveals barriers to booking appointments. Negative emotions surrounding the booking process suggest that dissatisfaction or distress can reduce individuals' motivation to act on their concerns, leading to delays or no bookings, as seen in Aisha and Alesha's personas, which evoked anxiety, concern, or caution around this process. Furthermore, it can be a contributing factor in patients' failure to attend appointments due to negative emotions (Lacy et al., 2004). Persons categorised within MST 3-10 are at an elevated risk of delayed diagnosis due to barriers they experience, according to the online survey. Harrison (2023) emphasises the safety issues surrounding patients with dark skin

tones due to HCPs lack of confidence in treating all ethnicities equally. Despite experiences, individuals want to trust HCPs; however, repeated poor appointment outcomes undermine this trust.

Conducting research before and/ or after appointments with HCPs are presented across several UJM; this could be to prepare, gather information, or reduce uncertainty (Kyriacou et al., 2019; Caiata-Zufferey et al., 2010) surrounding appointments. These behaviours, observed in Aisha and Hannah's personas, are widely reported among females and MST 5-10 in the online survey and can influence how patients perceive the quality of care and their appointment outcomes, periodically leading to discontentment when expectations are unmet (Li et al., 2014). Hannah implied that an appointment is satisfactory when outcomes align with independent research, illustrating how patient research behaviours impact healthcare experiences and trust.

Several personas demonstrated widespread acceptance and use of mHealth and DHT. While these tools provide support for patient's or a connection between patient's and HCP, Perry and Alesha, highlight the barriers, 'some of this app is not made all to your taste...', 'hard to tell what skin condition it might be' and, 'wasn't clear as I wasn't sure of the situation I was dealing with but it helped a little'. Patients matching these personas comprise of MST 5-10, indicating a lack of inclusivity and representation. There are prominent concerns surrounding data privacy and security among both individuals who have adopted mHealth and those who have not; these struggles are also discussed by Shojaei et al. (2025) and Zhou et al. (2019). A key focus on transparency in data use is needed for the future development of mHealth. Esmaeilzadeh (2019) found that perceived transparency builds trust, makes patients feel in control, and reduces risk perceptions.

Conclusion

This work contributes a user-centred perspective on how a variety of patients navigate skin health, mHealth, and DHT. If patient satisfaction and negative emotions can be managed through inclusive resources, existing datasets should be enhanced with high-quality images and validated metadata of skin

conditions, especially melanoma, and of POC, to support the development of future bias-free technology. However, the path to creating these datasets remains unclear, given the fragmented care experienced by POC in healthcare systems and their concerns about equitable treatment.

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Who deploys it and how: Organisational context and public trust in health AI

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Citation

Boyle, B. Who deploys it and how: Organisational context and public trust in health AI.

Aims

This study examines how organisational context shapes public trust in, and adoption intention toward, AI-enabled healthcare tools. It suggests that regulatory approaches alone may be insufficient to build confidence in health AI, particularly in a post-COVID context where institutional credibility has been strained, and that the grammar of how AI is presented in public health contexts may itself shape how it is understood, trusted, and engaged with.

Background

The rapid integration of AI into healthcare has prompted significant regulatory activity globally. The EU AI Act is now being implemented, with many provisions applying from August 2026 (European Union, 2024), and the European Health Data Space entered into force in March 2025 (European Union, 2025), creating new infrastructure for data-driven health innovation across the EU. Most recently, Brodeur et al. (2026) reported that a large language model outperformed physicians on diagnostic reasoning tasks in real-world emergency department conditions, underscoring that questions

of public trust and organisational context are becoming more urgent as AI clinical capability advances rapidly. WHO/Europe warned in November 2025 that AI in health is advancing without sufficiently developed legal and ethical safeguards across much of the region, and called for stronger public engagement alongside governance frameworks (World Health Organization Regional Office for Europe, 2025).

In this context, public trust in health AI remains fragile. A January 2026 national survey by the Coalition for Health AI, conducted by NORC at the University of Chicago, found that while 75% of respondents reported using AI in healthcare contexts, only 13% felt very comfortable with it, and 51% reported that AI makes them trust healthcare less (Coalition for Health AI, 2026). A 2025 study in JAMA Network Open found that patients with low system trust report substantially lower expectations of healthcare AI (Nong and Platt, 2025), and a large multinational study also in JAMA Network Open, surveying over 13,000 patients across 43 countries, found that attitudes varied significantly by demographic characteristics and that patients consistently preferred explainable AI and physician-led decision-making (Busch et al., 2025).

These developments suggest that regulation, while necessary, may not alone secure public confidence. Van Kolfschooten (2026) argues that the EU AI Act rests on a 'no trust, no use' logic, yet compliance may be insufficient to build trust, particularly where medical exceptions risk normalising practices the Act otherwise prohibits. Less attention has been paid to how the communicative and organisational context of AI deployment shapes public understanding. Drawing on consumer behaviour theory and the Technology Acceptance Model (Davis, 1989), this study examines whether the institutional package in which AI is presented, including organisational type, clinical oversight, regulatory status, and data practices, shapes trust and adoption intention.

Methods

An anonymous online survey (N = 66) was conducted with a convenience sample of adults recruited across Europe. Participants were presented with

two contrasting scenarios. Scenario A depicted an AI tool developed by a public hospital, operating under clinical guidelines, reviewed by medical professionals, and not sharing data with third parties. Scenario B depicted an AI health app developed by a private technology company, free to use but collecting data for marketing and partnership purposes, and explicitly not subject to public health regulation. The scenarios therefore differed across organisational type, clinical oversight, regulatory status, and data practices simultaneously. After each scenario, participants rated their perceptions on Likert scales across trust, perceived risk, adoption intention, and transparency, using validated instruments including the AIAS-4 (Grassini, 2023) and GAAIS (Schepman and Rodway, 2020). Demographic variables were also collected. Between-condition differences were assessed using paired and independent samples t-tests, and a sequential regression model examined predictors of adoption intention.

Results

Organisational context and associated governance cues were strongly associated with differences in trust, adoption intention, and concern (all $p < .001$). Participants in this sample rated the public hospital scenario substantially higher on trust (+1.81) and adoption intention (+1.18), and reported lower concern (+1.06), compared to the private commercial scenario. Trust was the dominant predictor of adoption intention in this dataset, accounting for 67% of variance and remaining the only significant predictor in regression models that included demographics and prior app use. Demographic variables were not major predictors in this sample, suggesting organisational context may shape responses to health AI more powerfully than individual characteristics, though some nuance was observed for gender and trust. An exploratory set of health consumer identity items performed poorly on reliability testing and was excluded from inference.

Conclusions

These indicative findings suggest that the institutional package in which health AI is presented, encompassing organisational type, clinical oversight, data practices, and regulatory status, appears to be a major influence on

public trust, showing a stronger association with responses than individual-level demographic and identity factors in this sample. Given the small convenience sample and the use of simplified experimental scenarios, these findings should be interpreted as indicative rather than generalisable, and as a basis for further research into how organisational and communicative cues shape public trust in health AI.

If trust is shaped not only by a technology's properties, but also by the organisational context in which it is introduced, this suggests a potential limit to regulation as a trust-building mechanism. Regulation can shape systems, standards, and accountability, but may not substitute for the communicative work through which trust develops at the level of lived experience.

These findings carry tentative implications at a critical regulatory moment. As the EU AI Act moves further into implementation, and international bodies stress that trust must be actively built, the gap between compliance and public confidence deserves closer scrutiny. Building trust in health AI may require attention to how tools are framed, by whom, and within what context, with visible human oversight, accessible communication, and engagement with public concerns.

This study forms part of ongoing dissertation research examining the communicative and market dynamics shaping public engagement with AI in public health.

Disclosure

A generative AI tool was used to assist with formatting and spelling/grammar checking when finalising this abstract text: specifically, Claude (Anthropic, claude-sonnet-4-6, 2026). All content, arguments, and conclusions are the author's own.

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Workshop Track

Human-Centred design for digital public health in LMICs: A persona-based workshop

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Kelly, L. Human-Centred design for digital public health in LMICs: A persona-based workshop.

Aims / Overview

Digital public health systems increasingly generate structured data through mobile tools, community health platforms, and primary care workflows. These systems capture symptoms, visit attendance, medication adherence, referral completion, and basic clinical indicators during routine care. While such data enable standardized protocol-driven responses, they often produce fragmented outputs in real-world settings. Frontline health workers, particularly in low- and middle-income country (LMIC) contexts, may encounter multiple independent alerts without clear guidance on how signals relate to one another or which issue should be prioritized.

This creates a gap between data capture and interpretable care guidance. A patient may be represented by missed appointments, incomplete referrals, declining adherence, and clinical risk indicators, yet these appear as separate signals rather than a coherent representation of the care situation. In practice, the challenge is not simply identifying individual conditions but understanding what type of care state the patient currently occupies and how to act accordingly within time and resource constraints. While existing systems primarily respond to discrete problems, care-state personas are designed to manage evolving care trajectories by supporting prioritization over time.

Background

Persona-based approaches in healthcare are widely used to support user-centered design, co-design processes, and implementation of digital health interventions. These approaches synthesize demographic, behavioral, and contextual characteristics to represent archetypal users, but are typically constructed as relatively stable representations and are not designed to reflect dynamically evolving care conditions or temporal changes in clinical and engagement status [1], [2].

At the same time, clinical decision support systems have highlighted the limitations of rule-based alerts, including fragmentation of information and the phenomenon of alert fatigue when multiple signals are presented independently without synthesis [3], [4].

Despite these parallel developments, there has been limited integration between persona-based design approaches and the structured digital health signals generated within care pathways. Existing methods either emphasize contextual understanding without dynamic care representation, or provide signal-level decision support without synthesizing signals into interpretable care situations.

Methods

This paper proposes care-state personas as a framework for interpreting dynamic care conditions derived from digital health signals. Care-state personas extend traditional persona methodology by representing the care situations individuals occupy as signals accumulate over time. Rather than describing who users are, care-state personas describe the evolving state of care within a clinical or public health pathway.

Care-state personas are constructed through the aggregation and interpretation of structured signals over time. Inputs may include visit attendance, referral completion, treatment adherence, symptom reports, and basic clinical measurements. These signals are organized into temporally relevant patterns and evaluated within an interpretation layer that identifies clinically meaningful configurations. Interpretation may be rule-based, pattern-based, or threshold-driven depending on system design and data availability.

Each care-state persona activates a small number of predefined intervention layers - such as restoring engagement, stabilizing basic physiology, or resolving care pathway gaps - that translate interpreted patterns into concrete, prioritized actions within existing system constraints.

Results

An illustrative example demonstrates the practical application of this framework. A child under five presents with recent diarrheal illness, caregiver-reported reduced appetite, slowed weight gain, and a missed vaccination visit. In conventional systems, these signals generate parallel alerts linked to different protocols.

Within a care-state framework, these signals are interpreted collectively as a pattern indicating early nutritional vulnerability. The child is therefore represented as the care-state persona *Emerging Nutrition Risk*. This representation enables a prioritized sequence of actions: first, assessment of nutritional status and feeding practices to prevent progression; second, management of recent illness; and third, completion of routine preventive care.

By synthesizing signals into a coherent care state, the system reduces fragmentation and supports more focused clinical decision-making.

Conclusions

The care-state persona framework can be understood as a four-layer interpretive architecture: (1) Signal Layer, consisting of structured digital inputs; (2) Interpretation Layer, where temporally related signals are aggregated into clinically meaningful patterns; (3) State Representation Layer, where patterns are translated into care-state personas; and (4) Action Layer, where personas are mapped to a small set of predefined, executable intervention layers that define prioritized actions.

This structure is designed to operate without additional data collection or new workflow requirements, enabling integration into existing digital and community health systems. The relevance of care-state personas is particularly pronounced in LMIC settings, where community health workers and mid-level providers manage complex care pathways with limited decision support [5].

Beyond individual encounters, aggregated care-state patterns may provide insight into system-level performance, identifying structural constraints such as missed follow-up or referral gaps. Care-state personas are intended as interpretive design tools rather than diagnostic categories, and their effectiveness depends on data quality and adaptation to specific contexts.

Preliminary modeling suggests that care-state personas may be applicable to a substantial proportion of routine cases, particularly those characterized by early instability, disengagement, or fragmented care, where existing approaches often lack clear prioritization.

By extending persona methodology to incorporate dynamic care conditions, care-state personas provide a practical framework for improving the interpretability and usability of digital public health signals in resource-constrained environments.

Conflict of Interest

LEK is Chief Executive Officer of Opal Health and holds equity in the company. This work was conducted independently and was not funded by Opal Health. The author reports no other conflicts of interest.

Author Contributions

All work done by LK.

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Second international workshop on Architecture-based IoT and Artificial Intelligence to support public health

B-COMPASS: An Architecture- Ready Explainable ML Framework for Segment-Tailored Treatment Adherence Support in Digital Public Health

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Overview

Improving treatment adherence is a persistent public health challenge. Yet adherence support is hindered by fragmented data. We present an architecture-ready, explainable ML tool that predicts non-adherence tendency from a compact set of psychological determinants plus age, grounded in the Subjective Health Experience (SHE) framework, which classifies patients into four segments [1]. The tool produces adherence risk profiles supporting tailored intervention across patient groups. Interoperability is enabled through alignment with the Observational Medical Outcomes Partnership Common Data Model (OMOP CDM), using a dedicated behavioural vocabulary and Extract, Transform, Load (ETL) workflow that maps questionnaire data into a digital platform compatible with real-world data (RWD) integration.

Background

Poor adherence has been characterised as a worldwide challenge with major clinical and economic consequences, particularly for chronic conditions [2]. However, adherence is frequently treated as uniform, despite evidence of meaningful heterogeneity. The SHE framework captures this using disease acceptance and perceived control, classifying individuals into four segments with distinct support needs. A persistent barrier is data fragmentation: behavioural determinants captured via surveys remain difficult to link with clinical data in reproducible, interoperable ways. OMOP CDM is a widely

adopted standard to harmonise observational health data [3], but requires structured extensions that preserve questionnaire semantics because adherence-related behavioural constructs are often underrepresented.

Methods

Design & data

We analysed a cross-sectional online survey (Jan–Apr 2024) of adults across six countries ($N \approx 4,081$) with varied chronic conditions. The outcome variable, perceived adherence, was measured using the Perceived Behaviour Scale from the Treatment Adherence Perception Questionnaire (TAPQ) [4], serving as the reference standard. Predictor variables comprised psychological determinants (health consciousness, health priority, treatment concerns, acceptance, perceived control) and age [5–7], selected through a theory-informed forward scheme balancing predictive performance and respondent burden.

Modelling strategy

We compared (i) regression on the continuous TAPQ adherence score with (ii) binary classification of adherence tendency. For classification, the continuous score was dichotomised at the sample median (score 28), yielding non-adherent (≥ 28) versus adherent (< 28) labels; additionally, SHE segments were used as a stratification context. Tested algorithms included regularised linear approaches and tree-based methods; model selection relied on five-fold cross-validation, applied to both training and held-out test splits, using R^2 and ROC AUC as primary metrics. The classification decision threshold was tuned to maximise F1 to optimise the precision-recall balance.

Explainability

We prioritised models with interpretable parameters to derive human-readable decision rules for targeted support within patient segments.

Architecture & interoperability

Behavioural questionnaire constructs were mapped to the OMOP CDM Observation domain via a structured ETL pipeline and a dedicated behavioural vocabulary preserving question-answer context. This constitutes a design capability enabling future linkage with observational datasets; clinical data linkage has not been performed in this study. The design provides a pathway to incorporate RWD as additional multimodal features within the same pipeline.

Results

Prediction framing

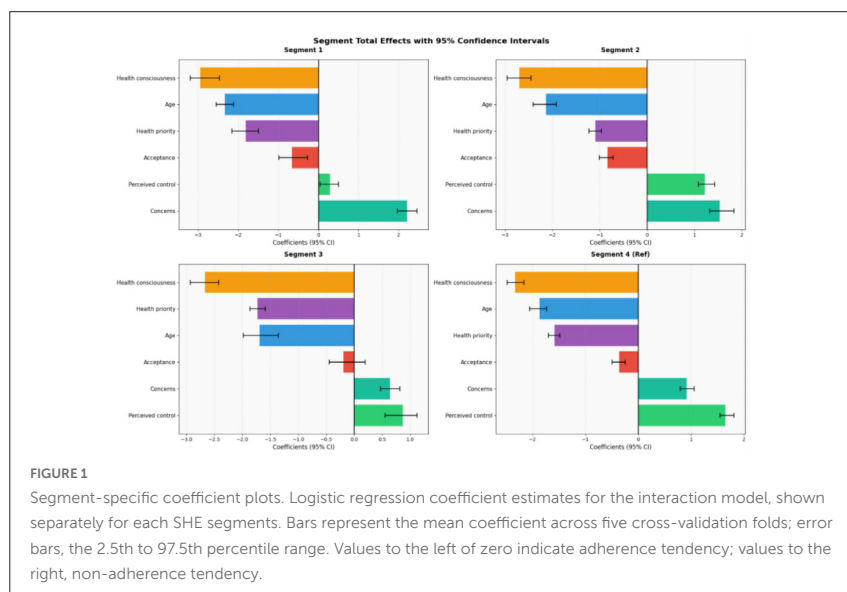
Regression of the continuous adherence score explained limited variance ($R^2 \approx 21\%$), reflecting the psychometric reality of predicting exact positions on a multi-item self-report scale, which is inherently sensitive to measurement noise. Binary classification of adherence tendency yielded stronger discrimination and greater actionability for triage and segment-guided support.

Performance vs burden

From the different variable sets evaluated, a compact feature set (psychological determinants + age) with logistic regression achieved ROC AUC ≈ 0.69 -0.72 across countries, consistent with acceptable discrimination for population (level triage and risk ranking, explicitly distinct from clinical diagnosis), reduced burden for patients and performed comparably to or better than more complex models such as random forest and XGBoost, while retaining superior transparency for operational decision-making.

Explainable, segment-aware insights

Coefficient patterns and segment interactions were stable and interpretable. Protective associations against non-adherence included acceptance, age, health priority, and health consciousness; treatment concerns and perceived



control showed positive association with non-adherence (Figure 1). These segment-aware adherence risk profiles constitute the empirical base for the B-COMPASS adherence groups, directly informing the provision of tailored recommendations and support from healthcare professionals.

Deployment readiness

The model is designed as a modular service embeddable in architecture-based pipelines where behavioural inputs arrive via surveys or apps and can be enriched through OMOP CDM-standardised linkage to clinical variables.

Ongoing Validation

Pilots are currently being conducted across 12 sites in 8 countries, involving patients under active treatment or health advice, spanning multiple disease

areas. These pilots are designed as randomised controlled trials and will provide external validation of the model, testing whether B-COMPASS segment-tailored recommendations translate into measurable improvements in adherence.

Conclusions

We present an explainable, segment-aware adherence prediction tool that balances discrimination and respondent burden while producing actionable, human-readable rules for tailored support across patient groups. Coupling interpretable ML with OMOP CDM-aligned interoperability illustrates how AI services and architecture-based standardisation can strengthen digital public health infrastructures.

Code availability

The model/code are publicly available at <https://github.com/BEAMERproject/B-COMPASS-model.git>.

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AI disclosure

The preparation of this manuscript was assisted by generative AI technologies. Microsoft 365 Copilot (Microsoft) was used during earlier drafting stages. Claude (Opus 4.6, Anthropic) was used to apply author comments and revisions to the document text. All substantive decisions regarding content, analysis, and interpretation were made by the authors.

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User-centred design of a remote virtual reality-supported physical activity program for older adults with chronic conditions

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Background

Regular physical activity is associated with reduced risk of non-communicable diseases, improved functional capacity, and better quality of life among older adults [1]. Despite these benefits, adherence to exercise programmes remains low due to barriers such as limited mobility, lack of access to supervised facilities, and reduced motivation [2]. Digital health technologies have emerged as promising tools to address these challenges by enabling remote monitoring and personalised interventions [3]. Virtual reality applications, in particular, provide immersive and gamified environments that can enhance motivation and social engagement during exercise [4]. However, the adoption of VR technologies among older adults is often constrained by usability issues, technological literacy gaps, and concerns regarding safety during unsupervised exercise at home. A user-centred design approach is therefore essential to ensure that digital rehabilitation systems respond to the needs, preferences, and capabilities of the target population [5]. “3D Community Aware Virtual Spaces as Smart Living Environments for Physical Activity and Rehabilitation” (VR2CARE) project aimed to develop and test an integrated platform for the promotion and monitoring of physical activity, based on VR solutions that enable immersive multi-user experiences for older adults [6].

Aims / Overview

This study explores the design and integration in service provision of Federico II University Hospital (F2UH) of a virtual reality (VR)–supported intervention aimed at promoting adapted physical activity (APA) among older adults with chronic conditions in home settings. The objective was to develop a user-centred service pathway that integrates VR technologies with tele-monitoring to enhance adherence, safety, and long-term engagement in physical activity programs.

Methods

A qualitative design methodology was adopted to define the service pathway for implementing a VR-supported APA programme within F2UH’s outpatient clinic for the prescription of APA to chronic disease patients.

A multidisciplinary focus group composed of clinicians, physical trainers, public health experts, and information technology specialists participated in a co-design process aimed at identifying clinical, usability, and engagement requirements for the intervention. Through a series of structured sessions, participants analysed the existing care pathway for physical activity prescription and identified opportunities for integrating VR-supported intervention. A user-centred design framework was applied, combining service pathway analysis with persona development to model the needs and experiences of representative end-users. The persona-based methodology enabled the creation of a realistic patient scenario that captures clinical characteristics, behavioural factors, social context, and digital literacy aspects relevant to the adoption of VR-supported physical activity program.

Results

The co-design process resulted in the development of an integrated service model for delivering adapted physical activity through a VR-enabled digital ecosystem. The proposed workflow begins with a clinical assessment conducted in the hospital setting to evaluate patient eligibility and define an individualised exercise prescription. Once enrolled, patients receive training on the setup and use of the home-based VR system, including environmental preparation and device operation. The core component of the intervention consists of a remote APA loop in which patients perform prescribed exercises within an immersive virtual environment. Motion tracking and monitoring technologies allow real-time collection of movement and clinical data. These data are transmitted to a clinical dashboard accessible to healthcare professionals and trainers, enabling continuous supervision and adjustments to exercise intensity. The persona scenario developed during the design phase illustrates a typical patient journey from hospital discharge to home-based rehabilitation. Within the virtual environment, patients interact through avatars and participate in structured exercise sessions that combine physical training with gamified tasks. The system also incorporates family support and community involvement to facilitate technology adoption and sustain long-term engagement. This integrated approach aims to bridge the

gap between hospital-based prescription and home-based physical activity by combining immersive technologies, telemonitoring tools, and coordinated professional supervision.

Conclusions

The proposed VR-supported model demonstrates the potential of immersive technologies to enhance the delivery of APA programmes for older adults with chronic conditions. By integrating clinical supervision, remote monitoring, and user-friendly interfaces, the system addresses key barriers related to safety, adherence, and technological accessibility. The user-centred design methodology proved valuable in identifying the functional requirements and service interactions necessary for implementing digital rehabilitation in real-world healthcare contexts. The resulting workflow highlights the importance of combining clinical expertise, technological innovation, and social support mechanisms to ensure sustainable adoption. Although the present study focuses on the design and conceptual validation of the service model, further research is required to evaluate its clinical effectiveness, economic sustainability, and scalability within healthcare systems, including LMICs.

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User-centric design in digital public health workshop: Challenges and opportunities in an Era of disruptive innovation

Tailored goals and individual choice: a field experiment on automated goal setting for individual exercise behavior

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Caro, J.C., Escolano, E., Arrigoni, F., Lipman, S. User-centric design in digital public health workshop: Challenges and opportunities in an Era of disruptive innovation.

Abstract

Physical inactivity is a major preventable cause of non-communicable disease and premature mortality. Mobile health interventions can enhance physical activity adherence via tailoring, but their effectiveness depends on the ability to adapt to user's preferences, agency and context. This study focuses on the role of individual choice in automated goal difficulty settings, designed to increase physical activity among university students using a mobile web app called *Apptivate* (<https://goals.apptivate.cl/>). The app uses fixed and time-variant characteristics to train a reinforcement learning model which predicts routine difficulty to enhance adherence over time. The primary aims are to determine: (1) whether reinforcement learning-based automated goal setting improves physical activity adherence, (2) how individual characteristics influence optimal workout schedules, and (3) how user autonomy (choice) impacts their engagement with the app. The intervention randomly assigns participants to one of three groups: (1) user choice without recommendation (choice-driven), (2) user choice with automated recommendation (data-recommendation), and (3) fully automated goal assignment (data-drive). The intervention is conducted simultaneously in three Universities for a period of six weeks. Primary outcomes are adjusted mean differences in activity metrics between intervention groups over time. Subgroup analyses will explore whether specific demographic or motivational profiles are associated with differential responses to automated recommendations.

Background

Physical inactivity is a leading risk factor for chronic disease and premature mortality globally, despite extensive public health efforts to increase engagement in regular exercise (Ding et al., 2016). Traditional mobile health interventions often fail to maintain sustained adherence because they rely on static tailoring based solely on baseline characteristics, overlooking key elements of user-centric design, the role of dynamic changes in user context and beliefs over time (Nahum-Shani et al., 2018; Tewari & Murphy, 2017). Adaptive interventions employing contextual bandit models (a type of reinforcement learning approach) permit them to continuously incorporate individual performance and contextual data to refine personalized

recommendations (Esposito & Sautmann, 2022; Mothilal et al., 2019). To provide theoretical grounds to the learning process, such interventions should also use a behavioral framework grounded on psychological theory (Caro et al., 2026). The Integrated Behavior Change (IBC) model addresses user-specific motivational and self-regulatory constructs alongside dynamic contextual factors in physical activity adherence (Hagger & Chatzisarantis, 2014; Hagger & Hamilton, 2020). This study proposes an adaptive intervention using contextual bandits, to test two hypotheses: (1) automated goal personalization decreases the intention-behavior gap for some users, and (2) the effectiveness of goal setting automation on behavior is mediated through individual choice.

Methods

The target population comprises undergraduate and graduate university students from three Universities, facilitating variability in region, language, and time zone. Eligibility criteria require participants to be at least 18 years old, physically healthy, and possess access to a compatible mobile device; individuals with significant exercise-limiting medical conditions or competitive athletes are excluded. Randomization to the three intervention groups is stratified by baseline demographic factors (household composition, income, age and gender) to ensure balanced allocation. For the main study, 330 participants were recruited to complete 10 rounds of three days each with a rest day in between, equivalent to 39 consecutive days using the app. Users will be allocated to each difficulty setting based on their baseline survey and the trained algorithm during the pilot stage. Regardless of their initial difficulty assignment, each participant will be randomly allocated with equal probability between three intervention groups (1) choice-driven, (2) data-recommended, and (3) data-driven. Participants will be compensated for their time, and there will be additional prizes using a lottery mechanism that favors (in probability) those users completing more consecutive workout routines over the course of the study. Participants stay engaged via direct email, push notifications, and social media campaign.

Appivate assigns users among three predefined workout difficulty levels that vary in exercise duration and rest intervals (Figure 1). Workouts are performed

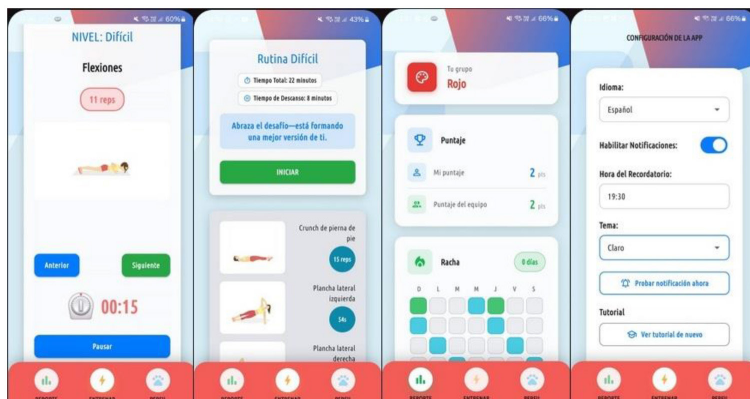


FIGURE 1

AppTivate mobile app main views. From left to right: (a) exercise view, (b) routine view, (c) report view, (d) profile view.

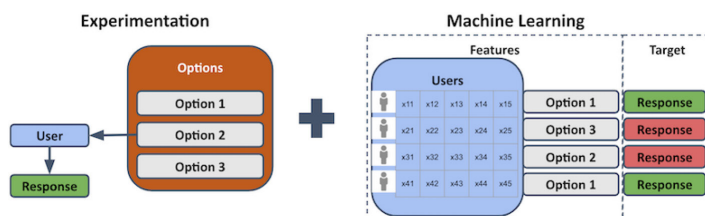


FIGURE 2

Adaptive experiment design (reinforcement learning model).

without equipment and include warm-ups, strengthening, and stretching exercises. Baseline survey captures demographic data and psychological constructs from the IBC model. Activity outcomes and app engagement metrics—goal completion rates, feedback and other interactions—are collected throughout the intervention period as input for the reinforcement learning model, which in turn provides recommended difficulty settings for the next 3-day round (Figure 2).

After the study is completed, pairwise contrasts estimate adjusted mean differences in activity outcomes such as workout completion rates and daily time engaged in exercise. To examine heterogeneity, cross-level interaction terms assess whether baseline motivational factors, demographics, and app feedback moderate the relationship between intervention conditions and activity outcomes. Statistical power was determined via simulation for adaptive experimental designs, to detect small-to-moderate differences between groups.

Results

The pilot study (n=558) was conducted in September 2025 to generate initial algorithm training data and app usability feedback (including focus groups with participants); data analysis is underway. The main study started in May 2026 with 330 participants participating consecutively for 39 days (including rest days). Primary outcomes are adjusted mean differences in activity metrics, effect size estimates, and interaction coefficients reflecting time-dependent changes. Subgroup analyses will explore whether specific demographic or motivational profiles are associated with differential responsiveness to autonomy versus automated recommendations. Main results will be available by June 2026.

Conclusions

This adaptive intervention rigorously evaluates the role of automated personalization and user autonomy within a mobile app to promote sustained physical activity, representing a key issue for user-centric design in digital health. By comparing autonomous user choice, algorithm-recommended choice, and fully data-driven assignments, the study quantifies how user-specific design features influence engagement and goal attainment. The integration of reinforcement learning with an explicit behavior change framework allows to provide comprehensive evidence on the interplay between user autonomy and algorithmic tailoring. Findings are expected to inform the optimization of user-centric digital health interventions that balance machine learning personalization with autonomy-supportive design to enhance adherence and public health impact.

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